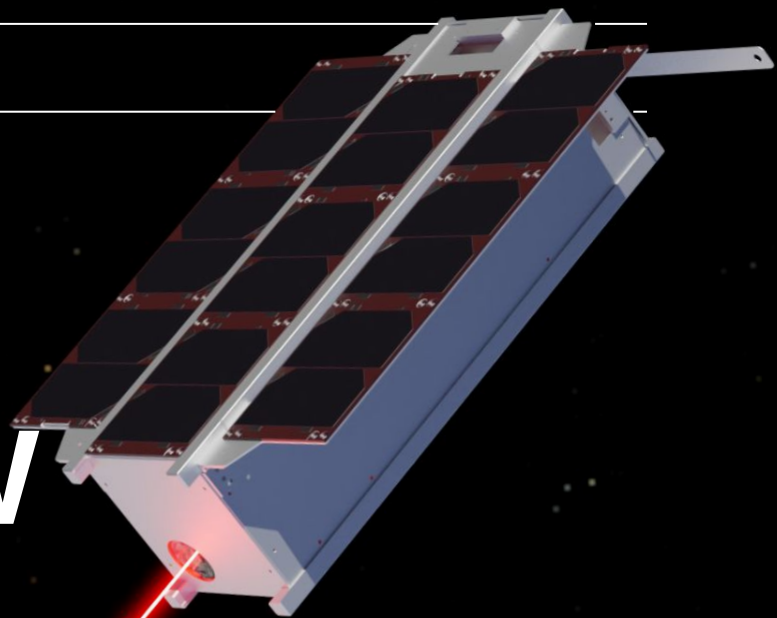

UCHICAGO SPACE PROGRAM

PULSE-A

INTERMEDIATE DESIGN REVIEW

OPTICAL SYSTEM

NOVEMBER 22 2025



Polarization-Modulated Laser Satellite Experiment

PULSE-A

Intermediate Design Review

Introduction

Presenter Introductions

Juan Ignacio Prieto Asbun – Project Director
and Lead Optical Ground Station Engineer

Robert Pitu –Lead Engineer

Sofia Mansilla – Lead Payload Engineer

Zumi Riekse – Lead Avionics Hardware Engineer

Henry Czarnecki – Lead Structures Engineer

Zane Ebel – Lead Thermal Engineer

Catherine Todd – Lead Flight Software Engineer

Vidya Suri – Lead Systems Engineer

Mason McCormack – Lead RF Engineer

Seth Knights – UCSP President

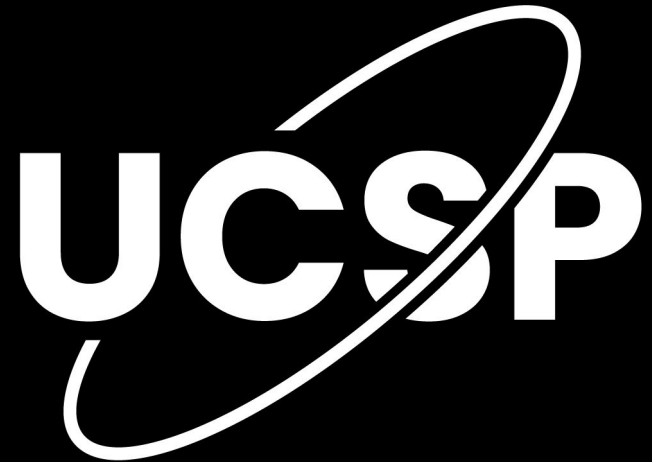
Logan Hanssler – UCSP Vice-President

Graydon Schulze-Kalt – UCSP Vice-President
and Avionics Hardware Member

Daniel Lee – Payload Optical Simulations
Engineer

University of Chicago Space Program (UCSP)

- ❑ The University of Chicago Space Program (UCSP) is a Registered Student Organization at the University of Chicago and a chapter of Students for the Exploration and Development of Space (SEDS)
- ❑ PULSE-A is UCSP's flagship project and the first mission of the organization's CubeSat Laboratory
- ❑ PULSE-A is the University of Chicago's first undergraduate-designed satellite and the sole opportunity for students interested in aerospace engineering to gain hands-on experience



A Brief History of UCSP

As the university's premier aerospace engineering organization, UCSP has grown from a small student group launching low-power rockets near campus to over 200 active members with a 82-student team working to develop PULSE-A.



*Rockets Launched on
Midway Plaisance Park
(circa 2021)*



*Hosted SpaceVision
2022 at the University
of Chicago
(Nov. 2022)*



*Developing PULSE-A
(Development Apr. 2023 – Present,
Team Photo from Oct. 2024)*



*Continuing Higher-Caliber
Rocketry Projects (Photo from May 2023)*

How the PULSE-A Team Got Here

- ❑ The project was founded in April 2023
- ❑ Accepted by Call 15 of NASA's CubeSat Launch Initiative in April 2024 – Defunded in August 2025
- ❑ Completed System Requirements Review on June 2, 2024
- ❑ Completed Preliminary Design Review on November 23, 2024



IDR Overview

IDR Purpose

- ❑ The Intermediate Design Review (IDR) demonstrates that progress on detailed design meets all system requirements with acceptable risk and within the program constraints and provides feedback for completion of detailed design.
- ❑ The IDR provides evidence of progress towards a completed detailed design for PULSE-A. It validates existing design and demonstrates correct choices in the direction of further detailed design.

IDR Entrance Criteria

1. Successful completion of the PDR.
2. A preliminary IDR agenda and agreed upon success criteria by the technical team and project director prior to the IDR.
3. Availability of IDR technical products listed below for both hardware and software system elements to the cognizant participants prior to the review:
 - a. Detailed subsystem design specifications for a variety of items, with supporting validation analyses and data.
 - b. Existing preliminary subsystem design specifications for all remaining items, with supporting analyses and data on design choices and direction of future detailed design.
 - c. Updated risk assessment and mitigations.
 - d. Applicable technical plans.
 - e. Applicable standards.

Not Included in the IDR

- ❑ Detailed cost estimates will be provided separately
- ❑ The following further detailed design performance analyses will be provided by Critical Design Review
 - ❑ Full-system polarization characterization
 - ❑ ISS tracking
 - ❑ Full PAT simulations
 - ❑ CPoISK Demonstration
 - ❑ Optical System Prototyping
 - ❑ Coarse and Fine Pointing Algorithms

Optical Systems Day IDR Agenda

1. Mission Overview
2. Optical Link Analysis
3. Optical Payload
 - 3.1. CPolSK Discussion
 - 3.2. Optical Design
 - 3.3. Structural Design
 - 3.4. Optical Analysis
4. Optical Ground Station
 - 4.1. Optical Design
 - 4.2. Optical Analysis
 - 4.3. Structural Design
 - 4.4. Electronics Design
 - 4.5. Electronics Analysis
 - 4.6. Ground Control
 - 4.6.1. Telescope Control
 - 4.6.2. Tracking Camera
 - 4.6.3. Graphical User Interface
5. Testing & Component Characterization
 - 5.1. Fiber Experiment
 - 5.2. Laser Driving
 - 5.3. QPD Characterization Plan
 - 5.4. Telescope Experiment
6. Pointing, Acquisition and Tracking (PAT)
 - 6.1. Coarse Pointing
 - 6.2. Fine Pointing
 - 6.3. Sequence Rundown
 - 6.4. Telescope Pointing & Tracking
 - 6.5. Telescope Tracking Analysis
 - 6.6. Sequence Simulations
7. Assembly, Integration, and Test (AIT)
8. Risk Assessment and Mitigation
9. Conclusion

Mission Overview

Principal Mission Objective

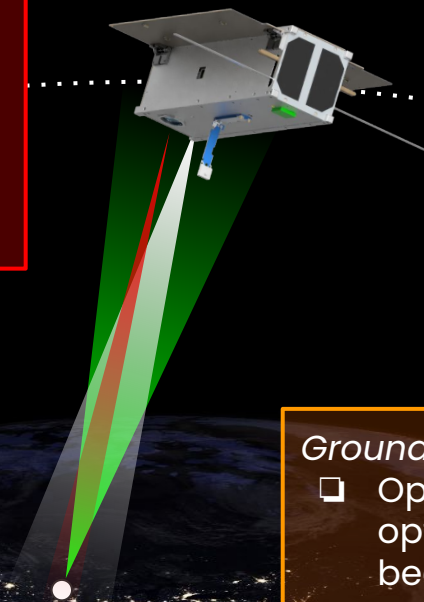
The mission shall execute Satellite-to-ground data transmission using circular polarization-modulated laser communications, qualified by:

- ❑ An uncoded data rate of 1 to 10 megabits per second
- ❑ A consecutive transmission time of 5 to 150 seconds
- ❑ A coded bit error rate of $1e-9$ to $1e-3$.
- ❑ An open-source, low cost system architecture

Mission Concept

Space Segment:

- ❑ Satellite transmits optical downlink and payload beacon laser
- ❑ Satellite receives OGS beacon laser
- ❑ Satellite performs closed-loop tracking to maintain optical link
- ❑ Satellite performs optical instrument calibrations



Ground Segment

- ❑ Optical Ground Station receives optical downlink and Payload beacon laser
- ❑ Optical Ground Station transmits beacon laser to Satellite

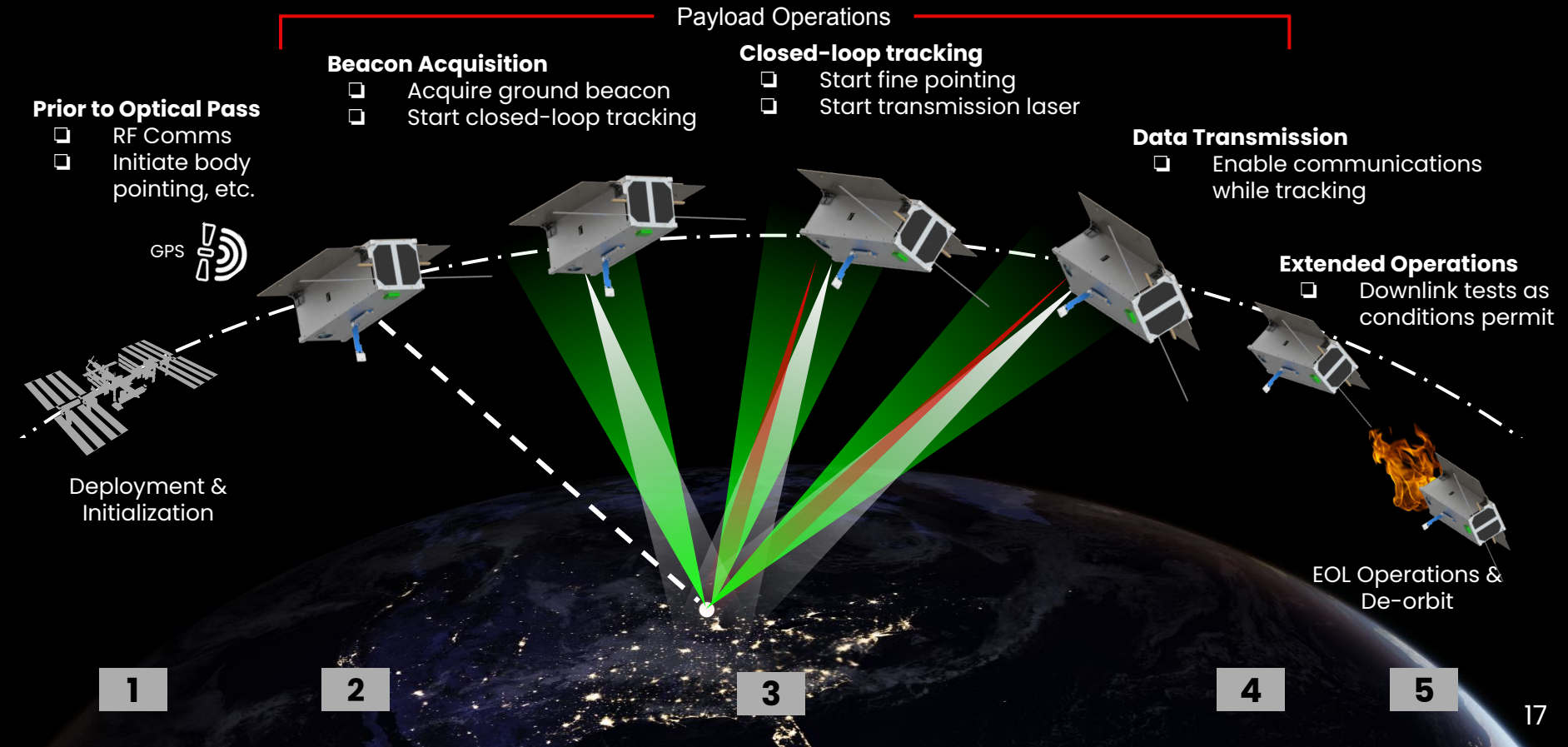
Follow-Up Mission Risk Reduction

PULSE-A functions as a risk reduction mission for a follow-up 3U CubeSat mission named PULSE-Q (Quantum). PULSE-Q will demonstrate space-to-ground quantum key distribution (QKD) using an optical downlink. The design for PULSE-A is intended to be repurposable with minimal modifications:

- ❑ Optical Payload: Must be modified for QKD transmission, but the foundation of an optical payload which modulates a laser transmission's polarization state remains the same.
- ❑ Bus: Repurposable after minor modifications for improved performance.
- ❑ Optical Ground Station (OGS): Must be modified to separate and detect 4 different polarization states, but the foundation of an OGS which tracks the CubeSat and interprets a laser downlink's polarization state remains the same.
- ❑ Radio Frequency Ground Station (RFGS): Repurposable as-is.
- ❑ Pointing, Acquisition, and Tracking (PAT) Sequence: Repurposable after minor modifications for improved performance.

The student knowledge base and documentation established during PULSE-A's development will significantly assist in the development of PULSE-Q and potential future missions.

Concept of Operations



Concept of Operations Considerations

1. Launch, Deployment, and Initialization

- ❑ Orbital parameters bounded by ~45 deg inclination through Sun Synchronous Orbit, ~90 deg inclination, to maximize launch opportunities.

2. Prior to Payload Operations

- ❑ GPS and RF communication must be established to enable accurate orbital propagation models for ground station tracking

3. Payload Operations

- ❑ Ideal launch timeline has payload operations primarily in summer (May–September) to maximize student availability and favorable weather conditions

4. Extended Operations

- ❑ Communications experiments are continued for as long as is feasible

5. End of Life Operations

- ❑ Natural orbit decay and re-entry, must happen within 5 years of mission end

Mission Success Status Determination

Success Status	Mission Result
Complete Success	Circular polarization-modulated downlink is successfully transmitted and received. All qualifications of mission success are achieved (raw data rate of 1 – 10 Mbps, link sustained for 5 – 150 seconds, $1e-3$ – $1e-9$ bit error rate).
Partial Success	Circular polarization-modulated downlink is successfully transmitted and received. At least one but not all qualifications of mission success are achieved (raw data rate of 1 – 10 Mbps, link sustained for 5 – 150 seconds, $1e-3$ – $1e-9$ bit error rate).
Limited Success	Circular polarization-modulated downlink is successfully transmitted and received, but no qualifications of mission success are achieved OR optical downlink is transmitted by the Satellite and detected by the OGS but cannot be interpreted.
Failure	Satellite is operational, but optical downlink cannot be transmitted by the Satellite OR Satellite is operational, but optical downlink cannot be received by the OGS.
Complete Failure	Satellite is not operational.

Secondary Project Objectives

1. Provide UChicago's undergraduates, who lack a traditional engineering program, an opportunity to design, build, assemble, and fly a fully-functioning CubeSat.
2. Prepare applied science students for careers in Aerospace, Electrical, Computer, and Mechanical engineering.
3. Design and build as much hardware in-house as possible.
4. Open-source all hardware, software, and relevant documentation.
5. Foster a co-working culture within our host student organization, UCSP

Optical Link Analysis

Laser Wavelength and Beacon Justification

- ❑ As a common telecom wavelength, components at 1550 nm are abundant.
- ❑ Thermal cameras required for detection of 1550 nm are outside of our budget .
- ❑ 638 nm is detectable by a CMOS camera.

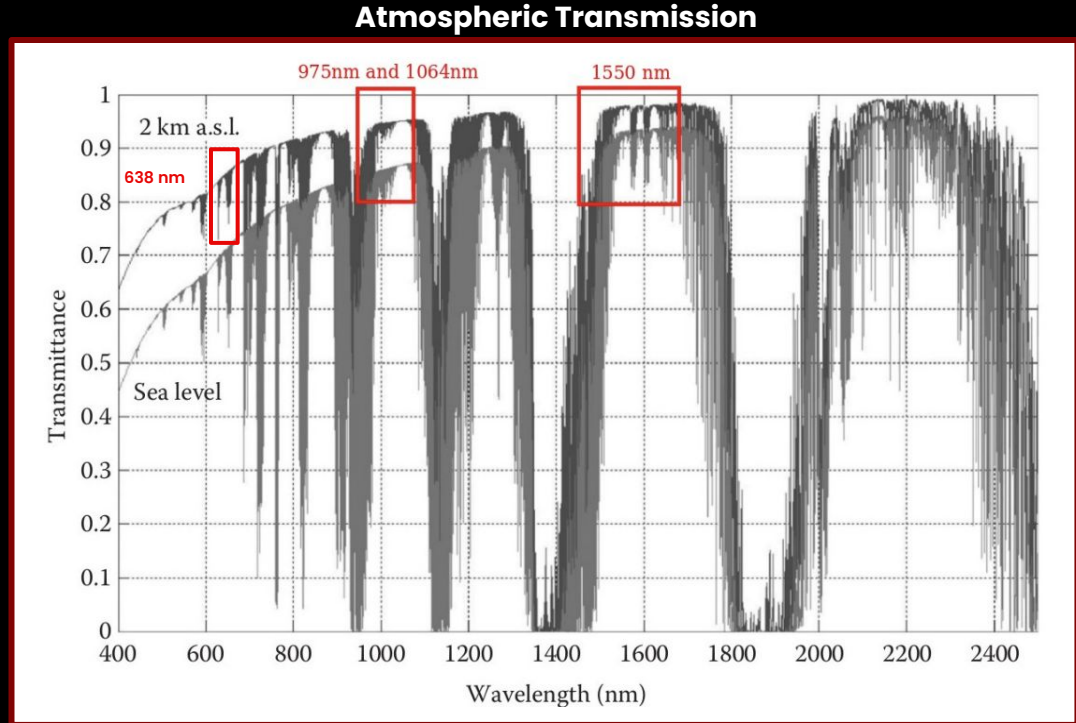


Figure adapted from Kingsbury, *Optical Communication for Small Satellites*, MIT 2015.

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All wavelengths used have a high atmospheric transmittance (> 0.8).

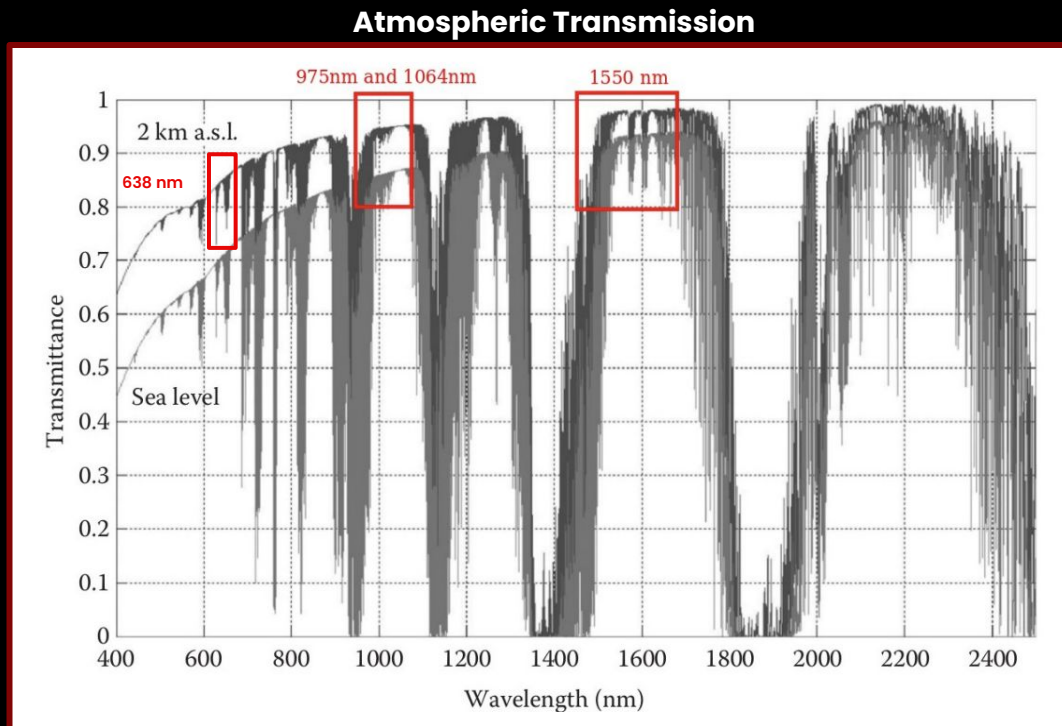


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Payload Transmission Link Budget

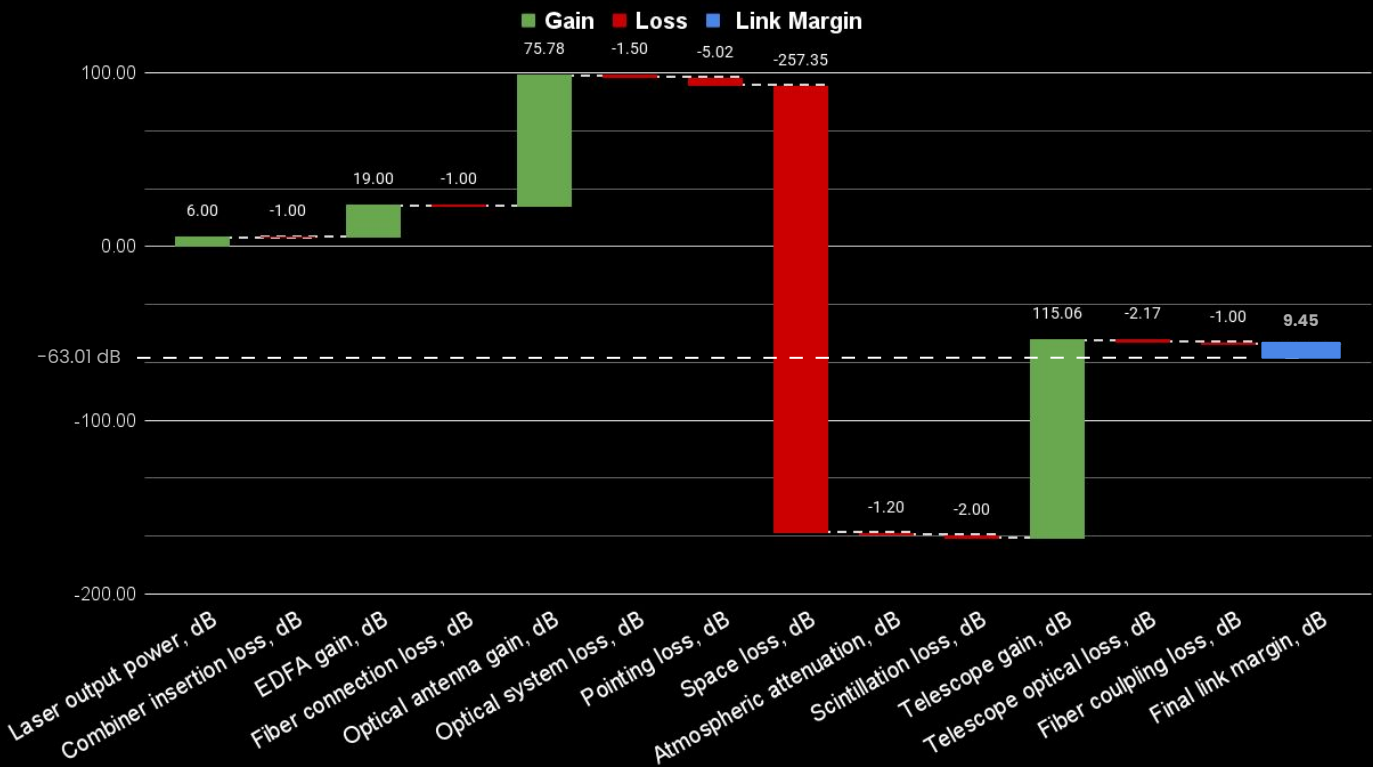
Transmission wavelength, nm	1550
Transmit power, dBm	6.00
Equivalent transmit power, W	3.9E-3
Received SNR, dB	9.45
Received power, dBm	-56.40
Equivalent received power, W	2.29E-9

Payload Transmission Link Budget

Transmission wavelength, nm	
Transmit power, dBm	6.00
Equivalent transmit power, W	3.9E-3
Received SNR, dB	9.45
Received power, dBm	-56.40
Equivalent received power, W	2.29E-9

SNR comfortably above 3 dB requirement, even at worst case 900 km link.

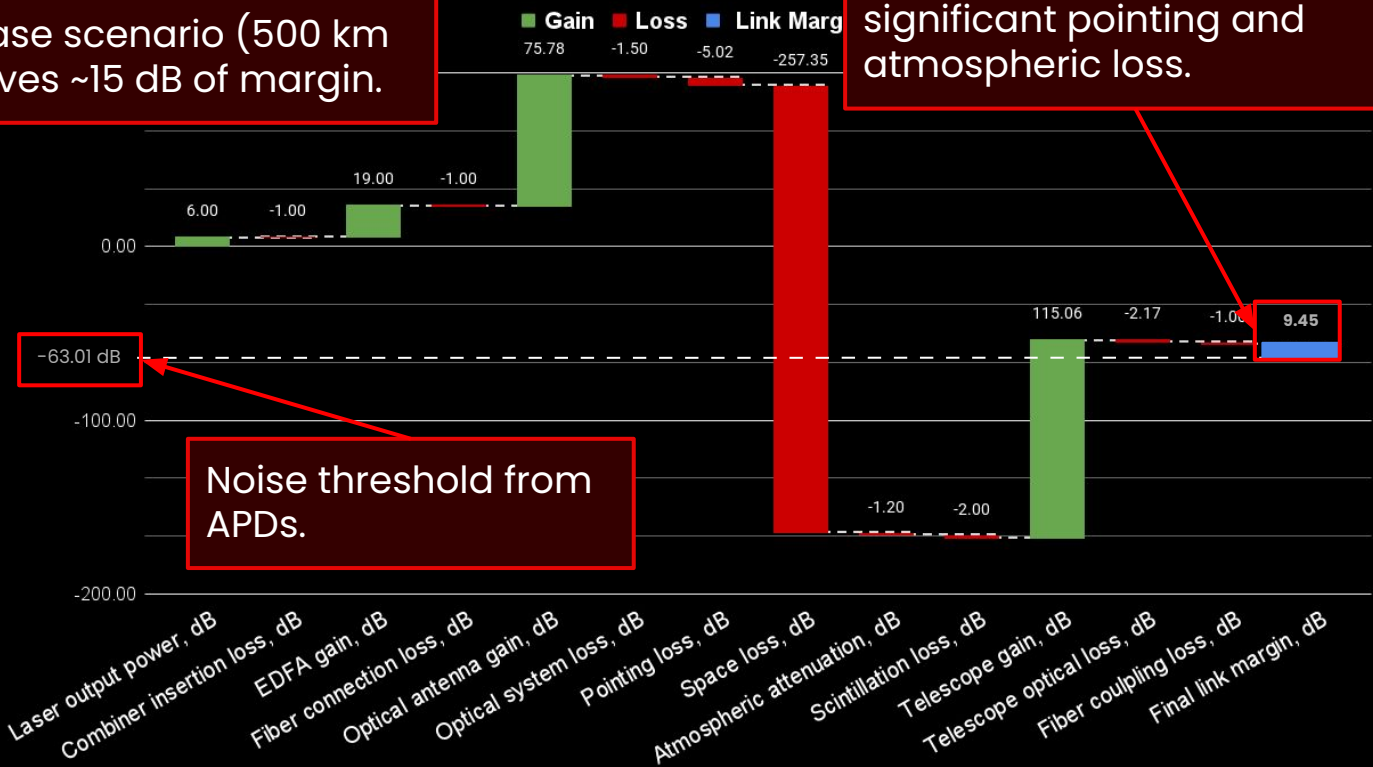
Payload Transmission Link Budget



Payload Transmission Link Budget

Best case scenario (500 km link) gives ~15 dB of margin.

>6 dB margin at worst-case (900 km) scenario, including significant pointing and atmospheric loss.



Noise threshold from APDs.

Payload Beacon Optical Link Budget Summary

Beacon transmit wavelength, nm	1064
Beacon transmit power, dBm	23.00
Received power, dBm	-87.20
Received SNR, dB	30.00
Equivalent received power, W	1.91E-12
Sirius App. Mag. (V)	-1.46
Flux of Sirius at Earth, W/m ²	1.00E-07
Flux of Beacon at Telescope, W/m ²	3.11E-11
Beacon Apparent Magnitude	7.31

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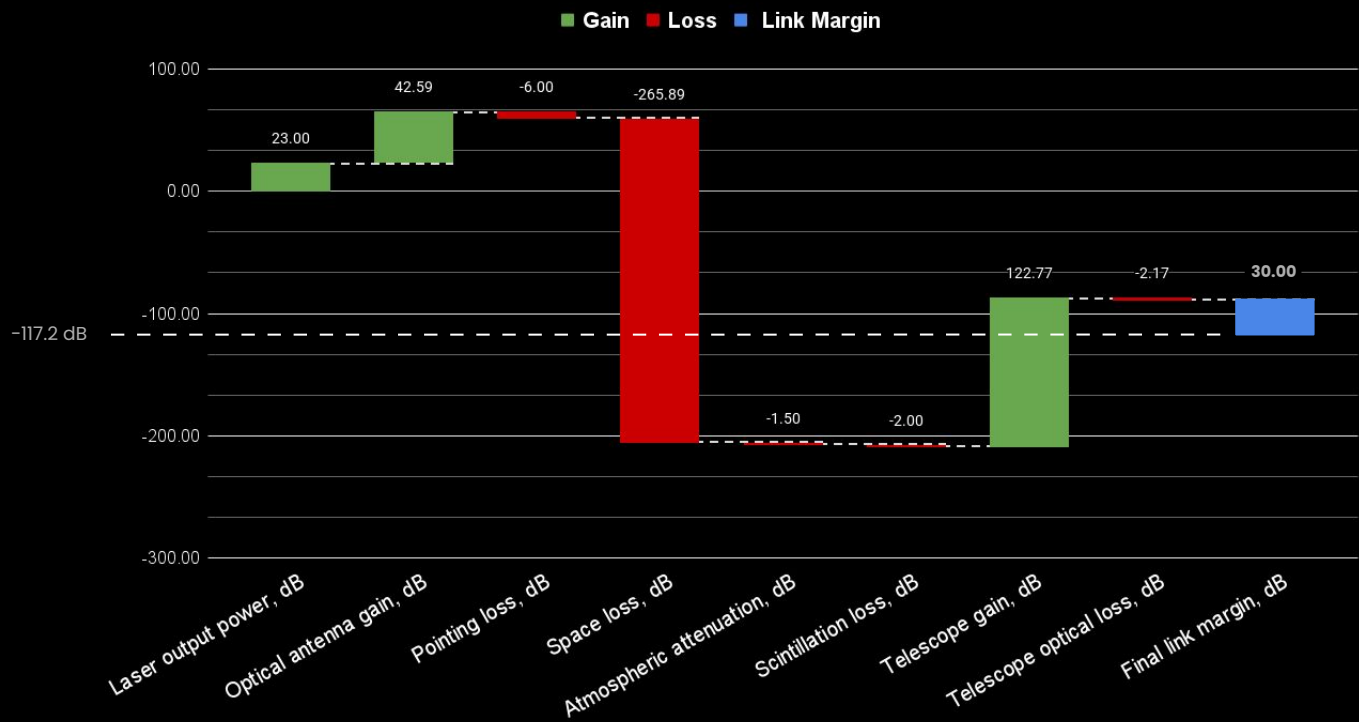
Using ~200mW output power with large divergence angle, designed for ~9dB of loss at 1 degree of satellite mispointing.

Payload Beacon Optical Link Budget Summary

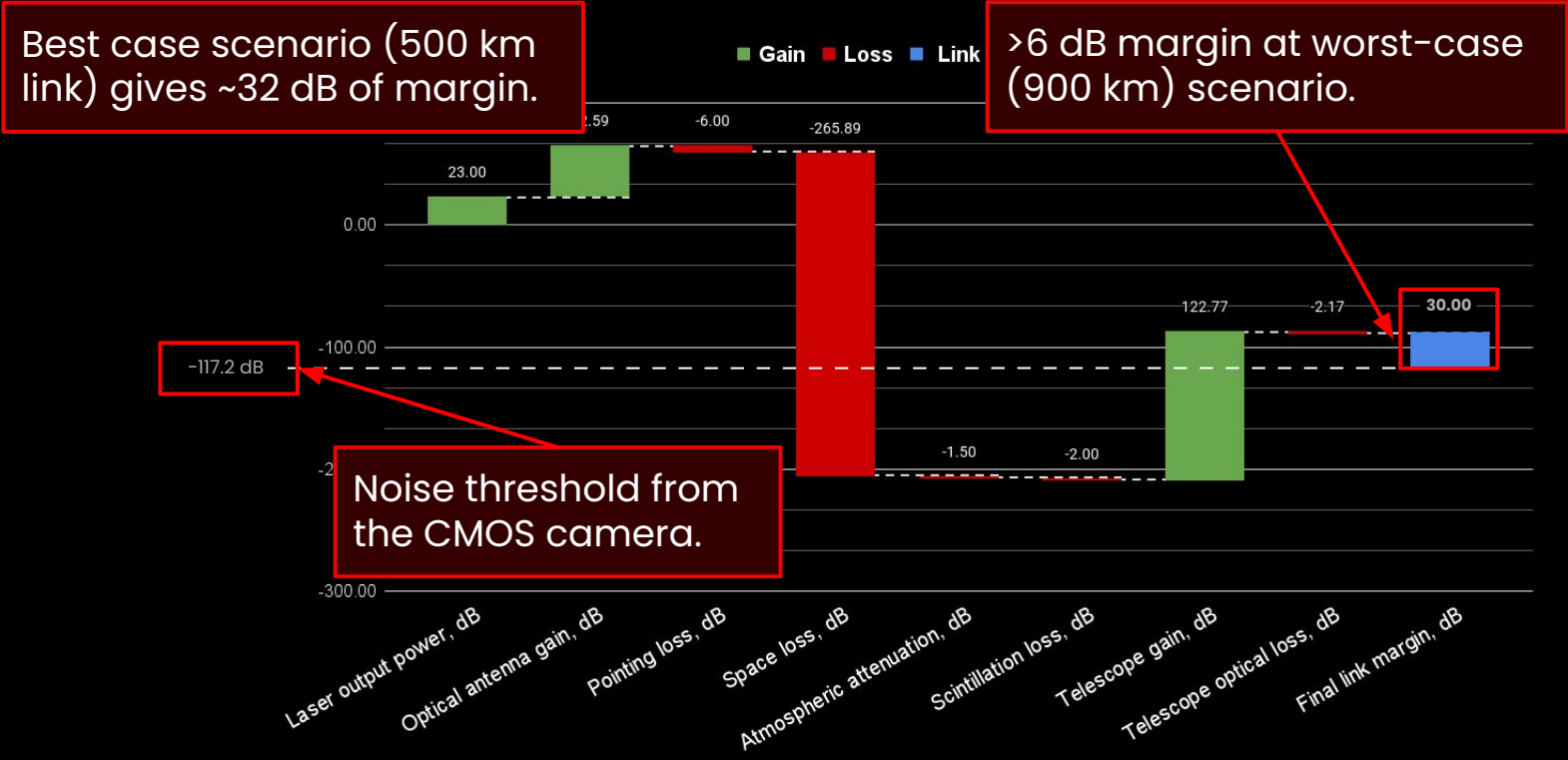
Link budgeting is more complex, but using flux calculations, we find worst-case ~7 visual magnitude (TBR). Integration time necessary on tracking camera will be experimentally determined.

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Payload Beacon Link Budget



Payload Beacon Link Budget

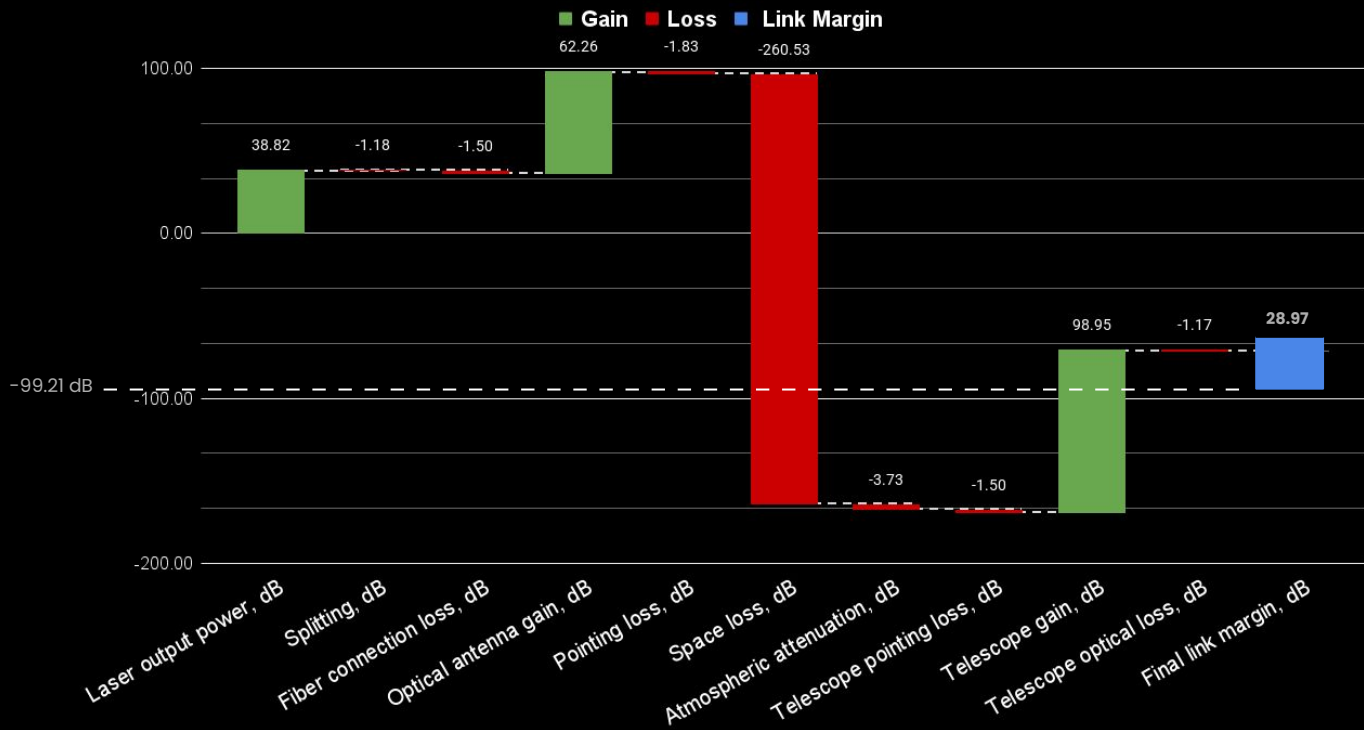


OGS Beacon Link Budget

- ❑ High transmit power means visibility (weather) is not a harsh limiting condition on transmission
- ❑ SNR significantly above 3 dB requirement even with small payload optical antenna (telescope assembly aperture)
- ❑ SNR values given at worst case 900 km link, without considering scintillation loss

Beacon wavelength, nm	1064
Beacon transmit power, dBm	38.82
Equivalent transmit power, W	7.62
Received SNR, dB	28.97
Received power, dBm	-70.24
Equivalent received power, nW	0.095

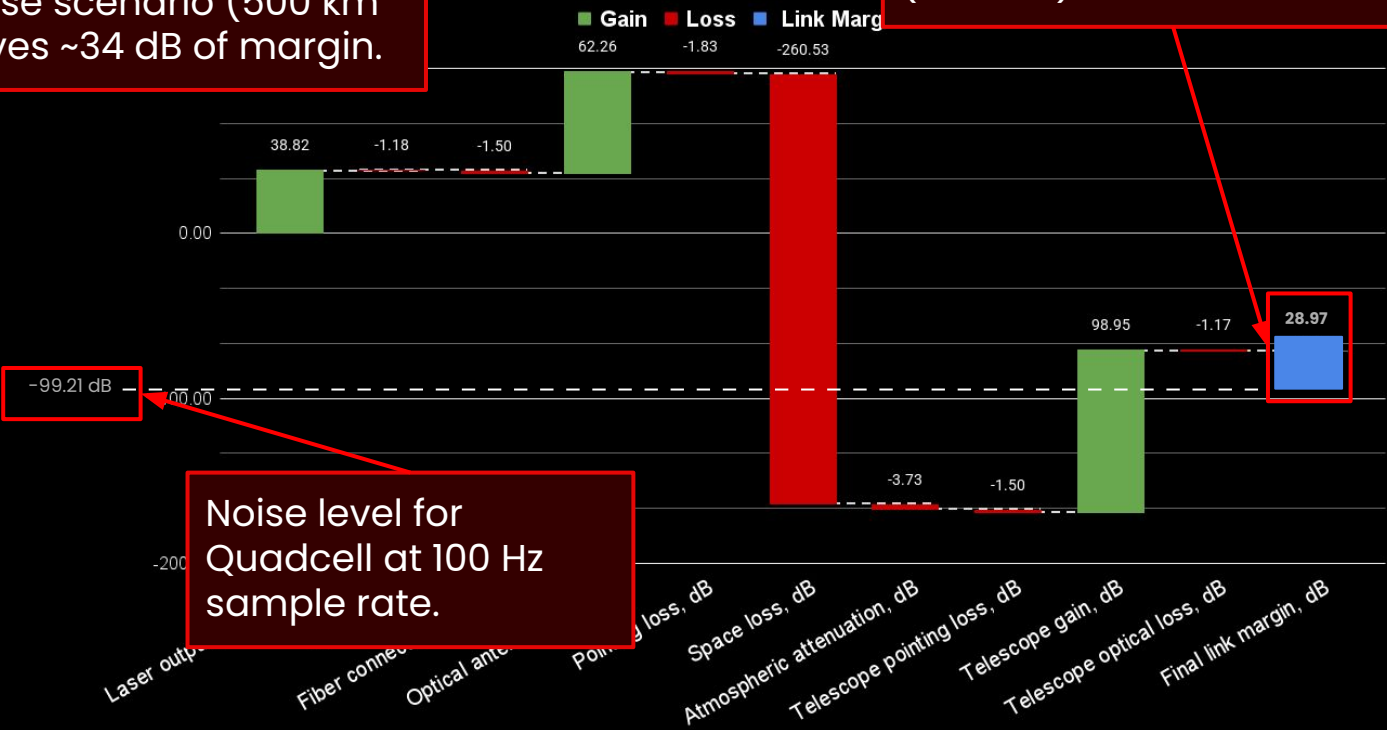
OGS Beacon Link Budget



OGS Beacon Link Budget

Best case scenario (500 km link) gives ~34 dB of margin.

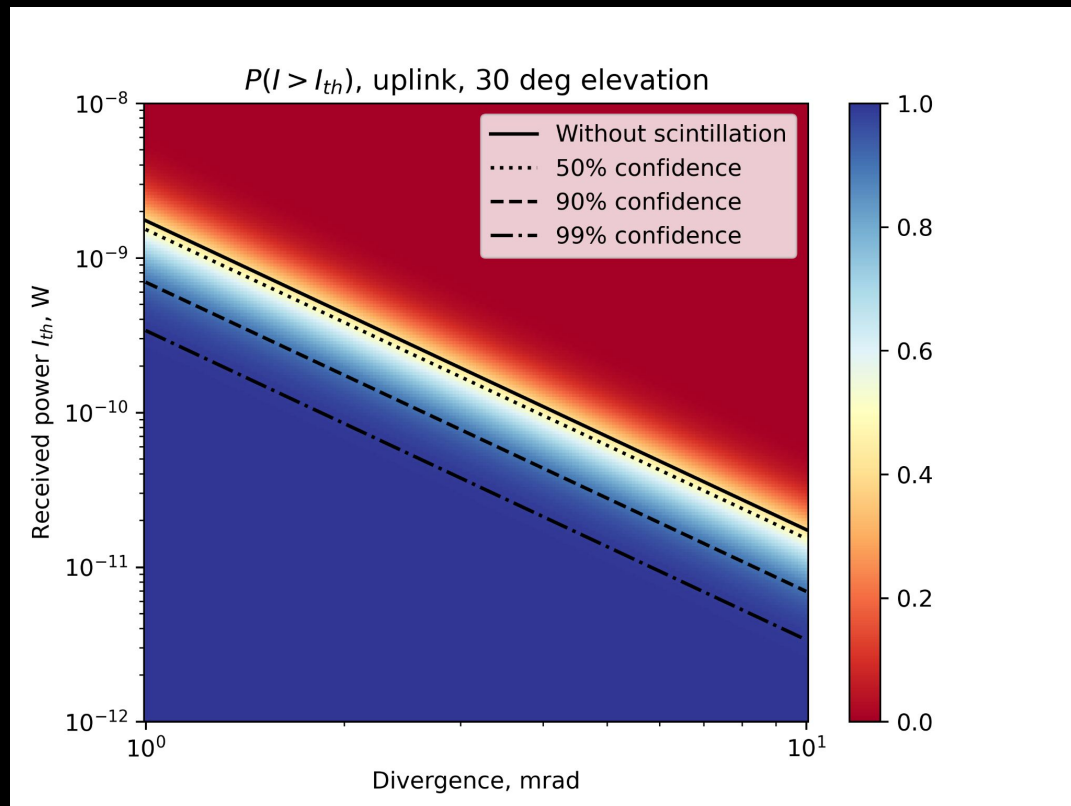
>6 dB margin at worst-case (900 km) scenario.



Noise level for Quadcell at 100 Hz sample rate.

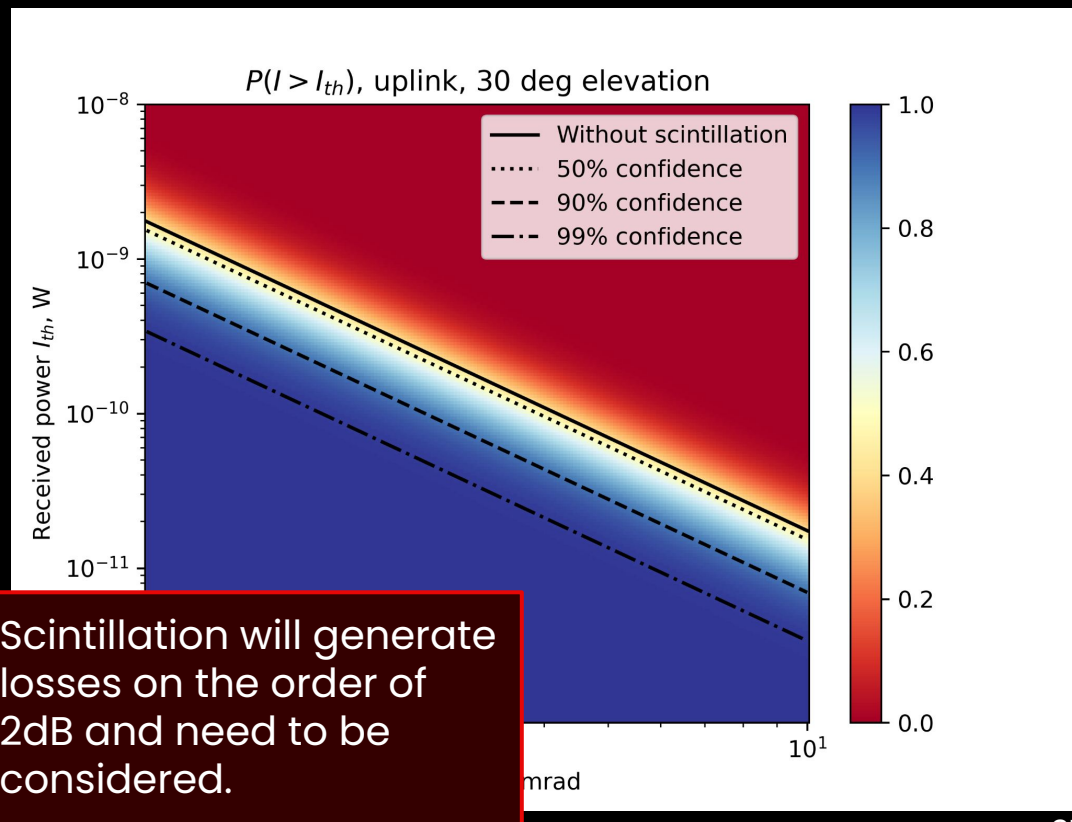
Scintillation

- ❑ Scintillation losses are on the order of 2 dB for the downlink transmission
- ❑ For the uplink beacon, which interacts with the earth's atmosphere before propagating through space, the losses are expected be larger. A statistical analysis of the effect can be simulated with MIT STARLab software (right)
- ❑ At a divergence of 4.36 mrad, worst case scenario is an additional drop of 10 dB, giving an SNR of 18.97 dB



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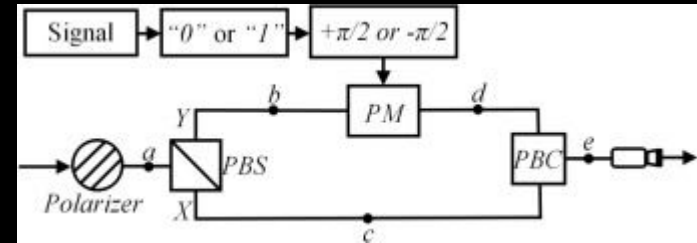


Optical Payload: **CPolsk Discussion**

CPolsk Designs from Literature

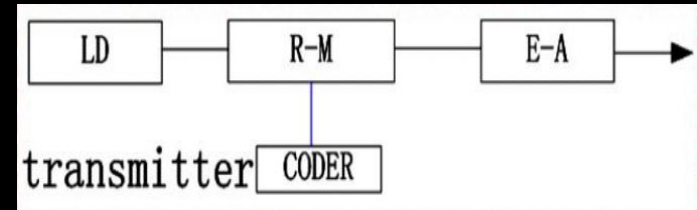
1. [Su et al](#):

Use one laser with polarizer. Beam splitter (PBS) separates X and Y component of polarized signal, Y component modulated using phase modulator (PM). Fiber combiner (PBC) recombines modulated Y component with X component into a single beam



2. [Zhao et al](#):

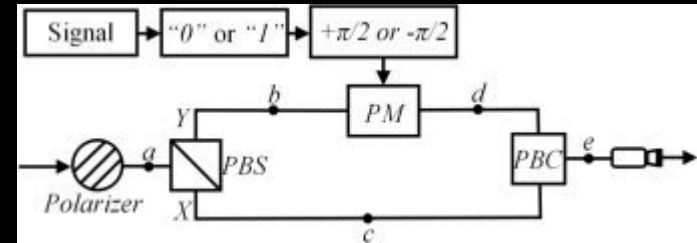
Use one laser with rotation modulator which seems to 1) circularly polarized light, 2) encode signal from coder onto polarized light. Emission antenna then emit circularly polarized light for data collection at receiver setup



CPolSk Designs from Literature

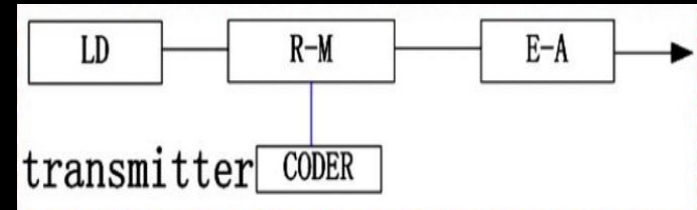
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Use one laser with rotation modulator which seems to 1) circularly polarized light, 2) encode signal from coder onto polarized light. Emission antenna then emit circularly polarized light for data collection at receiver setup



Literature design shows a single laser source split in free space.

Comparison to PULSE-A CPolSK setup

- ❑ PULSE-A's setup combines the modulation techniques of CPolSK with the simplicity of on-off keying
- ❑ Instead of a modulator (previous design iterations), PULSE-A is using two orthogonally polarized lasers, where each handedness corresponds to a 0 or 1 bit
- ❑ This setup is more space efficient and cost effective, using two customized laser drivers with bias-T's and laser diodes, instead of a space hardened modulator
- ❑ Retains the effectiveness of a CPolSK modulation scheme while being cost effective
- ❑ Might lead to greater SNR due to more circuit elements

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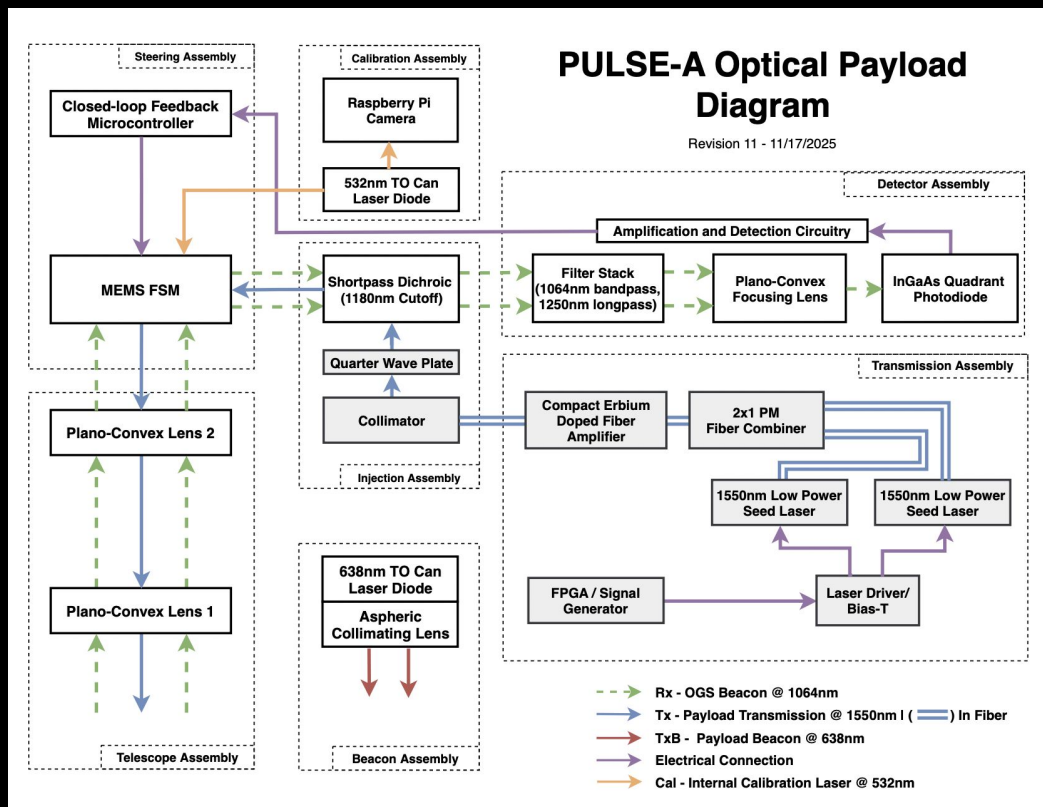
We opted for a simple on-off keying choice with two lasers.

Optical Payload: **Optical Design**

Payload Design Overview

Optical Payload Design is made of 7 Assemblies:

- ❑ Beacon Assembly
- ❑ Telescope Assembly
- ❑ Calibration Assembly
- ❑ Steering Assembly
- ❑ Detection Assembly
- ❑ Transmission Assembly
- ❑ Injection Assembly

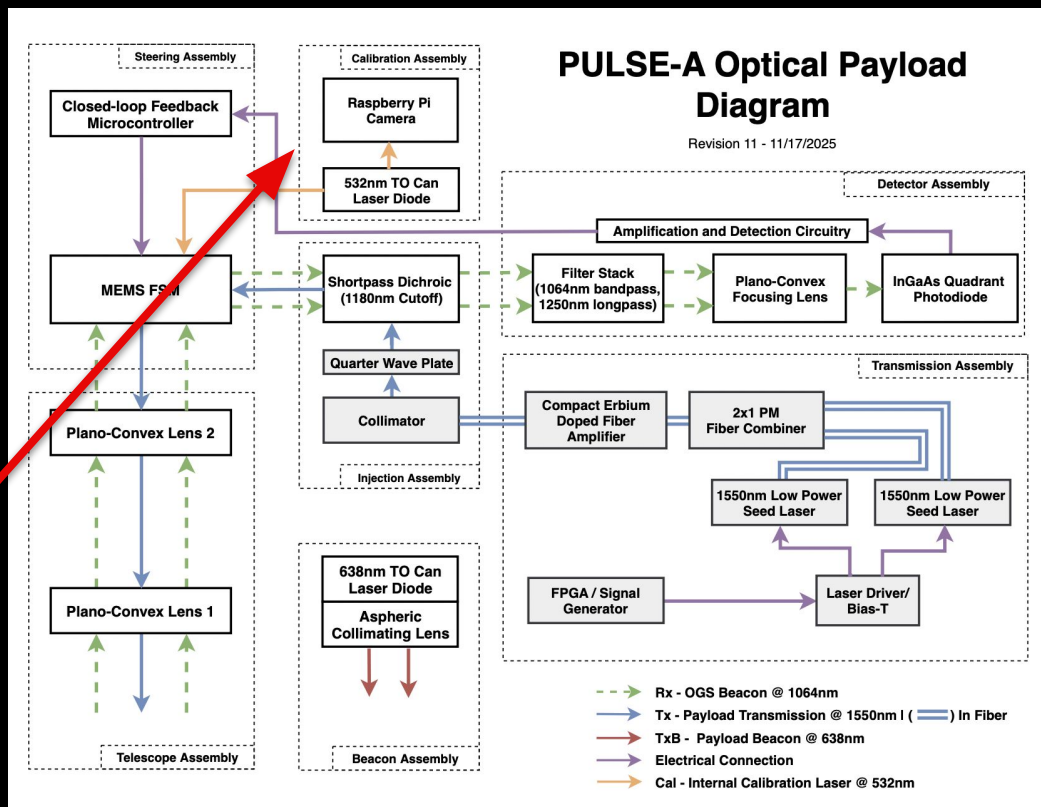


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New Assembly: Calibration

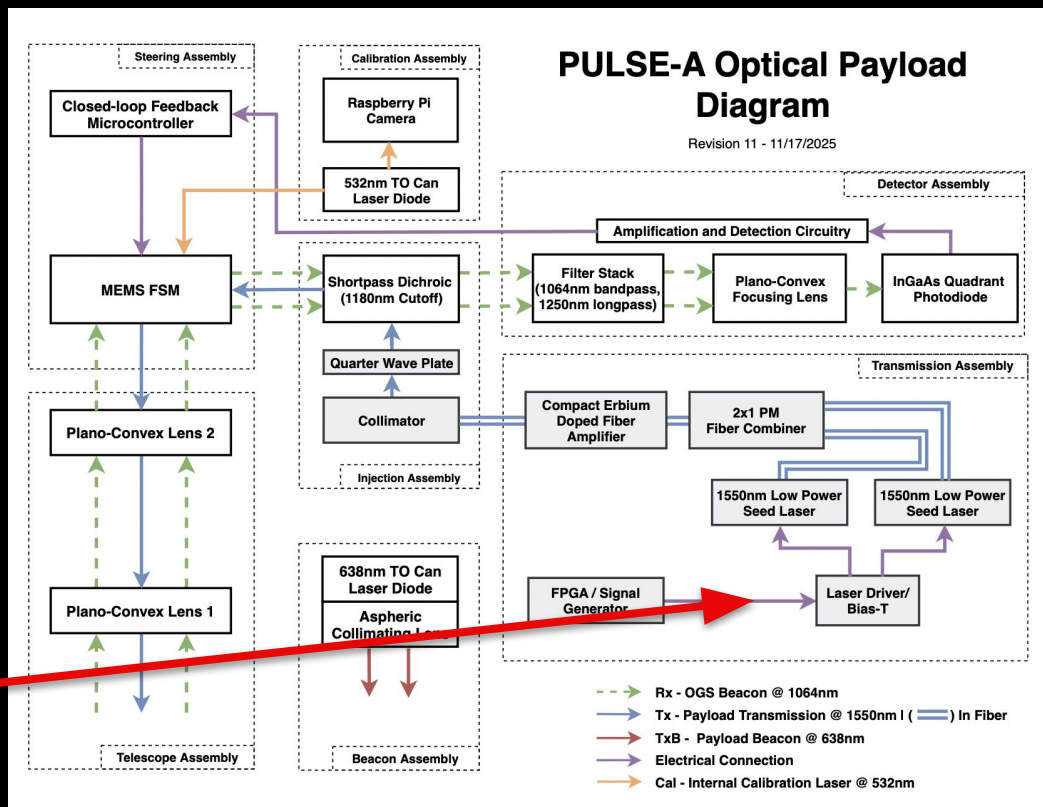


Payload Design Overview

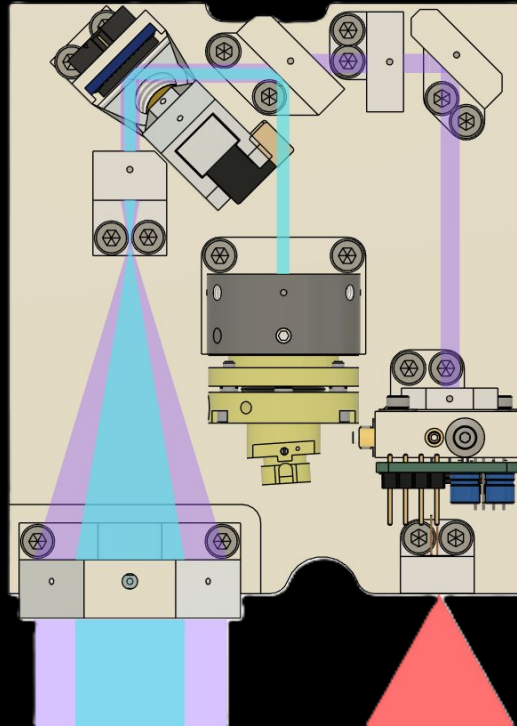
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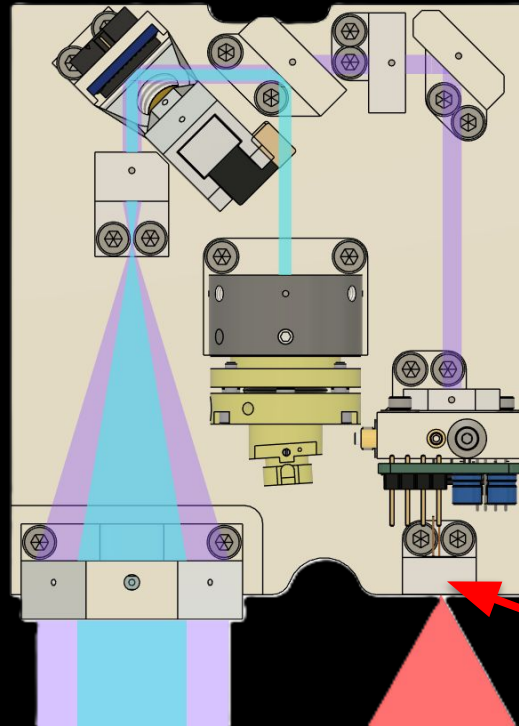
New modulation scheme:
ON/OFF Laser driving.



Payload Layer

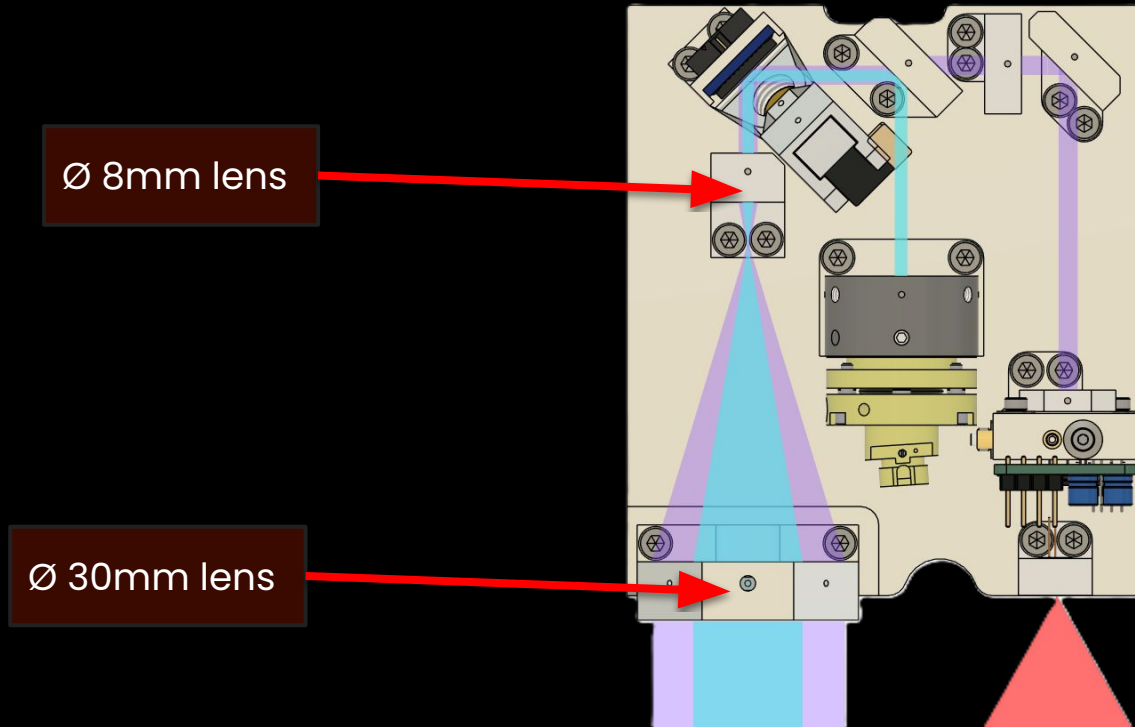


Payload Layer: Beacon Assembly

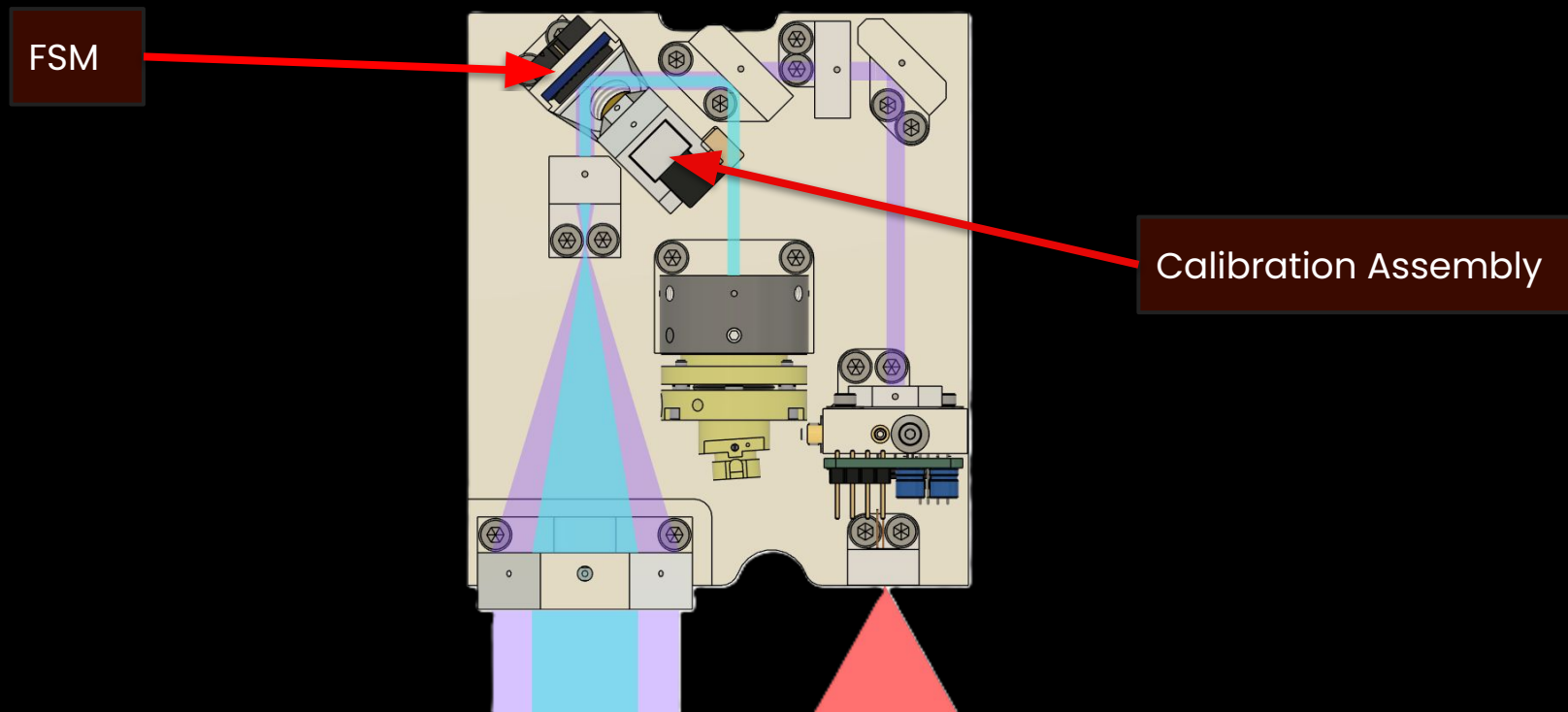


Downlink beacon

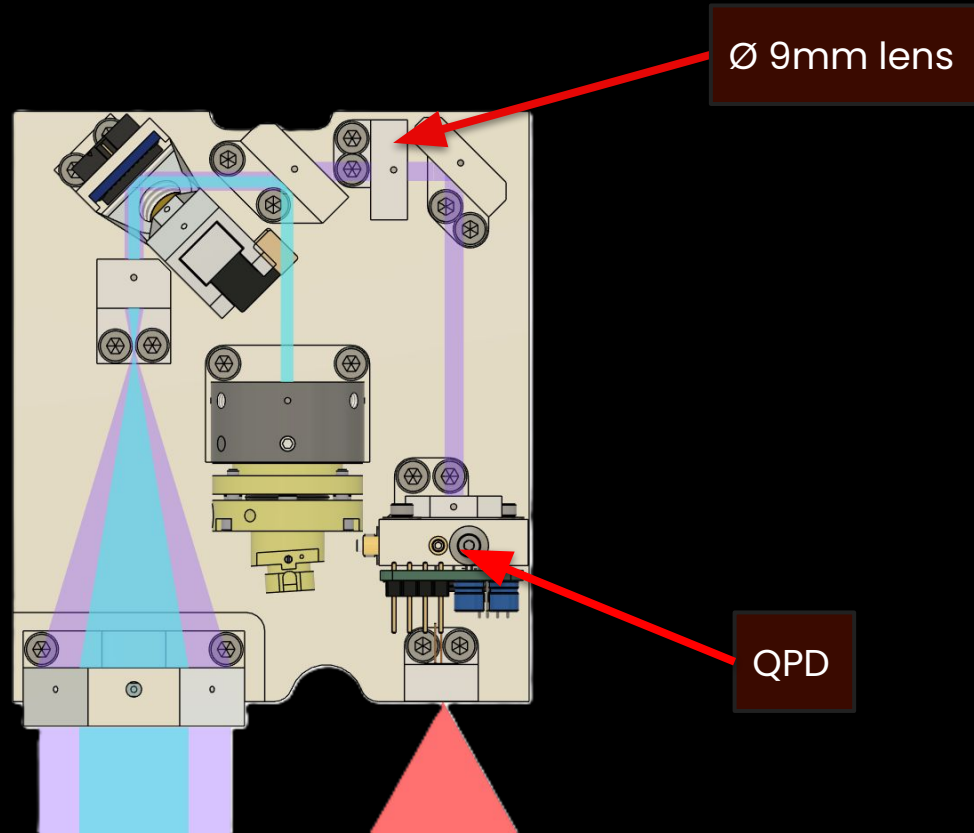
Payload Layer: Telescope Assembly



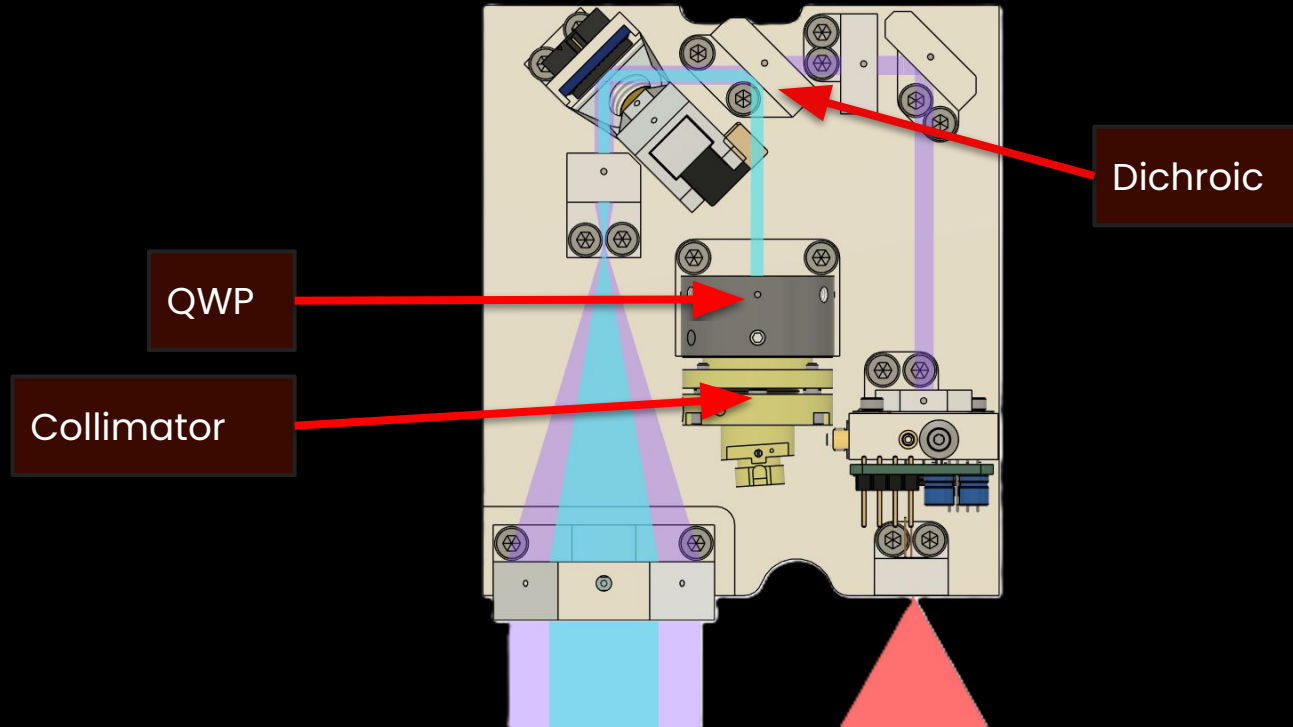
Payload Layer: Steering & Calibration Assembly



Payload Layer: Detection Assembly



Payload Layer: Transmission Assembly



Payload Beacon Assembly: Beacon Laser

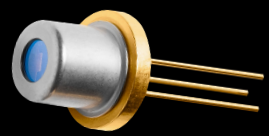
- ❑ **Function:** serves as a beacon source to aid in coarse pointing alignment between the satellite and the ground station.
- ❑ Beacon will be turned ON/OFF with laser driver (more on laser driver design in Transmission assembly).



Beacon Model	Ushio HL63022DG 638nm
Wavelength (nm)	639
Operating Temperature (C°)	-40 to 60
Max Beam divergence Parallel to the junction (degrees)	13
Max Beam divergence Perpendicular to the junction (degrees)	18
Optical output power (1) (Tc=25 oC) (mW)	210
Optical output power (2) (Tc=60 oC) (mW)	125

Payload Beacon Assembly: Beacon Laser

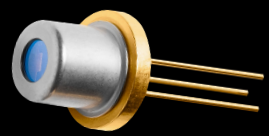
- ❑ **Function:** serves as a beacon source to aid in coarse pointing alignment between the satellite and the ground station.
- ❑ Beacon will be turned on only when needed. Significant drop in power with increasing temperatures.



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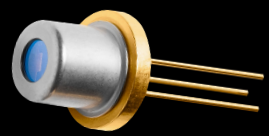
- ❑ **Function:** serves as a beacon source to aid in coarse pointing alignment between the
- ❑ Beam divergence above the optimized value of 2.4 degrees determined by link budget analysis.
- ❑ design in transmission assembly).



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Payload Beacon Assembly: Beacon Laser

- ❑ **Function:** serves as a beacon source to aid in coarse pointing alignment between the [unclear] test [unclear]
- ❑ Will require a collimation mechanism to achieve 2.4 degree divergence.
- ❑ [unclear] design in transmission assembly).

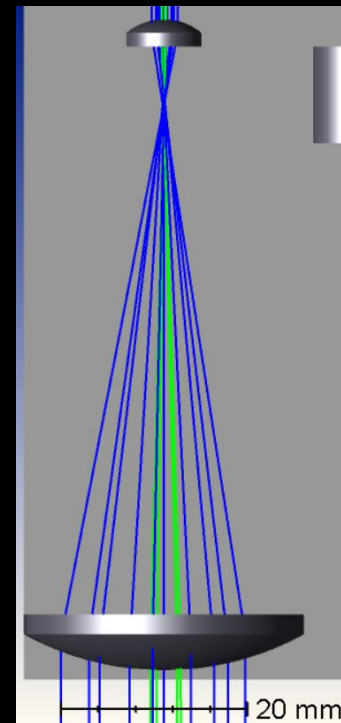


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Payload Telescope Assembly

- ❑ **Type:** Plano Convex
- ❑ **Function:** The Payload Keplerian Telescope is designed to simultaneously focus the OGS uplink beacon along the path to the QPD, while expanding the Payload transmission beam on its way to the OGS.
 - ❑ Transverse/Spot Magnification: 6.1
 - ❑ Angular Magnification: 7.4

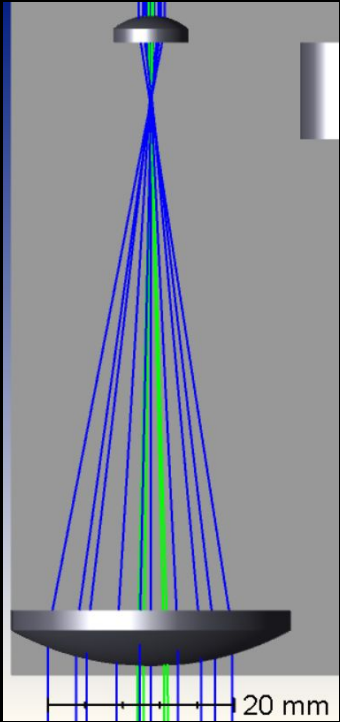
Lens 1 Model	EO #45-241
Radius	30mm
Effective Focal Length	60.00mm
Back Focal Length	56.03mm
Substrate	N-BK7
Lens 2 Model	EO #45-241
Radius	8mm
Effective Focal Length	8.00mm
Back Focal Length	6.40mm
Substrate	UV Fused Silica



Payload Telescope Assembly

- ❑ **Type:** Plano Convex
- ❑ **Function:** The Payload Keplerian Telescope is designed to simultaneously focus the OGS uplink beacon along the path to the QPD, while expanding the Payload transmission beam on its way to the OGS.
- ❑ Transverse/Spot Magnification: 6.1
- ❑ Angular Magnification: 7.4

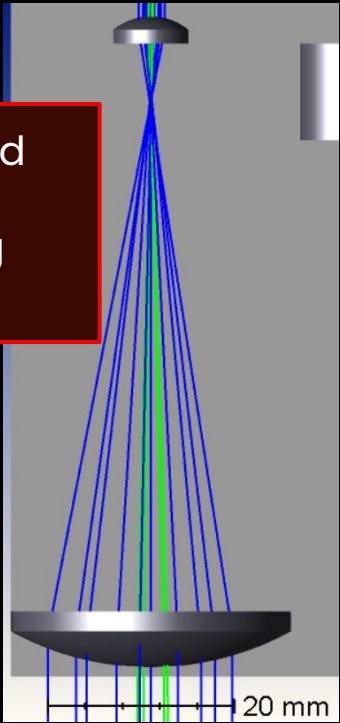
	Lens 1 Model	EO #45-241
	Radius	30mm
	Effective Focal Length	60.00mm
	Back Focal Length	56.03mm
Back Focal Lengths most relevant for simulations.		N-BK7
		EO #45-241
	Radius	8mm
	Effective Focal Length	8.00mm
	Back Focal Length	6.40mm
	Substrate	UV Fused Silica



Payload Telescope Assembly

- ❑ **Type:** Plano Convex
- ❑ **Function:** The Payload Keplerian Telescope is designed to simultaneously focus the OGS uplink beacon along the path to the QPD, while expanding the Payload transmission beam on its way to the OGS.
 - ❑ Transverse/Spot Magnification: 6.1
 - ❑ Angular Magnification: 7.4

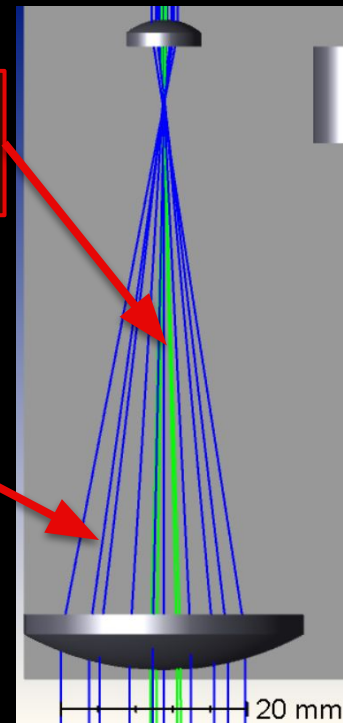
Lens 1 Model	EO #45-241
Radius	30mm
Effective Focal	Receiving OGS beacon and expanding transmission beam beam to OGS along same path.
Back Focal Len	
Substrate	
Lens 2 Model	EO #45-241
Radius	8mm
Effective Focal Length	8.00mm
Back Focal Length	6.40mm
Substrate	UV Fused Silica



Payload Telescope Assembly

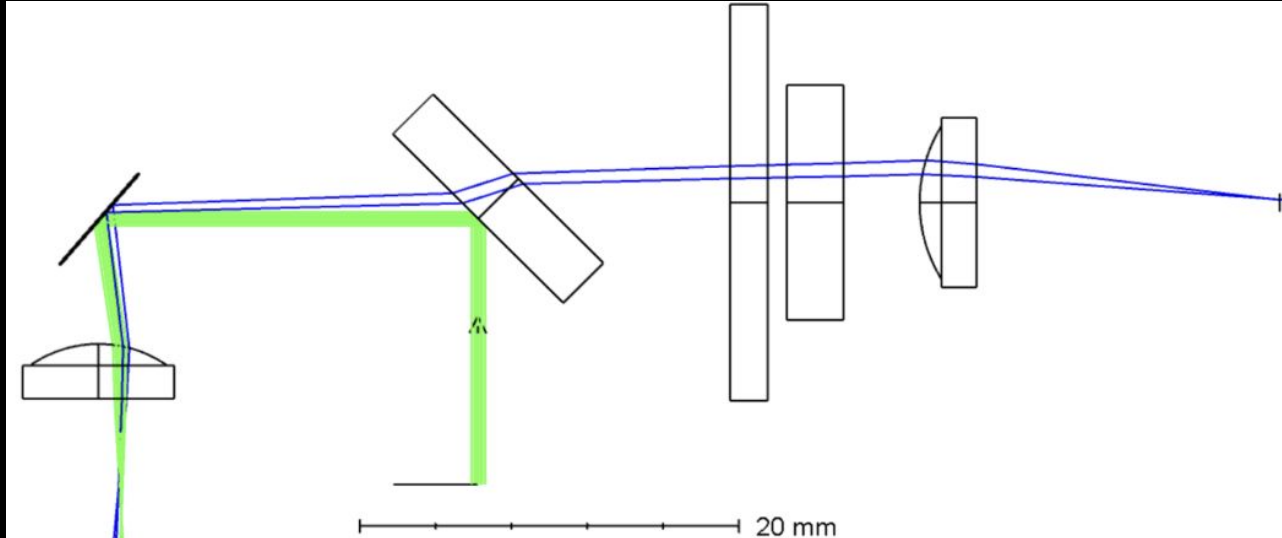
- ❑ **Type:** Plano Convex
- ❑ **Function:** The Payload Keplerian Telescope is designed to simultaneously focus the OGS uplink beacon along the path to the QPD, while expanding the Payload transmission beam on its way to the OGS.
 - ❑ Transverse/Spot Magnification: 6.1
 - ❑ Angular Magnification: 7.4

Lens 1 Model	EO #45-241
Radius	Transmission Laser (1550nm)
Effective Focal Length	
Back Focal Length	56.03mm
Substrate	N-BK7
Lens 2 Model	OGS Beacon (1064nm)
Radius	
Effective Focal Length	
Back Focal Length	
Substrate	UV Fused Silica



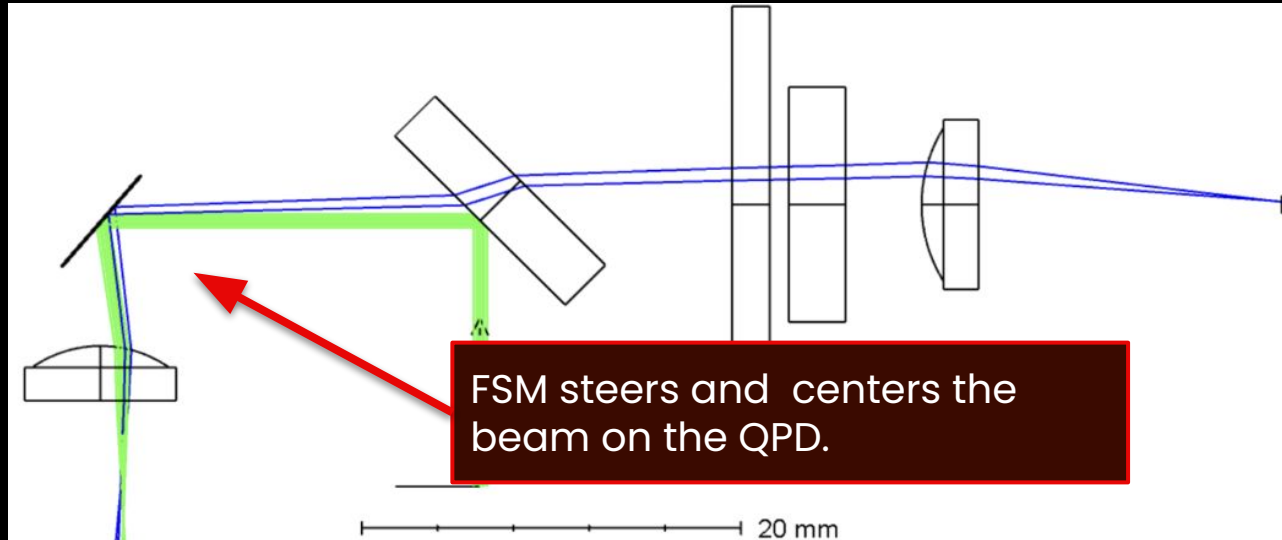
Payload Steering Assembly

Fine-steering mechanism (FSM) corrects uplink beacon misalignments caused by body-pointing, thermal drift, or vibrations. For angular errors up to 0.7° , the FSM adjusts the optical path to center the beam on the quadrant photodiode (QPD). A closed-loop control uses the QPD signal to compute the FSM mirror tilt via a voltage-transfer function.



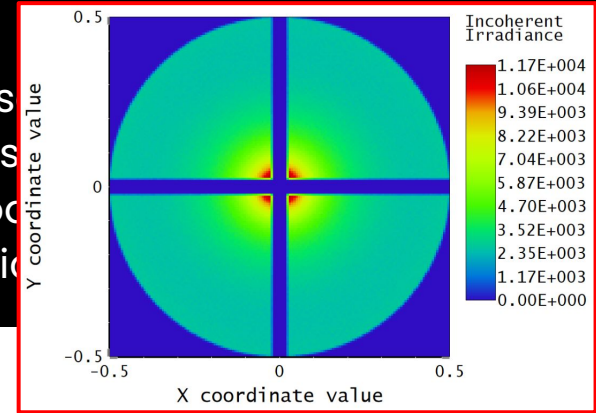
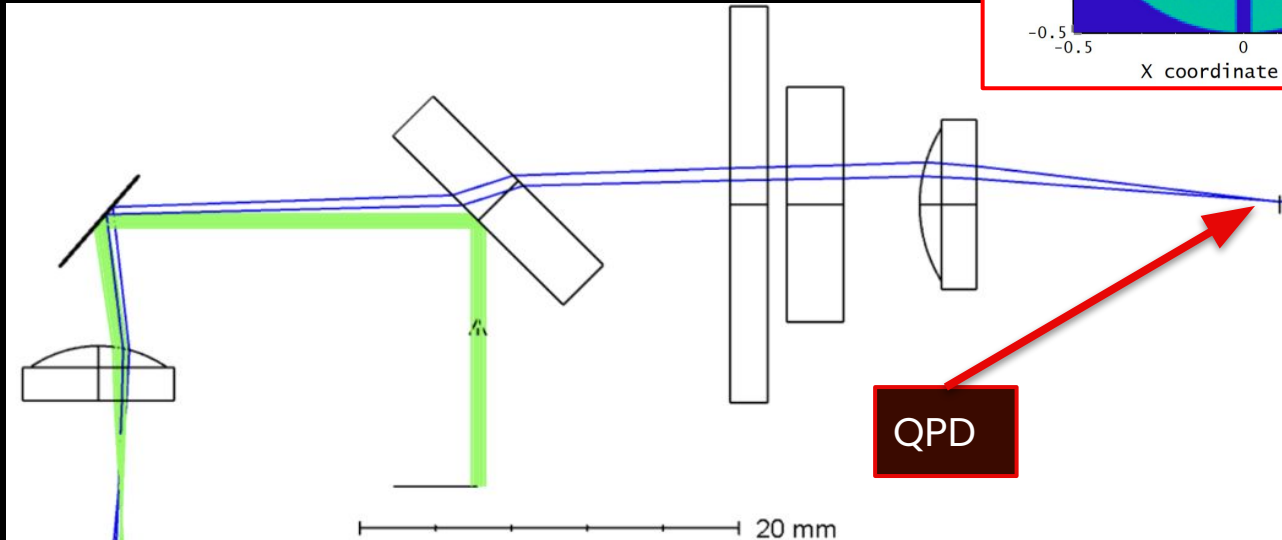
Payload Steering Assembly

Fine-steering mechanism (FSM) corrects uplink beacon misalignments caused by body-pointing, thermal drift, or vibrations. For angular errors up to 0.7° , the FSM adjusts the optical path to center the beam on the quadrant photodiode (QPD). A closed-loop control uses the QPD signal to compute the FSM mirror tilt via a voltage-transfer function.



Payload Steering Assembly

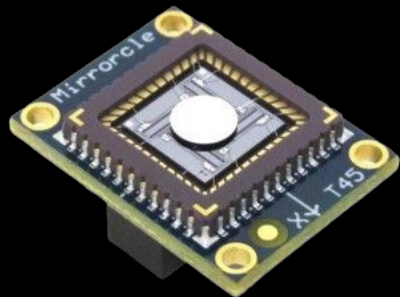
Fine-steering mechanism (FSM) corrects uplink beacon misbody-pointing, thermal drift, or vibrations. For angular errors the optical path to center the beam on the quadrant photodiode control uses the QPD signal to compute the FSM mirror tilt via a control function.



Payload Steering Assembly: FSM

Function:

- ❑ Steering OGS uplink beacon to detection assembly.
- ❑ Reflects Payload downlink transmission to exit aperture.
- ❑ Tilt based on closed-control feedback loop with QPD.

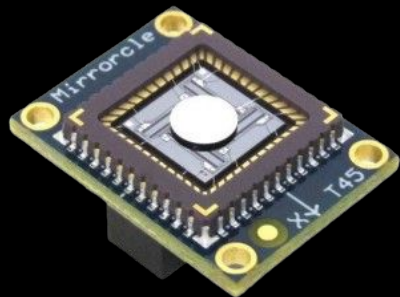


FSM Model	Mirrorcle MEMS Mirror (A5L3.3(C2)-6400AU-TI NY48.4-C/W/EP)
Diameter (mm)	6.4
Max. Mech. Angle (x axis)	+/- 4.25°
Max. Mech. Angle (y axis)	+/- 4.25°
Mirror Coating	Gold (AU)
Optical Window (coating)	1040 - 1600 nm
Mirror Reflectance (350-2000nm)	95%

Payload Steering Assembly: FSM

Function:

- ❑ Steering OGS uplink beacon to detection assembly.
- ❑ Reflects Payload downlink transmission to exit aperture.
- ❑ Tilt based on closed-control feedback loop with QPD.

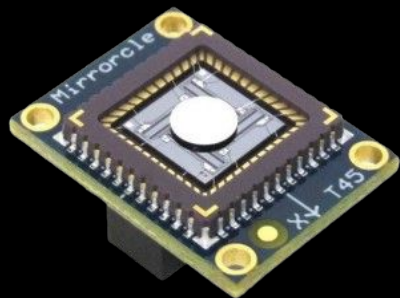


FSM Model	Mirrorcle MEMS Mirror (A5L3.3(C2)-6400AU-TI NY48.4-C/W/EP)
Diameter	6.4mm
Max. Mech. Angle (x axis)	+/- 4.25°
Relevant Pointing Specs.	+/- 4.25°
	Gold (AU)
Optical Window (coating)	1040 - 1600 nm
Mirror Reflectance (350-2000nm)	95%

Payload Steering Assembly: FSM

Function:

- ❑ Steering OGS uplink beacon to detection assembly.
- ❑ Reflects Payload downlink transmission to exit aperture.
- ❑ Tilt based on closed-control feedback loop with QPD.

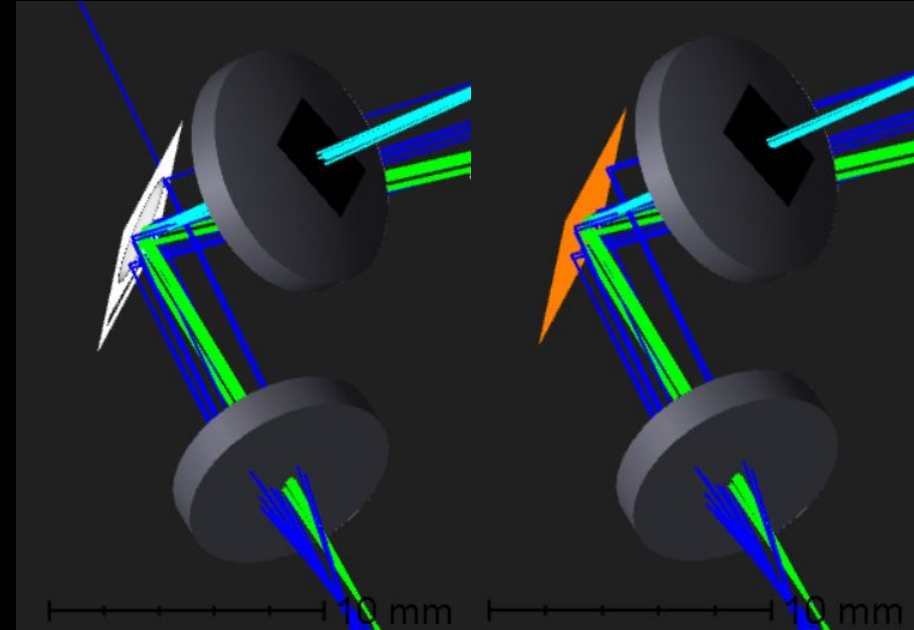
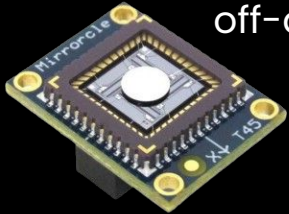


FSM Model	Mirrorcle MEMS Mirror (A5L3.3(C2)-6400AU-TI NY48.4-C/W/EP)
Diameter	6.4mm
Tilt (X axis)	+/- 4.25°
Tilt (Y axis)	+/- 4.25°
Coating	Gold (AU)
Wavelength	1040 - 1600 nm
Mirror Reflectance (350-2000nm)	95%

Using maximum tilt can overdrive and damage mirror. Our expected range is <3.4 degrees.

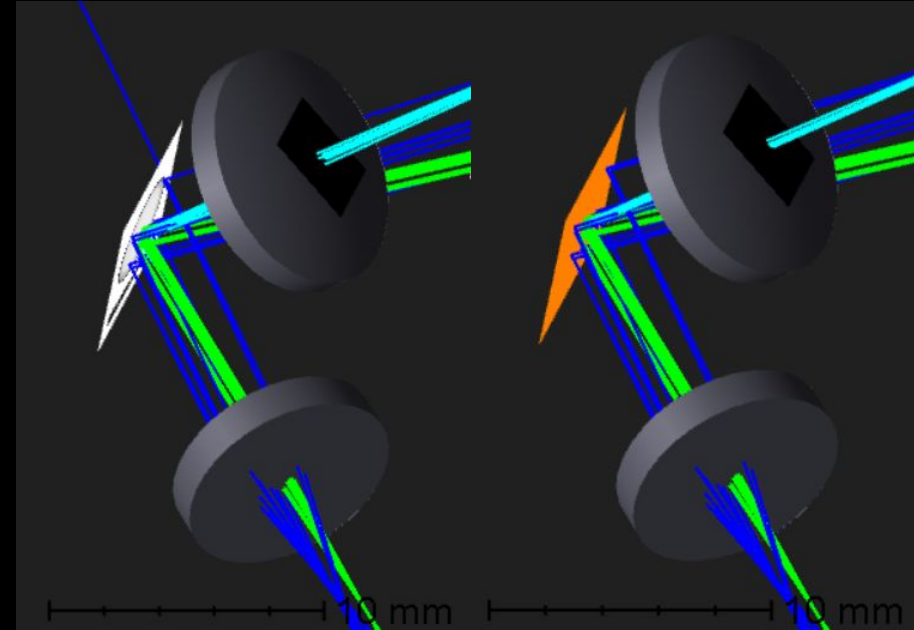
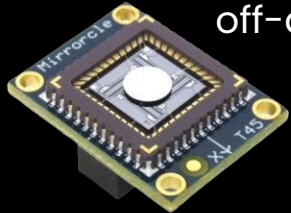
Payload Steering Assembly: FSM Choice Considerations

- ❑ FSM Diameter:
 - ❑ Capture entire OGS uplink beacon
 - ❑ Mirror clipping concern for products $< 6.4\text{mm}$
- ❑ FSM Tilt:
 - ❑ Tilting Range considered with coarse-pointing error (Optical Analysis)
 - ❑ Preliminary choice: 4.25° tilt range ensured to correct 1.0° coarse-pointing off-angle error
 - ❑ Validation: Effective tilt range will cover 0.7° coarse-pointing off-angle error



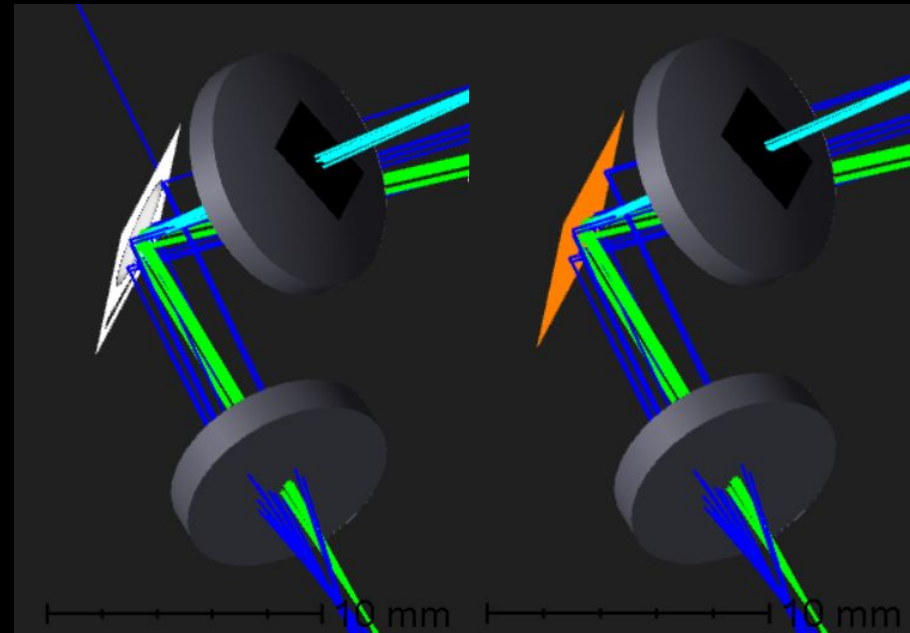
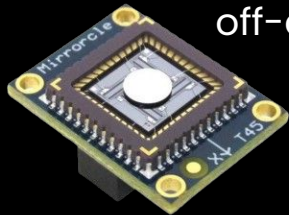
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Payload Steering Assembly: FSM Choice Considerations

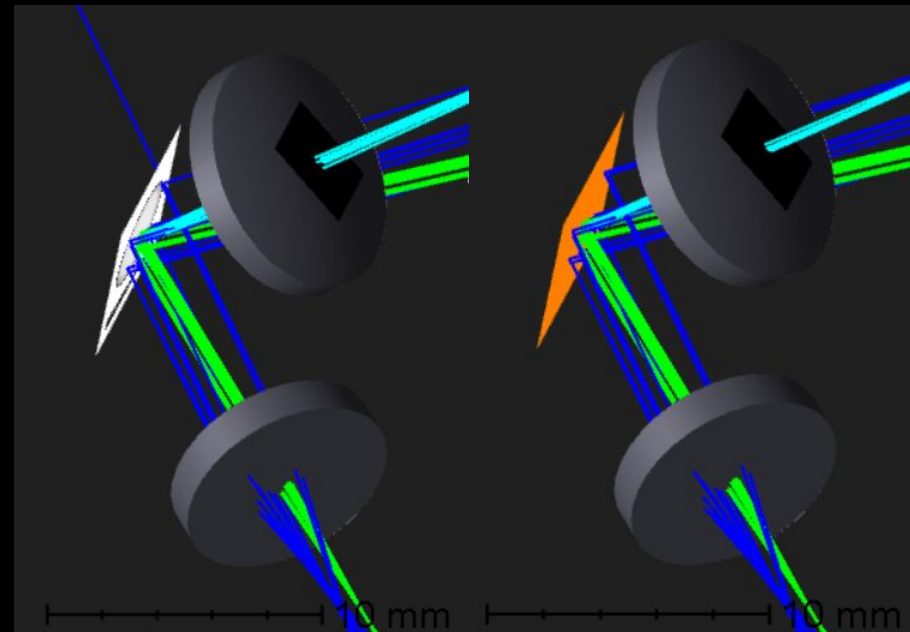
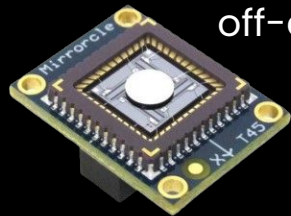
- ❑ FSM Diameter:
 - ❑ Capture entire OGS uplink beacon
 - ❑ Mirror clipping concern for products $< 6.4\text{mm}$
- ❑ FSM Tilt:
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 - ❑ Validation: Effective tilt range will cover 0.7° coarse-pointing off-angle error



5mm Diameter
Product; Potential
clipping

Payload Steering Assembly: FSM Choice Considerations

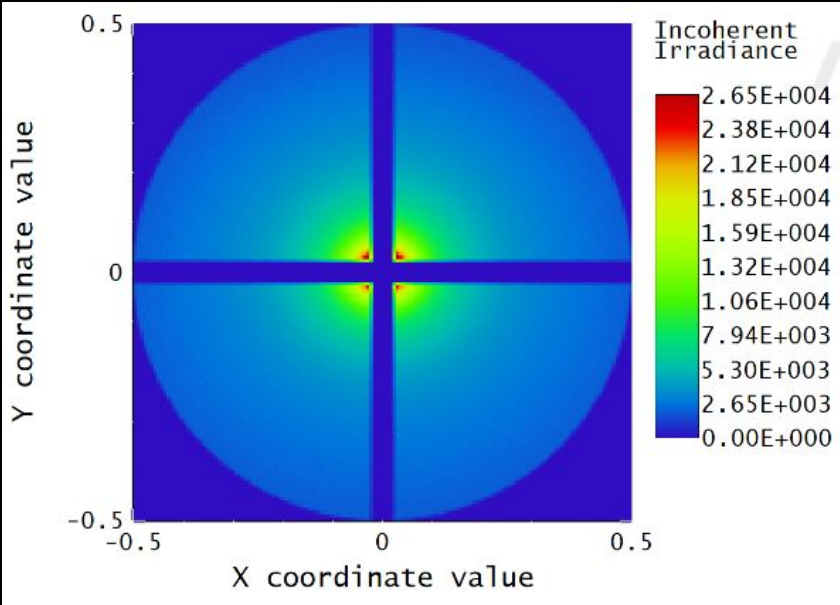
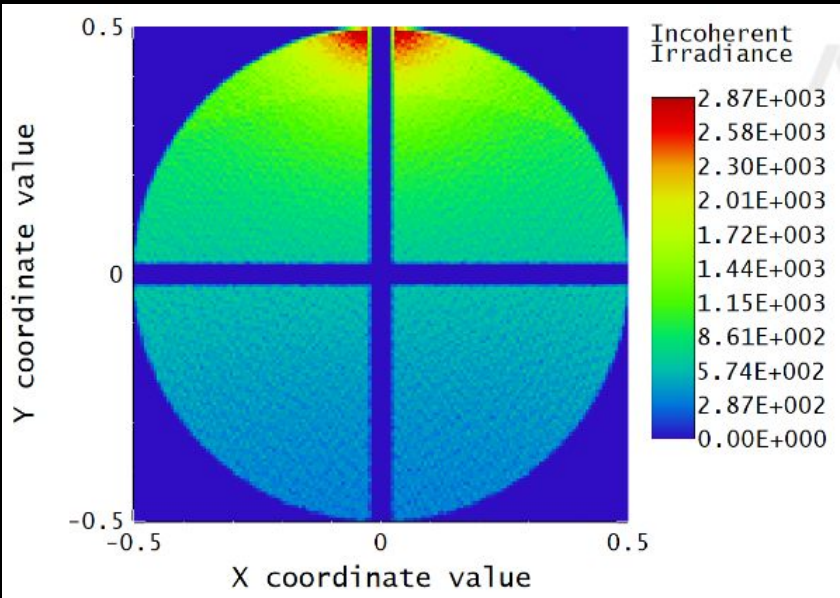
- ❑ FSM Diameter:
 - ❑ Capture entire OGS uplink beacon
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- ❑ FSM Tilt:
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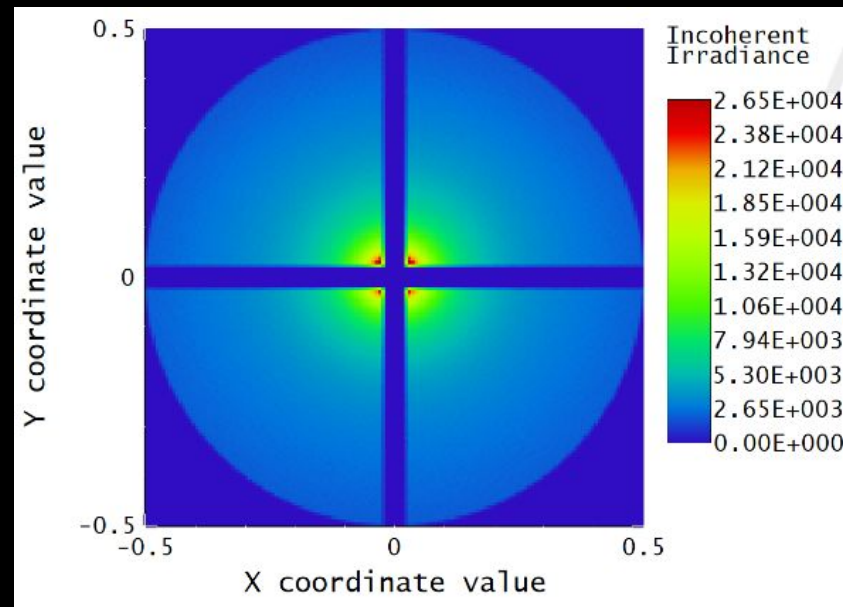
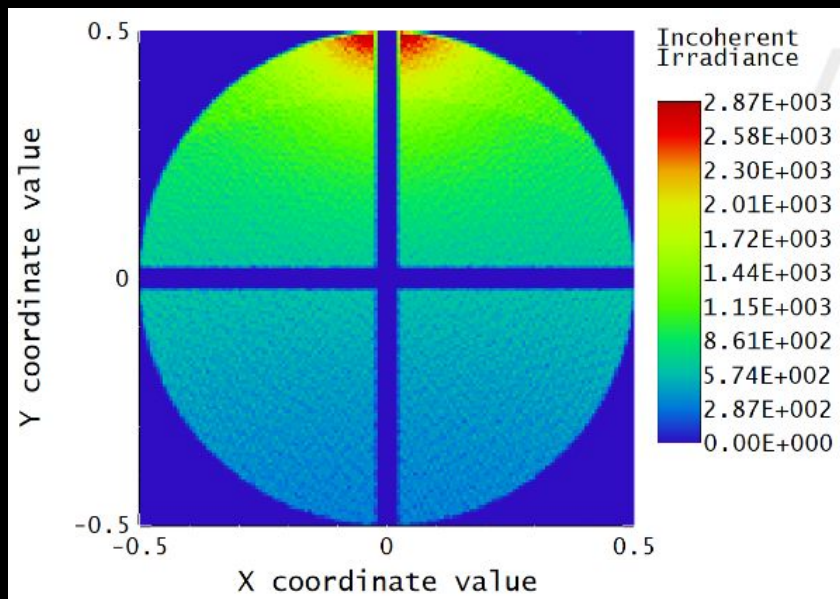
5mm Diameter
Product; Potential
clipping

6.4mm Diameter
Product; no
clipping

Payload Steering Assembly: FSM Choice Considerations

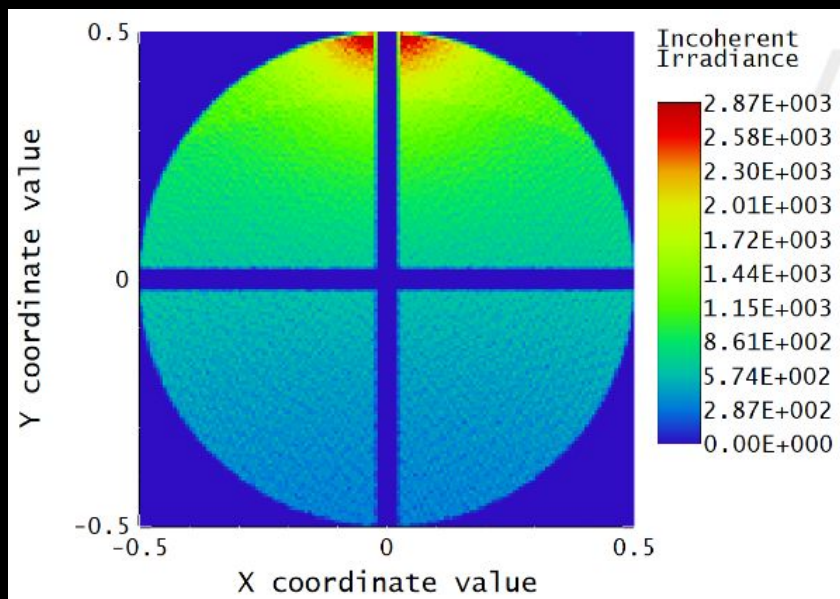


Payload Steering Assembly: FSM Choice Considerations

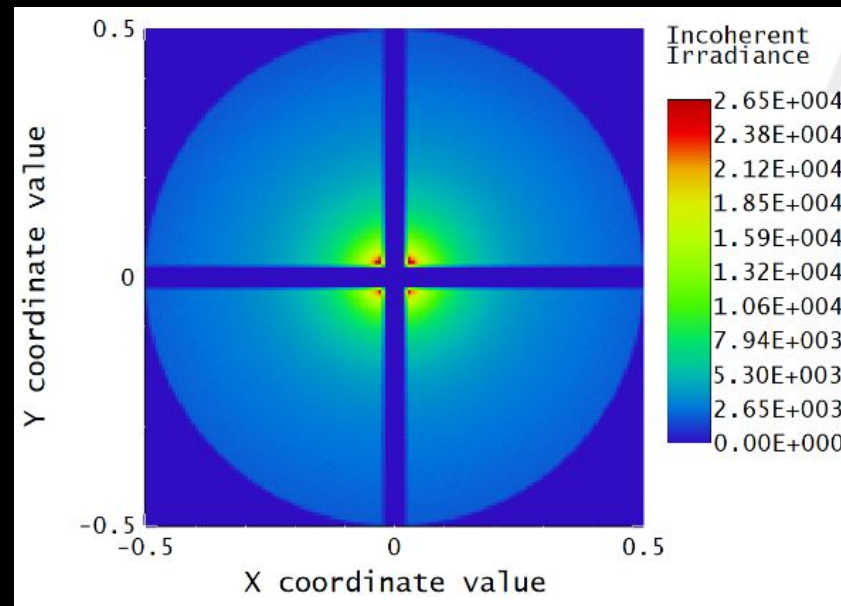


0.7° Coarse-Pointing
Error on QPD

Payload Steering Assembly: FSM Choice Considerations

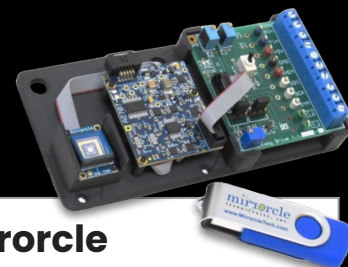


0.7° Coarse-Pointing
Error on QPD



2.4° FSM correction,
recenter on QPD

Payload Steering Assembly: FSM Driver

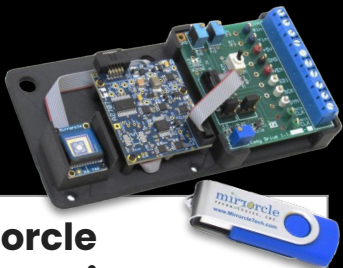


❑ Function:

- ❑ Drive supply voltage → FSM tilt
- ❑ **Mirror Compatibility:** Driver is vendor-matched to the Mirrorcle MEMS mirror.
- ❑ **SPI Control & Input Scaling:** Easy Drive 1.x potentiometers limit ± 10 V analog inputs to the mirror's maximum angular range (does not linearize response).
- ❑ **FCLK Frequency:** Potentiometer sets the update rate of high-voltage signals to ensure safe, accurate mirror operation.

FSM Driver Model	Mirrorcle MEMS Driver – Digital Input (DR-10-056-00)
Power Consumption (mW)	~100–150mW
Signal Bandwidth (Hz)	< 25kHz
Output Voltage (V)	0–200V
Supply Voltage (V)	5
Size (mm)	35x35x9

Payload Steering Assembly: FSM Driver

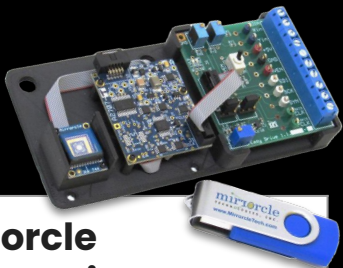


- ❑ **Function:**
 - ❑ Drive supply voltage → FSM tilt
 - ❑ **Mirror Compatibility:** Driver is vendor-matched to the Mirrorcle MEMS mirror.
- ❑ **Amplitude Scaling:** Easy Drive limit ± 10 V analog mirror's maximum (does not linearize response).
- ❑ **FCLK Frequency:** Potentiometer sets the update rate of high-voltage signals to ensure safe, accurate mirror operation.

FSM Driver supplies voltage for tilting the mirror.

FSM Driver Model	Mirrorcle MEMS Driver – Digital Input (DR-10-056-00)
Power Consumption (mW)	~100–150mW
Signal Bandwidth (Hz)	< 25kHz
Output Voltage (V)	0–200V
Supply Voltage (V)	5
Size (mm)	35x35x9

Payload Steering Assembly: FSM Driver



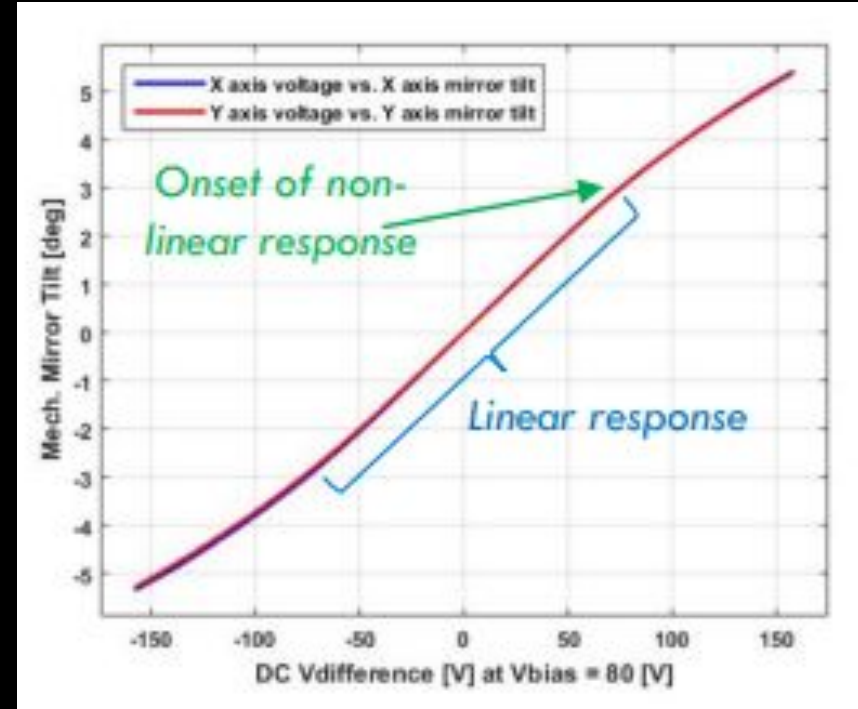
- ❑ **Function:**
 - ❑ Drive supply voltage → FSM tilt
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 - ❑ **SPI Control & Input Scaling:** Easy Drive 1.x potentiometers limit ± 10 V analog inputs to the mirror's maximum angular range (does not linearize response).
 - ❑ **FCLK** ...ter sets the up ...ge signal ...ate mirror opera

Size is appropriate for Payload layer.

FSM Driver Model	Mirrorcle MEMS Driver – Digital Input (DR-10-056-00)
Power Consumption (mW)	~100–150mW
Signal Bandwidth (Hz)	< 25kHz
Output Voltage (V)	0–200V
Supply Voltage (V)	5
Size (mm)	35x35x9

Payload Calibration Assembly

- ❑ FSM voltage response is linear within a 60% of the tilting range
- ❑ Non-linear fit for <80% tilt range ($\pm 3.4^\circ$)
- ❑ Calibrate FSM before PAT Sequence
 - ❑ Mapping the response while in orbit before doing a pass
 - ❑ Non-linear range expected to change with temperature and vibration

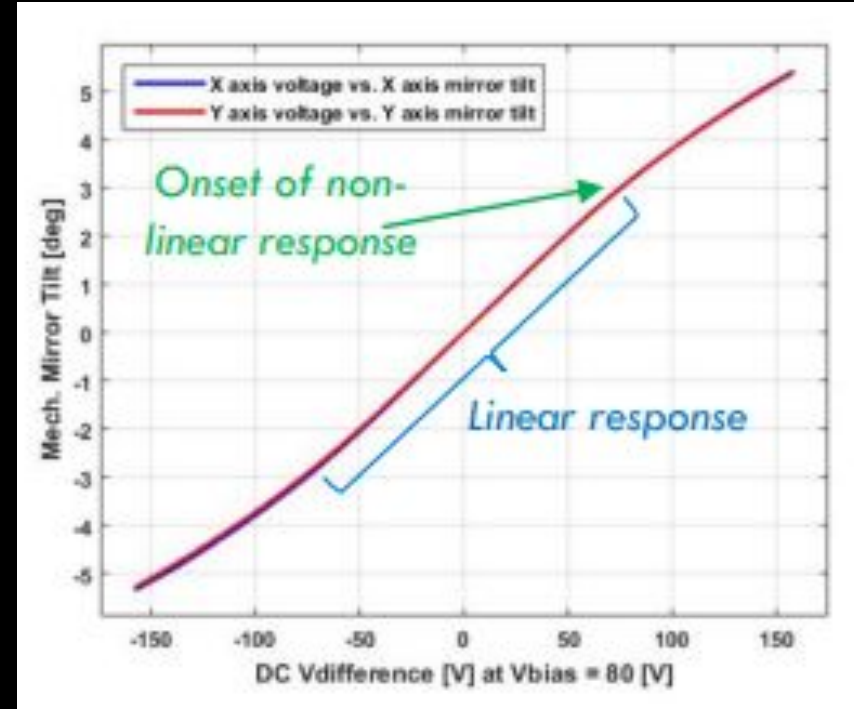


FSM Mechanical Tilt vs. Voltage Response

Payload Calibration Assembly

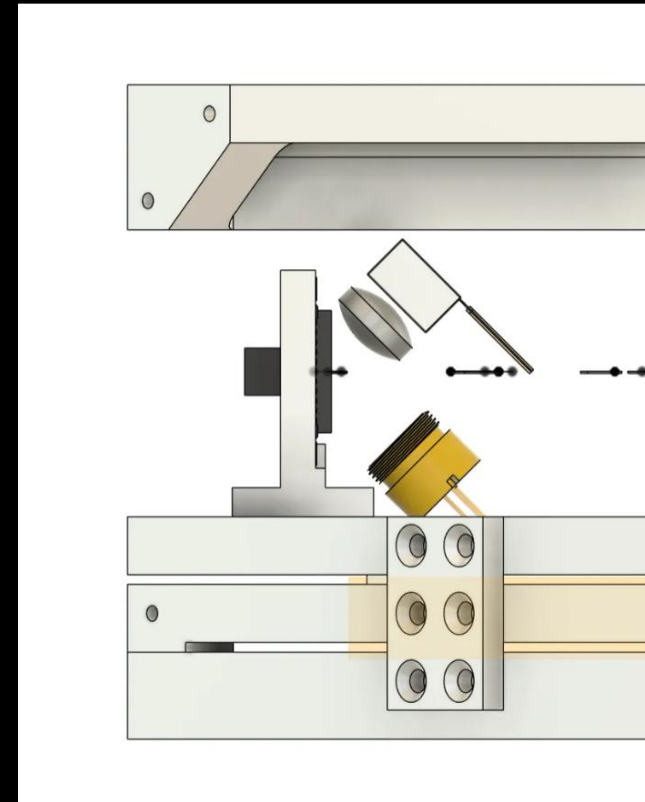
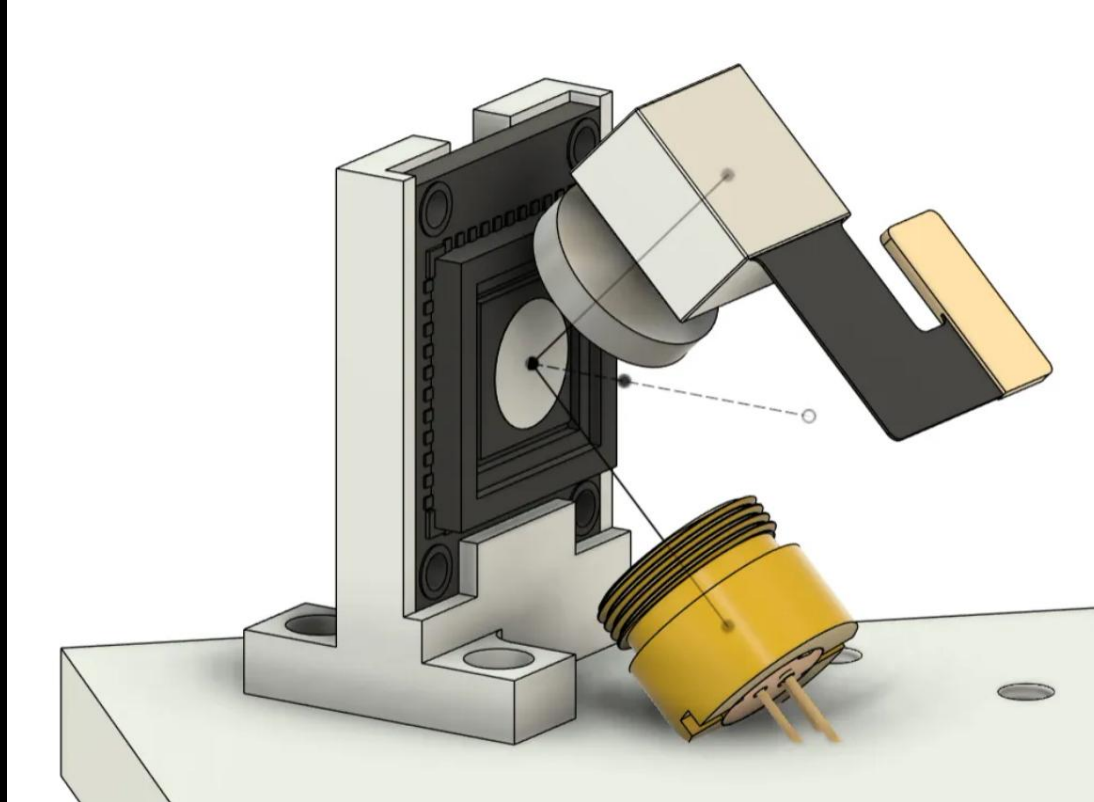
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- ❑ Non-linear fit for <80% tilt range ($\pm 3.4^\circ$)
- ❑ Calibrate FSM before PAT Sequence
 - ❑ Mapping the response while in orbit before doing a pass
 - ❑ Non-linear range expected to change with temperature and vibration

Calibration will map mechanical tilt to DC voltage difference in orbit before a pass.

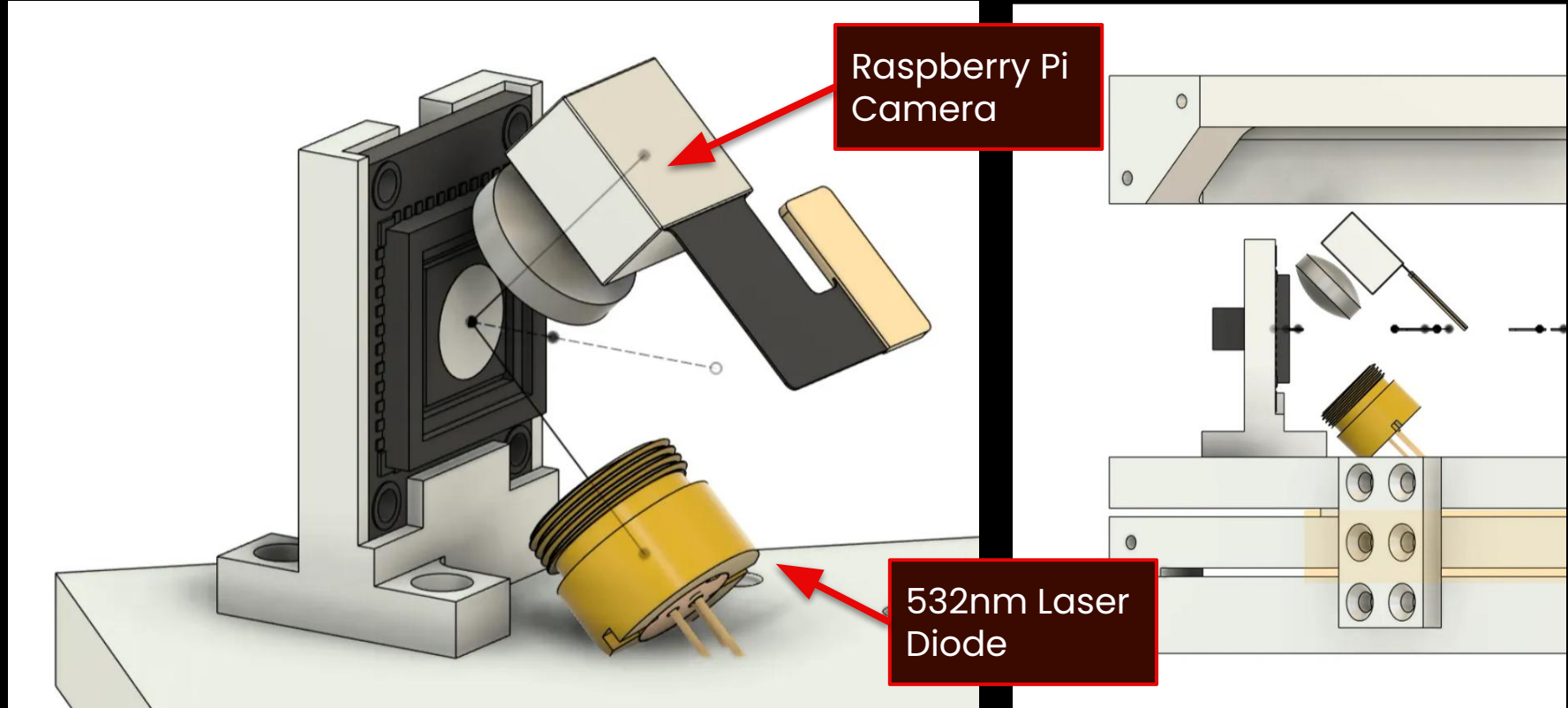


FSM Mechanical Tilt vs. Voltage Response

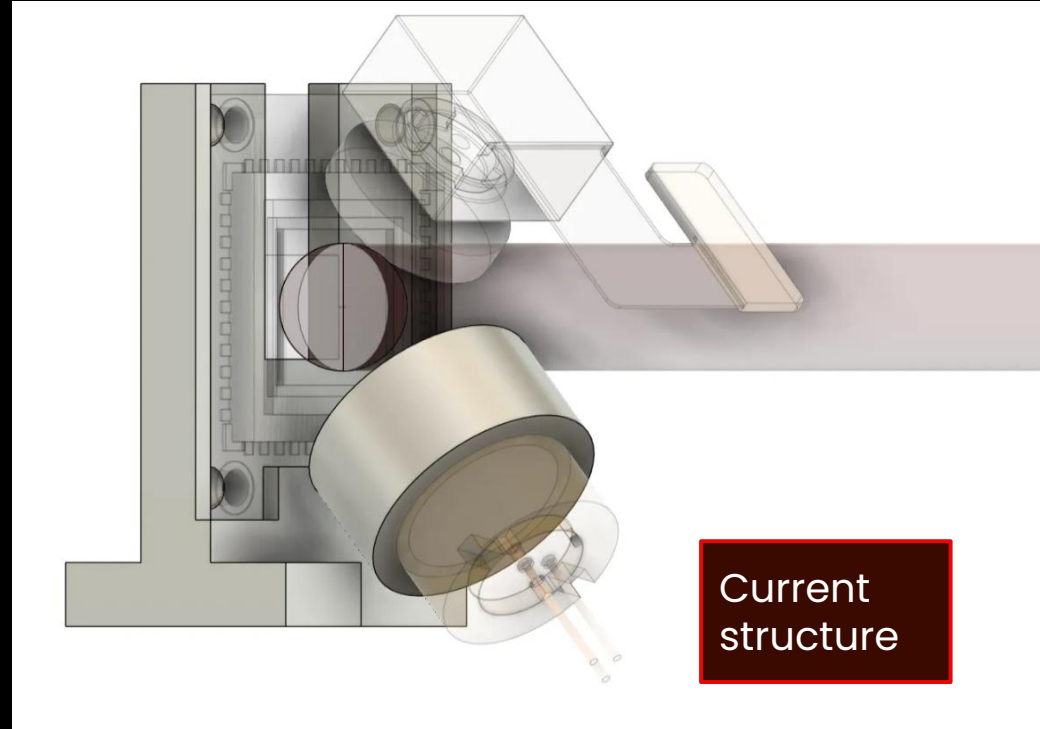
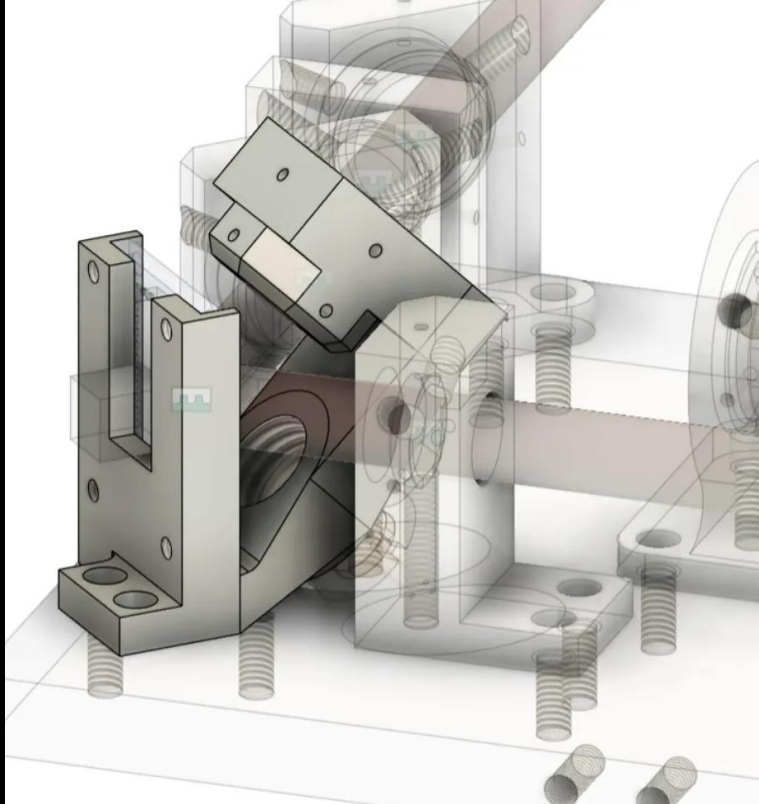
Payload Calibration Assembly



Payload Calibration Assembly

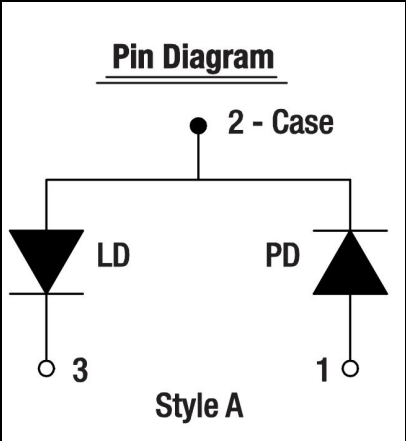


Payload Calibration Assembly



Payload Calibration Assembly: Laser Source

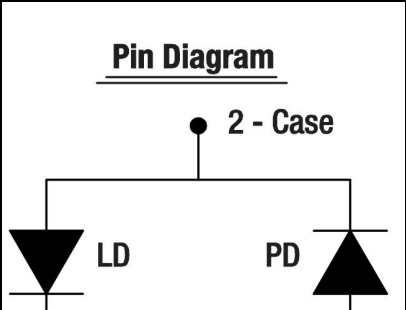
- ❑ **Function:** Reflect off FSM toward calibration camera, detect FSM mechanical tilt for some voltage response.



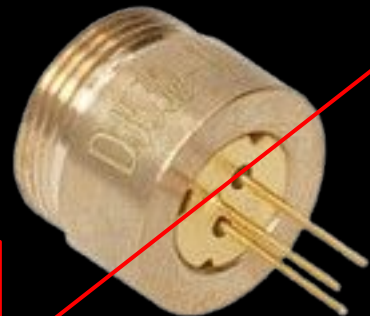
FSM Driver Model	Thorlabs DJ532-10
Wavelength	532nm
Full-Angle Divergence	12 mrad
Power	10mW

Payload Calibration Assembly: Laser Source

- ❑ **Function:** Reflect off FSM toward calibration camera, detect FSM mechanical tilt for some voltage response.



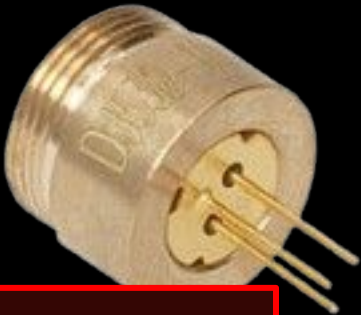
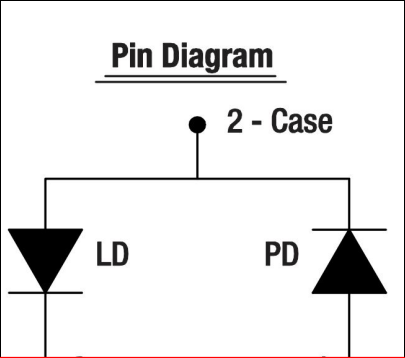
Chosen wavelength for calibration purposes only. Works for camera peak responsivity.



FSM Driver Model	Thorlabs DJ532-10
Wavelength	532nm
Full-Angle Divergence	12 mrad
Power	10mW

Payload Calibration Assembly: Laser Source

- ❑ **Function:** Reflect off FSM toward calibration camera, detect FSM mechanical tilt for some voltage response.

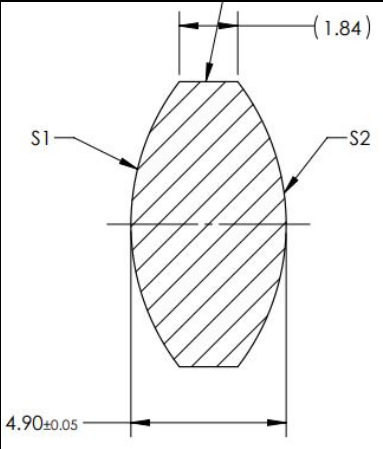


Challenges: operating temperatures are too strict (20 - 25 °C) and style of pins does not connect to selected laser driver.

FSM Driver Model	Thorlabs DJ532-10
Wavelength	532nm
Full-Angle Divergence	12 mrad
Power	10mW

Payload Calibration Assembly: Lens & Camera

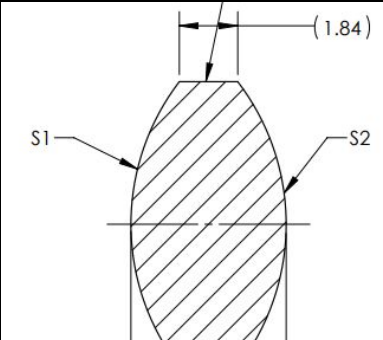
- ❑ **Biconvex Lens Function:** Control Calibration Beam & convert angular tilt → linear translation across calibration camera
- ❑ **Camera Function:** Obtain calibration beam centroid to translate to FSM mechanical tilt angle



Lens Model	EO #32-016
Radius	9mm
Effective Focal Length	13.5mm
Back Focal Length	12.56mm
Material	UV Fused Silica

Payload Calibration Assembly: Lens & Camera

- ❑ **Biconvex Lens Function:**
Control Calibration Beam & convert angular tilt → linear translation across calibration camera



- ❑ **Camera Function:**
calibration beam
translate to FSM mechanical
tilt angle

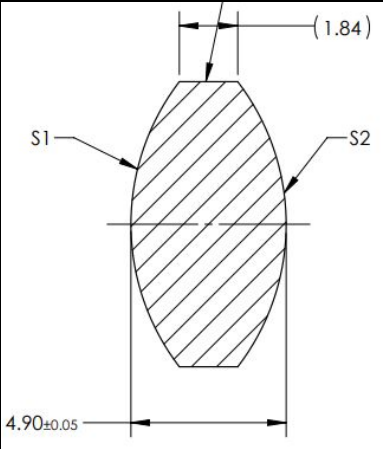
Lens focal length sufficient to circumvent FSM FOV & satisfy Payload size constraints



Lens Model	EO #32-016
Radius	9mm
Effective Focal Length	13.5mm
Back Focal Length	12.56mm
Material	UV Fused Silica

Payload Calibration Assembly: Lens & Camera

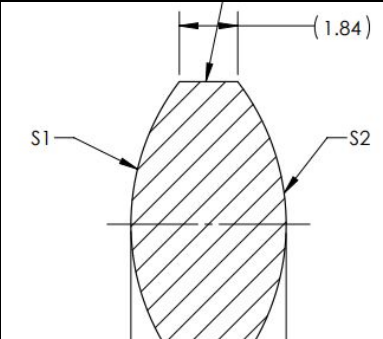
- ❑ **Biconvex Lens Function:** Control Calibration Beam & convert angular tilt → linear translation across calibration camera
- ❑ **Camera Function:** Obtain calibration beam centroid to translate to FSM mechanical tilt angle



Camera Model	Raspberry Pi Camera Module v1
Pixels	5 megapixels
Sensor	Omnivision OV5647
Sensor Image Area	3.76mm x 2.74mm
Pixel Size	1.4 μ m
Responsivity (at 532nm)	~90%
Focal Length	3.60±0.01mm

Payload Calibration Assembly: Lens & Camera

- ❑ **Biconvex Lens Function:**
Control Calibration Beam & convert angular tilt → linear translation across calibration camera



- ❑ **Camera Function:** Obtain calibration beam center, translate to FSM mechanical tilt angle

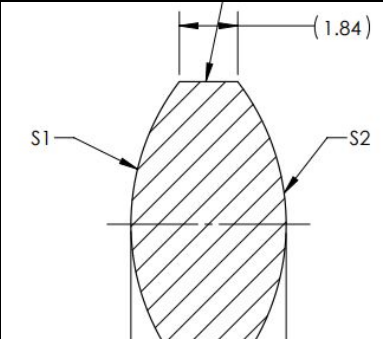
Camera Compatible with Calibration Source Wavelength



Camera Model	Raspberry Pi Camera Module v1
Pixels	5 megapixels
Sensor	Omnivision OV5647
Sensor Image Area	3.76mm x 2.74mm
Pixel Size	1.4 μ m
Responsivity (at 532nm)	~90%
Focal Length	3.60 $\pm$ 0.01mm

Payload Calibration Assembly: Lens & Camera

- ❑ **Biconvex Lens Function:** Control Calibration Beam & convert angular tilt → linear translation across calibration camera



- ❑ **Camera Function:** Obtain calibration beam center FSM tilt range (covered in optical analysis) translate to FSM mechanical tilt angle

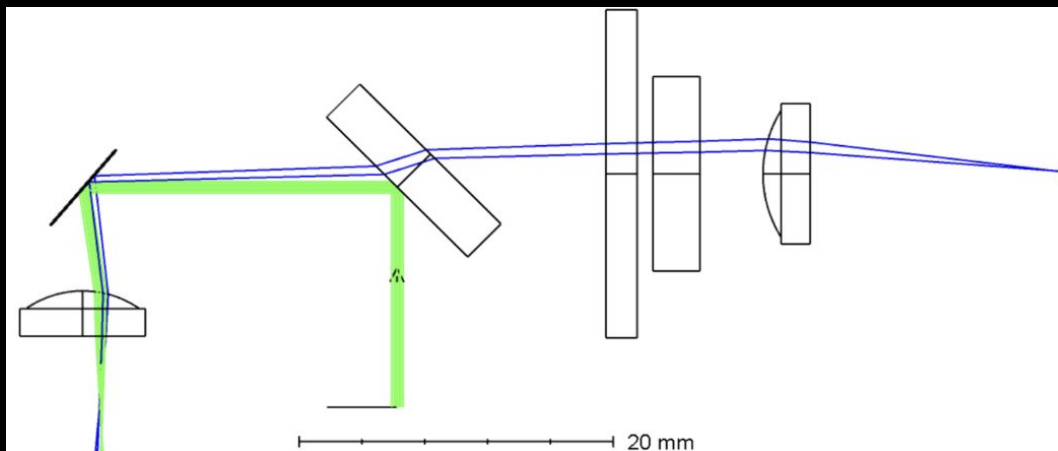
Camera capable of capturing full effective FSM tilt range (covered in optical analysis)



Camera Model	Raspberry Pi Camera Module v1
Pixels	5 megapixels
Sensor	Omnivision OV5647
Sensor Image Area	3.76mm x 2.74mm
Pixel Size	1.4 μ m
Responsivity (at 532nm)	~90%
Focal Length	3.60 $\pm$ 0.01mm

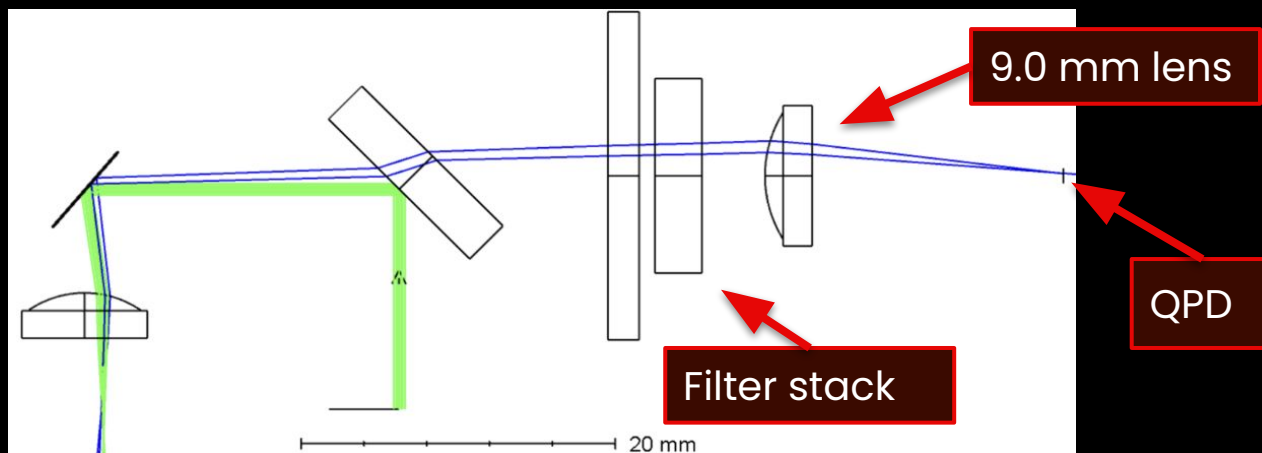
Payload: Detection Assembly

- ❑ The OGS beam (1064 nm) passes through a 1064 nm bandpass and 1250 nm shortpass filter stack, which transmits the beacon while rejecting out-of-band and 1550 nm light.
- ❑ A 9-mm plano-convex lens focuses the beam on the quadrant photodiode.
- ❑ The outputs are routed to a transimpedance amplifier board, where they are converted to voltages, filtered, and digitized.
- ❑ The digitized quadrant signals are sent to the Payload computer.



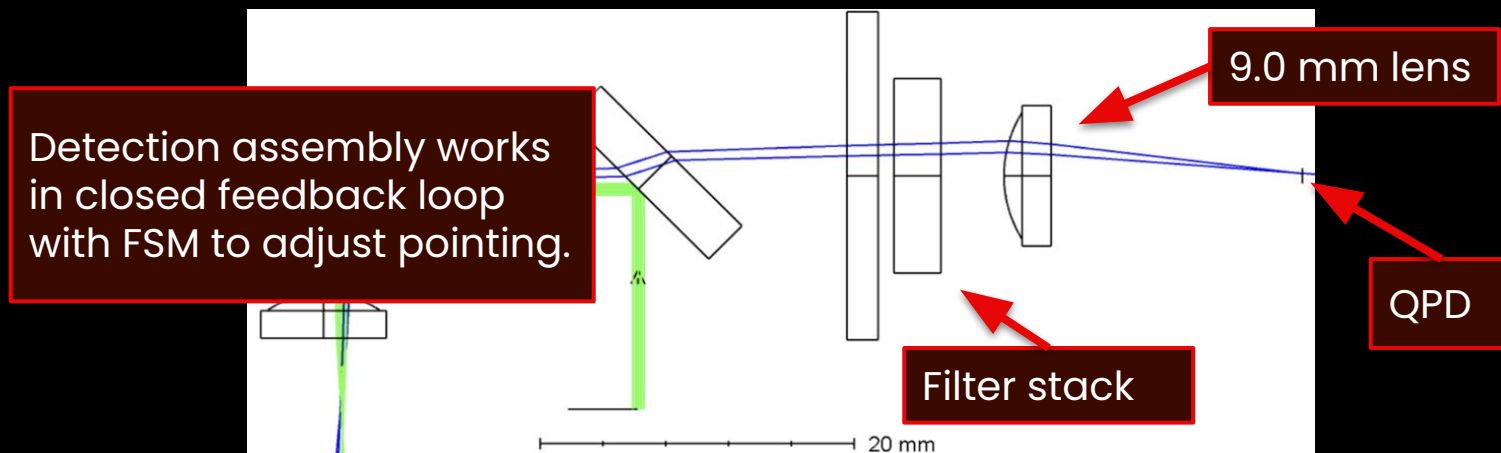
Payload: Detection Assembly

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- ❑ The outputs are routed to a transimpedance amplifier board, where they are converted to voltages, filtered, and digitized.
- ❑ The digitized quadrant signals are sent to the Payload computer.



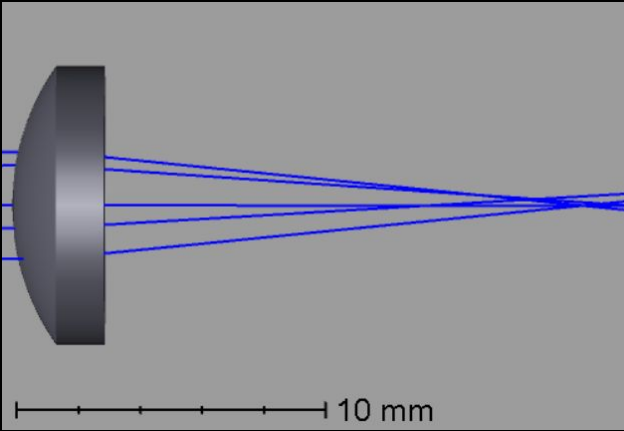
Payload: Detection Assembly

- ❑ The OGS beam (1064 nm) passes through a 1064 nm bandpass and 1250 nm shortpass filter stack, which transmits the beacon while rejecting out-of-band and 1550 nm light.
- ❑ A 9-mm plano-convex lens focuses the beam on the quadrant photodiode.
- ❑ The outputs are routed to a transimpedance amplifier board, where they are converted to voltages, filtered, and digitized.
- ❑ The digitized quadrant signals are sent to the Payload computer.



Payload Detection Assembly: Lens

Lens Model	Edmund Optics #49-876
Type:	Plano-convex
Diameter (mm):	9.0 mm
Effective Focal Length EFL (mm):	15.00 @ 587.6nm
Coating:	Uncoated



Function: Focus OGS beacon spot within smaller QPD active area.

Payload Detection Assembly: Quadrant Photodiode

Quadrant Photodiode (QPD):

Four pixel InGaAs detector used for ground station beacon laser detection to achieve fine pointing. The center of the four pixels represents correct optical path alignment.

Proved performance: Tested on GRACE for inter-satellite laser interferometry.

Certified Robustness: OSI Q1000 meets industrial standards including thermal cycling, physical analysis, and integrity checks

QPD Model	OSI; AFG91 FCI-INGAAS-Q1000
Active Area Diameter (mm)	1
Responsivity @1310n (A/W)	0.90
Responsivity @1550nm (A/W)	0.95
Element Gap (mm)	0.045
Crosstalk (1550 nm, 5.0V) (%)	1
NEP @1550nm (W/√Hz)	1.20E-14
Rise Time (ns)	3
Operating temperature (C°)	-40 to 70



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Responsivity @1550nm (A/W)	0.95	
Element Gap (mm)	0.045	
Crosstalk (1550 nm, 5.0V) (%)	1	Low crosstalk (1550 nm, 5.0 V) so quadrant cells are well-isolated.
NEP @1550nm (W/√Hz)	1.20E-14	
Rise Time (ns)	3	
Operating temperature (C°)	-40 to 70	





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Responsivity @1550nm (A/W)	0.95	
Element Gap (mm)	0.04	
Crosstalk (1550 nm, 5.0V) (%)	1	
NEP @1550nm (W/√Hz)	1.20E-14	Rise time compatible with high speed applications.
Rise Time (ns)	3	
Operating temperature (C°)	-40 to 70	



Payload Detection Assembly: Quadrant Photodiode

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Four pixel InGaAs detector used for ground station beacon laser detection to achieve fine pointing. The center of the four pixels represents correct optical path alignment.

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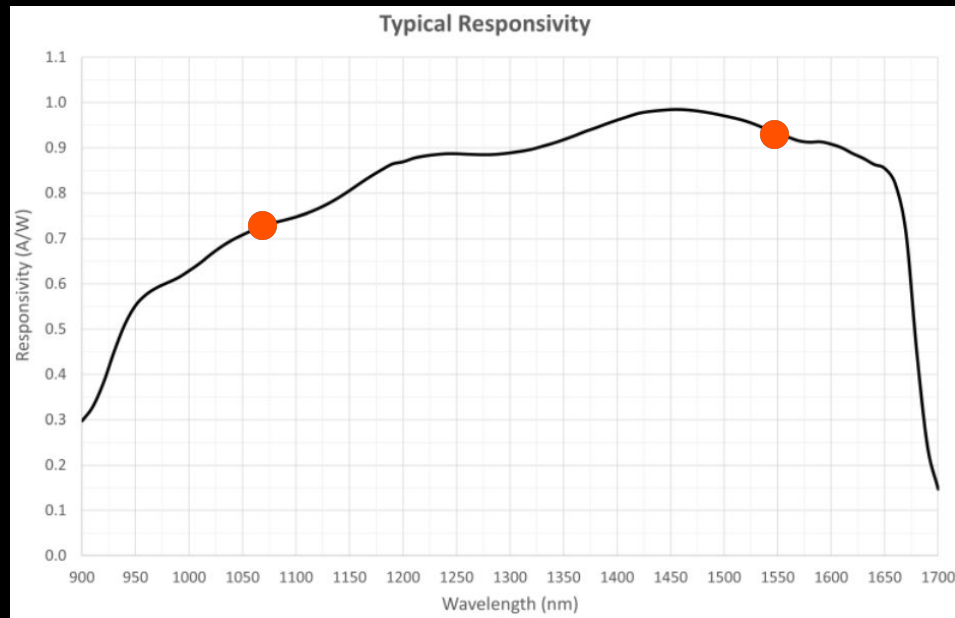
QPD Model	OSI; AFG91 FCI-INGAAS-Q1000
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Wavelength (nm)	0.90
Element Gap (mm)	0.95
Crosstalk (1550 nm, 5.0V) (%)	0.045
NEP @1550nm (W/√Hz)	1
Rise Time (ns)	1.20E-14
Operating temperature (C°)	3
	-40 to 70

More specifications on these requirements needs to be provided.



Payload Detection Assembly: QPD Responsivity Graphs

- ❑ QPD has good responsivity at 1064 nm (~ 0.75 A/W)
- ❑ Responsivity is higher near 1550 nm (~ 0.95 A/W) which is downlink wavelength so any 1550 nm back-reflections would also produce a strong signal on the QPD

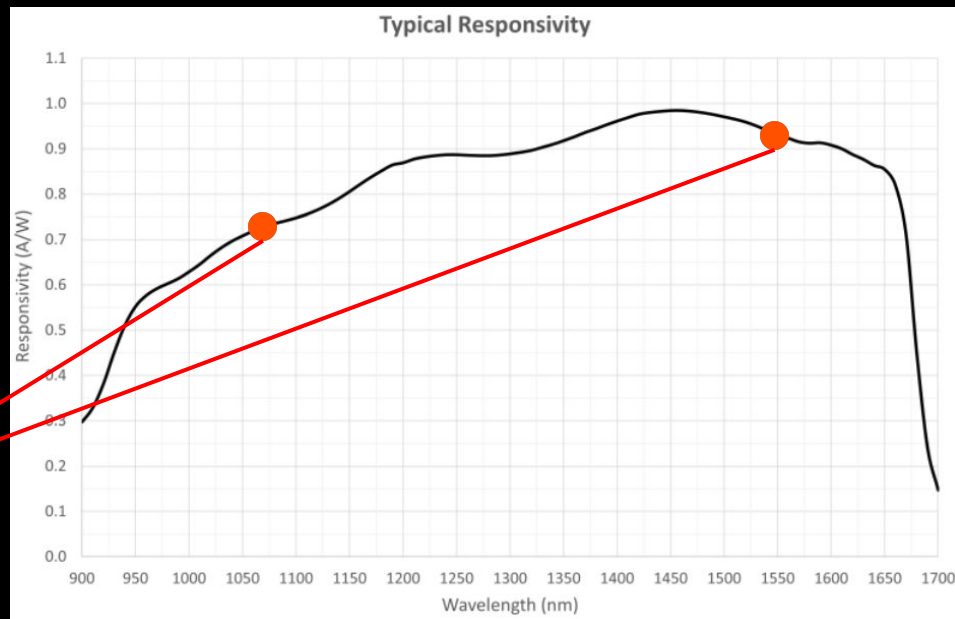


typical responsivity profile for OSIO model
FCI-InGaAs-Q1000 provided by OSI
optoelectronics

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The QPD exhibits peak responsivity near 1064 nm and 1550 nm; as a result, back-reflections can introduce ambiguity in wavelength detection.



typical responsivity profile for OSIO model FCI-InGaAs-Q1000 provided by OSI optoelectronics

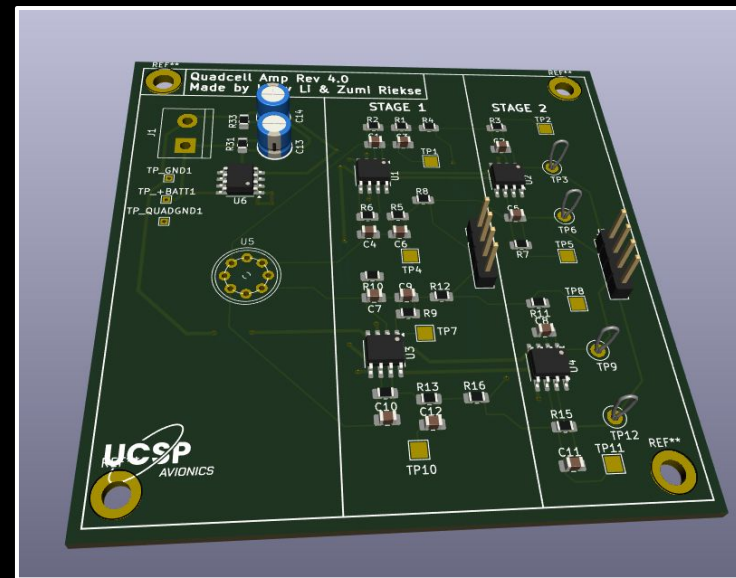
Payload Detection Assembly: QPD Amplifier Design

The amplifier is made based off of a design by MIT's CLICK.

The design consists of a two-stage transimpedance amplifier from the quadcell's signal.

- ❑ The first stage is purely a transimpedance amplifier
- ❑ The second stage provides further amplification and a low pass filter to avoid noise in the reading

An ADC with an I2C interface relays the amplified voltages to the PLCM



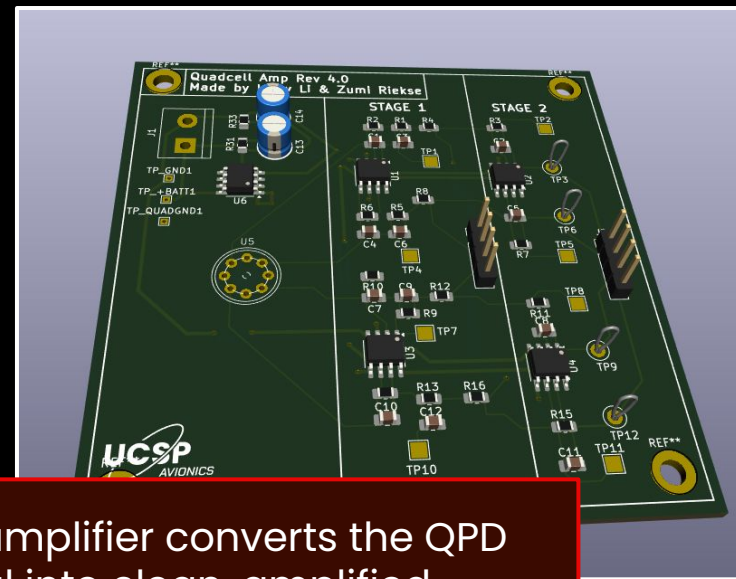
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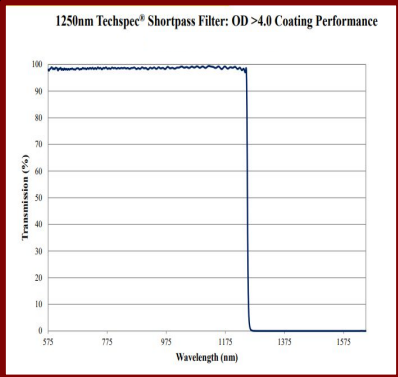
The amplifier converts the QPD signal into clean, amplified voltages with filtering.

Payload Detection Assembly: Filters

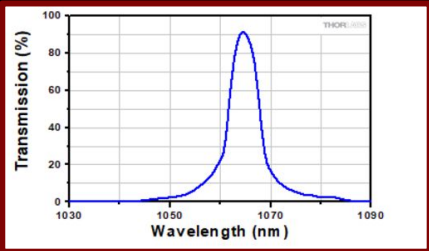
Bandpass Filter: FLH1064-8

Company	Thorlabs
Center Wavelength (nm)	1064 (T > 90%)
FWHM Bandwidth (nm)	8
Blocking Regions (nm)	200-1039, 1089-1200
TWE	$\lambda/4$
Outer Diameter (mm)	25.0
Surface Quality (SD)	60-40

Shortpass Transmission



Bandpass Transmission



Shortpass Filter: #89-671

Company	Edmund Optics
Cutoff Wavelength (nm)	1250
Rejection Wavelength (nm)	1290-1650
Transmission Wavelength (>91%)	575-1235
Angle of Incidence	0°
Diameter (mm)	12.5 +/- 0.2
Surface Quality (SD)	60-40 SD==

Payload Detection Assembly: Filters

Bandpass Filter: FLH1064-8

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Center Wavelength (nm)	1064 (T > 90%)
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TWE	$\lambda/4$
Outer Diameter (mm)	25.0
Surface Quality (SD)	60-40

Shortpass Transmission

1550 nm Technical Shortpass Filter QD-S-L0 Coating Performance

Passes only ground station beacon, capturing majority of laser bandwidth

Bandpass wavelength does not extend to light above 1200 nm. Blocks 1550 nm light to filter out potential back-reflections from transmission beam.

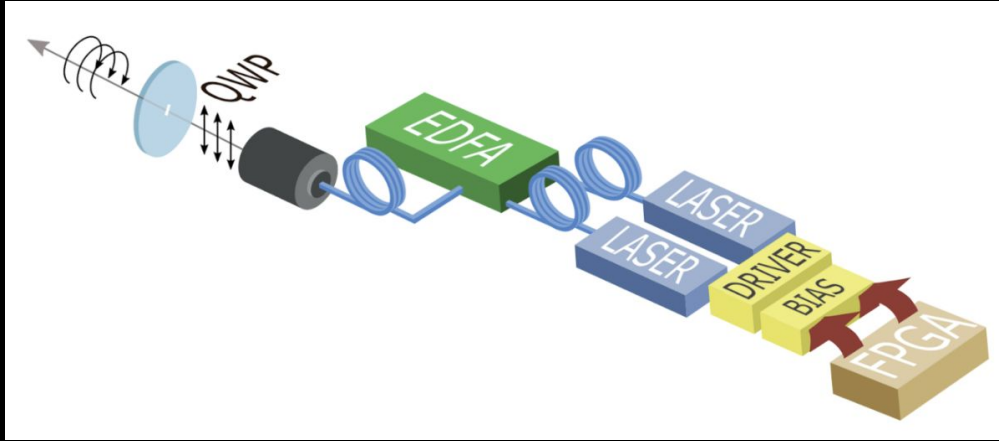
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Transmission Wavelength (>91%)	575-1235
Angle of Incidence	0°
Diameter (mm)	12.5 +/- 0.2
Surface Quality (SD)	60-40 SD==

Payload Recap: Steering and Detection Assemblies

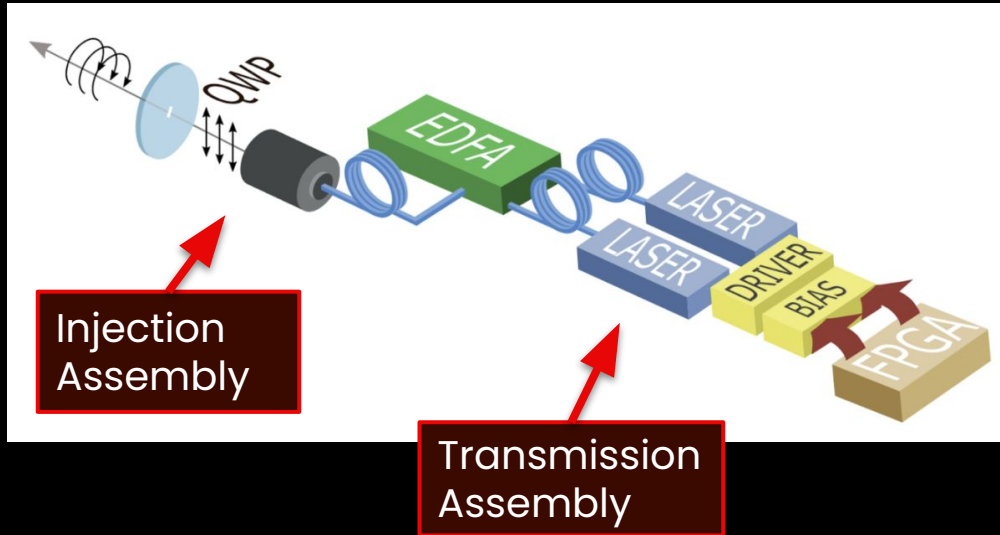
- ❑ Steering and Detection together are responsible for fine-pointing capabilities of the Payload.
- ❑ Steering the beam to the center of the QPD ensures that the minimal angle between the OGS beacon and Payload transmission lasers can be achieved given an initial misalignment.
- ❑ More information on pointing analysis is the Optical Analysis and PAT sequence.

Payload Transmission Assembly



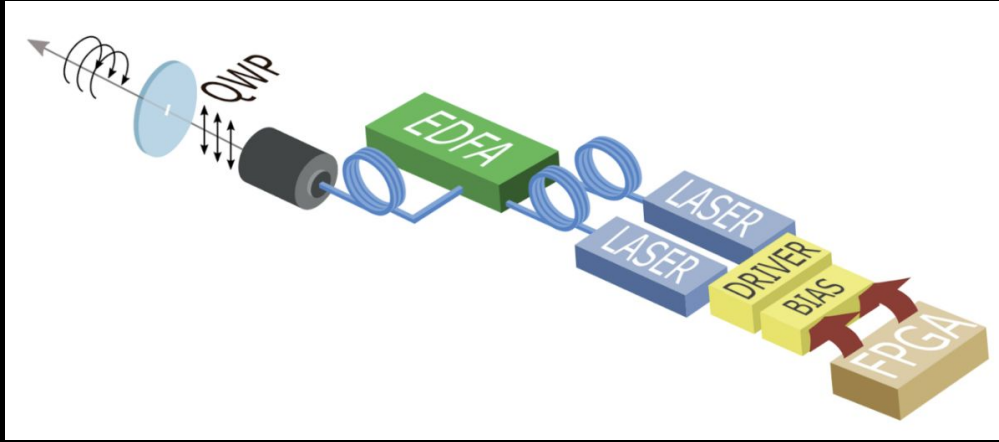
- ❑ The Payload transmits data using CPoSK.
- ❑ The final design has two sources of 1550 nm seed lasers with orthogonal linear polarizations.
- ❑ An FPGA drives one laser at a time via a bias circuit, producing a modulated linearly polarized signal at 10MHz, which is amplified by an erbium-doped amplifier (EDFA).
- ❑ Lasers outputs are connected via a fiber combiner.

Payload Transmission Assembly



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Payload Transmission Assembly



Generates ON/OFF signals of orthogonally linear polarized light at 10 MHz.

- ❑ The Payload transmits data using CPoSK.
- ❑ The final design has two sources of 1550 nm seed lasers with orthogonal linear polarizations.
- ❑ An FPGA drives one laser at a time via a bias circuit, producing a modulated linearly polarized signal at 10MHz, which is amplified by an erbium-doped amplifier (EDFA).
- ❑ Lasers outputs are connected via a fiber combiner.

Payload Transmission Assembly: FPGA

The Payload FPGA SoC serves as the payload controller, handling the low-latency outputs for laser driving, as well as the closed feedback loop for the QPD FSM.

A Mercury XU5 SoC module suits our needs due to the following features:

- ❑ ARM microprocessor (Cortex-A53).
- ❑ Small form factor.
- ❑ 5V single power supply.
- ❑ Design-in kit available for more efficient development.
- ❑ Communication with OBC via CAN.

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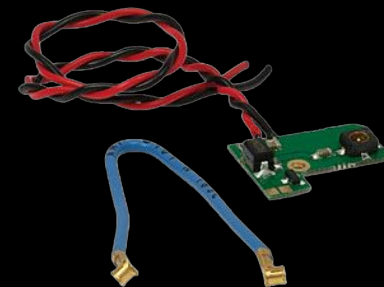
The Mercury XU5 FPGA SoC controls lasers outputs and QPD FSM feedback.

Payload Transmission Assembly: LD1101 + T1G Bias-T

- ❑ We use this commercial pair (LD1101 + T1G) as a **validated development reference** while we prototype our custom board.
- ❑ Thorlabs designed these two modules to **work as a matched system**
 - ❑ Interface impedance, grounding, and protection networks are all mutually compatible between the LD1101 and T1G.
- ❑ Using this off-the-shelf configuration gives us:
 - ❑ Known-good operating conditions
 - ❑ A stable platform to characterize seed lasers
 - ❑ Benchmark data for our integrated design



LD1101 Constant Power Laser Diode (LD)
Driver



T1G Bias-T

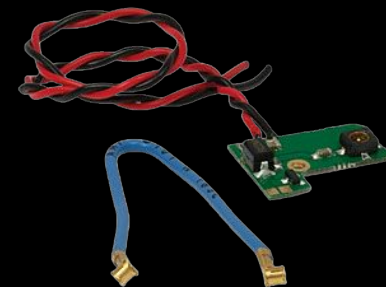
Payload Transmission Assembly: LD1101 + T1G Bias-T

These boards are a test reference, not flight hardware. We are redesigning them into a compact form-factor suitable for our spacecraft.

- ❑ Interface impedance, grounding, and protection networks are all mutually compatible between the LD1101 and T1G.
- ❑ Using this off-the-shelf configuration gives us:
 - ❑ Known-good operating conditions
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LD1101 Constant Power Laser Diode (LD) Driver



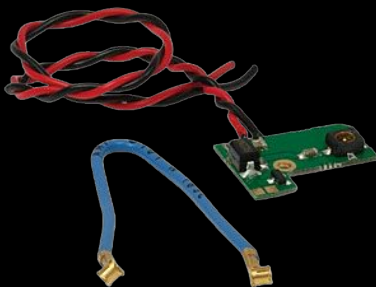
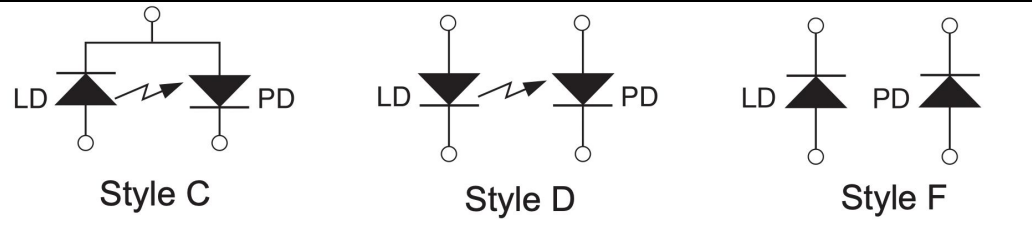
T1G Bias-T

Payload Transmission Assembly: LD1101 + T1G Bias-T

Laser Driver Model	Thorlabs LD1101
Operating Voltage	8 to 12 VDC
Operating Mode	Constant-Power (photodiode feedback)
Output Current (mA)	0 to 250
Style Configurations	D,C,F



LD1101 Constant Power Laser Diode (LD) Driver



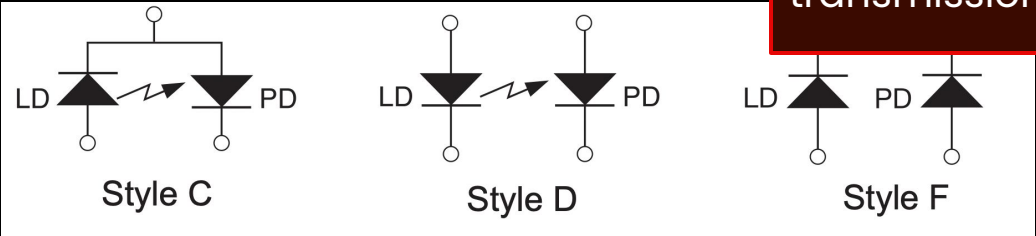
T1G Bias-T

Payload Transmission Assembly: LD1101 + T1G Bias-T

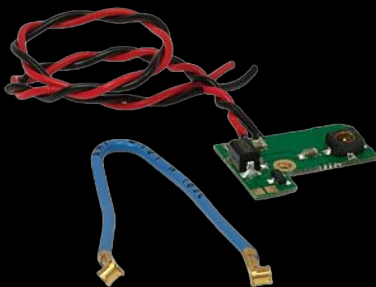
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Supports pin style of transmission laser (F)



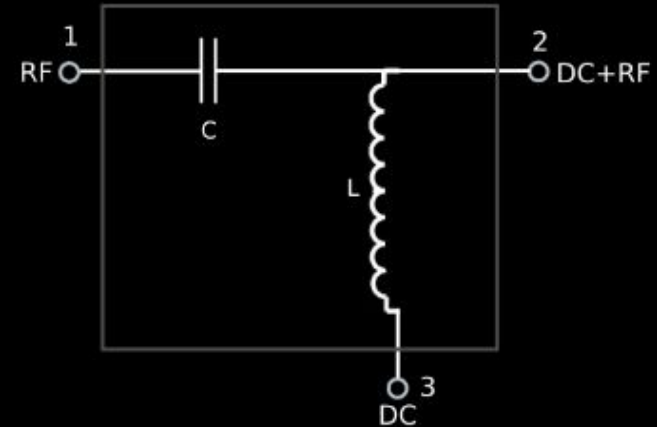
LD1101 Constant Power Laser Diode (LD) Driver



T1G Bias-T

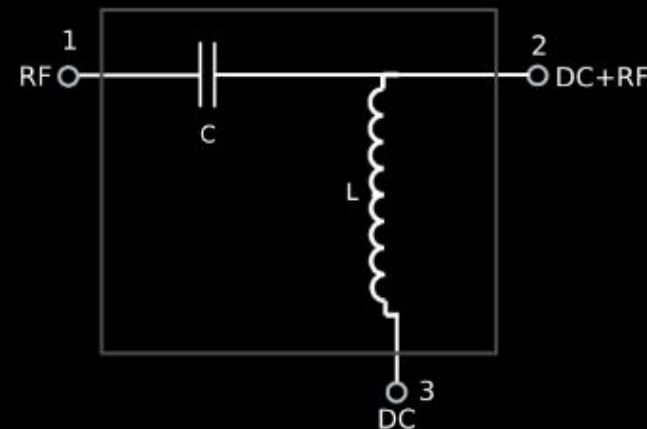
Payload Transmission Assembly: LD1101 + T1G Bias-T

- ❑ **Model:** Thorlabs T1G – Laser Diode Bias-T PCB
- ❑ **Function:** Provides modulation current that will be superimposed on laser driver current, allowing lasers to be modulated on/off at 10kHz–1GHz. Laser driver alone cannot provide modulation, so bias-T is required to modulate at desired frequency.
- ❑ **Testing:** Connected in parallel with laser driver and laser diode, with OSMT (blue wire) connected to function generator (FPGA).



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Bias-T injects high-frequency modulation onto the LD's DC bias without disturbing the constant-power loop.

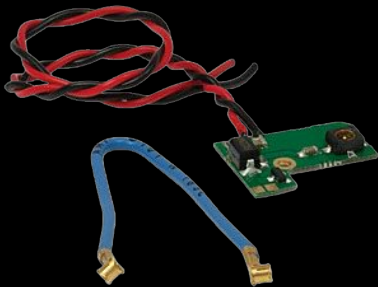


Payload Transmission Assembly: LD1101 + T1G Bias-T

Bias-T Model	Laser Diode Bias-T PCB
DC Current (Max)	110 mA
RF Current (Max)	50 mA (RMS)
Modulation speed	10kHz - 1GHz



LD1101 Constant Power Laser Diode (LD) Driver



T1G Bias-T

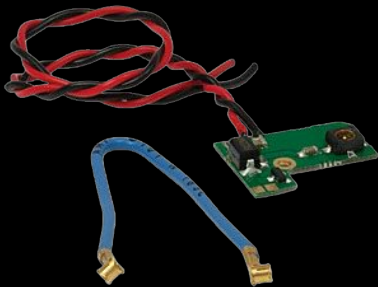
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DC Current (Max)	110 mA
RF Current (Max)	50 mA (RMS)
Modulation speed	10kHz - 1GHz

Supports modulation speed of 10MHz.



LD1101 Constant Power Laser Diode (LD) Driver



T1G Bias-T

Payload Transmission Assembly: Issues with Set Up

❑ **Form factor**

- ❑ Unnecessary duplicated circuitry across the COTS boards
- ❑ Physical footprint does not meet chassis constraints

❑ **Connections**

- ❑ OSMT connectors are fragile, hard to route, and difficult to interface with other hardware
- ❑ Excess wiring increases assembly complexity and failure risk

❑ **Telemetry Limitations**

- ❑ Temperature
- ❑ Drive Current

Payload Transmission Assembly: Issues with Set Up

❑ **Form factor**

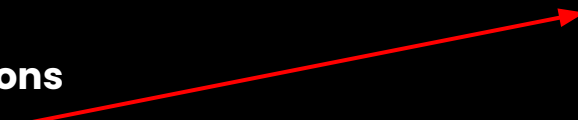
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❑ **Telemetry Limitations**

- ❑ Temperature
- ❑ Drive Current



Laser power decreases with rising diode temperature. If output falls below the EDFA input threshold, the transmitter may fail to produce enough optical power for the downlink.

Payload Transmission Assembly: Issues with Set Up

❑ **Form factor**

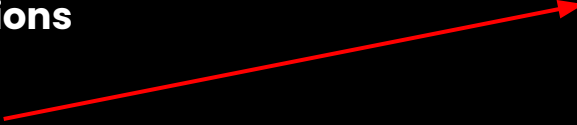
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- ❑ Excess wiring increases assembly complexity, failure risk

❑ **Telemetry Limitations**

- ❑ Temperature
- ❑ Drive Current



Drive current monitoring detects overcurrent events and acts as a proxy for diode health and aging, which enables early fault detection.

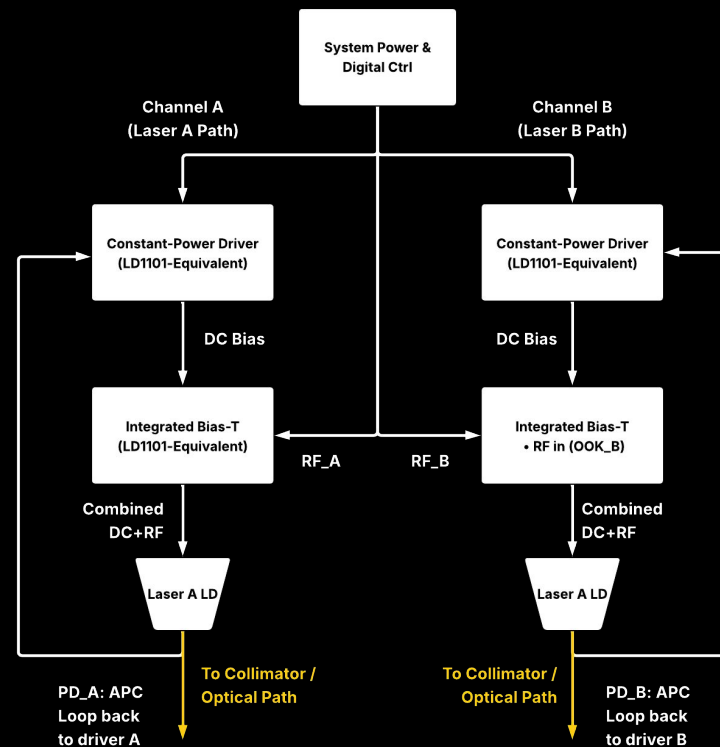
Payload Transmission Assembly: Planned Revisions

Goal: We aim to integrate LD1101 + T1G functionality into a single, flight-ready board with reduced SWaP, reduced complexity, and controlled interfaces.

Next Steps

- ❑ Simplify and remove unnecessary duplicated circuitry
- ❑ Map form-factor constraints
- ❑ Scope control: no new features until basic functionalities are proven

Block-level architecture (conceptual)



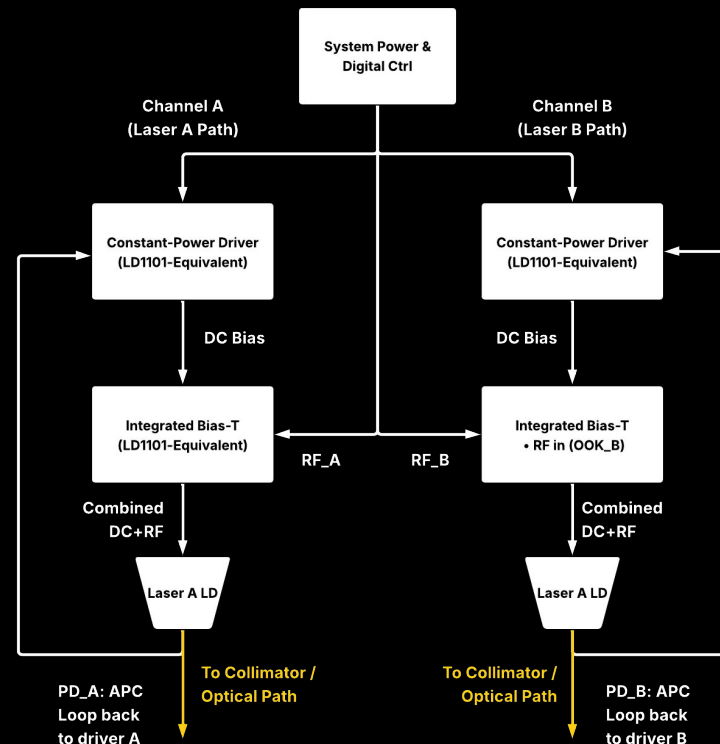
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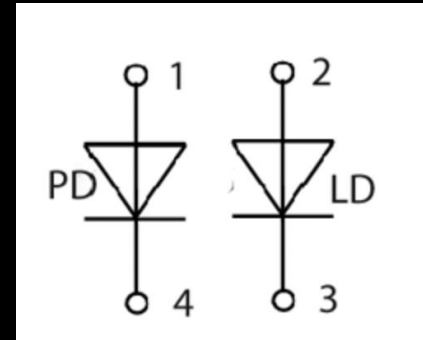
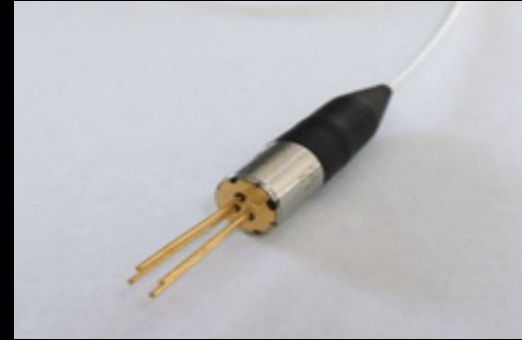
- ❑ Simplify and remove unnecessary duplicated circuitry
- ❑ Map form-factor constraints
- ❑ Initial scope control: no new features until basic functionalities are proven
- ❑ Ideal system: conglomerated board with temperature and current monitoring telemetry

Block-level architecture (conceptual)



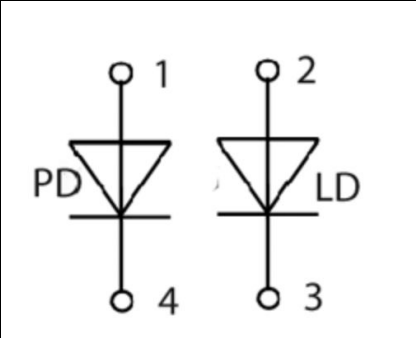
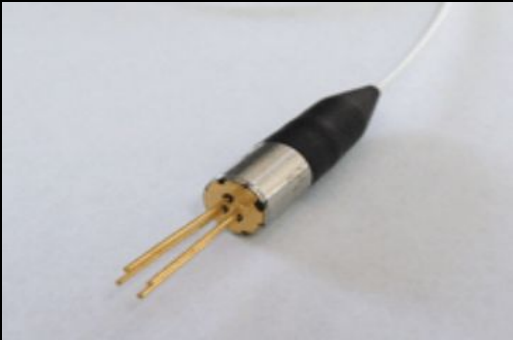
Payload Transmission Assembly: Lasers

- ❑ **Model:** QPhotonics QDFBLD-1550-5AX (Wavelength stabilized PM mode fiber coupled laser diode 5mW @ 1550nm)
- ❑ **Function:** Outgoing orthogonally linearly polarized fiber coupled transmission lasers at 1550nm, to be received at OGS.
- ❑ **Testing:** Pins connected to laser driver (LD1101) and bias-T (T1G). Fiber connected to APD, which is connected to an oscilloscope to monitor modulation frequency and power output.



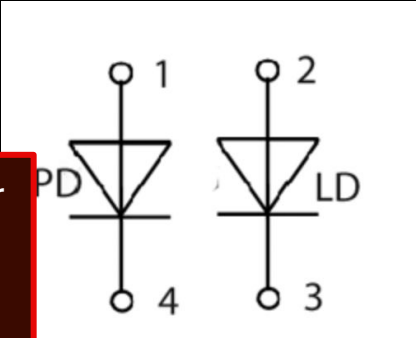
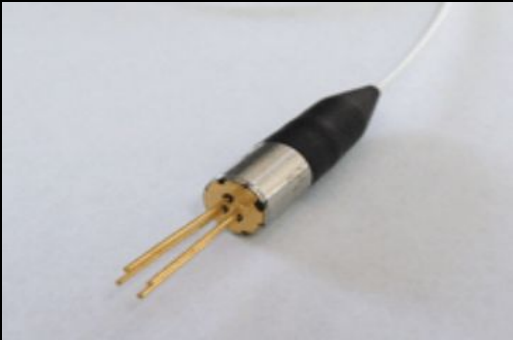
Payload Transmission Assembly: Lasers

Transmission Laser Model	QPhotonics QDFBLD-1550-5AX
Wavelength (nm)	1550
Operating Case Temperature (°C)	0 to 60
Wavelength v/s temperature coefficient (nm/°C)	0.2
Optical power from pigtail	5
Pin Style	F
Fiber type	PM



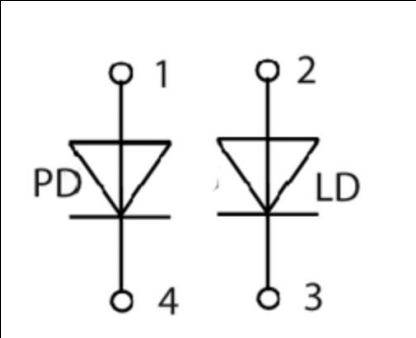
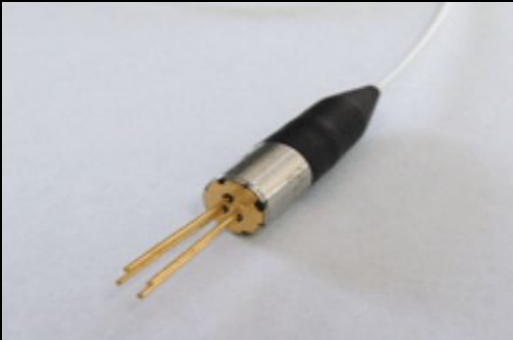
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Transmission Laser Model	QPhotonics QDFBLD-1550-5AX	
Wavelength (nm)	1550	
Operating Case Temperature (°C)	0 to 60	
Wavelength v/s temperature coefficient (nm/°C)	0.2	
Optical power from pigtail (mW)	5	Meets the minimal power required to be amplified by the EDFA.
Pin Style	F	
Fiber type	PM	



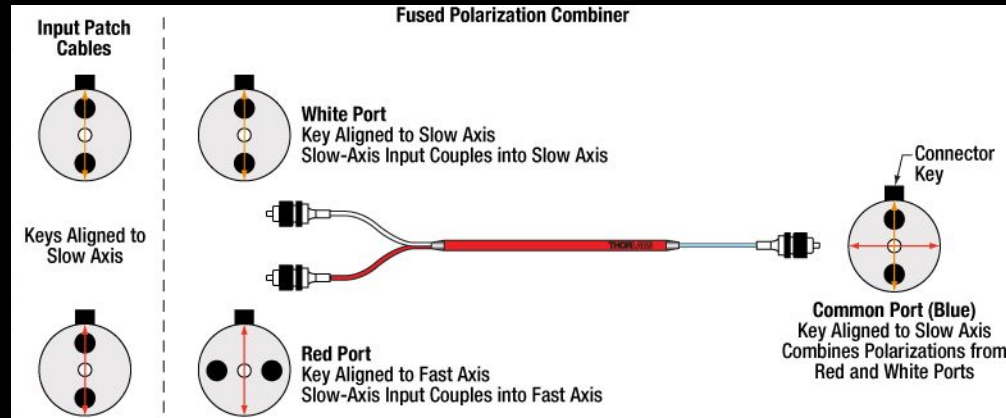
Payload Transmission Assembly: Lasers

Transmission Laser Model	QPhotonics QDFBLD-1550-5AX	
Wavelength (nm)	1550	
Operating Case Temperature (°C)	0 to 60	
Wavelength v/s temperature coefficient (nm/°C)	0.2	
Optical power from pigtail	5	
Pin Style	F	
Fiber type	PM	PM fiber coupled.



Payload Transmission Assembly: 2x1 PM Combiner

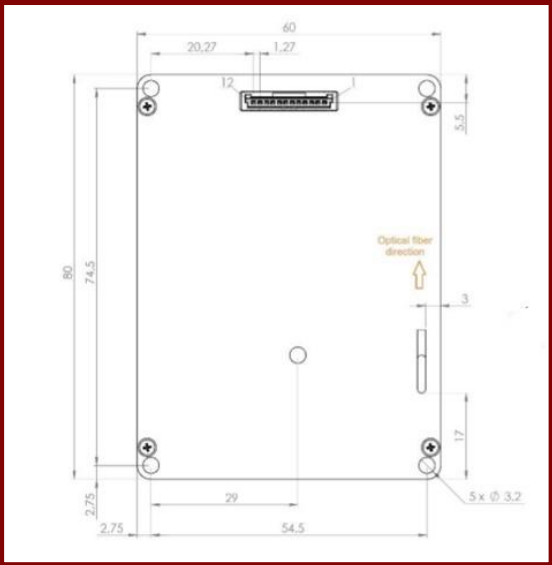
- ❑ **Model:** Thorlabs PFC1550A Polarization Fiber Combiner
- ❑ **Function:** Combines signal from both transmission lasers into a single output.
- ❑ The input patch cables from the laser have their keys aligned to the slow axis. At the combiner, the white port supports a slow-axis to slow-axis coupling, whereas the red port supports a slow-axis to fast-axis coupling.



Payload Transmission Assembly: EDFA

Amplifies power generated by seed laser to ~250 mW such that it can be detected by the APDs at the OGS.

Company	BKtel Photonics
Model	Compact Booster Amplifier Module
Wavelength (nm)	1540-1565
Power Output (dBm)	24
Polarization Dependent Gain (PDG)	3-5%
Noise figure (dBm)	8.5
Polarization	Linear
Return Loss (dBm)	40
Dimensions (mm)	60 x 80 x 12

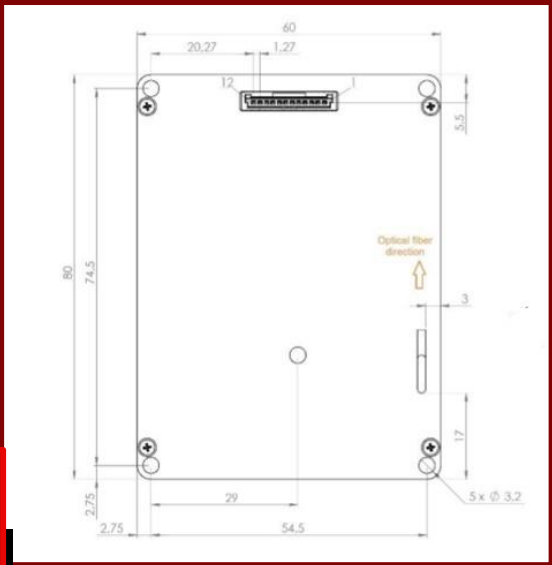


Payload Transmission Assembly: EDFA

Amplifies power generated by seed laser to ~250 mW such that it can be detected by the APDs at the OGS.

Company	BKtel Photonics
Model	Compact Booster Amplifier Module
Wavelength (nm)	1540-1565
Power Output (dBm)	24
Polarization Dependent Gain (PDG)	3-5%
Noise figure (dBm)	8.5
Polarization	Linear
Return Loss (dBm)	40
Dimensions (mm)	60 x 80 x 12

We have requested a product of PER > 20 dB. We have been warned it may be <20 dB PER.



Payload Transmission Assembly: PM Fiber Space Connectors

Connectors Model	TBD
Low loss ferrules	0.1 dB Grade ferrules with diameter $< 0.2 \mu\text{m}$
ACA (active core alignment) with ultra low eccentricity	$< 0.15 \mu\text{m}$
Exit angle	$< 0.6^\circ$
Active Polarization Orientation (APO)	$< \pm 2^\circ$

- ❑ For PM fibers connections between EDFA, lasers, combiners:
- ❑ Currently looking into models.
 - ❑ DIAMOND PM Fiber connectors:
 - ❑ Polarization maintaining devices/applications.
 - ❑ Space conditions.



Payload Transmission Assembly: PM Fiber Space Connectors

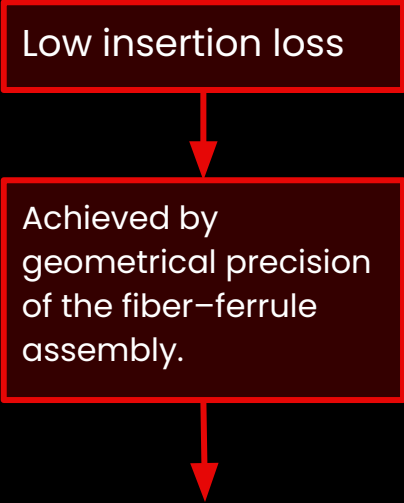
Connectors Model	TBD
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Exit angle	< 0.6°
Active Polarization Orientation (APO)	< $\pm 2^\circ$

Low insertion loss



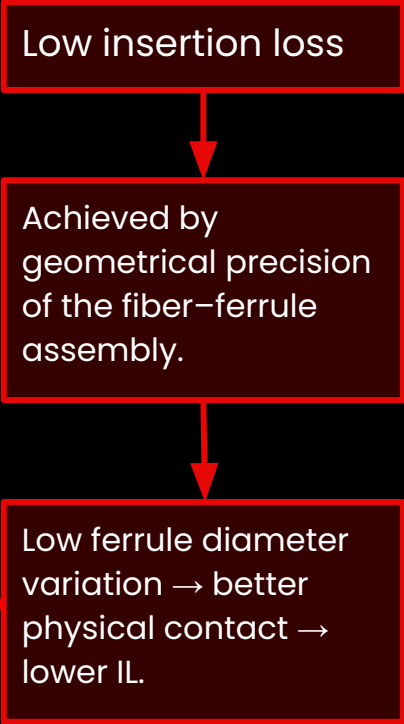
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Low insertion loss

Achieved by geometrical precision of the fiber-ferrule assembly.

Fiber core is centered relative to the connector axis. Minimizes mode mismatch → **reduces insertion loss.**



Payload Transmission Assembly: PM Fiber Space Connectors

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Active Polarization Orientation (APO)	< $\pm 2^\circ$

High Extinction Ratio.



How well the connector aligns the **slow axis** of the PM fiber.

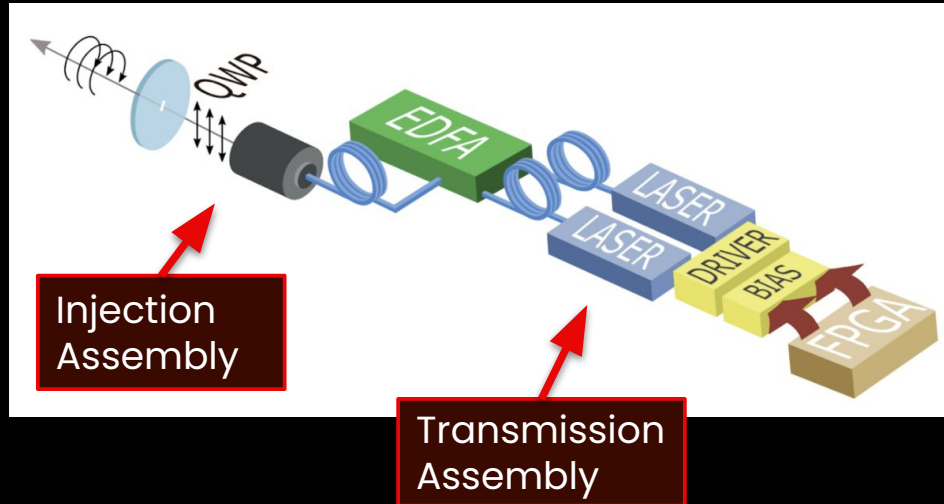


A small orientation error.



Payload Injection Assembly

The transmission beam passes through a collimator, which controls beam divergence and enables the transition of the beam from fibre to free-space. Following this, a quarter-wave plate (QWP) converts the incoming linearly polarized light into circularly polarized light, with each linear polarization state becoming either left or right hand circularly polarized light.



The beam is finally sent to the 1180 nm short-pass dichroic, where the 1550 nm light is reflected toward the FSM and telescope.

Payload Injection Assembly : Collimators

Currently investigating the models for collimators.

- ❑ Required divergence optimized from link budget of approximately 0.8 mrad.



Company	60SMS-1-4A7.5-03-Ti (Schäfter + Kirchhoff)
Focal Length (mm)	7.5
Optics Type	Asphere, Monochromatic
AR-Coating	1050 - 1650 nm
Connection	FC-APC

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Optics Type	Asphere, Monochromatic
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Connection	FC-APC

Used for coupling applications. Tested in Zemax simulations and can work for collimating our transmission beam.

Payload Injection Assembly : Collimators

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Focal Length (mm)	7.5
Optics Type	Asphere, Monochromatic
AR-Coating	1050 - 1650 nm
Connection	FC-APC

More information is needed on the vendor's customized collimators intended for space use.

Payload Injection Assembly : Collimators

Other attempts:

- ❑ **Pigtail collimators:** Can be customized to include vent holes. However, the hytrel jacket and acrylate coating on the PM fiber would outgas (Thorlabs).
- ❑ **90-degree Reflective Collimators:** Can be used and customized for vent holes. However, due to the parabolic mirror, the polarization would not be preserve (elliptical). (Thorlabs)



Payload Injection Assembly : Collimators

Other attempts:

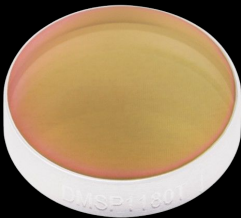
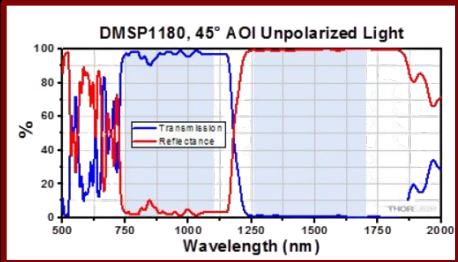
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Challenge: finding a collimator for space applications and polarization maintaining applications.

Payload Injection Assembly: Dichroic

Dichroic mirror, DMSP 1180T, serves to spectrally separate light by transmitting and reflecting light as a function of wavelength. It is highly transmissive below the cutoff wavelength and above the cut-on wavelength, while highly reflective between the cutoff and cut-on.



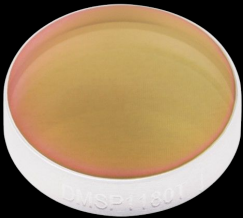
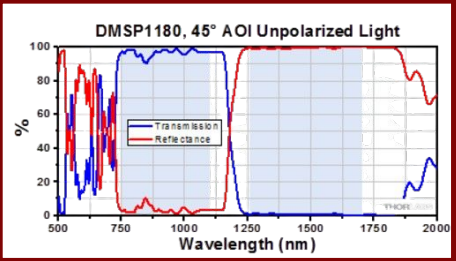
Company	Thorlabs
Cutoff Wavelength (nm)	1180
Transmission Band (nm)	750 – 1100
Reflection Band (nm)	1260 – 1700
Size (in)	Ø 1/2"
Clear Aperture (mm)	Ø 11.4
TWE	$\lambda/4$ (at 633 nm)
Incident Angle	45°

Payload Injection Assembly: Dichroic

Dichroic mirror, DMSP 1180T, serves to spectrally separate light by

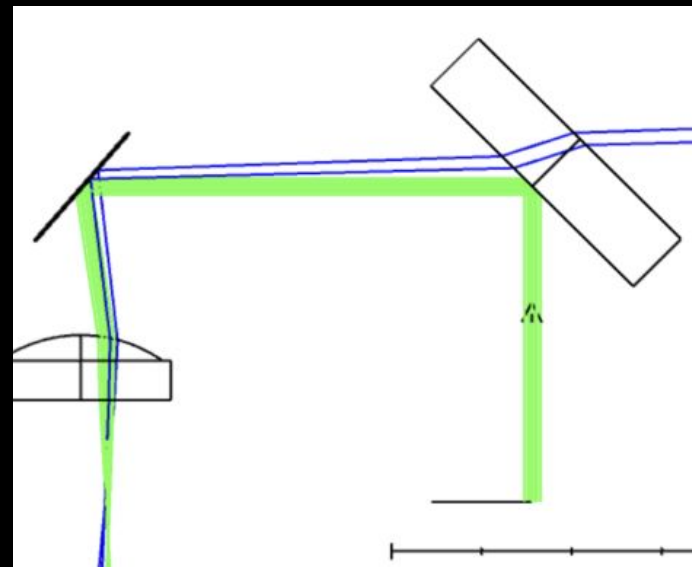
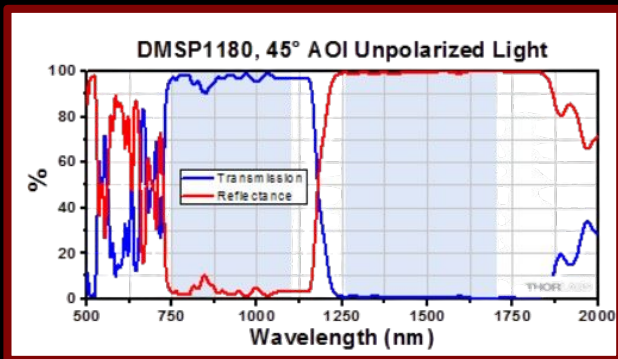
Directs incoming ground beam and outgoing transmission beam onto the same optical path. Passes the ground beam and reflects the transmission beam.

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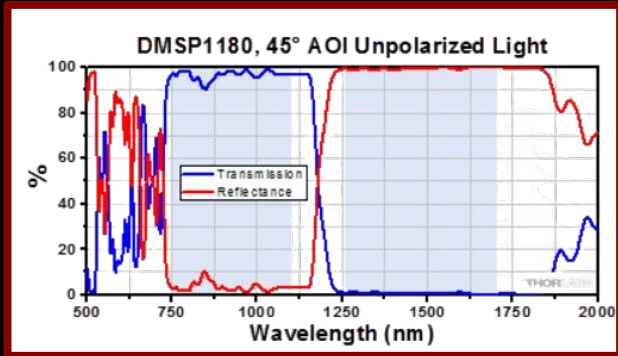
Payload Injection Assembly: Dichroic AOI and Polarization effects

- ❑ Zemax data suggests the AOI on the dichroic might fall within $\pm 0.3^\circ$ of target value at 45°
- ❑ We expect that this angle will not significantly shift the cutoff such that it will let 1550nm transmit, or block 1064nm from transmitting.
- ❑ To be tested: polarization-distorting effects.

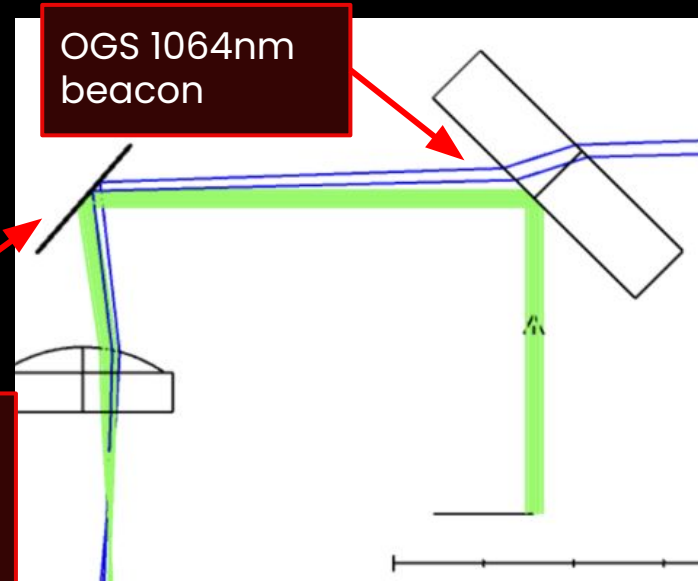


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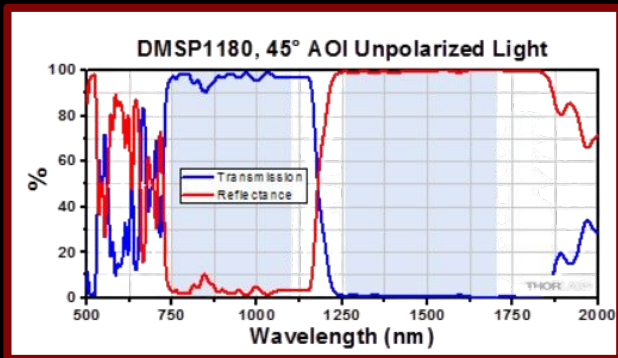


OGS beacon AOI on the dichroic is not 45 degrees due to steering.

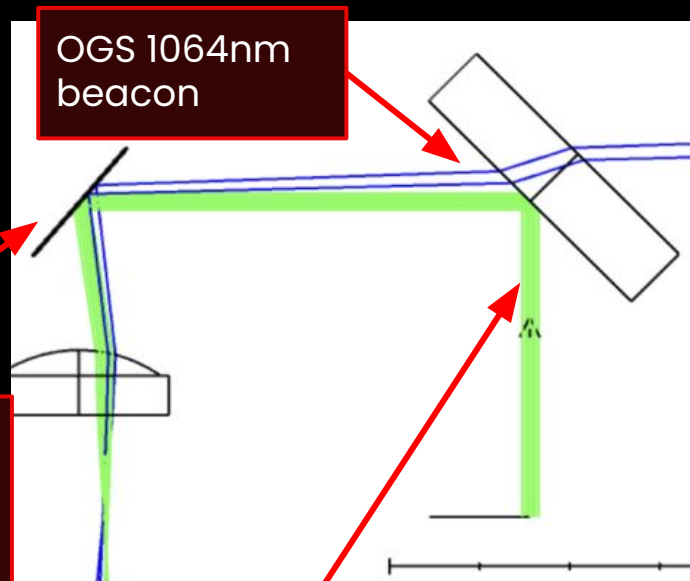


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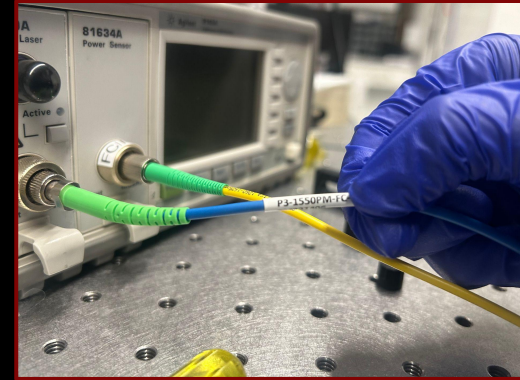


Polarization-distortions caused by dichroic can affect transmission beam.

Next Steps for Payload Design

In terms of component-level choices:

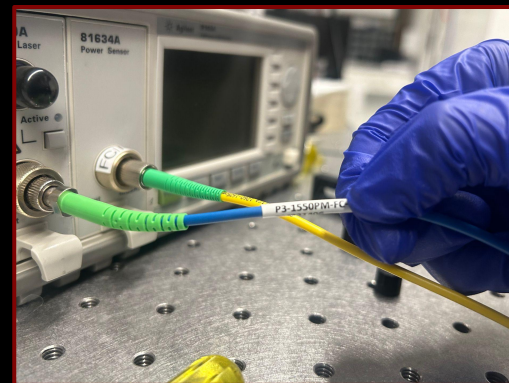
- ❑ Confirmation of collimator choice (vendors).
- ❑ Confirmation of PM space connector choice (vendors).
- ❑ Confirmation of EFDA polarization maintaining capabilities (vendors).
- ❑ Confirmation of calibration laser source, camera, lens for new calibration assembly. (Zemax optical analysis + CAD).
- ❑ Preliminary laser driving board needs to be made.
- ❑ Confirmation of QPD amplification circuitry capabilities (testing optical table).



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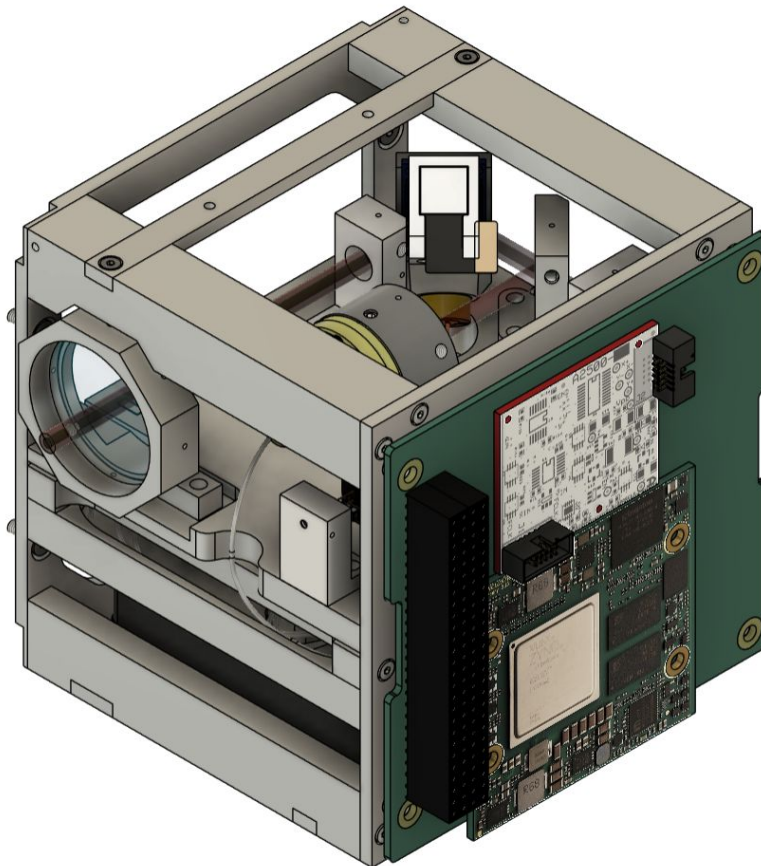
Testing of various components will be later discussed.

Optical Payload: **Structural Design**

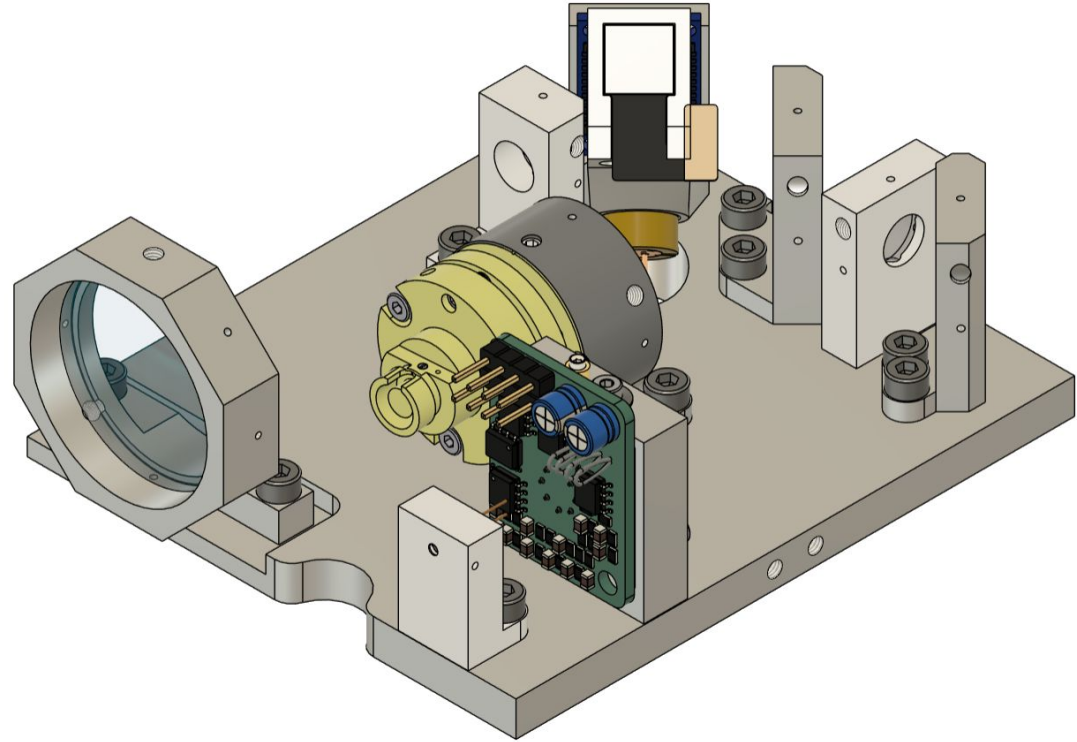
The Optical Payload is composed of 4 main parts:

- ❑ Optomechanical Layer
- ❑ Generation layer
- ❑ Amplification layer
- ❑ Payload controller board (PLCL)

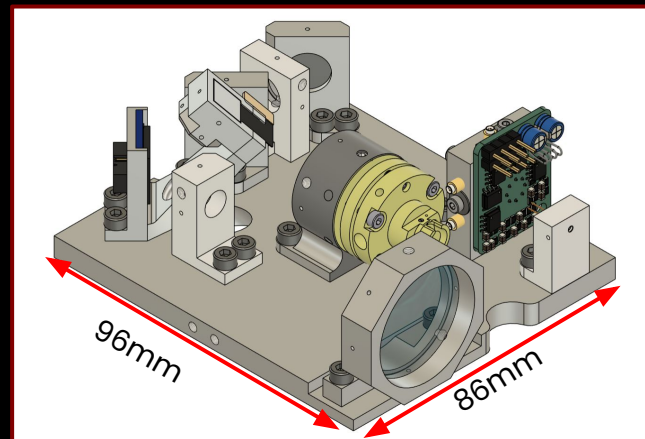
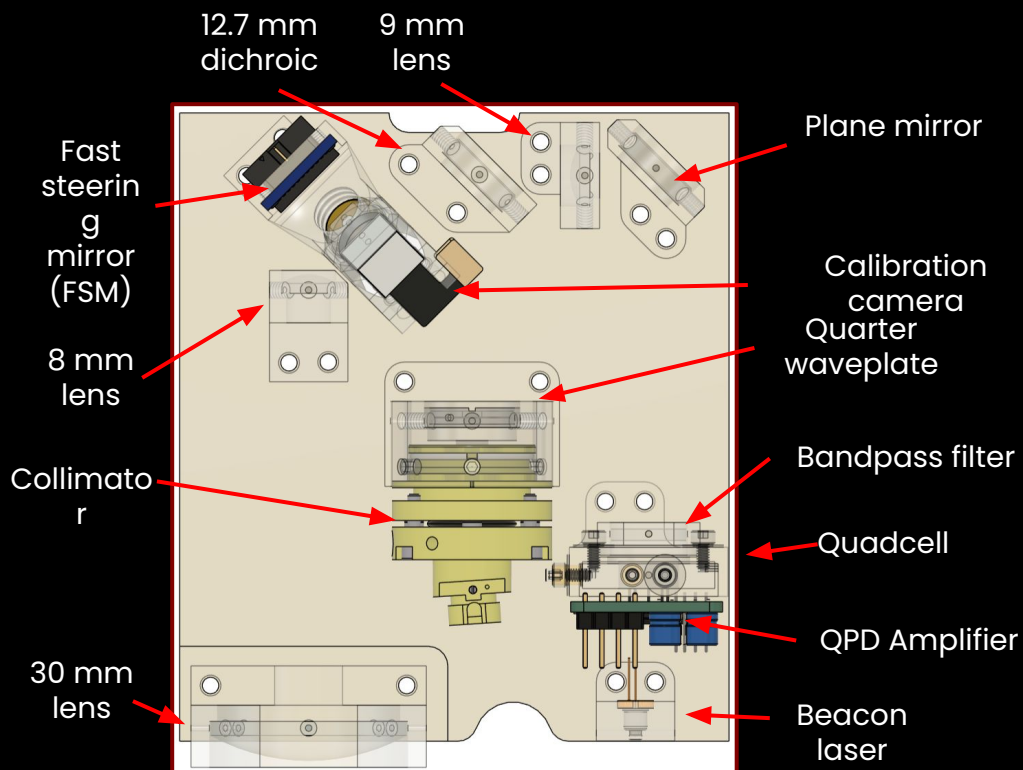
The design allows for easy assembly and integration.



Optomechanical layer



Payload Optical Mounting Overview



Combination of adjustable and static mounts

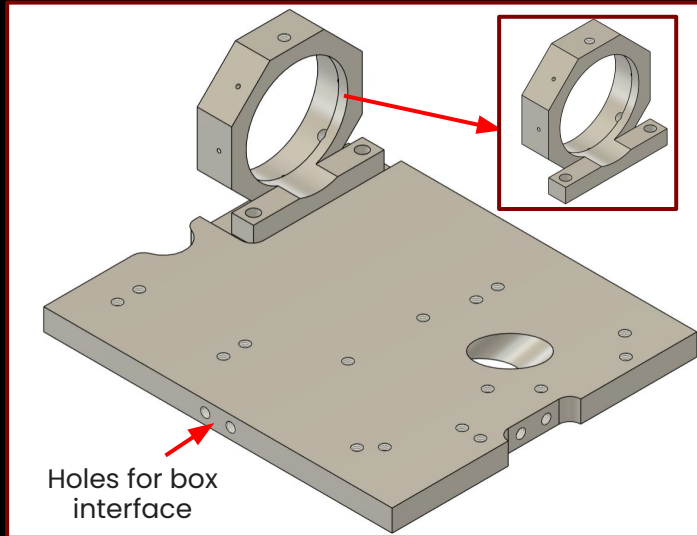
- Allows precise beam adjustments in lab

Manufactured out of aluminum:

- Lightweight
- Easy to manufacture and iterate
- Conductivity, heat capacity

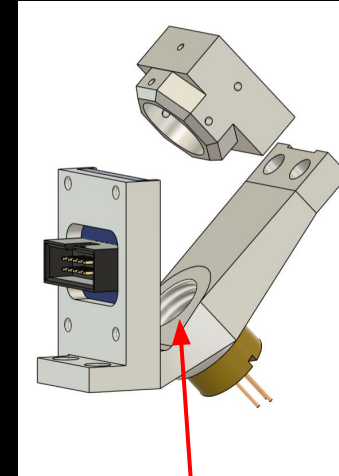
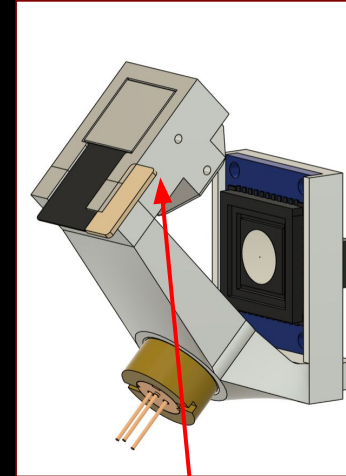
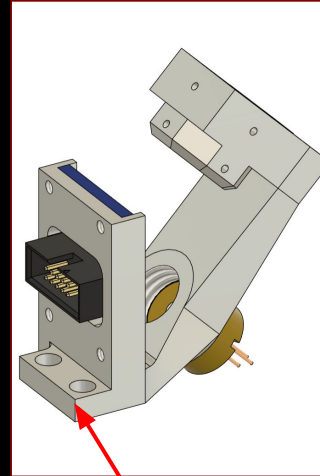
Breadboard, Static Mounts

PULSE-A Optical Breadboard

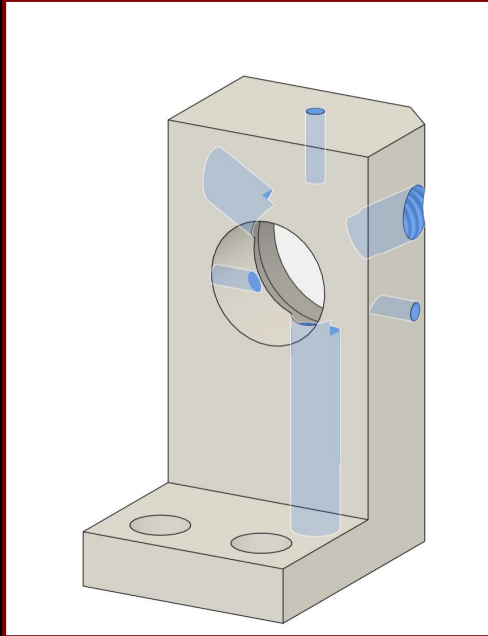


Modular 30 mm lens mount attaches to breadboard via staking pin, countersunk fastener, and side pin

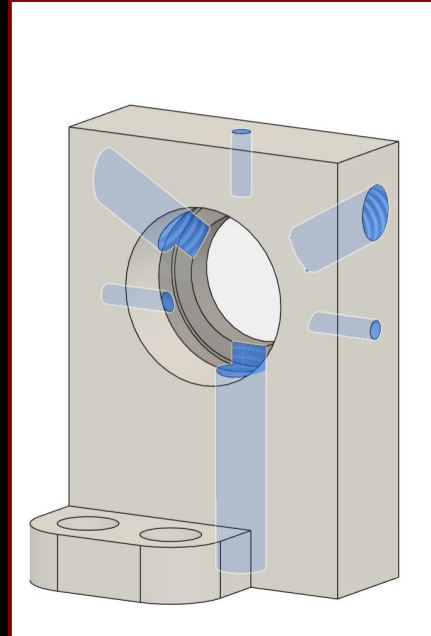
FSM mount w/ calibration



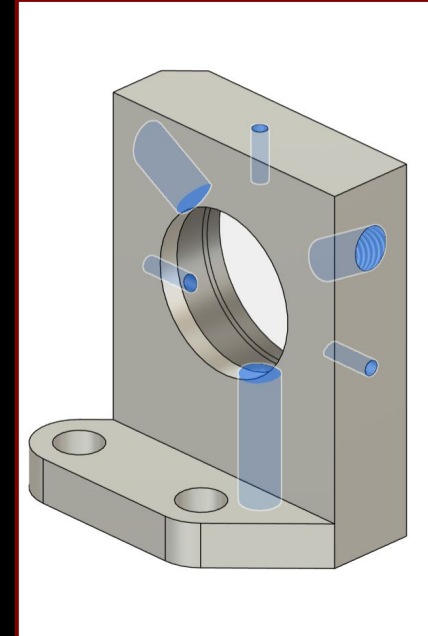
Static Mounts cont.



8 mm lens
mount



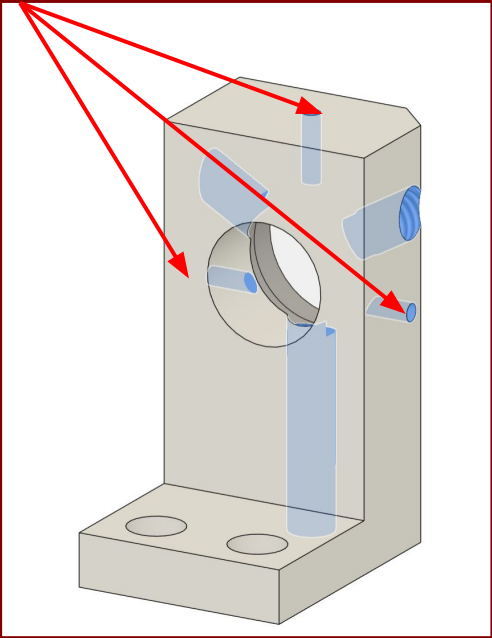
9 mm lens
mount



Dichroic
mount

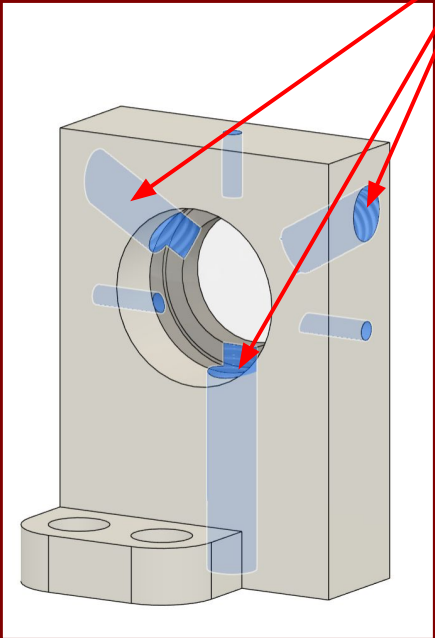
Static Mounts cont.

Epoxy injection ports

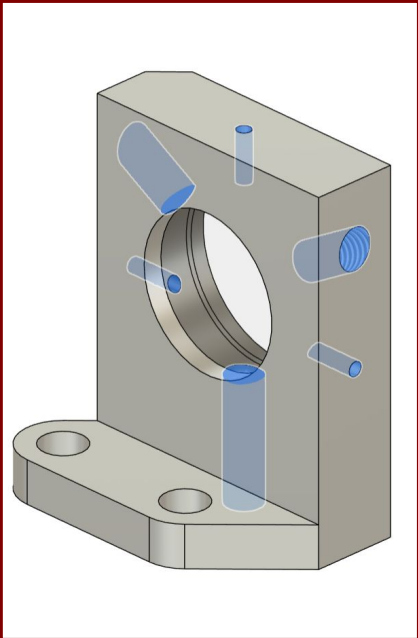


8 mm lens
mount

Ground Support Hardware
Set Screw Holes (M3)

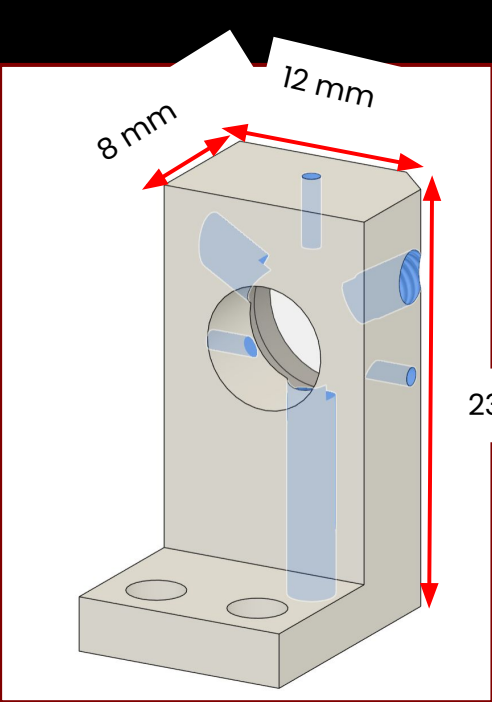


9 mm lens
mount

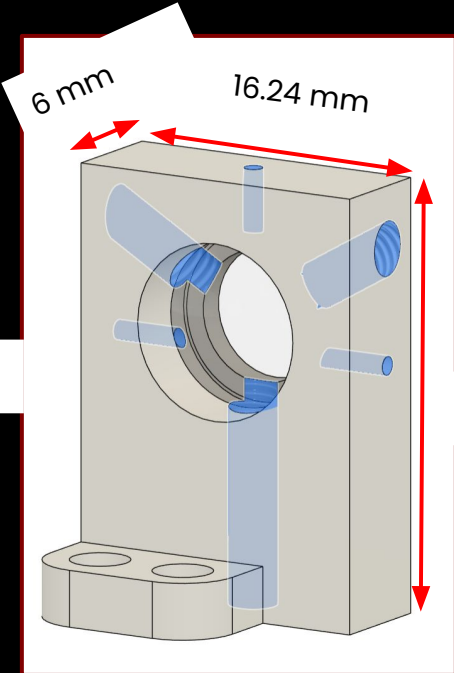


Dichroic
mount

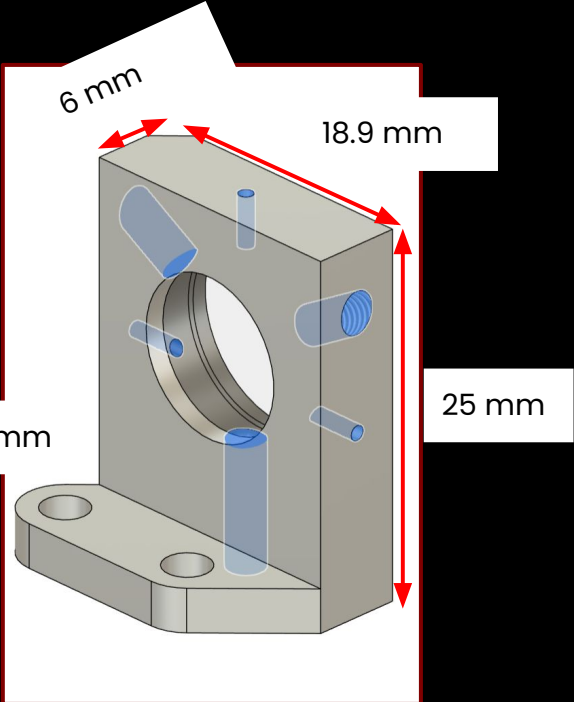
Static Mounts cont.



Ø8 mm lens
mount

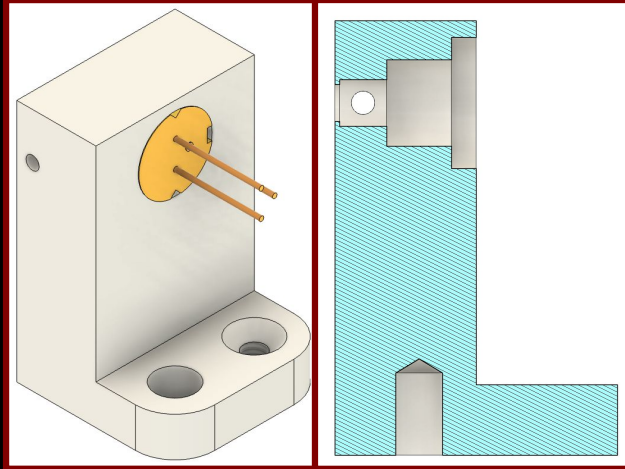


Ø 9 mm lens
mount



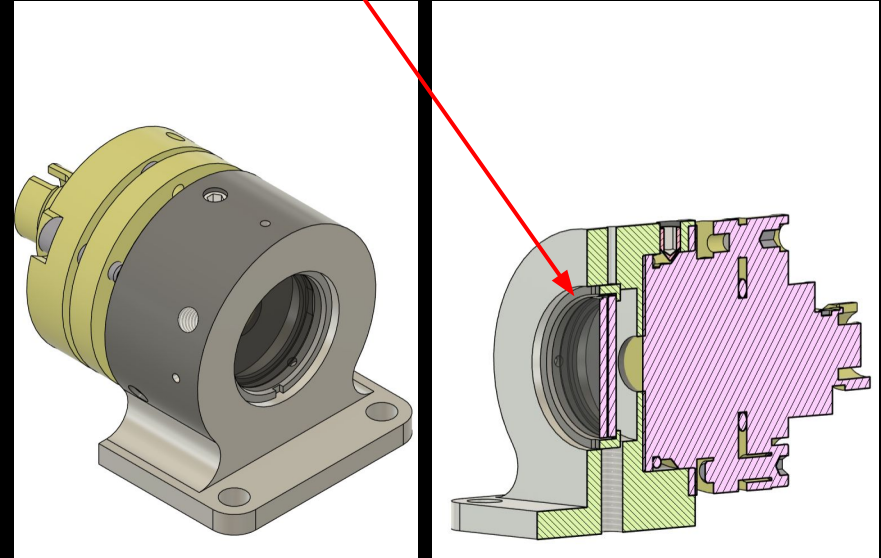
Dichroic / Plane Mirror
mount

Static Mounts cont.



Beacon laser mount

Rotating mount allows in-lab adjustment

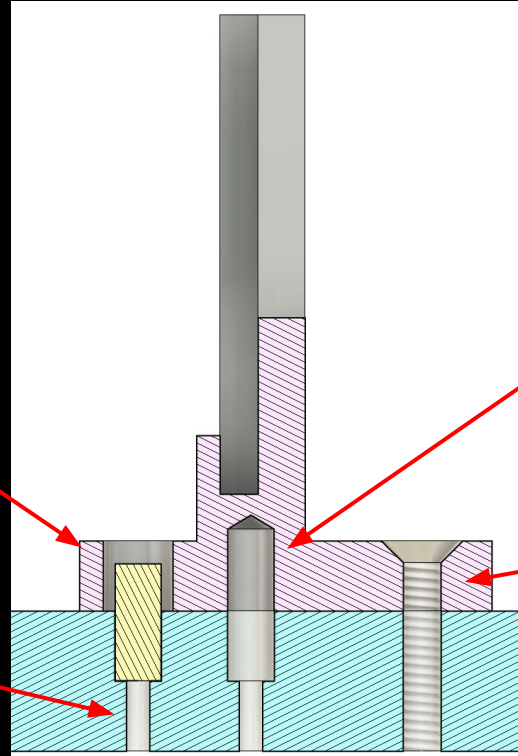


QWP and Collimator mount

3 Part Mounting Method

2 mm staking pin is placed in a slightly oversized hole and filled with epoxy - liquid set mounting reduces vibration

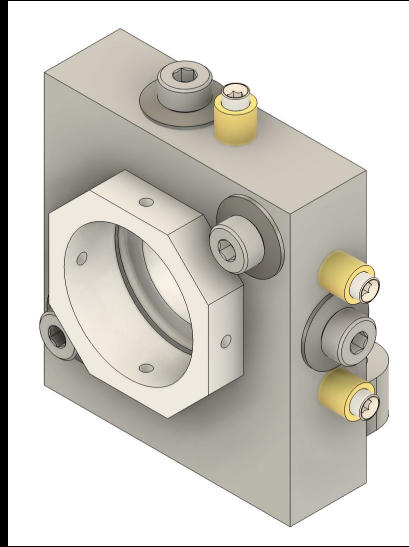
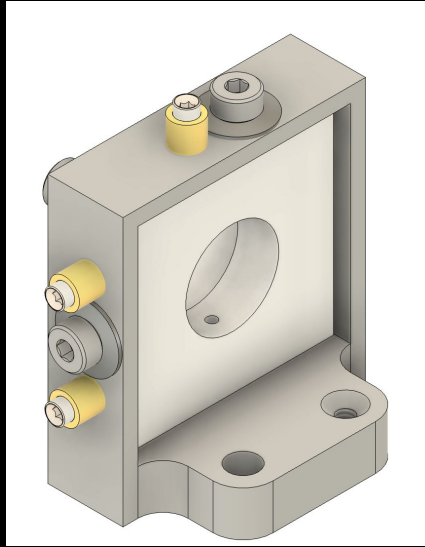
Clearance holes for outgassing?



Hole for press-fit alignment pin - precisely locates mount on breadboard

Countersunk fastener constrains rotation about the central pin, and reduces vibration

Adjustable Mounts

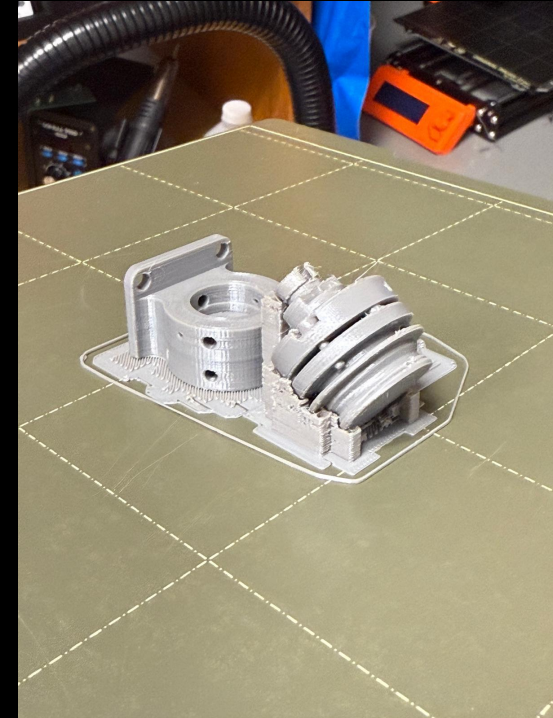
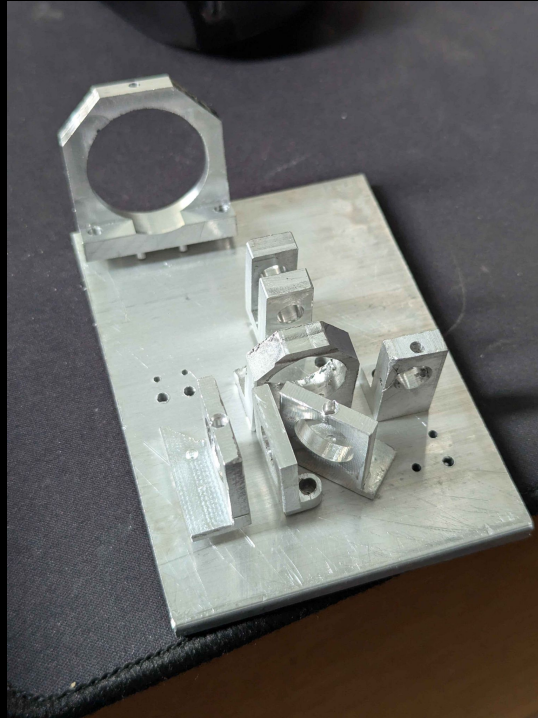


The QPD mount is translationally adjustable within the support mount normal to the direction of light it senses. To achieve this, bolts securing the QPD within the mount support are used in tandem with Belleville washers, which provide a preload against adjustment screws to reduce backlash and increase rigidity.

Fabrication

In order to reduce cost of fabrication, all CAD designs were created to be manufacturable at our on-campus machine shops, using simple 3-axis milling and CNC machines. All parts have been manufactured within a tolerance of 0.13 millimeters and were made using aluminum 6061 stock material.

Prototyping to verify fit is done using FDM 3d printing



Next steps

- ❑ Complete our STOP analysis (Structure-Thermal-Optomechanical)
- ❑ Complete our system sensitivity analysis
- ❑ Full optomechanical layer assembly
- ❑ Vibration Simulation (Ansys Mechanical)

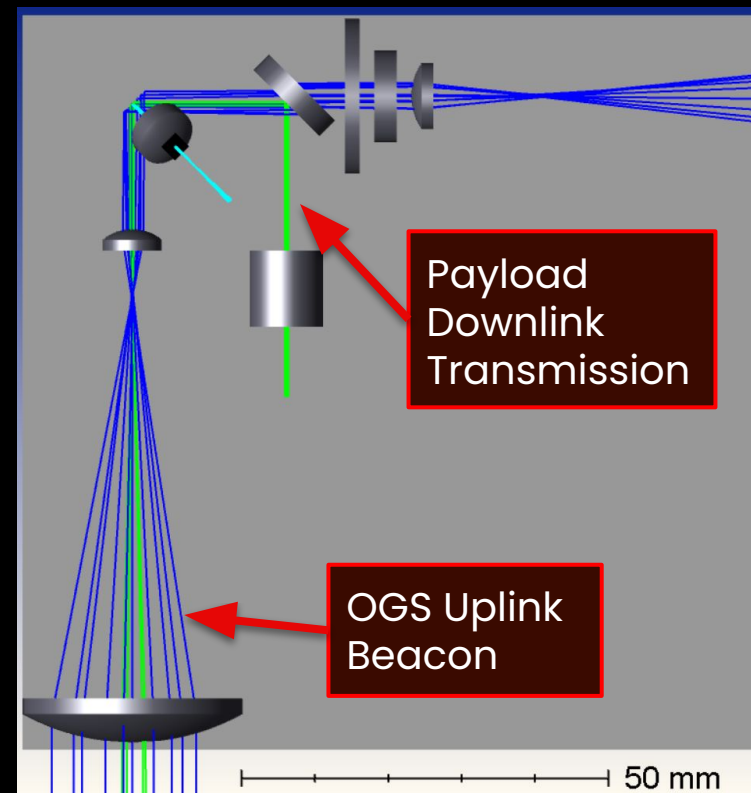
Optical Payload: **Optical Analysis**

General Overview

- ❑ Optical Analysis using Ansys Zemax OpticStudio (Zemax)
- ❑ **Simulate Entire Optical Payload design**
- ❑ **Analyze QPD Signal**
 - ❑ Preliminary studies of QPD Centroid calculation for Closed-Control Feedback Loop
- ❑ **Optical Tolerancing**
 - ❑ Body-/Coarse-Pointing Requirement, given FSM product's tilt range
 - ❑ Thermal Expansion of Plates impacting optical path
 - ❑ Back-reflected light disturbing QPD signal
 - ❑ FSM Calibration system to address non-linear FSM Voltage Response

Steering and Detection Assemblies

- ❑ Two main optical paths: OGS Uplink Beacon (1064nm), Payload Downlink Transmission (1550nm)
 - ❑ **Uplink:** shrunk by Keplerian Telescope, reflected by FSM, filtered & focused onto QPD
 - ❑ **Downlink:** Reflected by Dichroic & FSM, expanded by Keplerian Telescope
- ❑ Keplerian Telescope: 30mm & 8mm Plano-convex Lenses
- ❑ 3rd Optical Path: **FSM Calibration**



Payload Optical Layout

Steering and Detection Assemblies

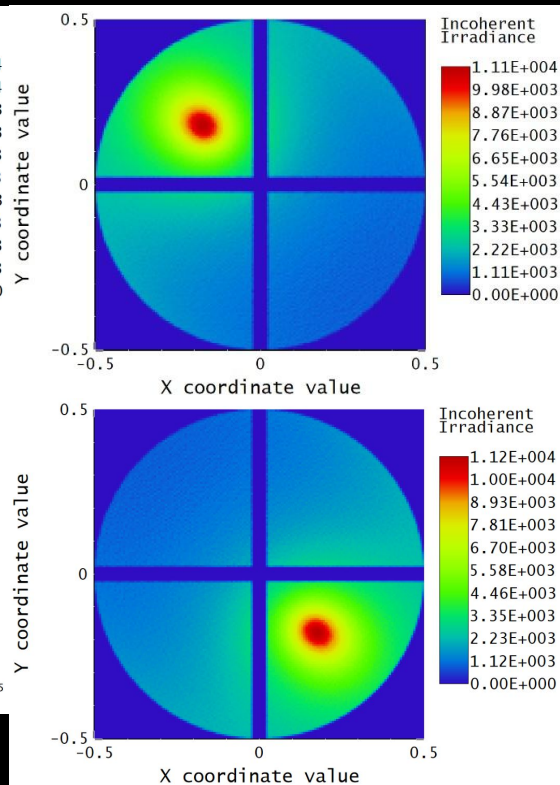
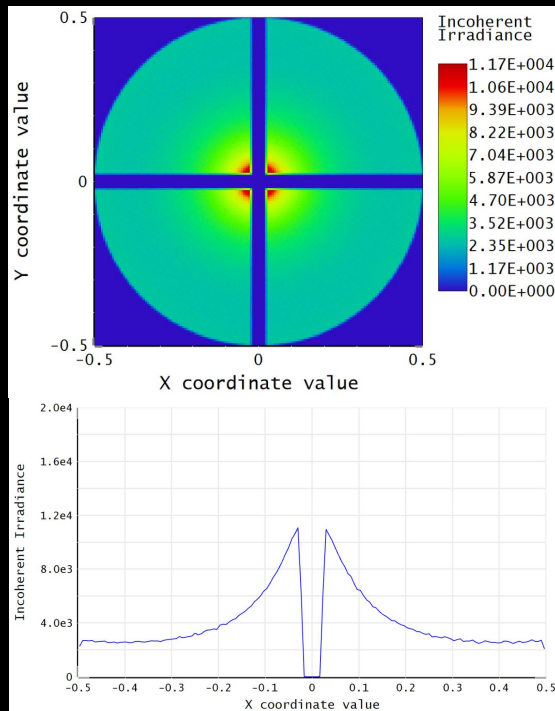
- ❑ Power drop across each optical component
 - ❑ Power drops differently across optical paths (Uplink & Downlink)

OGS Uplink Beacon Path								
Optical Component	30mm Lens	8mm Lens	FSM	SWP Dichroic	Bandpass Filter	Shortpass Filter	9mm Lens	Total Loss
Power Drop from Previous Component (%)	0.49	0.41	1.39	6.53	16.52	17.19	39.51	61.81

Payload Downlink Transmission Path					
Optical Component	SWP Dichroic	FSM	8mm Lens	30mm Lens	Total Loss
Power Drop from Previous Component (%)	0.24	3.01	0.38	0.80	4.39

Steering and Detection Assemblies

- QPD Detector's unique profile:
 - four quadrants separated by two dead zones

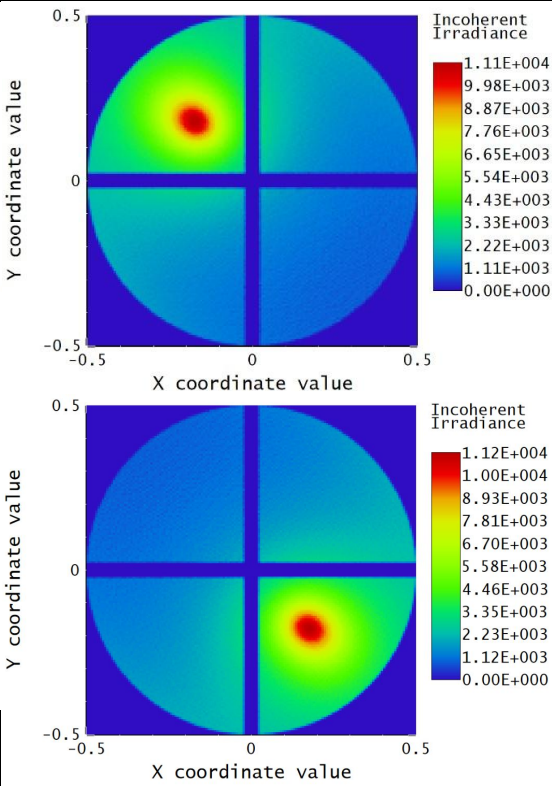
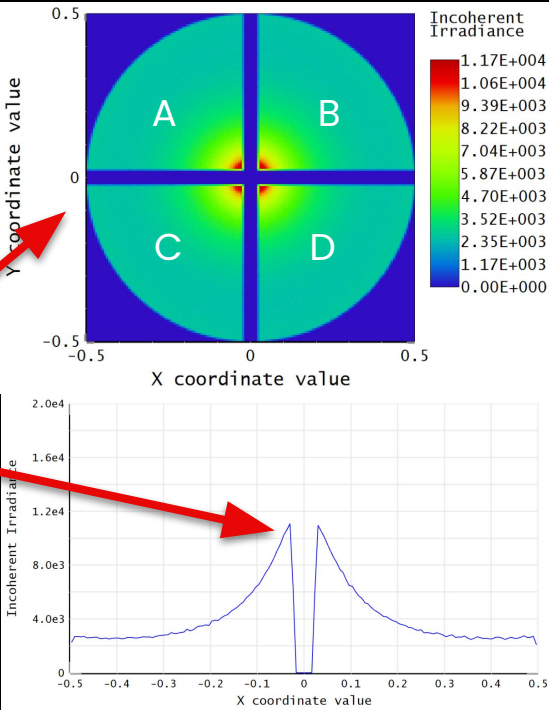


Steering and Detection Assemblies

- QPD Detector's unique profile:
 - four quadrants separated by two dead zones

Beacon Wavefront
Centered on QPD

QPD Cross-Section
over Dead Zone

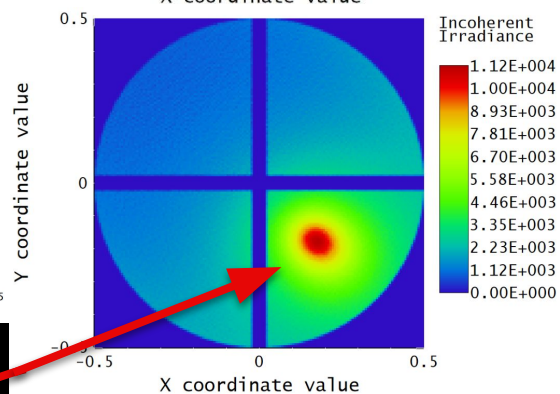
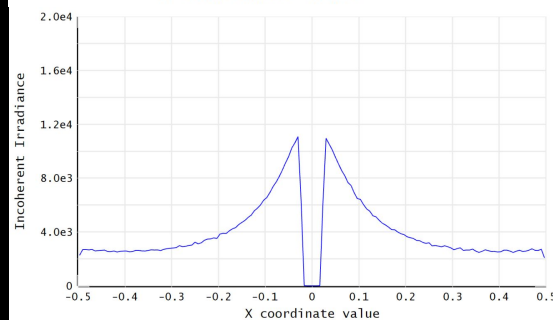
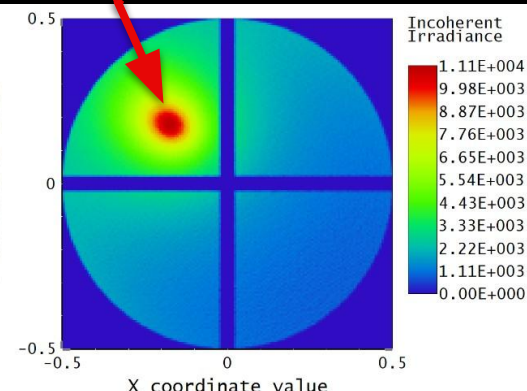
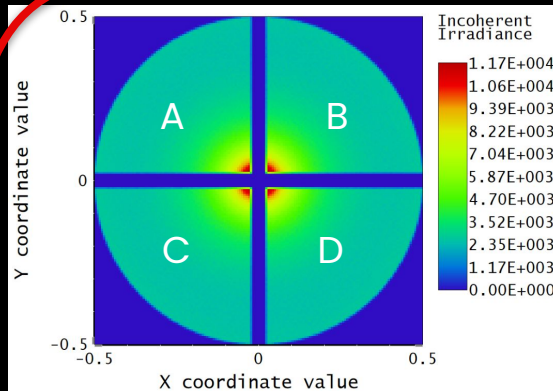


Steering and Detection Assemblies

- ❑ QPD Detector's unique profile:
 - ❑ four quadrants separated by two dead zones

Beacon Wavefront
on QPD Quadrant A
(Top Left)

Beacon Wavefront
on QPD Quadrant D
(Bottom Right)

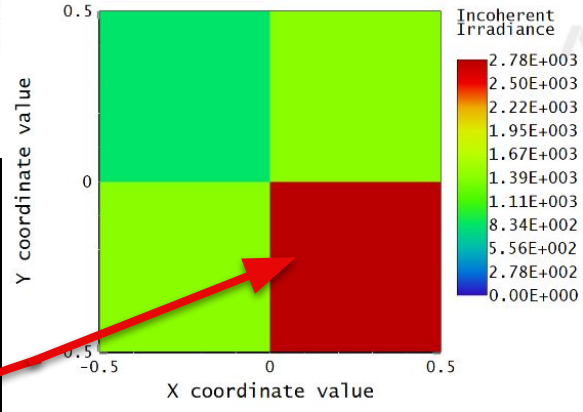
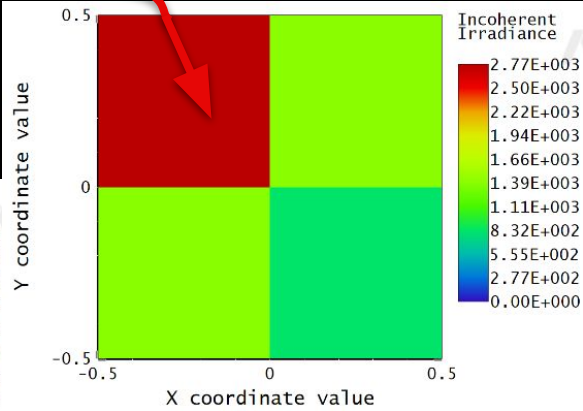
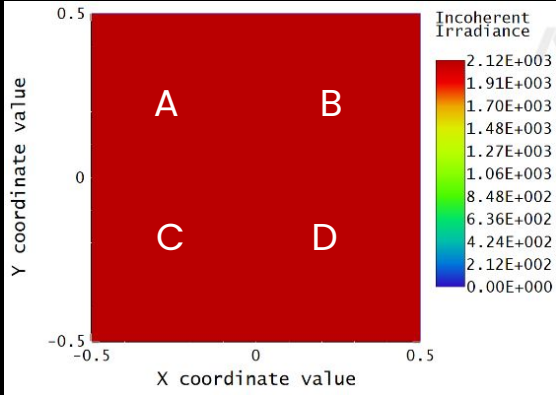


Steering and Detection Assemblies

- What we actually see: four quadrant signals

Beacon Wavefront on QPD Quadrant A (Top Left)

Beacon Wavefront on QPD Quadrant D (Bottom Right)



Steering and Detection Assemblies

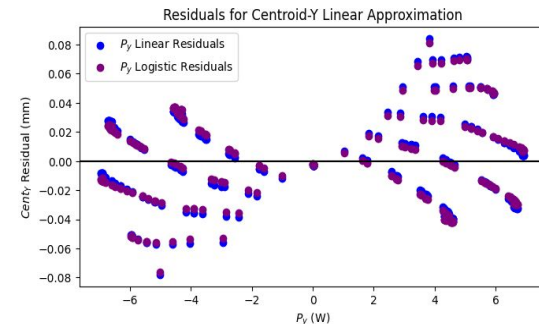
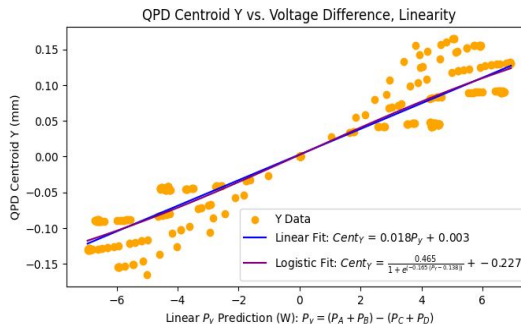
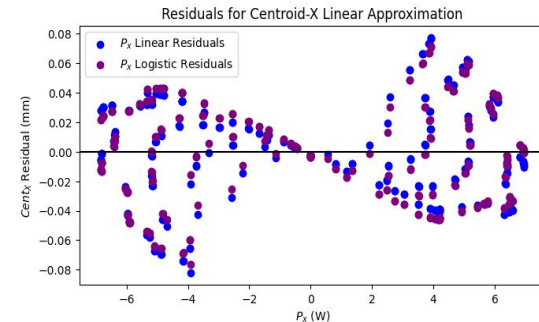
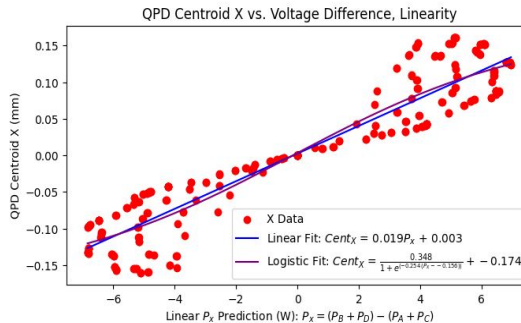
- ❑ Motivation: **Obtain algorithm for conversion from four QPD Voltage Signals → Actual Beam Centroid**
 - ❑ Useful for FSM-QPD Closed-Control Feedback Loop and NEA calculations
- ❑ Literature Formula: Linear Approximation of Voltage Signal Difference
 - ❑ Linear Approximation → Proportional to real beam centroid

$$V_X = \frac{(V_B + V_D) - (V_A + V_C)}{V_A + V_B + V_C + V_D},$$
$$V_Y = \frac{(V_A + V_B) - (V_C + V_D)}{V_A + V_B + V_C + V_D}$$

Steering and Detection Assemblies

- ❑ Fine-pointing Feedback Loop: translates between QPD Signal & FSM Tilt
- ❑ Translate QPD Signal → Beam Centroid
 - ❑ Also useful to set noise thresholds (NEA)

Linear Signal-Difference Approximation: **may require more accurate centroid prediction**



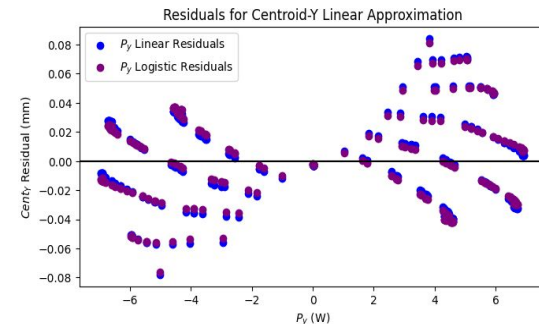
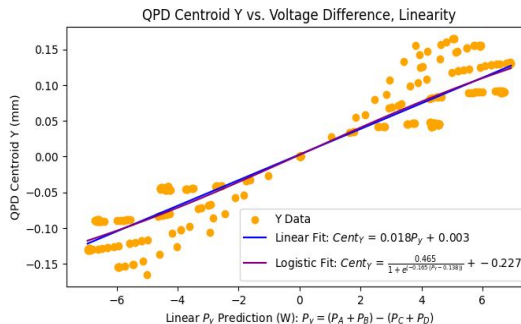
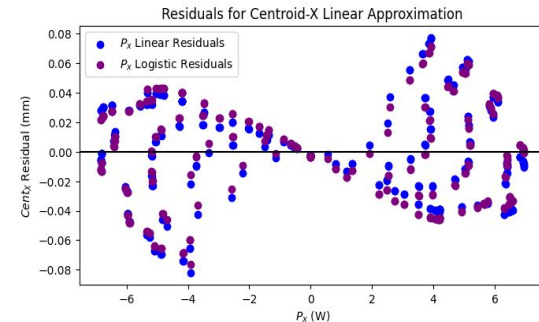
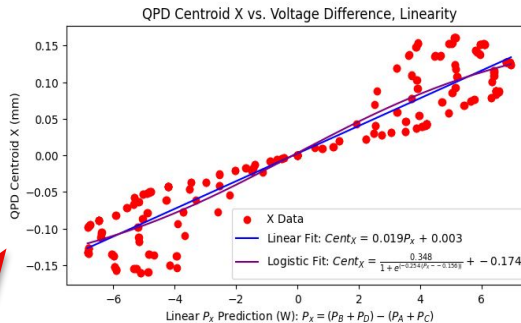
QPD Simulated Centroid Data vs. Linear Power Prediction

Steering and Detection Assemblies

- ❑ Fine-pointing Feedback Loop: translates between QPD Signal & FSM Tilt
- ❑ Translate QPD Signal → Beam Centroid
 - ❑ Also useful to set noise thresholds (NEA)

Linear & Logistic Fitting
for QPD Centroid X Data

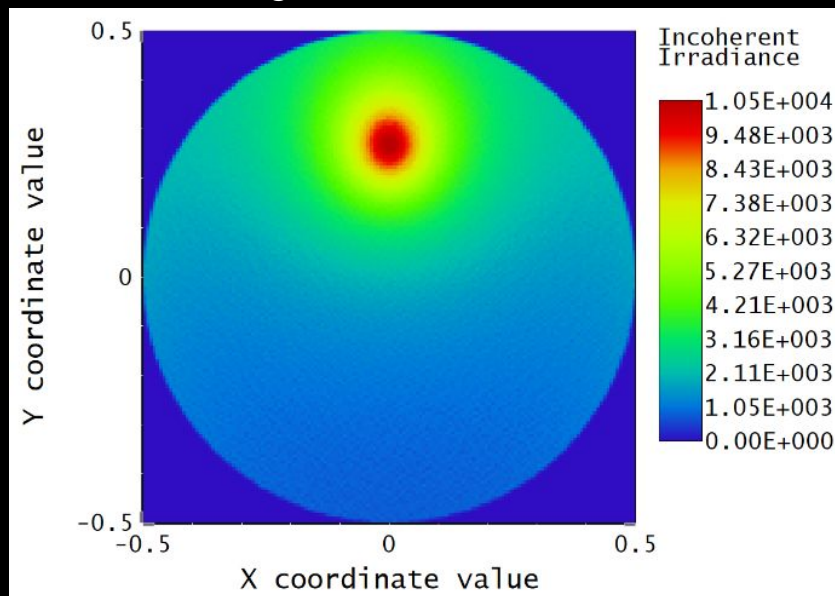
Linear & Logistic Fitting
for QPD Centroid Y Data



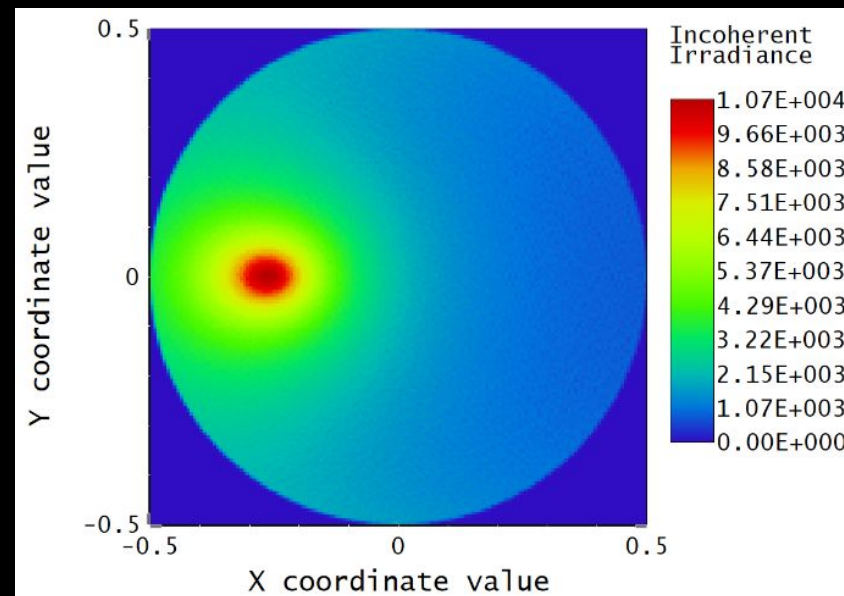
QPD Simulated Centroid Data vs. Linear Power Prediction

Steering and Detection Assemblies

- FSM Field-of-View (FOV) can distort the beam, though **little real effect is observed**



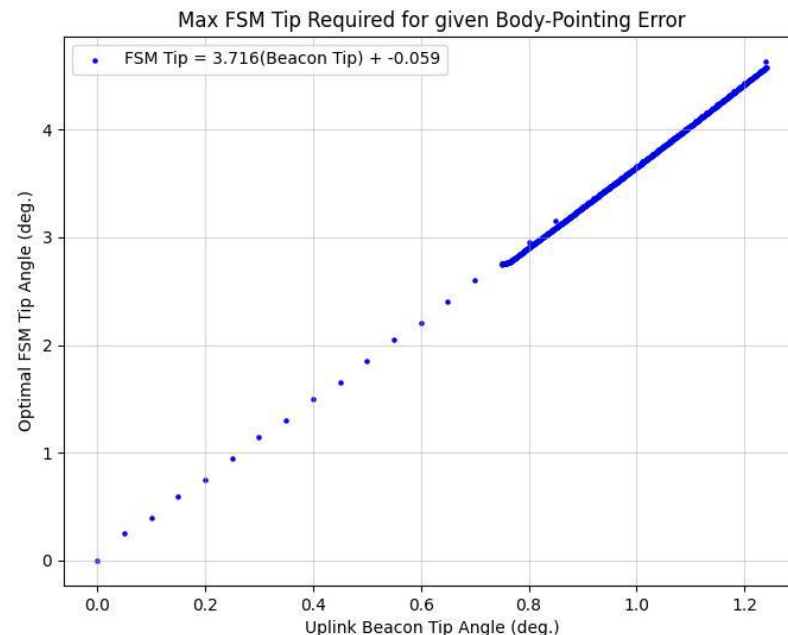
Beacon Wavefront on QPD at 0.3° Beacon Tilt (not considering dead zones)



Beacon Wavefront on QPD at 0.3° Beacon Tip (not considering dead zones)

Coarse-Pointing Tolerancing

- ❑ Coarse Pointing Requirement limited by Fine Pointing Capabilities
- ❑ Consider FSM tilt range limited to ~60% full tilt range $\rightarrow 2.8^\circ$
- ❑ Strongly Linear Relationship between Beacon Tip Angle & FSM Tip Angle
 - ❑ **Ideal coarse-pointing error: 0.7°**
 - ❑ Updated data currently being collected to obtain more accurate optimal FSM tilt angle

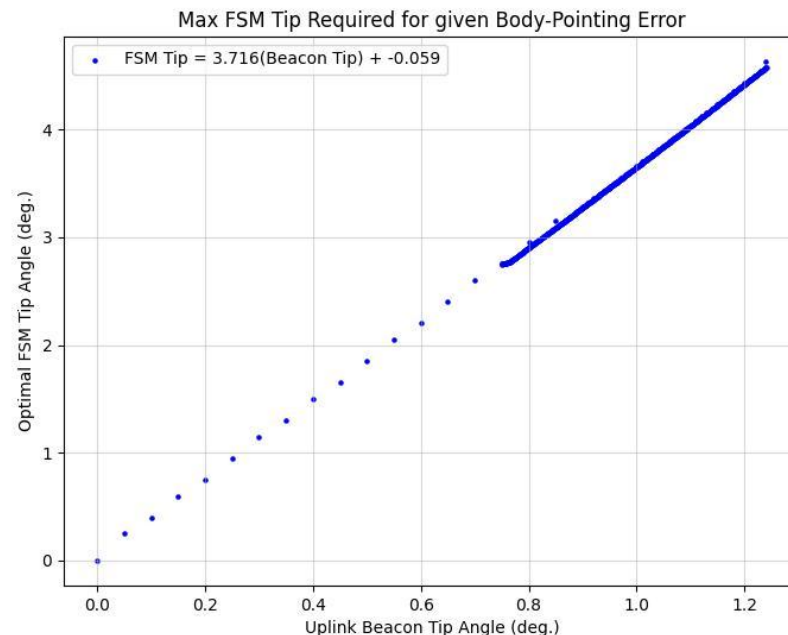


Optimal FSM Tilt vs. ADCS Coarse-Pointing Error

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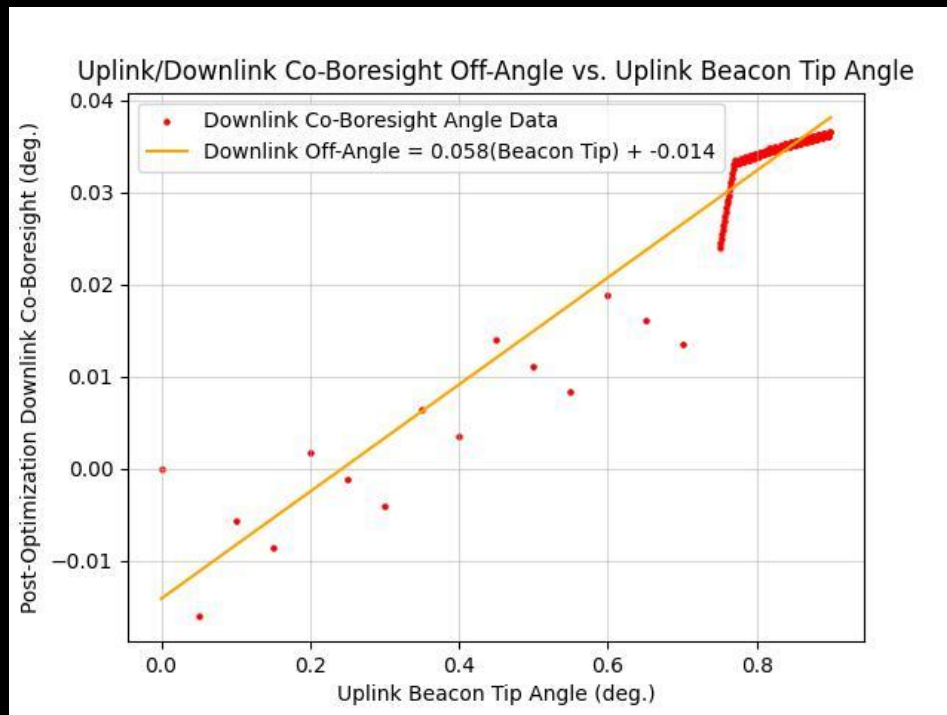
We can extend into non-linear range $> 80\%$, but coarse-pointing error of 0.7° will ensure we stay within linear response



Optimal FSM Tilt vs. ADCS Coarse-Pointing Error

Coarse-Pointing Tolerancing

- ❑ Feedback Loop Algorithm Validation: Centering uplink beacon on QPD aligns transmission
- ❑ Though some linear correlation, only need to consider ADCS coarse-pointing error
- ❑ → **Co-boresight angle between uplink & downlink is low**
 - ❑ (~1 order of magnitude below coarse-pointing error)

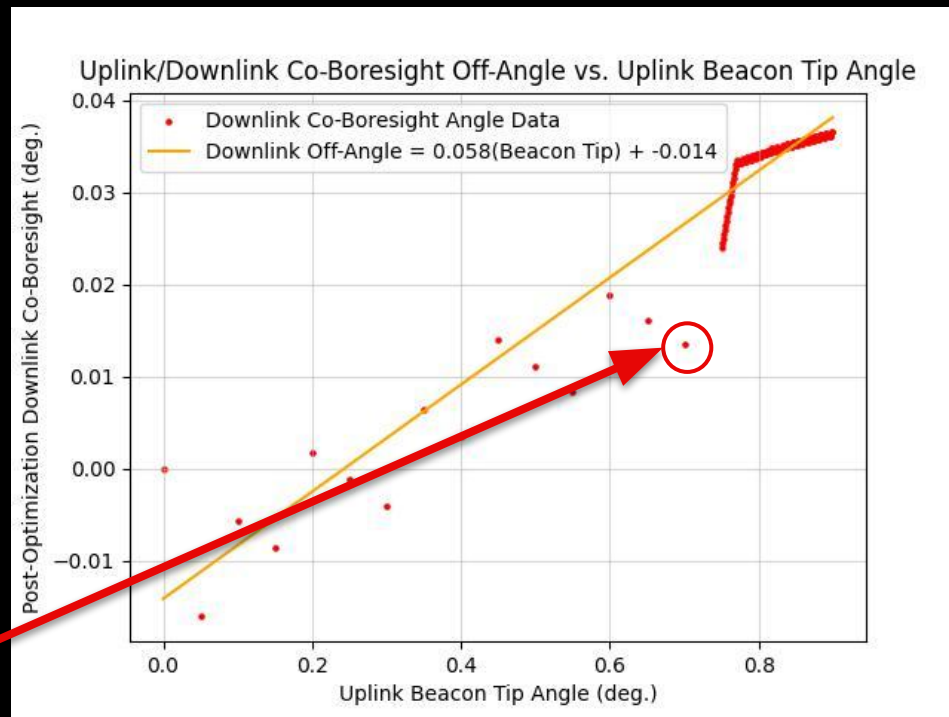


Downlink-Uplink Off-Angle after recentering Uplink on QPD

Coarse-Pointing Tolerancing

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At 0.7° coarse-pointing angle, off-angle between uplink & downlink is $\sim 0.01^\circ$

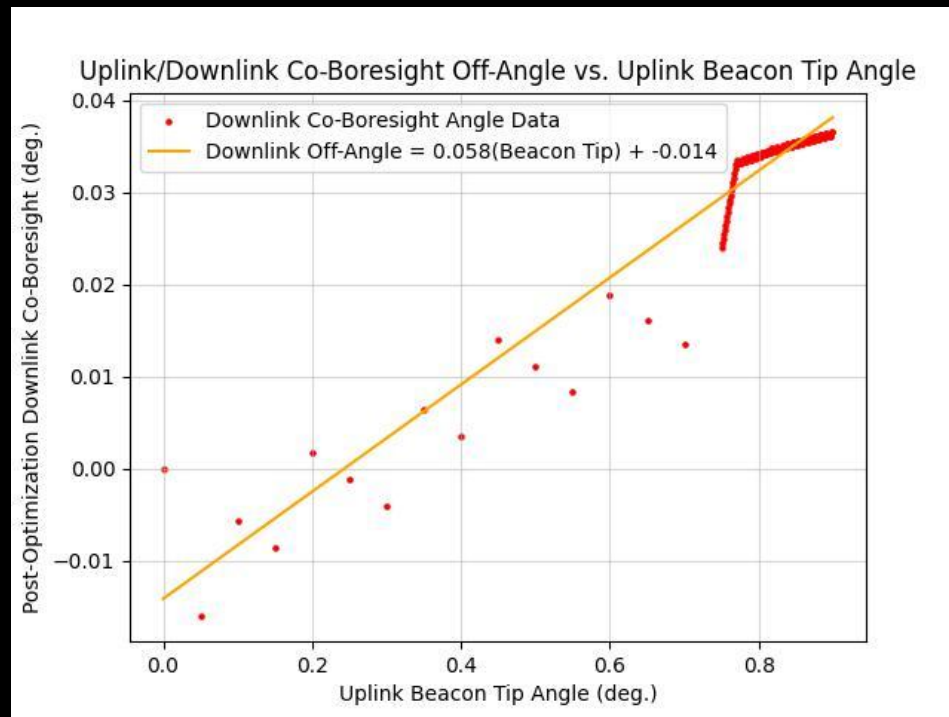


Downlink-Uplink Off-Angle after recentering Uplink on QPD

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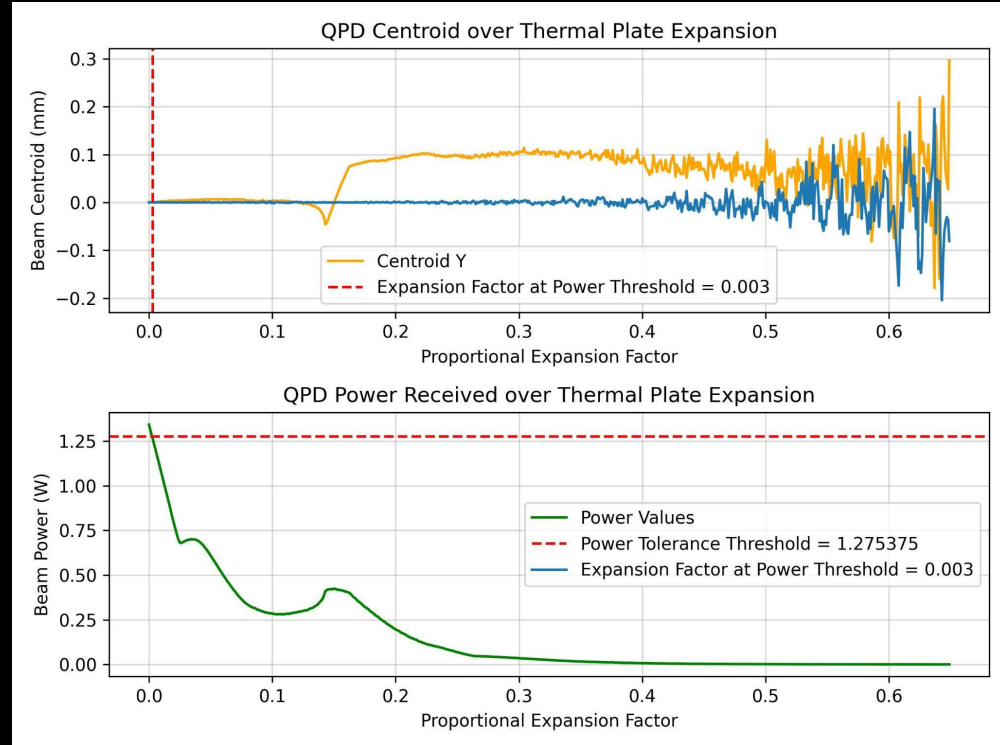
Takeaway: Centering on QPD sufficiently allows us to align downlink & uplink pointing



Downlink-Uplink Off-Angle after recentering Uplink on QPD

Optical Thermal Tolerancing

- ❑ Thermal Expansion of 1U Plating (Al)
- ❑ Assuming Linear Plate Expansion
- ❑ Power is our limiting factor
- ❑ Physical Expansion Factor of 0.003 →
130° C range to maintain at least 90% Maximum Power on QPD

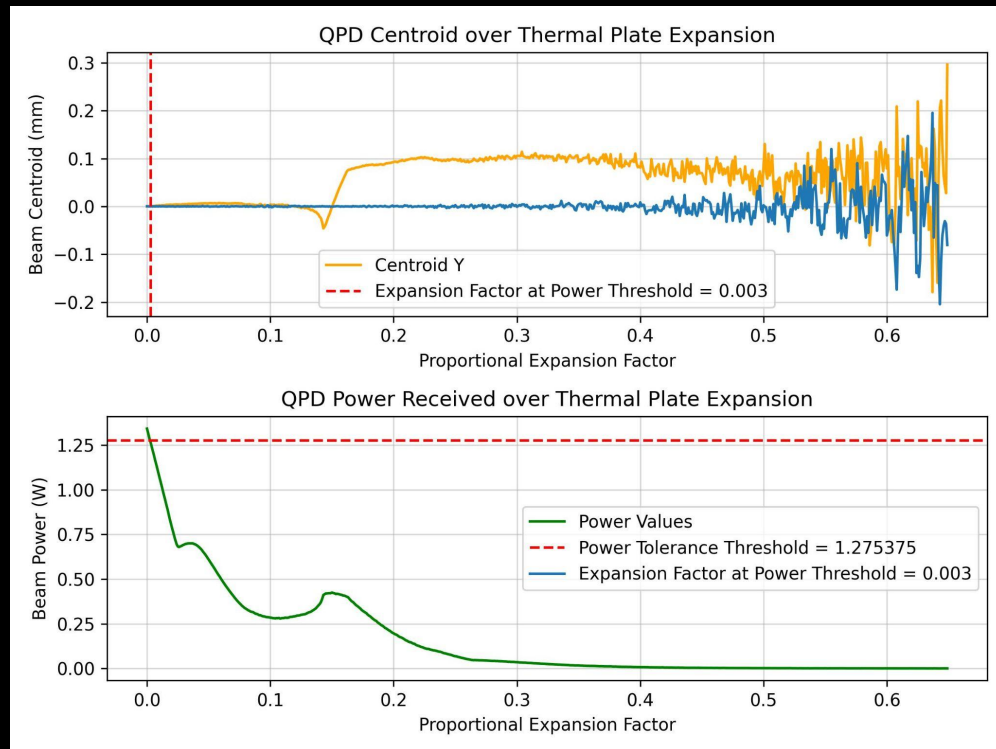


Centroid & Power Results for Thermal Expansion of 1U

Optical Thermal Tolerancing

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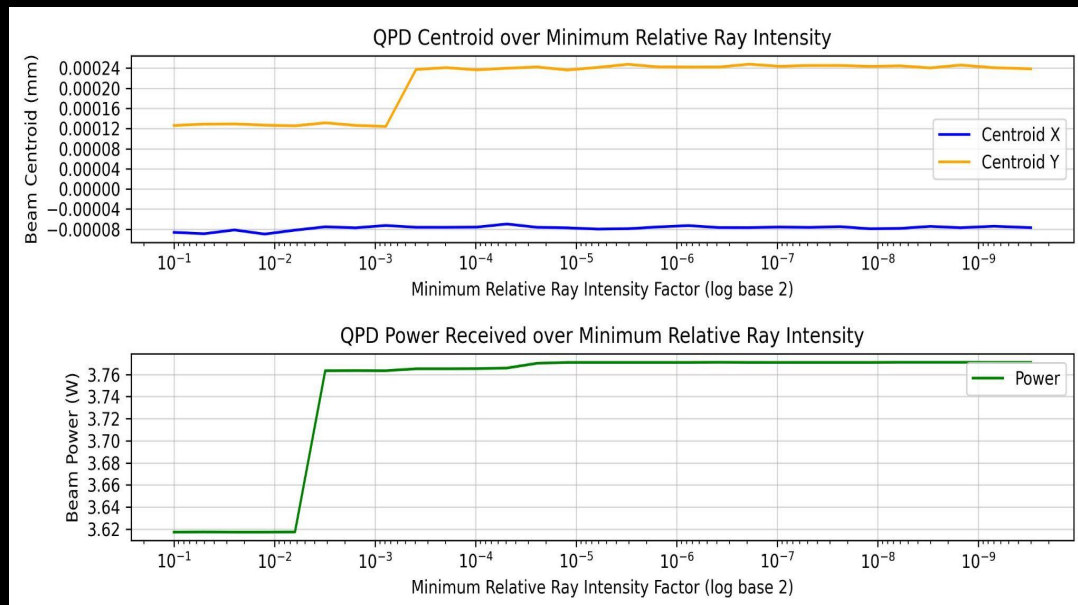
Tolerance above known
ADCS operating temperature
(~1 order of magnitude)



Centroid & Power Results for Thermal Expansion of 1U

QPD Back Reflections Tolerancing

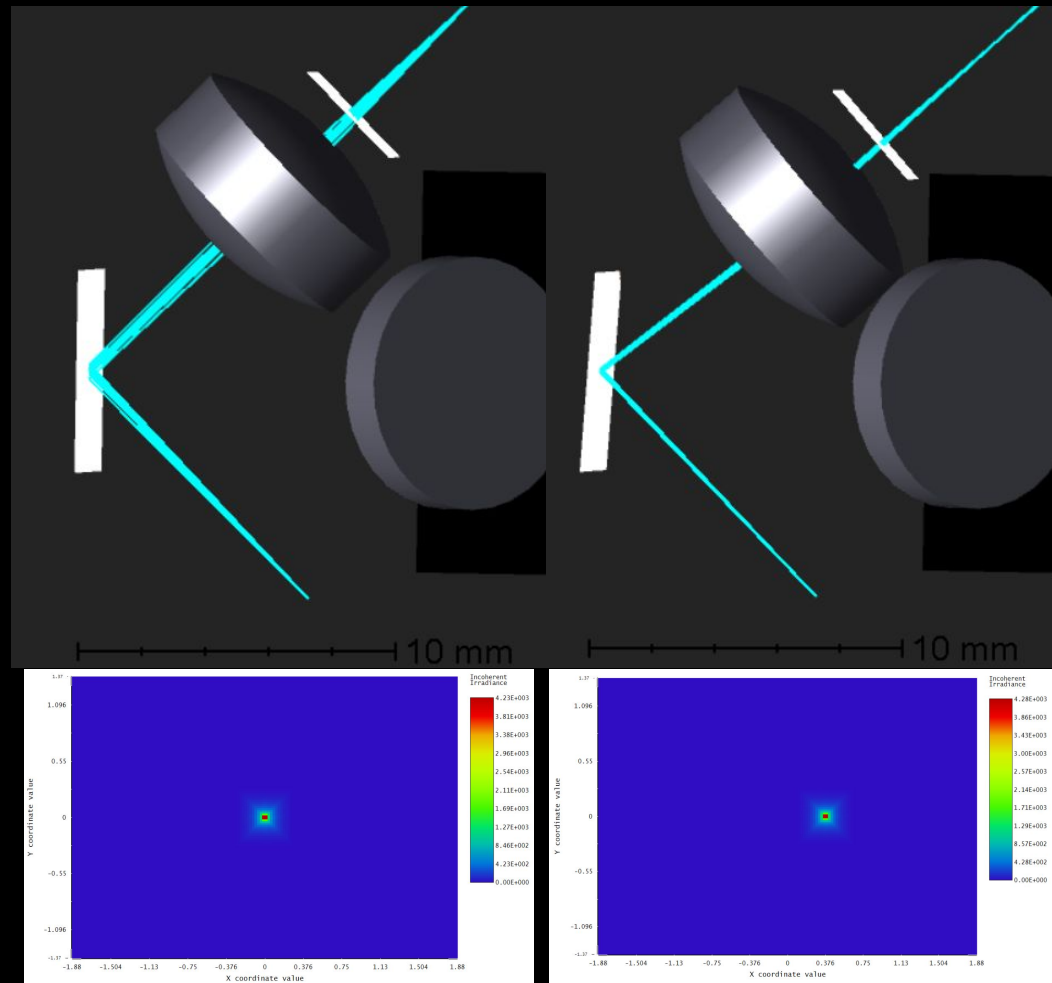
- ❑ Back-reflected light from beam propagation imperfections analyzed at QPD
- ❑ Considering Back-reflections: **3% increase in expected power received on QPD**



Back-Reflection Impacts on Received Centroid & Power

FSM Calibration System

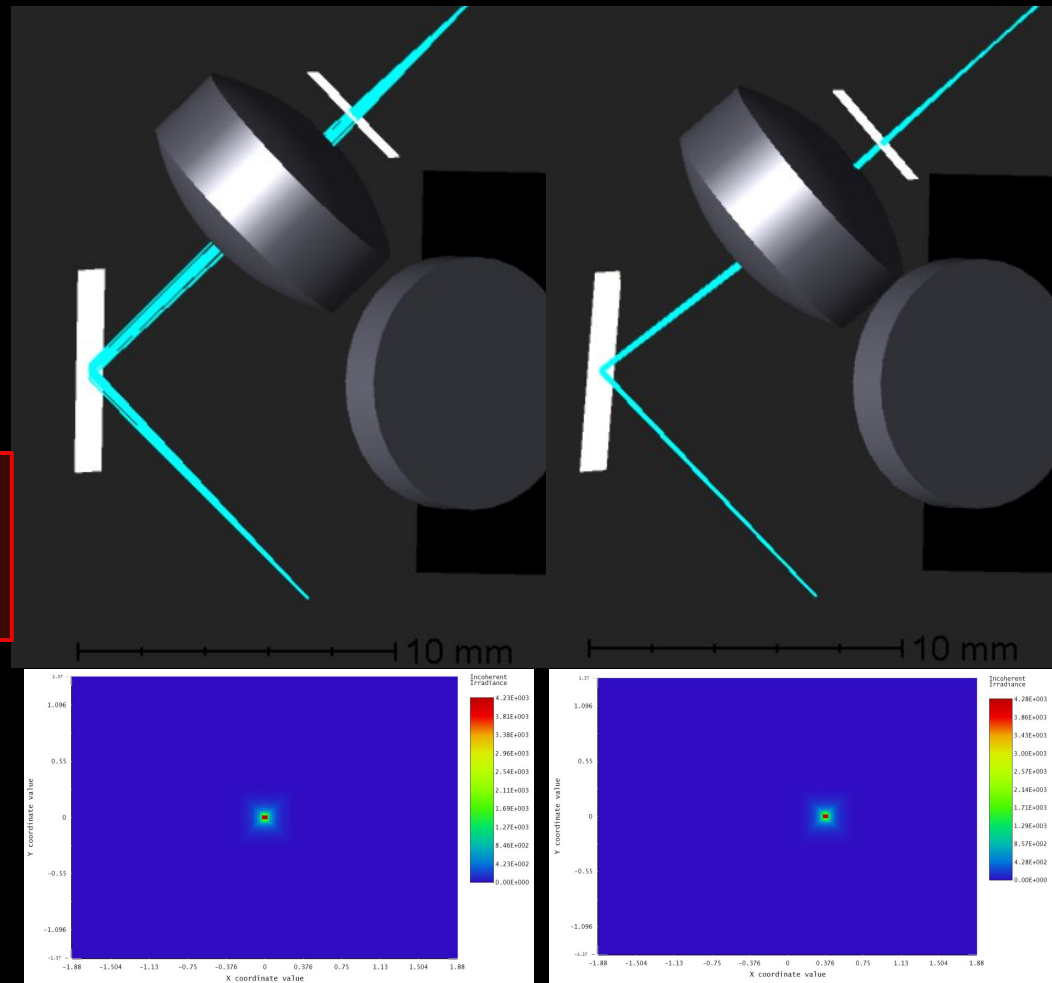
- FSM Tip Angle $\rightarrow$ Translation across Raspberry-Pi Camera
- Obtain FSM Tip Response from Beam Centroid



FSM Calibration System

- FSM Tip Angle $\rightarrow$ Translation across Raspberry-Pi Camera
- Obtain FSM Tip Response from Beam Centroid

Calibration
Optical Path
for 0° & 3.4°
FSM Tip

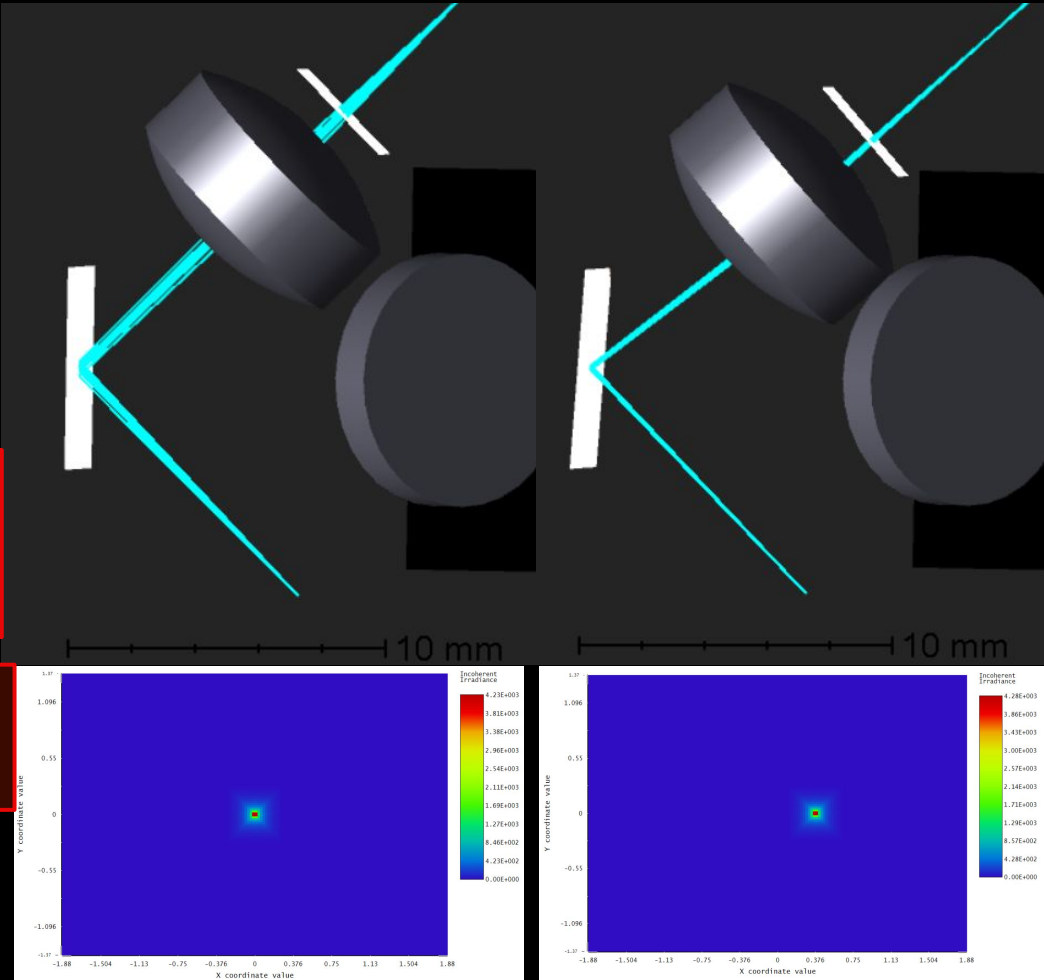


FSM Calibration System

- FSM Tip Angle → Translation across Raspberry-Pi Camera
- Obtain FSM Tip Response from Beam Centroid

Calibration
Optical Path
for 0° & 3.4°
FSM Tip

Raspberry Pi
Signal for 0° &
3.4° FSM Tip

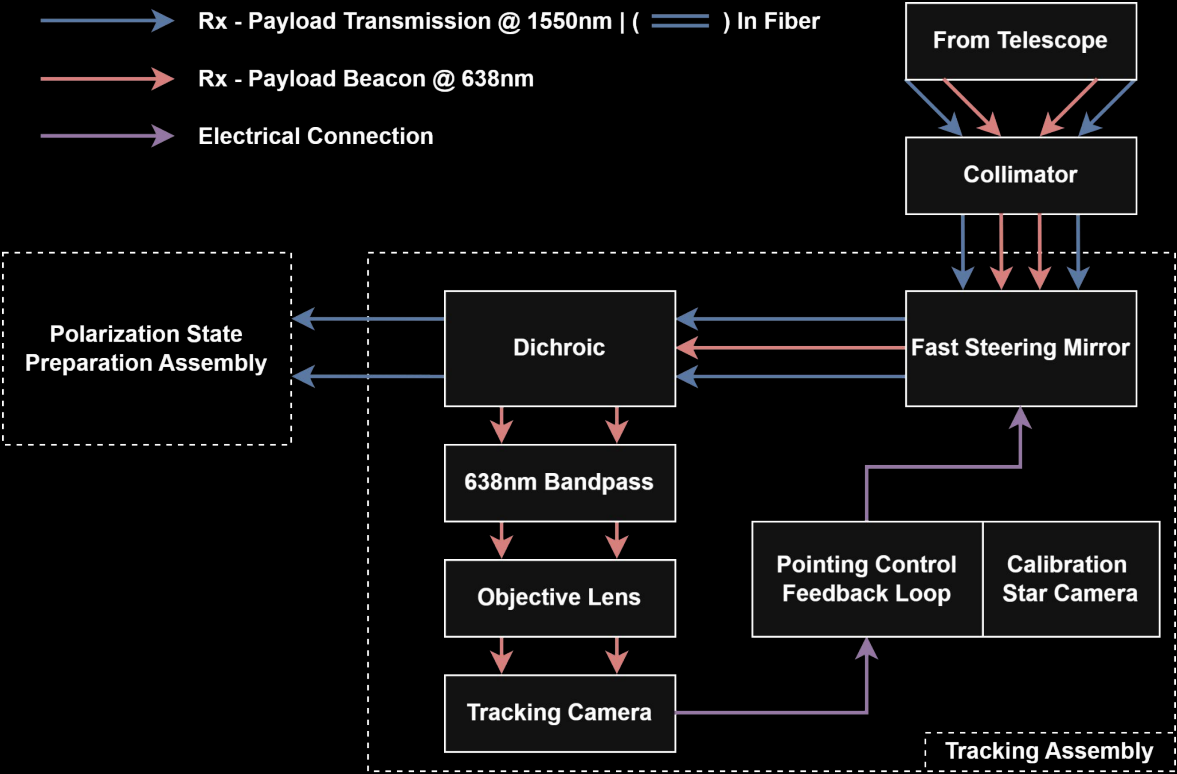


Next Steps for Payload Optical Analysis

- ❑ **General Design:**
 - ❑ Determine optimal distances between optical components
 - ❑ Beam Polarization Analysis
- ❑ **Closed-Loop Feedback Algorithm:**
 - ❑ Obtain improved simulation fit of QPD PSF with FSM Tilt and Uplink Beacon off-angle
- ❑ **Optical Tolerancing:**
 - ❑ Further Validation of current results: Optical Power losses, coarse-pointing, thermal effects
 - ❑ Next optical tolerancing simulation procedures:
 - ❑ Vibrational effects
 - ❑ Thermal effects on individual optical components
 - ❑ Predict FSM tilt response through translation across FSM calibration camera

Optical Ground Station: **Optical Design**

OGS Optical Design - Tracking Assembly



OGS Optical Design – Tracking Assembly

The Celestron CPC 1100 telescope includes a fully computerized Altitude–Azimuth mount and internal pointing model. The telescope is responsible for the coarse tracking of the Payload beacon and will collect both the beacon and data transmission downlinks.

Telescope Type	Aperture	Focal Ratio	Max Slew Rate	Total Mass (kg)	Secondary Mirror Obstruction by Area
Schmidt-Cassegrain Telescope (SCT)	11"	f/10	5° / second	38.1	12%



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Schmidt-Cassegrain Telescope (SCT)	11"	f/10	5° / second		

Max slew rate sufficient for satellite tracking. Large aperture means higher flux received for optical link



OGS Optical Design – Tracking Assembly

The CMOS star camera is mounted on top of the telescope and is paired to a lens which will determine its field-of-view and plate scale. The star camera is used to calibrate the telescope’s pointing model prior to each optical pass by taking images across the sky for star identification.

Model	Mass (kg)	Dimensions (mm)	Resolution (pixels)	Exposure Range
ASI120 MINI	0.06	Ø 36 x 61	1280 x 960	64 µs to infinity



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ASI120 MINI	0.06	Ø 36 x 61	1280 x 960	64 μ s to infinity

Simple and cheap system with widely available software for plate scale.



OGS Optical Design – Tracking Assembly

The fast steering mirror directs the downlink beam with protected aluminum coating mounted on rotary structures with motion tracking sensors. This allows the FSM to calibrate itself to redirect incoming signal beam.

Model	Mass (kg)	Mirror head size (mm)	Clear Aperture (Ø mm)	Angular range (degrees)	Angular Resolution (urad rms)
OIM 5001	0.60	40.64 x 40.64 x 41.402	23.876	+/- 1.5 to +/- 3.0	<0.6 to <1.2



OGS Optical Design – Tracking Assembly

The CMOS tracking camera is mounted onto the telescope and continuously takes images throughout the entire optical pass. It is responsible for determining the switch from open to closed loop tracking (Acquisition), where it then operates in a closed loop with the FSM and telescope for fine pointing.

Model	Mass (kg)	Dimensions (mm)	Resolution (pixels)	Pixel size (μm)	Exposure Range	Plate scale
ASI294MM-P	0.41	73.5 x Ø 78	4144 x 2822	4.63	32 μs to infinity	1 pixel ~ 0.19 arcsec



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Near 0.15 degree FOV

OGS Optical Design – Tracking Assembly

Component	Model	Mass (kg)	Outer Diameter (Ø mm)	Clear Aperture (Ø mm)	Wavelength Range (nm)
PCX Lens	LA4380	0.05	25.4	22.9	185 – 2100
PCX Lens	LA4052	0.05	25.4	22.9	185 – 2100
Dichroic	DMLP900	0.03	25.4	22.86	1550
638 nm Bandpass	F810FC-1550	0.07	24	18.6	1550

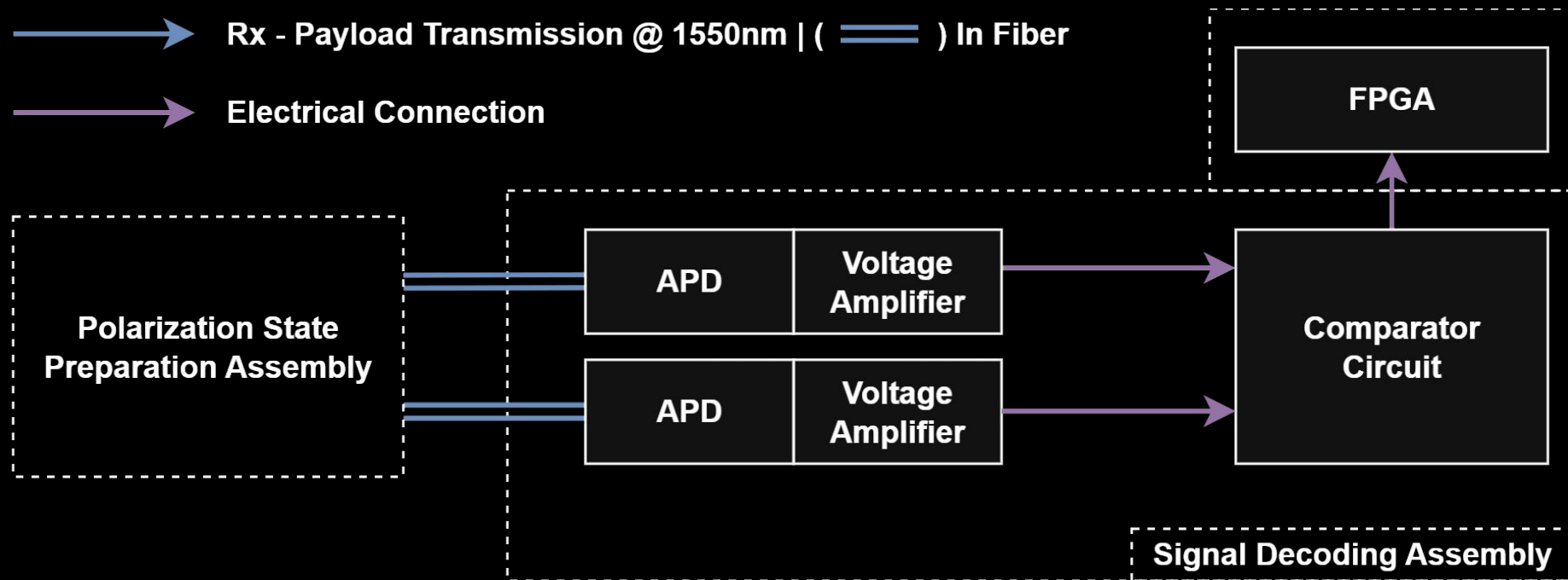
OGS Optical Design – Polarization State Prep. Assembly

Component	Model	Mass (kg)	Outer Diameter (Ø mm)	Clear Aperture (Ø mm)	Wavelength Range (nm)
1550 Bandpass	FBH1550-12	0.05	25	21.1	1550 (T ≥ 90%)
Quarter Waveplate	WPQ10M-1550	0.05	25.4	22.6	1550
Polarizing Beamsplitter	PBSW-1550	0.03	25.4	22.86	1550
Fiber Collimator	F810FC-1550	0.14	24.0	18.11	1550

OGS Optical Design - Signal Decoding Assembly

→ Rx - Payload Transmission @ 1550nm | (==) In Fiber

→ Electrical Connection



OGS Optical Design – Signal Decoding Assembly

The two APD units will capture the laser output from the collimators.

Model	Nominal Bandwidth	Noise Equivalent Power (NEP)	Responsivity at 1550 nm	Wavelength (nm)
LLAM-1550-R 2AH	50 MHz	130 fW/ $\sqrt{\text{Hz}}$	90 kV/W	1550



OGS Optical Design – Signal Decoding Assembly

The two avalanche photodiodes (APD) units will capture the laser output from the collimators to convert directly into an analog electrical signal for the comparator.

Model	Nominal Bandwidth	Noise Equivalent Power (NEP)	Responsivity at 1550 nm	Wavelength (nm)
LLAM-1550-R 2AH	50 MHz	130 fW/ $\sqrt{\text{Hz}}$	90 kV/W	1550



Low NEP and high responsivity supported by signal to noise ratio calculation. Nominal bandwidth within the requirement for modulation, and allows for faster than modulation sampling.

OGS Optical Design – Signal Decoding Assembly

Voltage amplifier will amplify the incoming electrical signal within the detection range of the comparators and FPGA while maintaining an appropriate signal to noise ratio.

Model	Gain	Noise Type	Noise	Cost per unit	Bandwidth
Pasternack PE15A1013	50 dB	Noise Figure	1.1 dB	\$730	10 MHz to 1 GHz



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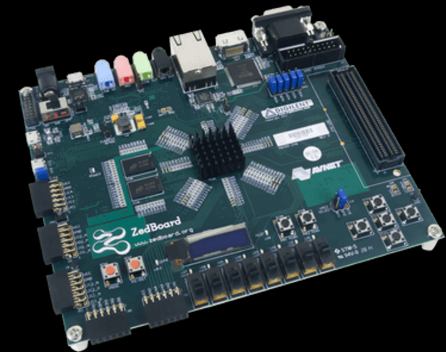


50dB gain results in ~150mV signal out in worst case scenario. Enough to perform comparison and digitization operations.

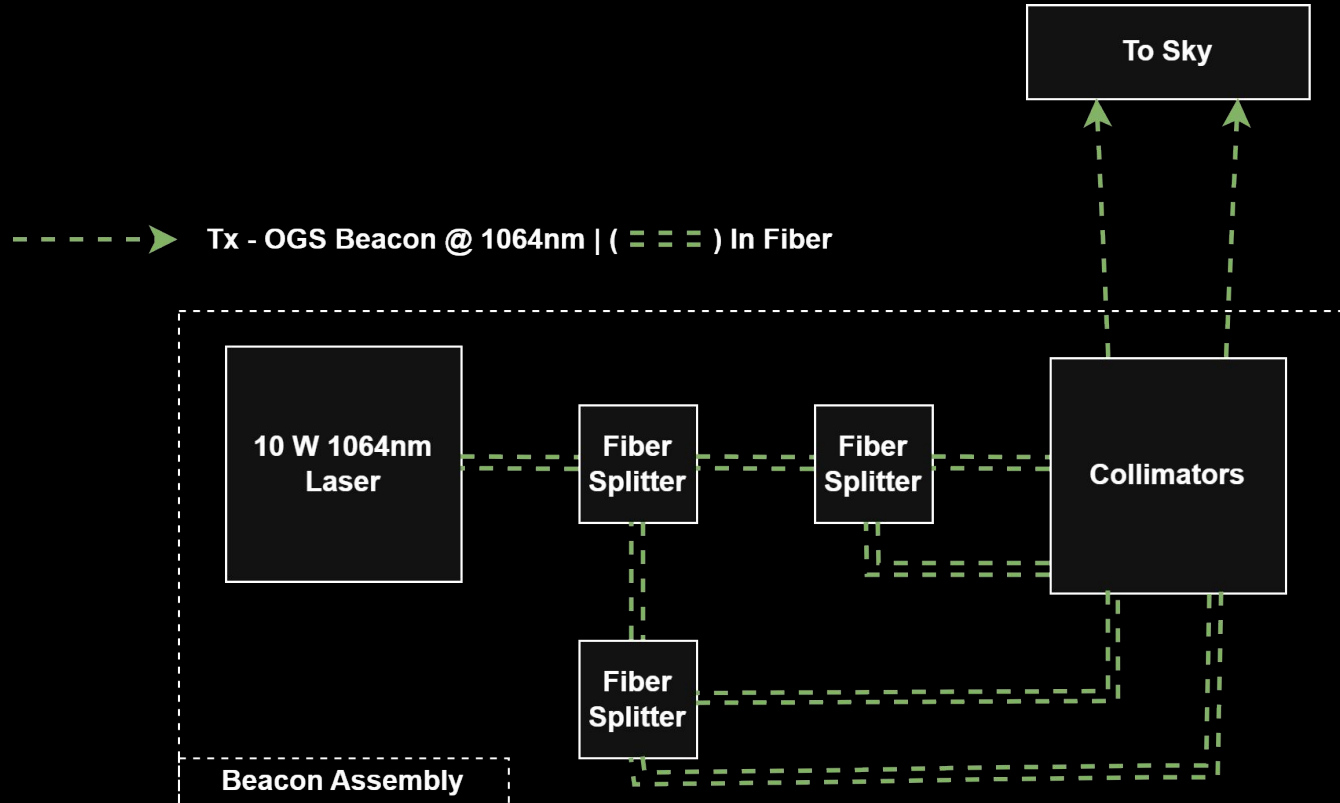
OGS Optical Design – Signal Decoding Assembly

- ❑ Zedboard w/ Zynq-7000 SoC is used for high-speed recording of incoming APD signals from a comparator circuit.
- ❑ The Zedboard has a theoretical maximum processing bandwidth of 100 MHz

Model	FMC Connector Bandwidth	Oscillators Frequencies	Ethernet Bandwidth
Zedboard Zynq-7000	2 Gbit/s per differential pin	33.333 MHz (PS) / 100 MHz (PL)	1 Ghz



OGS Optical Design – Beacon Assembly



OGS Optical Design – Beacon Assembly

Subassembly is powered by a singular 10 Watt narrow linewidth laser at 1064 nm. The fiber output will enter a 50/50 fiber splitter and each output will then enter another 50/50 fiber splitter. Each fiber output will be connected to a lens fiber collimator.

Model	Outer Power	Linewidth (100μs integration)	Output	Beam Quality	Center Wavelength
FL-SF-106 4-10-CW	10 W	<5 kHz	PM fiber with FC/APC connector	M2 < 1.1	1064 nm



OGS Optical Design – Beacon Assembly

Component	Model	Power (W)	Splitting Ratio	Fiber Length (m)	Center Wavelength (nm)
1 x 2 Fiber Splitter	HPFC-1-6-A-1-8-2-2-3	10	50/50	0.5	1060

Component	Model	Insertion Loss (dB)	Beam Diameter (mm)	Beam Divergence (degree)	Center Wavelength (nm)
Single Mode Fiber Collimator	FCHP-1-01-0-5-1-1-3	<0.3 dB	≤ 0.5	≤ 0.25	1064

Optical Ground Station: **Optical Analysis**

Overview of Zemax

The OGS Zemax simulation is a model of the OGS system which uses ray tracing analysis to validate the optical system and component selections. The simulation is built on non-sequential mode because scattering and splitting must be supported.

The incoming source is gaussian and has its wavelengths, divergence and size defined.

The components are modeled using specific coatings to mimic the appropriate reflectivity and transmission ratios.

Constraints:

- ❑ **Polarization**

- ❑ The current model does not have polarization implemented. There are minimal tools for analyzing polarization in non-sequential mode.

- ❑ **Wave Properties**

- ❑ Zemax focuses on geometric ray tracing, so it does not inherently simulate any diffraction or interference patterns during ray propagation.

Overview of Zemax

The OGS Zemax simulation is a model of the OGS system which uses ray tracing analysis to validate the optical system and component selections. The simulation is built on non-sequential ray tracing, which includes scattering and surface roughness.

The incoming source is defined by its wavelengths, position, and direction.

The components are modeled using specific coatings to mimic the appropriate reflectivity and transmission ratios.

Constraints:

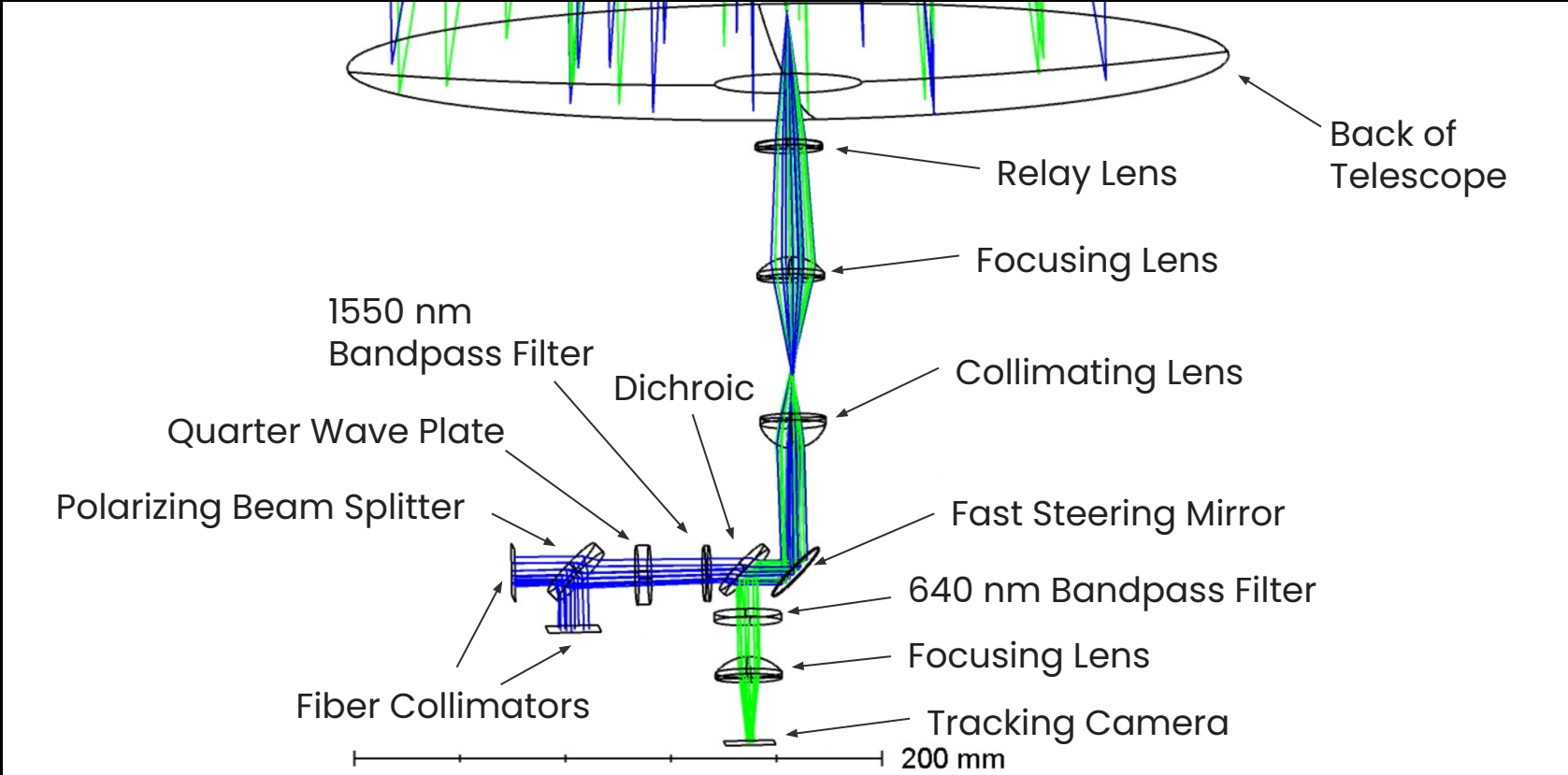
❑ **Polarization**

- ❑ The current model does not have polarization implemented. There are no options for analyzing polarization.

The OGS Zemax simulation is modeled in a way that demonstrates proper gaussian source behavior and component performance. The system limitations do not affect any of the current component analysis and tolerancing results.

Since Zemax is based on geometric ray tracing, so it does not inherently simulate any diffraction or interference patterns during ray propagation.

Overview of Zemax



Optical System Analysis (Zemax)

❑ **Collimation/Focusing**

- ❑ The collimating lens allows the beam to travel through the entire system without interfering with the edges of the optics, ensuring that it reaches the detectors uniformly.
- ❑ The focused beam at the tracking camera also ensures that the centroid does not stray off the detector, even as the incoming incidence angle varies.

❑ **Numerical Aperture Constraints**

- ❑ Since the beam is truncated to a smaller size, it consistently approaches the fiber collimators within the acceptance angle. Therefore, there is no concern about numerical aperture because the beam remains precisely collimated.

Optical System Analysis (Zemax)

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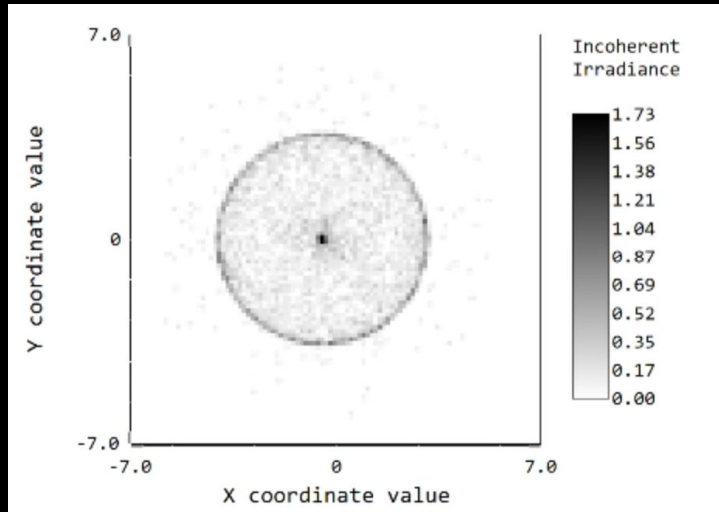
- ❑ The foc
centro
incide

The usage of collimation and focusing lenses allows the beam to maintain the size necessary to fully reach the fiber collimators and tracking camera, while also minimizing divergence and reducing stray light.

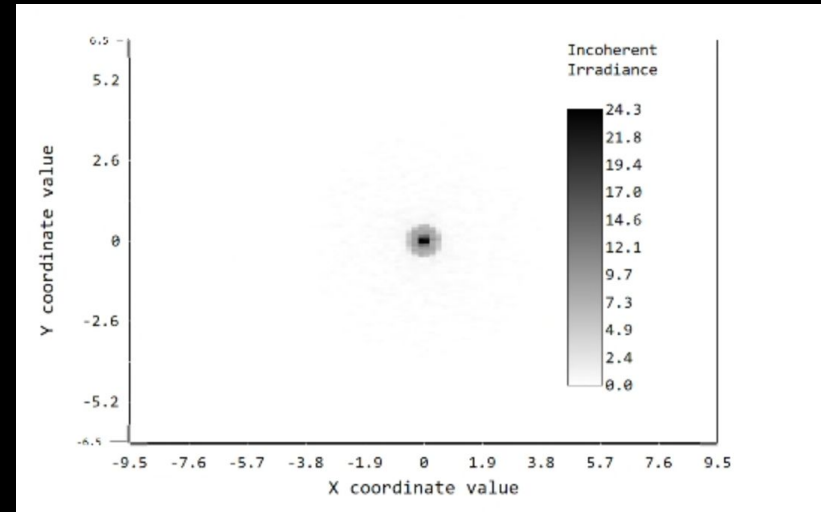
❑ Numerical

- ❑ Since
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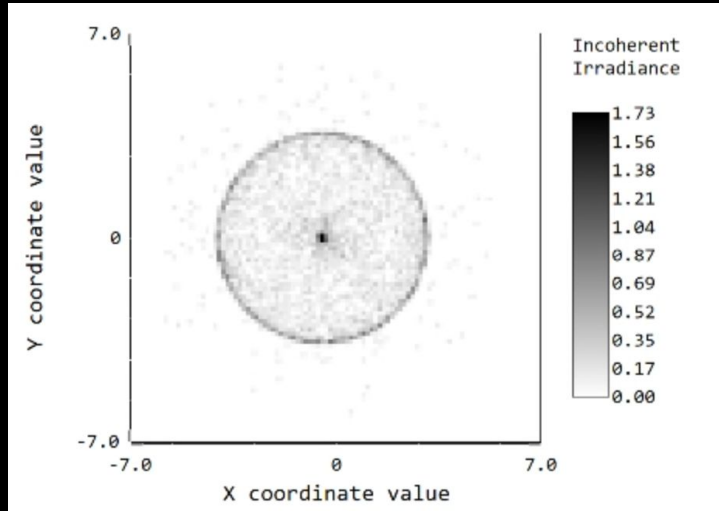


Collimated beam at the fiber collimators

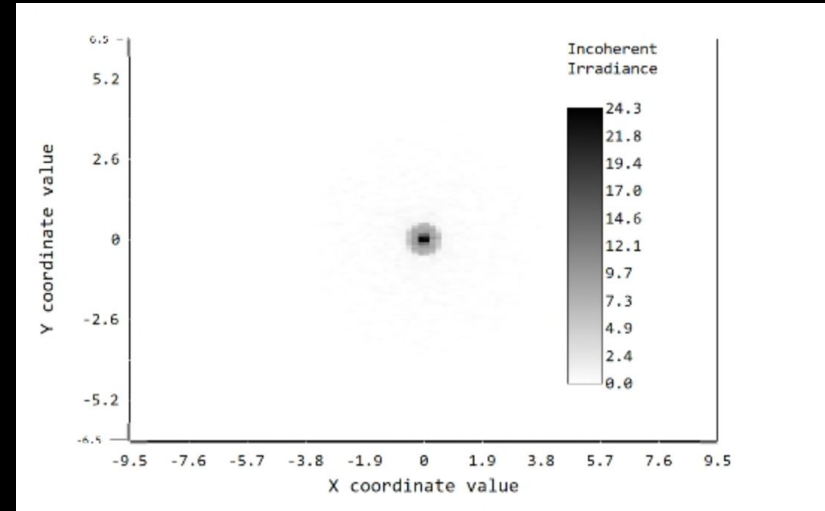


Focused beam at the tracking camera

Optical System Analysis (Zemax)



Collimated beam at the fiber collimators



Focused beam at the tracking camera

The beams are well within the apertures for the fiber collimators and tracking camera

Optical System Analysis (Zemax)

The FSM needs to be specifically analyzed because it must correct incoming beam deviations without significantly affecting how the beam is received at the fibers and the tracking camera.

❑ **Angle shift**

- ❑ The 1-inch FSM allows for a relatively large range of angular adjustment because the beam diameter is small.
- ❑ A larger mirror is unnecessary because any tolerable angle deviations fall within the steering range of the current FSM (± 1.5 to ± 3.0 degrees).

❑ **Physical constraint**

- ❑ Additionally, increasing the mirror size would cause it to interfere with the dichroic and bandpass filter, which are positioned within 20 mm of the FSM.

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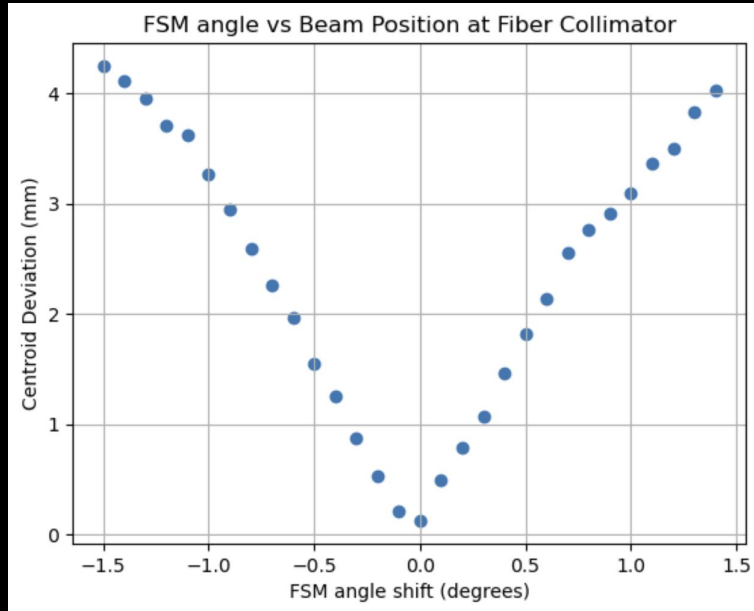
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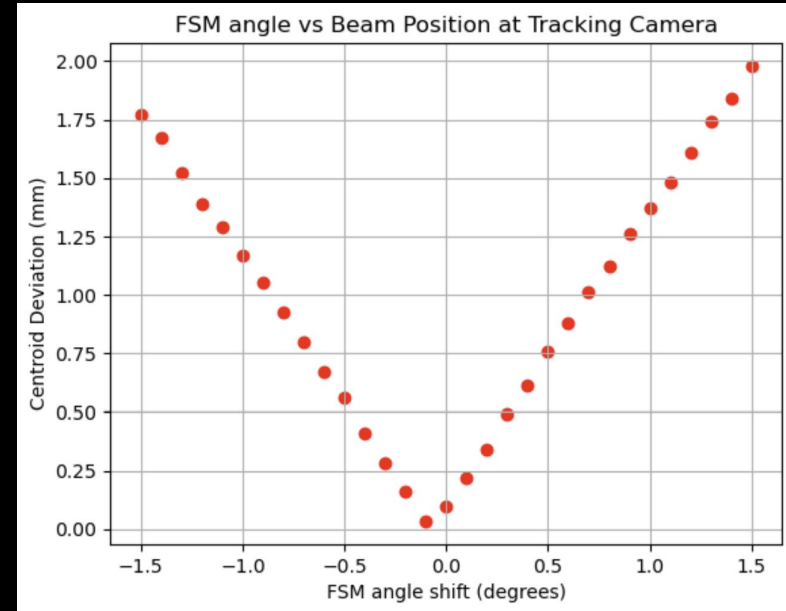
The chosen FSM is the best option for our system because it meets the physical spacing requirements and can shift over a wide range while still keeping the beam aligned on the path to the fibers and tracking camera.

Optical System Analysis (Zemax)

FSM Angle Shift vs. Centroid Deviation



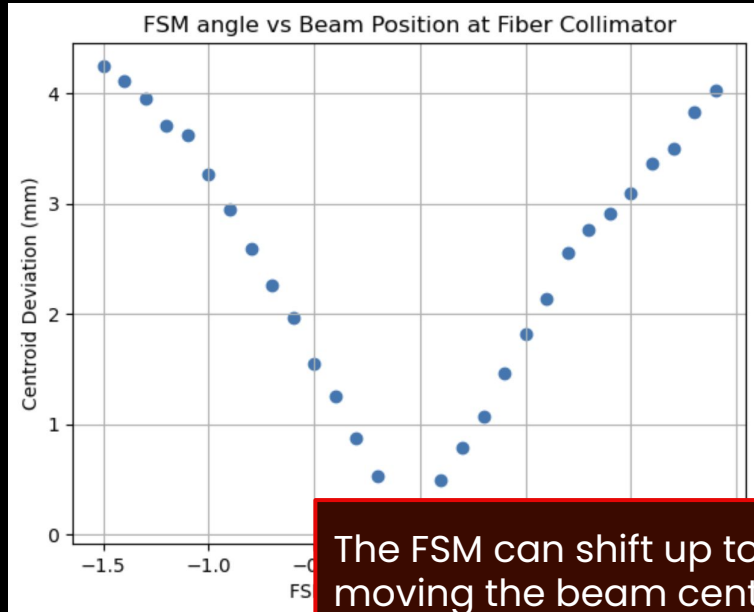
Fiber Collimator



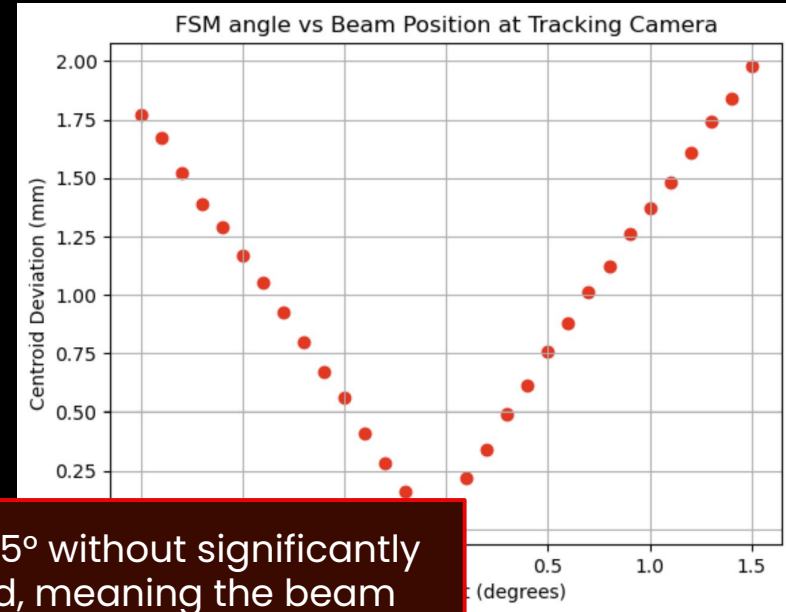
Tracking Camera

Optical System Analysis (Zemax)

FSM Angle Shift vs. Centroid Deviation



Fiber



Camera

The FSM can shift up to $\pm 1.5^\circ$ without significantly moving the beam centroid, meaning the beam stays within the tracking camera's field and continues along the correct optical path.

Telescope Model (Zemax)

The 11in Schmidt-Cassegrain Telescope is modeled so that the behavior of the beam is as accurate as possible.

- ❑ A Zemax model of an 8-inch telescope in sequential mode was scaled up to represent an 11-inch telescope.
- ❑ The components were translated into non-sequential objects so that they could be added to the OGS simulation.
- ❑ The only part missing within the model is the corrector plate, which isn't currently important because aberrations are not being measured.
- ❑ Relay lens and beam expander were added directly behind the telescope

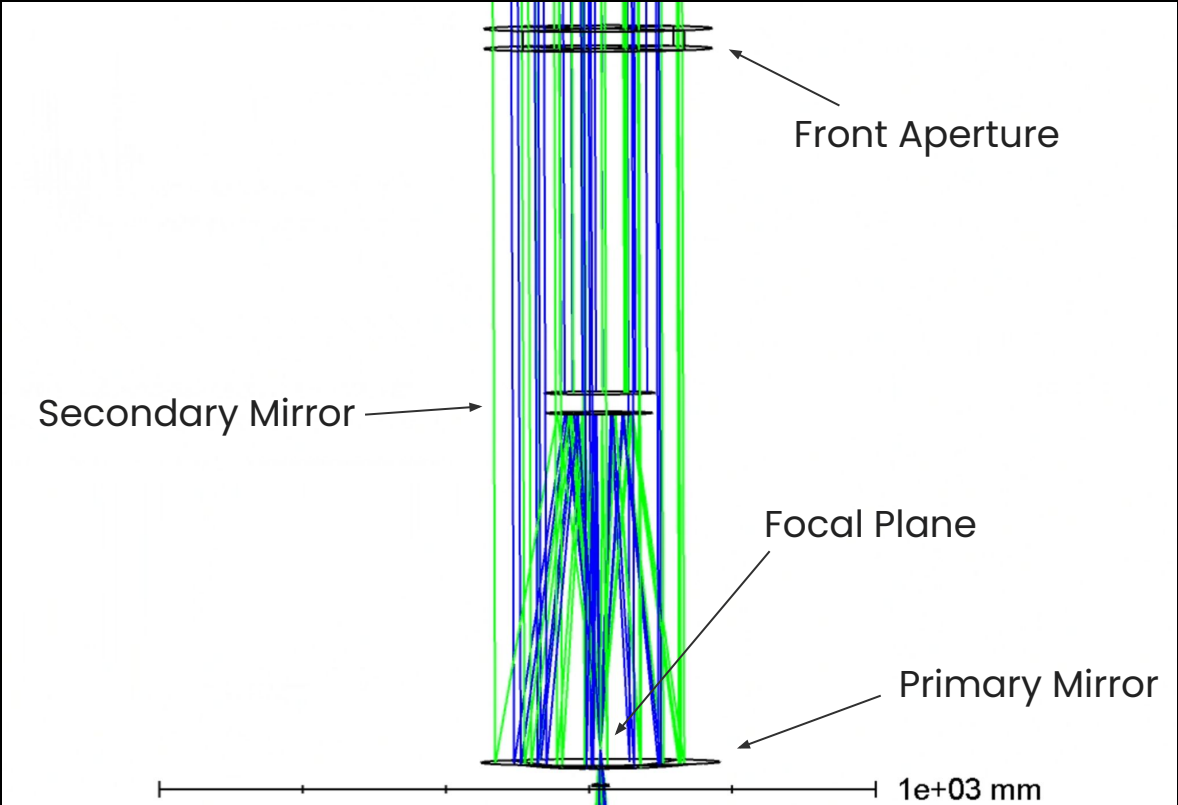
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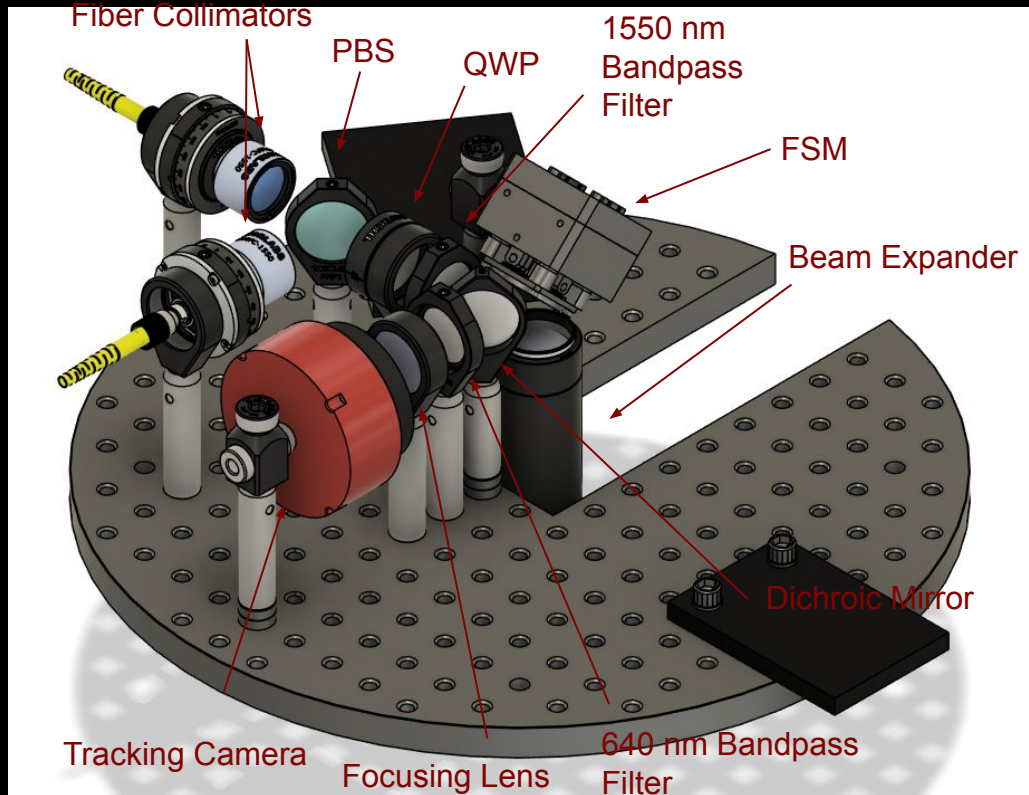
The simulated telescope isn't an exact replica of the optics within the actual model; however, it includes the main components needed to demonstrate how the beam will propagate through the entire system at the output.

Telescope Model (Zemax)



Optical Ground Station: **Structural Design**

CAD Assembly Model



- ❑ All optical components have a similar diameter of $\varnothing 1''$ to accommodate the large beam waist.
- ❑ FSM is mounted at 45° to take incident light into the plane of the optical breadboard.
- ❑ An attachable 3D printed encasement will be used to cover optical components from environment.
- ❑ Total Mass: 4.232 kg

CAD Telescope Model

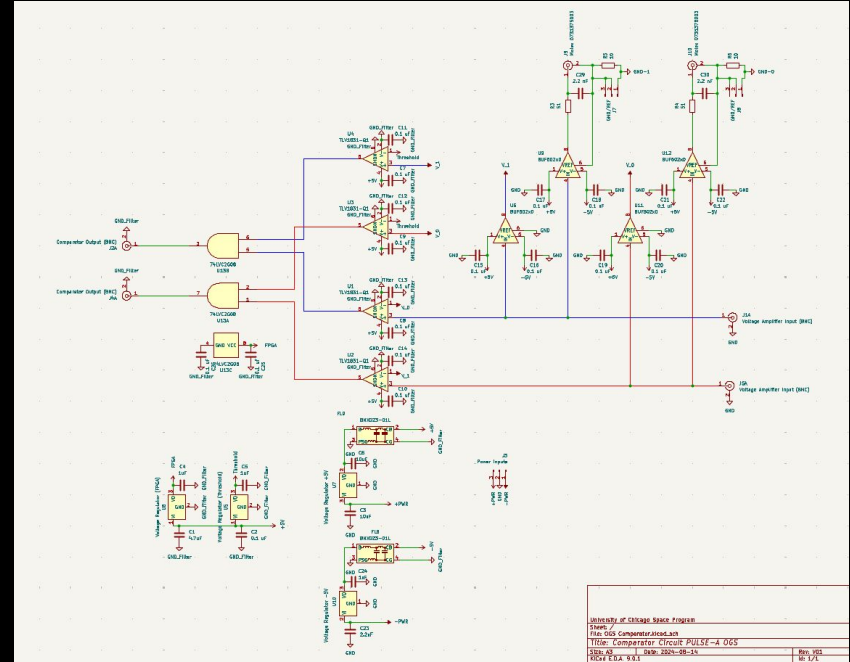
The entire layout will be attached to the telescope using custom brackets compatible with the telescope's pre-existing mounting holes.



Optical Ground Station: **Electronics Design**

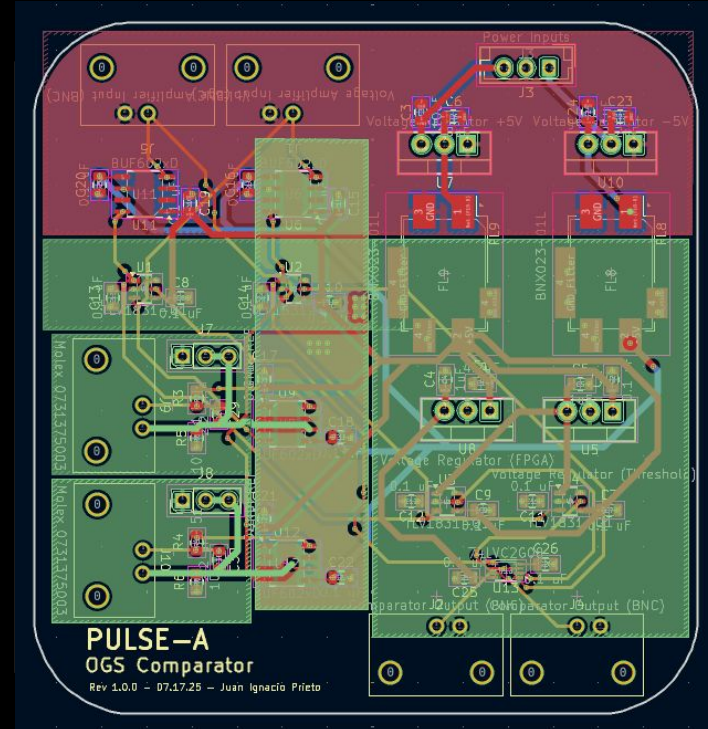
Comparator Circuit

- ❑ A comparator circuit is used to reduce noise and quantize analog photodetector signals to digital LOW/HIGH signals.
- ❑ We assume only one photodetector signal may be high at a time.
- ❑ Both channels are first compared against a minimum voltage threshold.
- ❑ If they meet the threshold, the higher voltage signal will be pulled HIGH, and the lower will be pulled LOW.



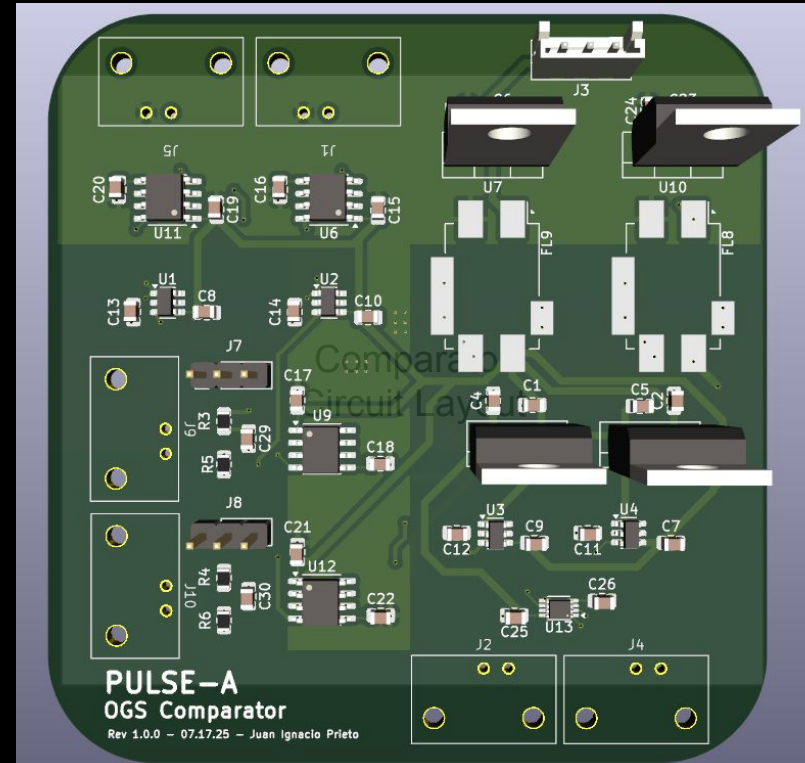
Comparator Circuit

- ❑ The comparator circuit output may be in the following states:
 - ❑ Both signals $<$ threshold
Result: Both channels LOW
 - ❑ Signal 1 $<$ threshold
Signal 2 $>$ threshold
Result: Channel 1 LOW, Channel 2 HIGH
 - ❑ Both channels $>$ threshold
Signal 1 $>$ Signal 2
Result: Channel 1 HIGH, Channel 2 LOW
 - ❑ Both channels $>$ threshold
Signal 1 = Signal 2
Result: Undefined Behavior



Comparator Circuit

- ❑ 4-layer design
 1. +PWR, PWR GND
 2. Signal GND
 3. Signal
 4. -PWR, PWR GND
- ❑ TLV3501 comparator IC with 500 μ V resolution at high bandwidth
- ❑ EMI filters + 4-layer mean low noise and low crosstalk
- ❑ 4 outputs
 - ❑ 2 for digitizing
 - ❑ 2 for data logging and laser acquisition verification
- ❑ May need op-amp to increase input voltage

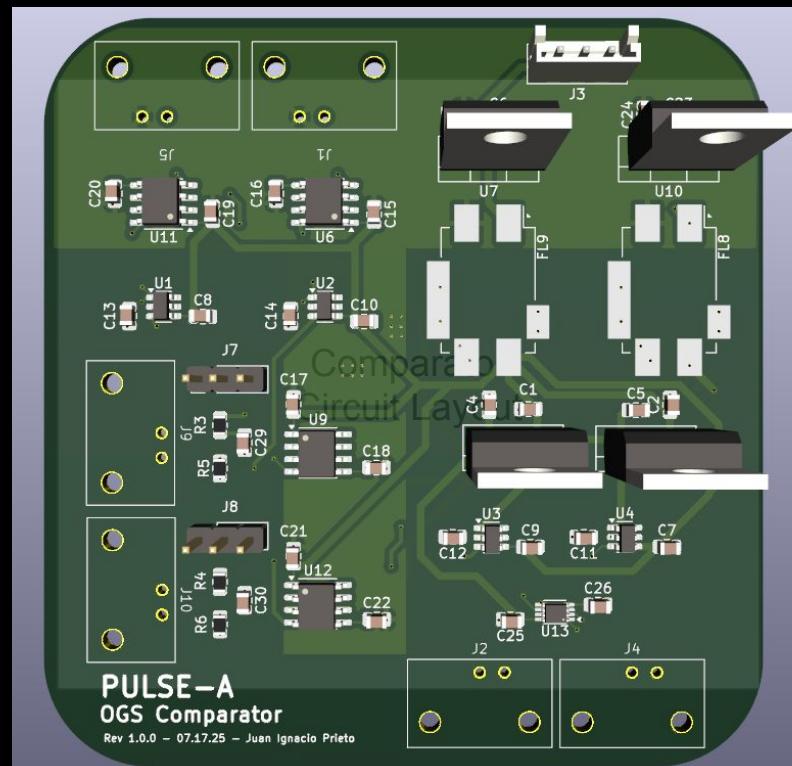


Comparator Circuit

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 1. +PWR, PWR GND
 2. Signal GND

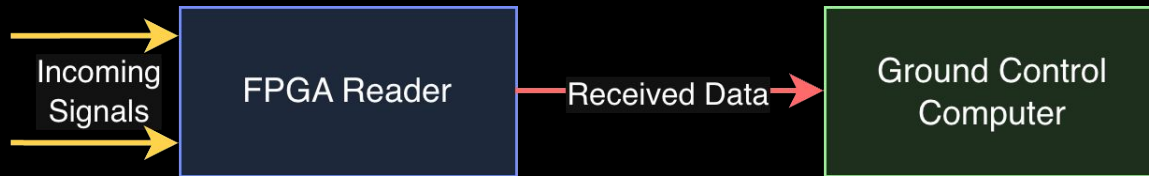
Low noise and high resolution two channel comparator can measure small differences in polarization states.

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- ❑ 4 outputs
 - ❑ 2 for digitizing
 - ❑ 2 for data logging and laser acquisition verification
- ❑ May need op-amp to increase input voltage



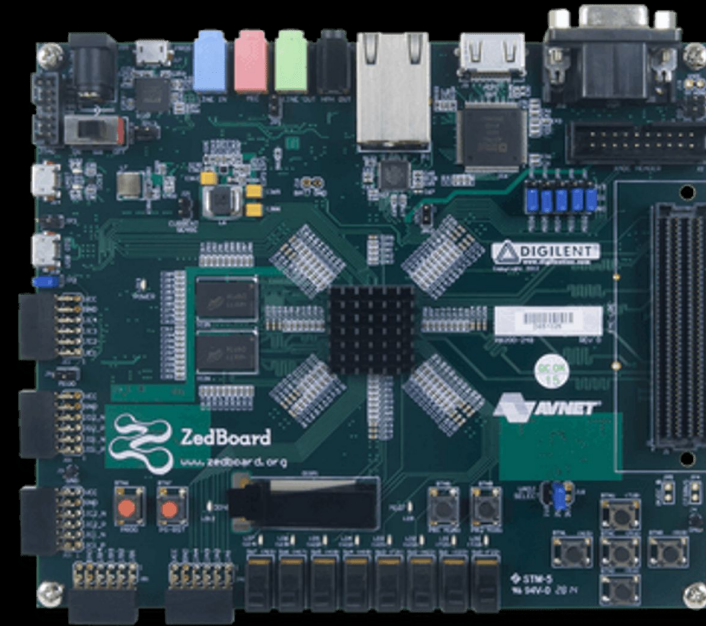
FPGA Assembly – Overview

- ❑ An FPGA-based board decodes the comparator circuit's quantized signals into a single data bitstream
- ❑ The FPGA is responsible for:
 - ❑ Decoding the transmission protocol
 - ❑ Applying error correction decoding
 - ❑ Transmitting the resulting data to the OGS control computer



FPGA Component Choice

- ❑ The Digilent Zedboard is a development board for the Xilinx Zynq-7000 SoC (combined FPGA and CPU system)
- ❑ We use the board's expansion connector (FMC) to connect the comparator circuits two digital output pins
- ❑ The board's Ethernet port will be used to output collected data to a computer

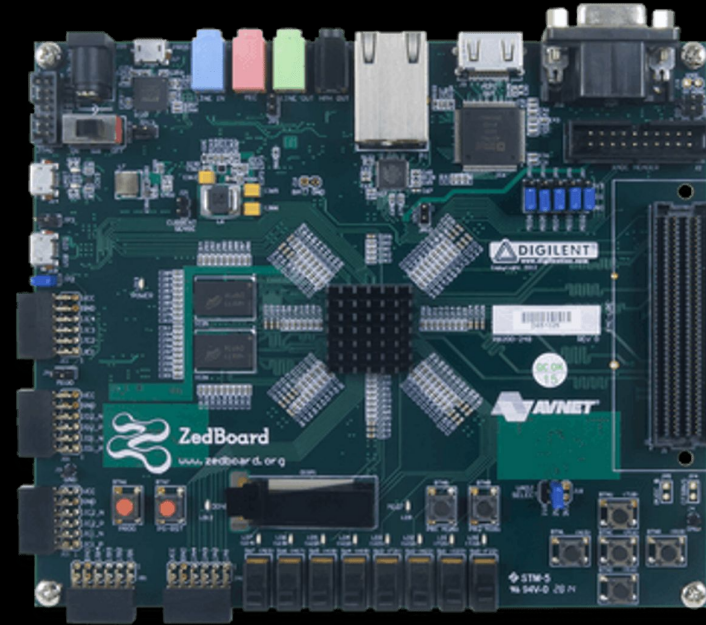


FMC Connection
to Comparator

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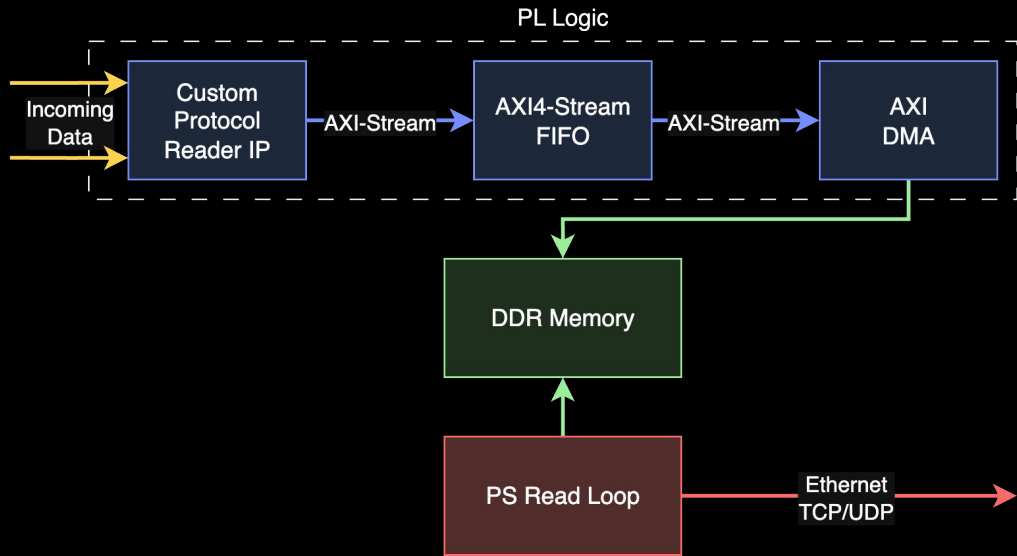
Zedboard's FPGA fabric can be clocked up to 100MHz



FMC Connection
to Comparator

FPGA Block Diagram

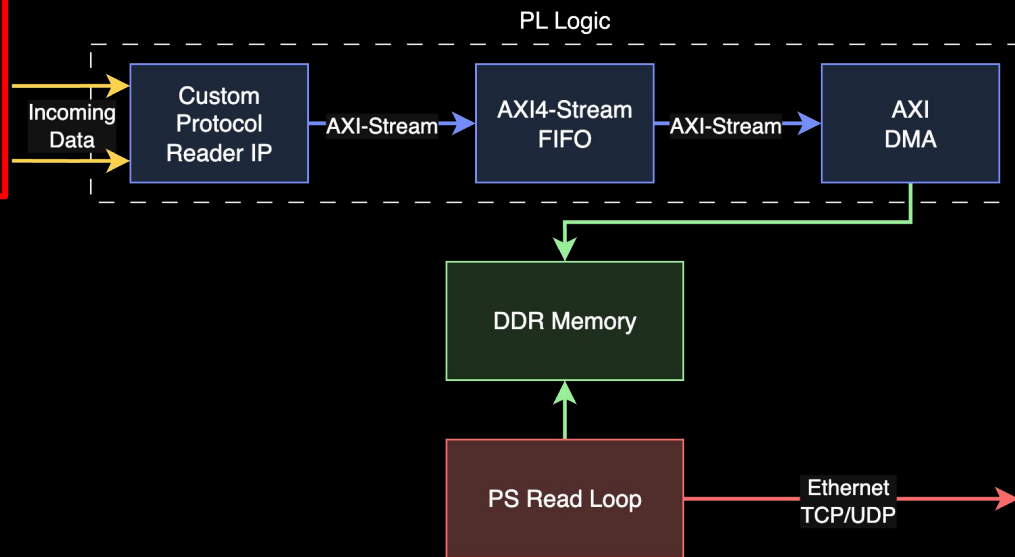
- ❑ Zynq-7000 allows parallelizing work between
 - ❑ FPGA - Programmable Logic (PL)
 - ❑ CPU - Processing System (PS)
- ❑ PL handles all high-speed operations at data rate
- ❑ PS does the slower transmission to the OGS computer
- ❑ PL and PS communicate by storing collected bits in a buffer in DDR memory



FPGA Block Diagram

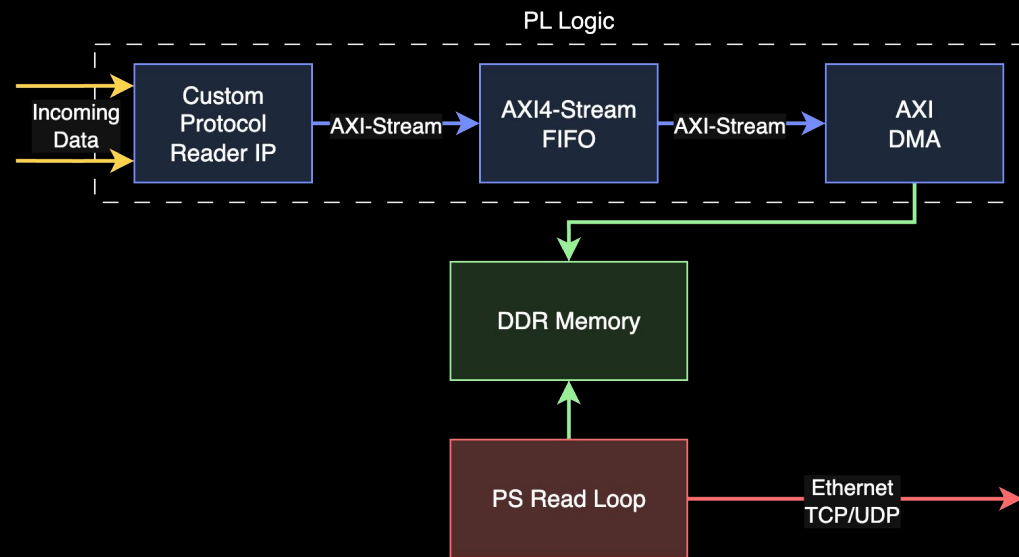
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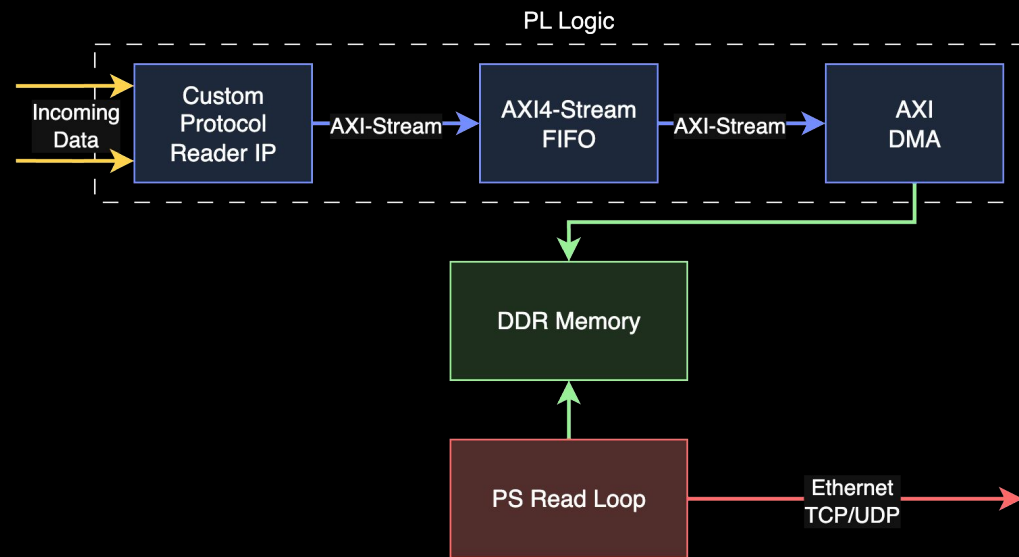
FPGA Block Diagram

- ❑ The FPGA fabric is clocked by the fastest clock available (100MHz on Zedboard)
- ❑ A custom protocol reader module will decode the transmission pins
- ❑ Data is stored in a FIFO buffer until it is ready to be written in memory
- ❑ A Direct Memory Access (DMA) module reads chunks of data out of the FIFO and stores them in one of two memory buffers
- ❑ DDR memory has two alternating storage buffers to allow the PS to transmit one while the PL writes to the other



FPGA Block Diagram

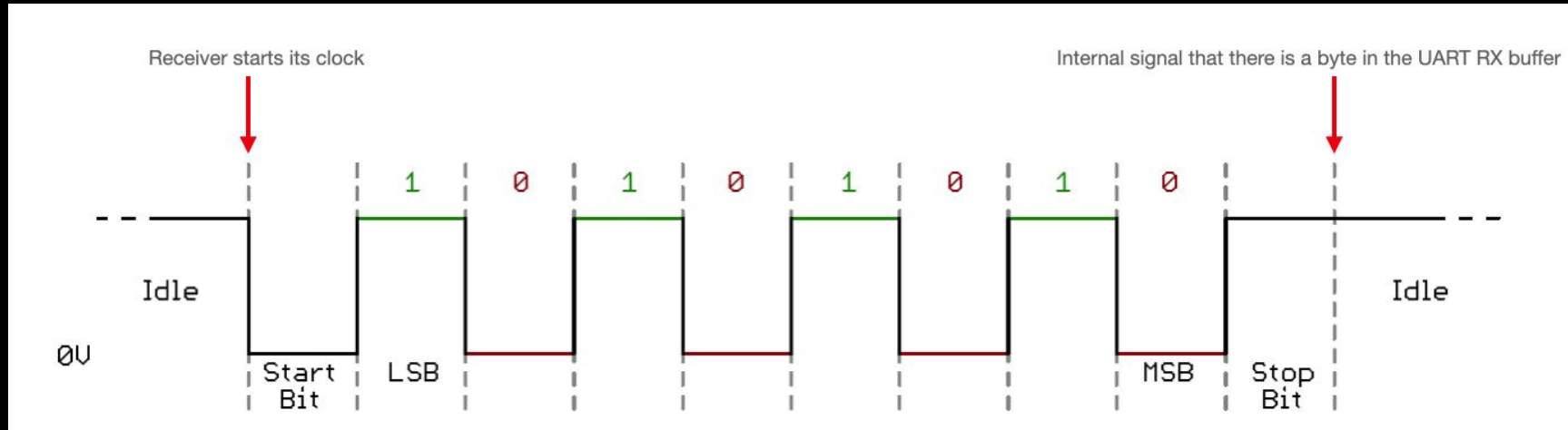
- ❑ The PS runs a TCP server (lwIP) accepting clients on the network connected to its Ethernet port
- ❑ When a buffer is filled, the PS sends a packet with the buffer contents to all connected clients



Optical Ground Station: **Electronics Analysis**

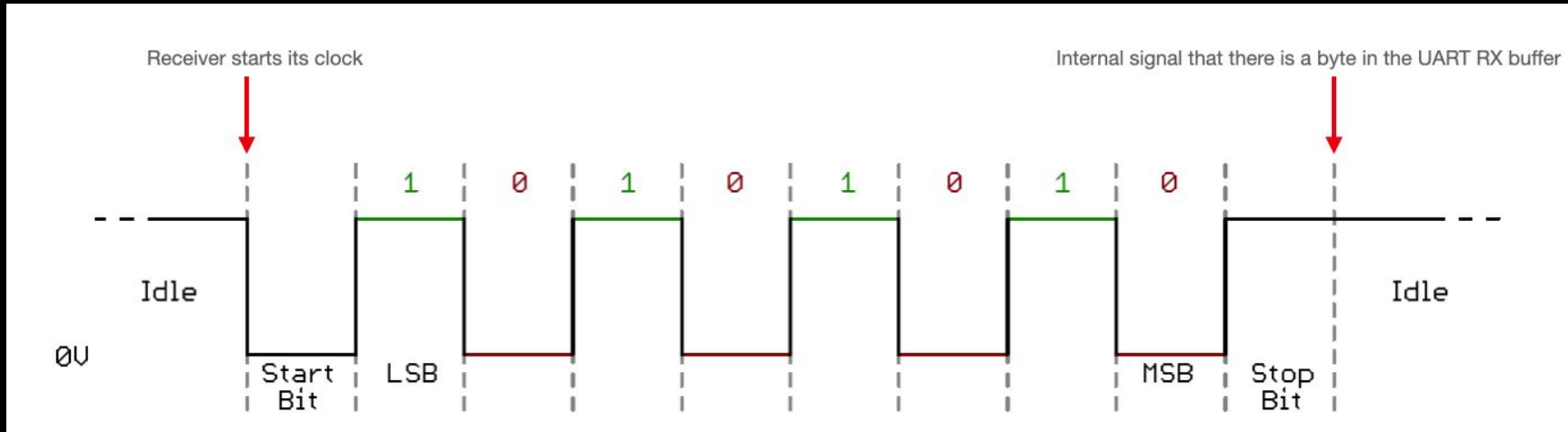
Transmit/Receive Clock Synchronization

- ❑ Since we cannot transmit the clock signal along with the data, we must use an asynchronous transmission protocol that allows us to recover the clock signal
- ❑ The FPGA's protocol reader detects the clock and outputs the data



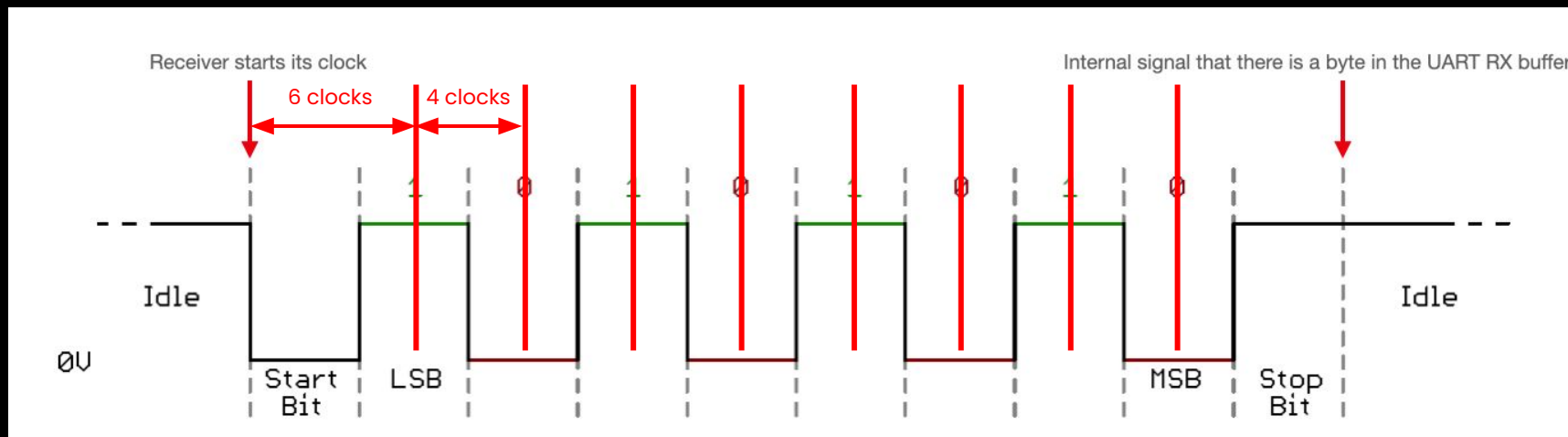
Transmit/Receive Clock Synchronization

- ❑ We intend to use UART, which transmits an idle HIGH state, then sends one start (LOW) bit to signal the start of a data frame, 8 data bits, and returns to idle for at least one bit



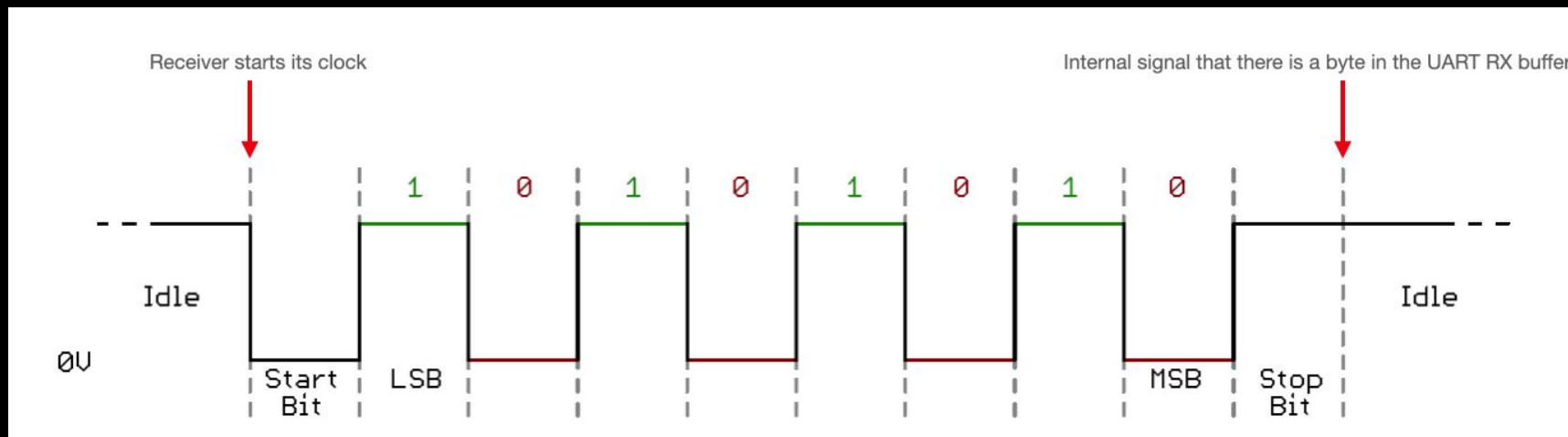
Transmit/Receive Clock Synchronization

- ❑ By running the receiver FPGA at a clock speed multiple times higher than the data rate, we wait until the start bit is detected.
- ❑ Assuming a transmission rate of 10Mhz and receiver clock of 40Mhz, we wait for 6 clock pulses to sample into the first bit, then sample a bit every 4 clocks



Transmit/Receive Clock Synchronization

- ❑ We are limited by the sampling rate of the photodetectors, which only allow us to sample at 50Mhz
- ❑ The more precisely we can detect the leading edge of the start bit, the more data bits we can transmit without needing to resync the clock



Error Correction – Signal Decoding Assembly

Source → Channel → Receiver: Raw sensor packets encoded on-board, transmitted through atmospheric channel, then decoded on ground.

Current Decoder Assembly:

- ❑ Hamming decoding (baseline): detects/corrects single-bit errors but insufficient under higher-noise regimes.
Reed–Solomon (RS) code attempt: Implemented RS using $GF(2^8)$ arithmetic
- ❑ Built new `source_encoder`, `channel_encoder`, `channel_decoder` modules to support symbol-level operations.
- ❑ Modular abstraction so Hamming, RS, and other codes can be swapped without rewriting pipeline.

Error Correction – Signal Decoding Assembly

Status: RS fully implemented but requires integration testing with real packet sizes and simulated burst-error patterns.

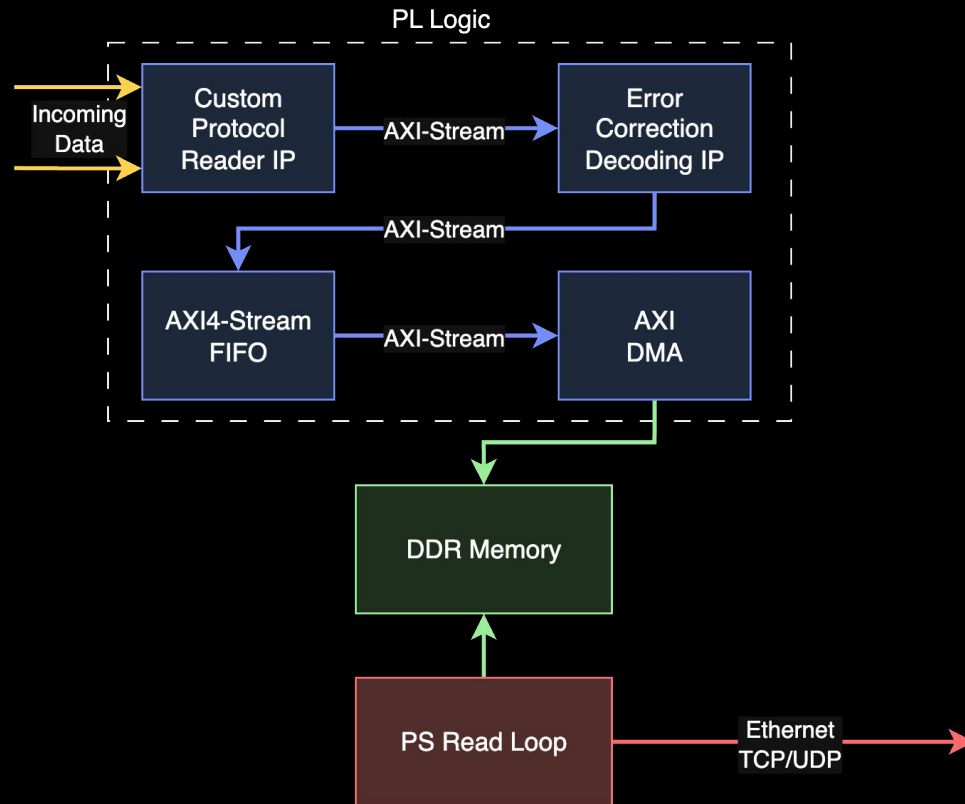
Preliminary Results

Hamming decoder: ~1-bit correction, ~low redundancy, struggles with burst errors typical in atmospheric channel.

RS prototype: corrects multi-symbol bursts and expected to outperform Hamming under atmospheric noise

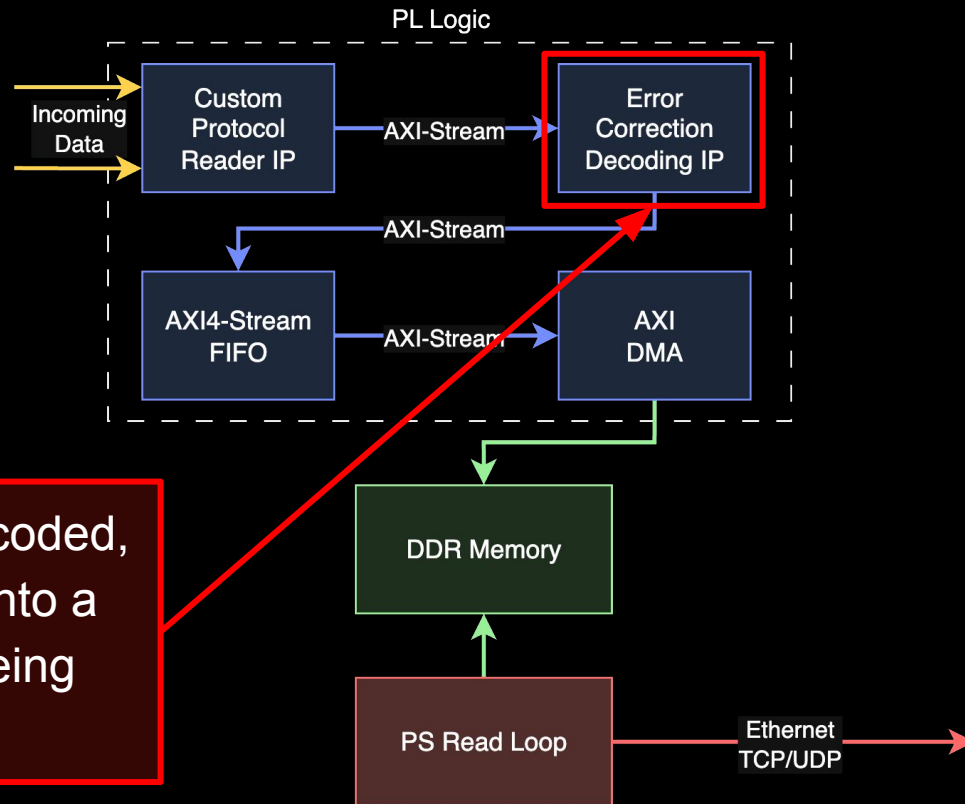
Error Correction – Signal Decoding Assembly FPGA

- ❑ FPGA allows for at-speed error correction decoding inline with data collection
- ❑ Xilinx SDK includes a standard Reed-Solomon decoder IP



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Error Correction – Statistical Analysis

- ❑ Based on preliminary testing with atmospheric noise approximations, we know that Reed Solomon (RS) decoding works well for sparse noise lower than approximately 10% distortion from the original transmission as well as noise coming from erasures.

Error Correction – Statistical Analysis

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We suspect that with the more destructive bursts of atmospheric noise, RS alone will not be enough to mitigate error.

Error Correction – Statistical Analysis

- ❑ Based on this, we plan to leave RS as the outer level of error correction and add an inner schema to lower the raw error before RS does the final decoding.

Error Correction – Statistical Analysis

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Moving forward we are considering Convolutional codes, Turbo codes, Polar codes, and Low Density Parity Check codes to test in tandem with RS

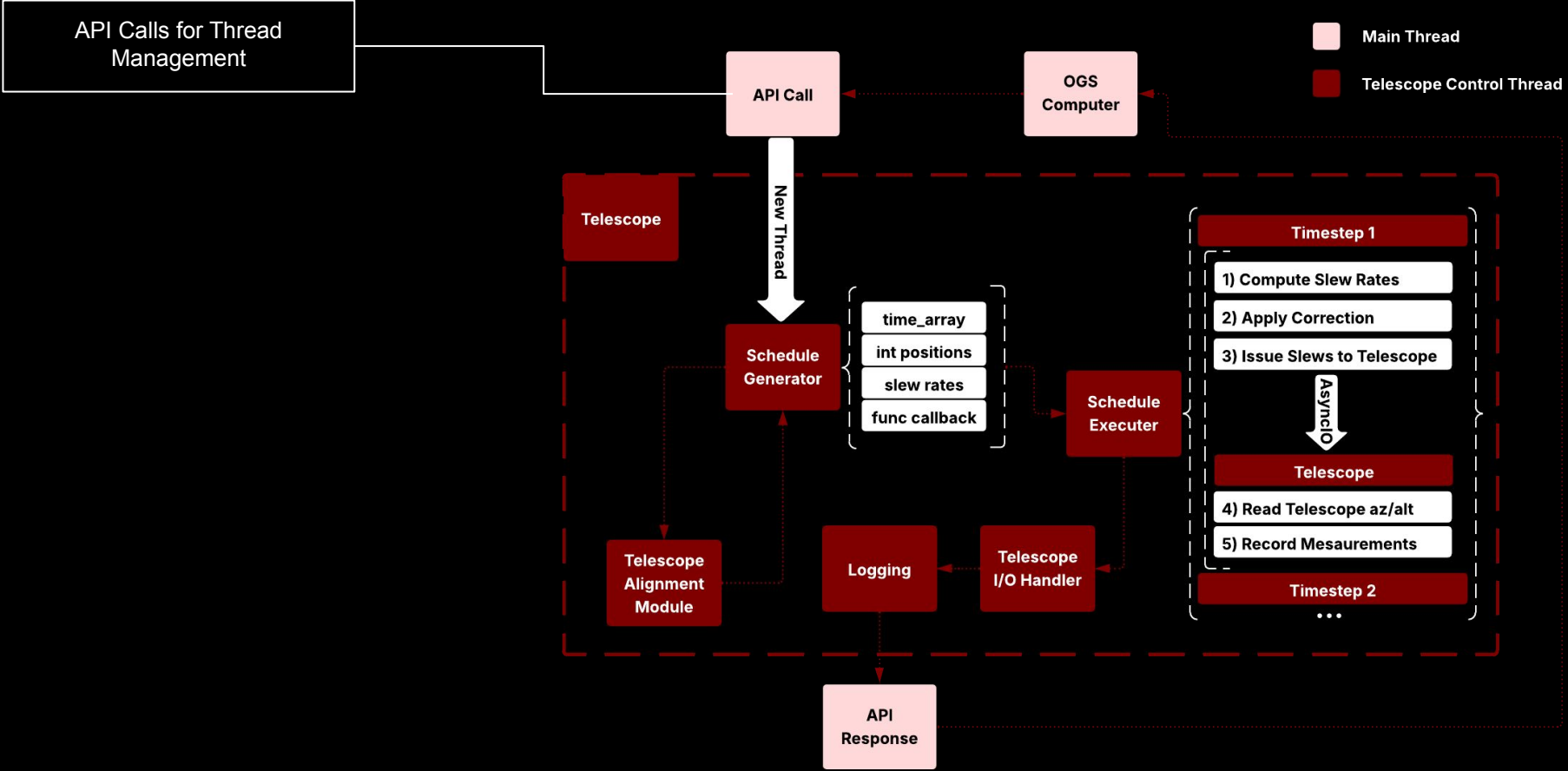
Optical Ground Station: **Ground Control**

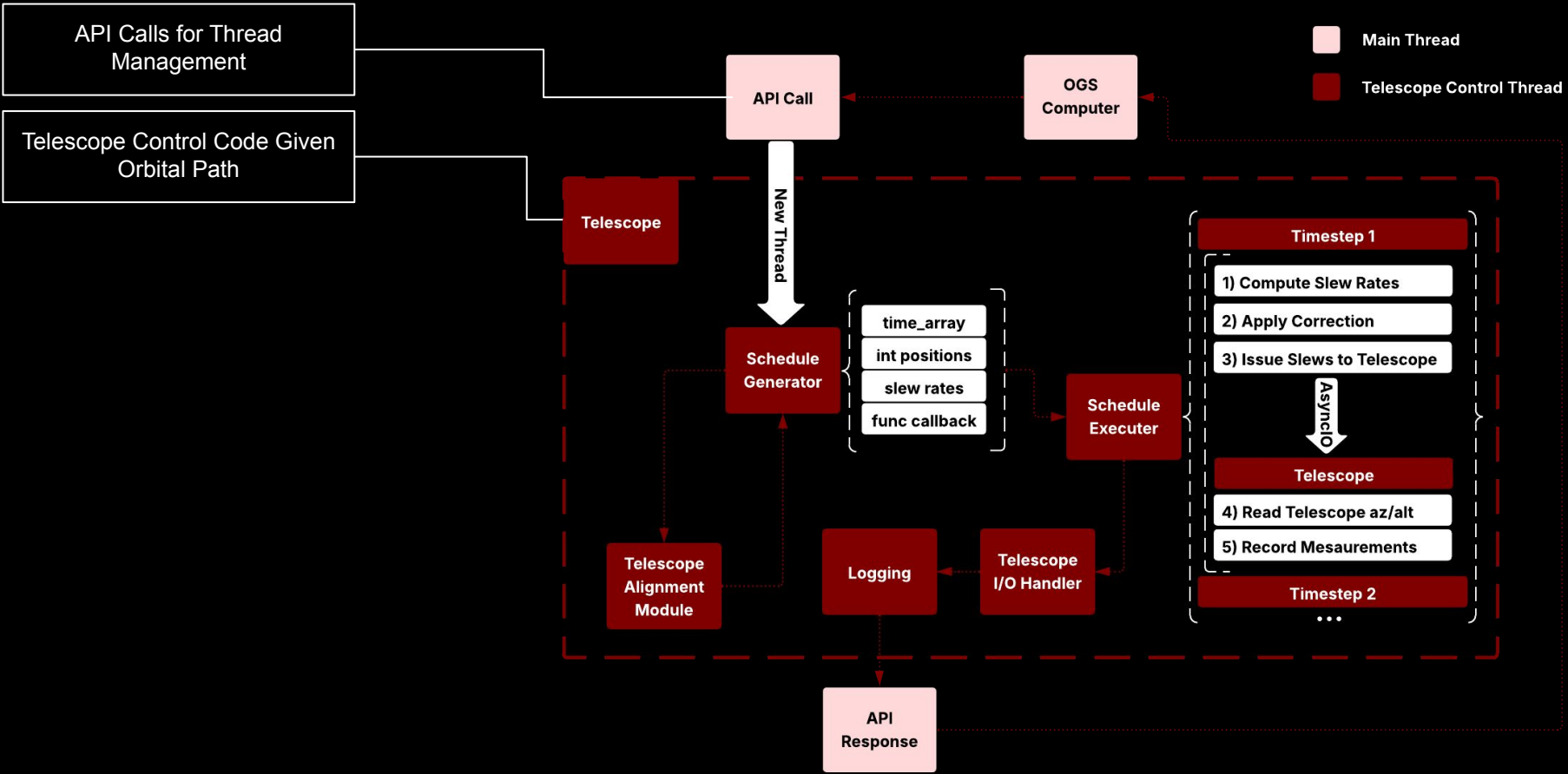
Ground Control: RFGS Integration

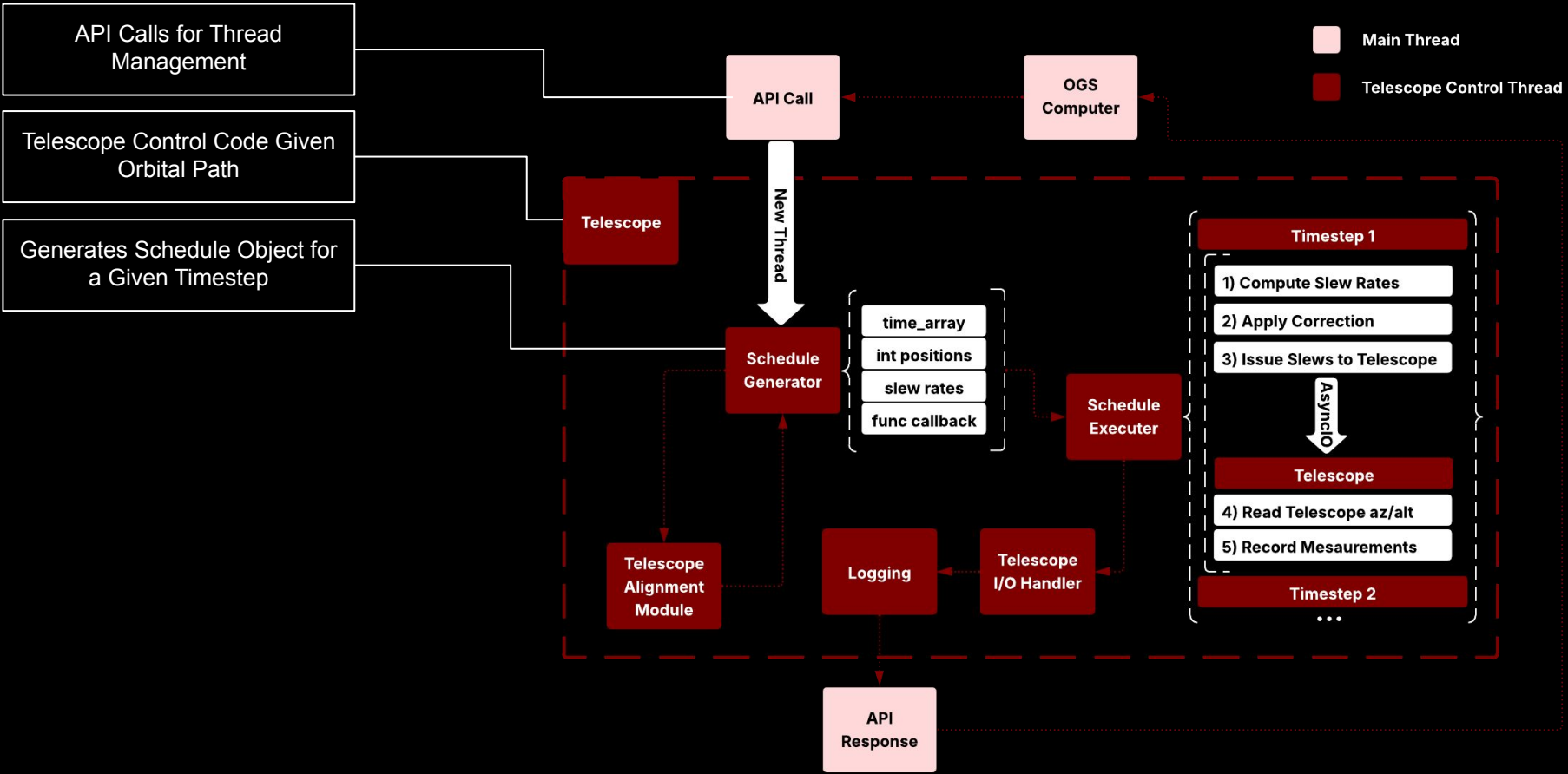
Assumption: RFGS in Chicago & OGS in Illinois. Ethernet available.

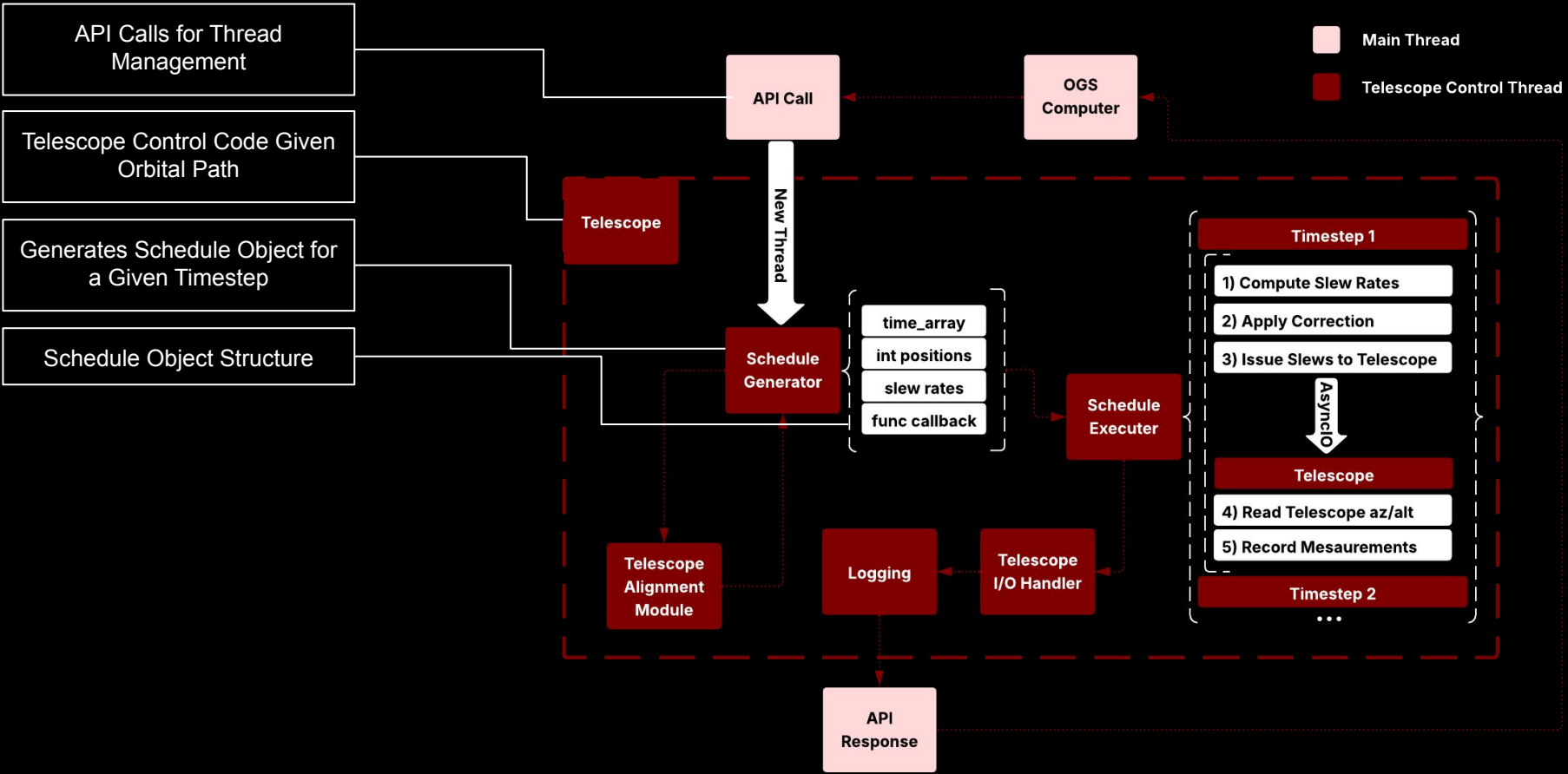
Requirements:	Solution:
RFGS to OGS Communication below ~1 minute latency for orbital path communications.	TLS over TCP (<i>approx. 10–40 ms</i>)
OGS to RFGS Communication for general data transfer. No latency limit.	TLS over TCP (<i>approx. 10–40 ms</i>)
OGS to RFGS Communication below 20ms latency for PAT Acquisition confirmation	UDP over WireGuard (<i>approx. 5–20 ms [one-way]</i>)

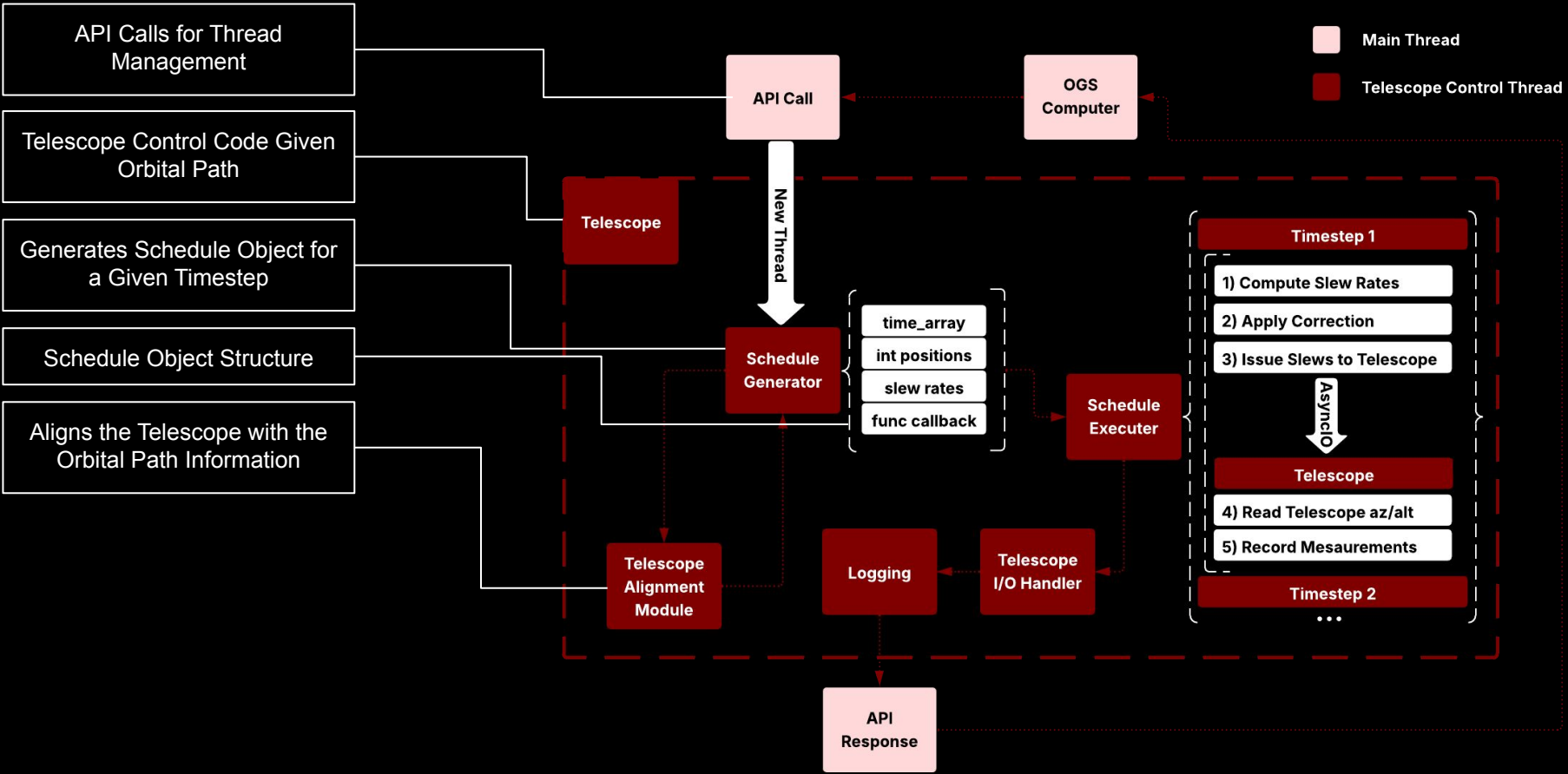
Ground Control: Telescope Control

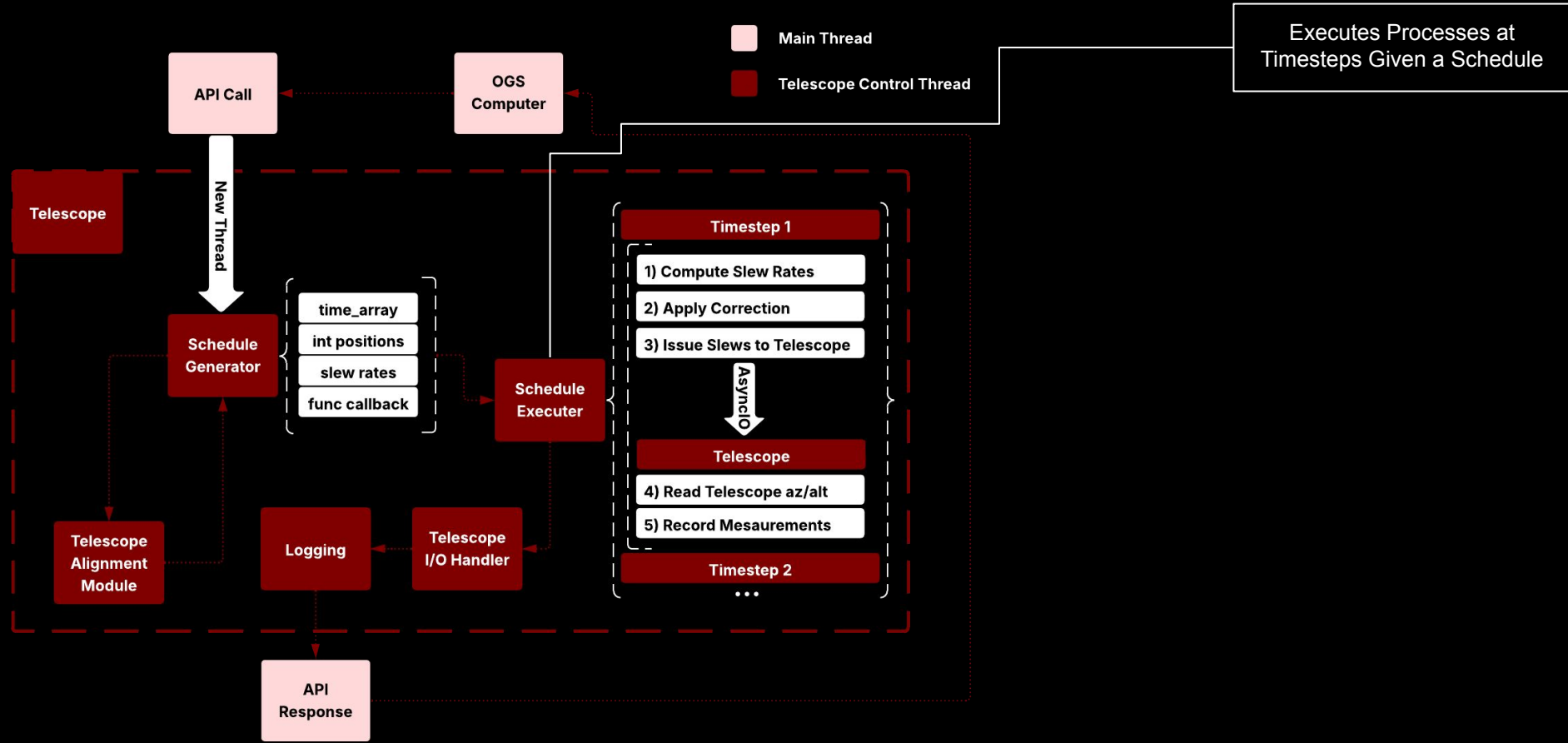


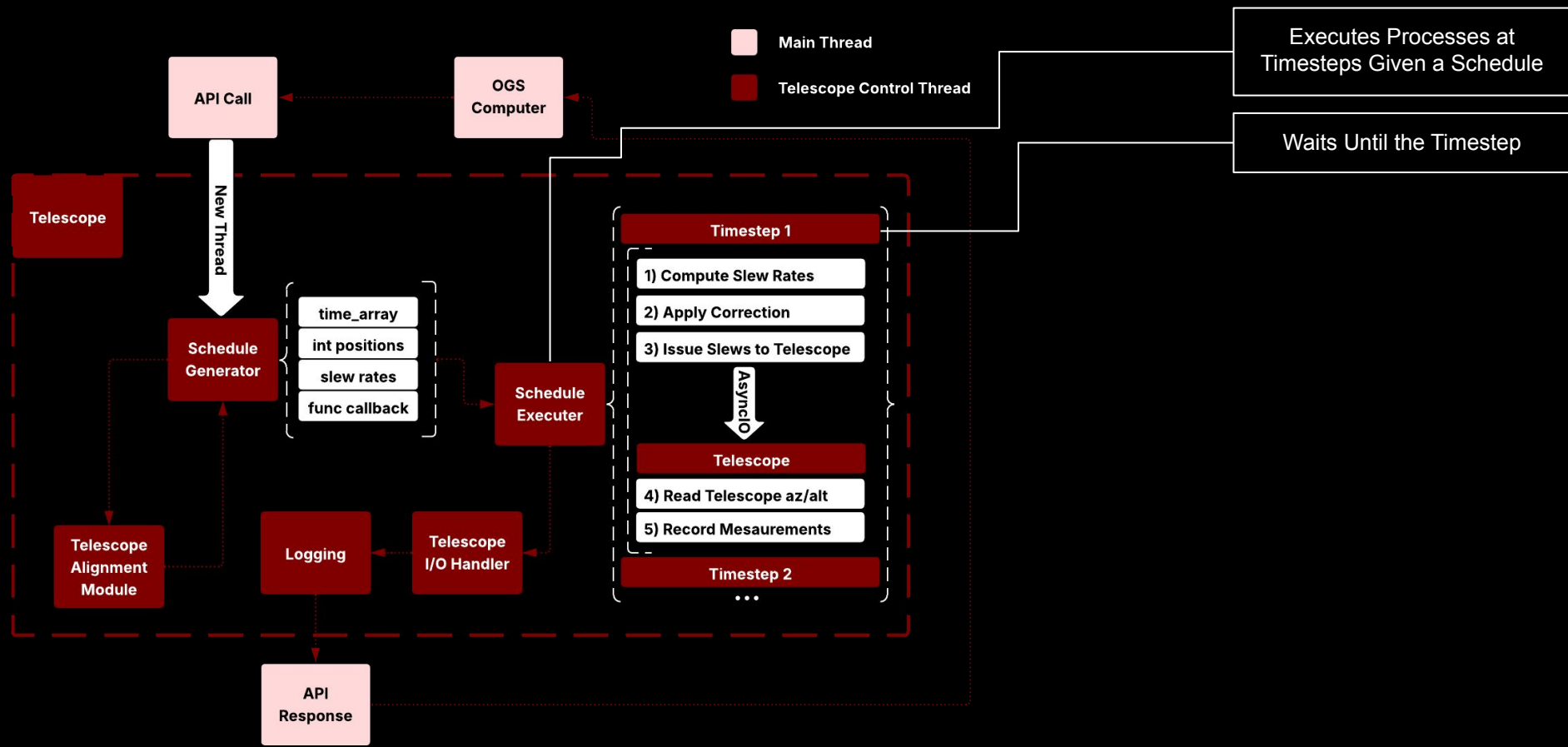


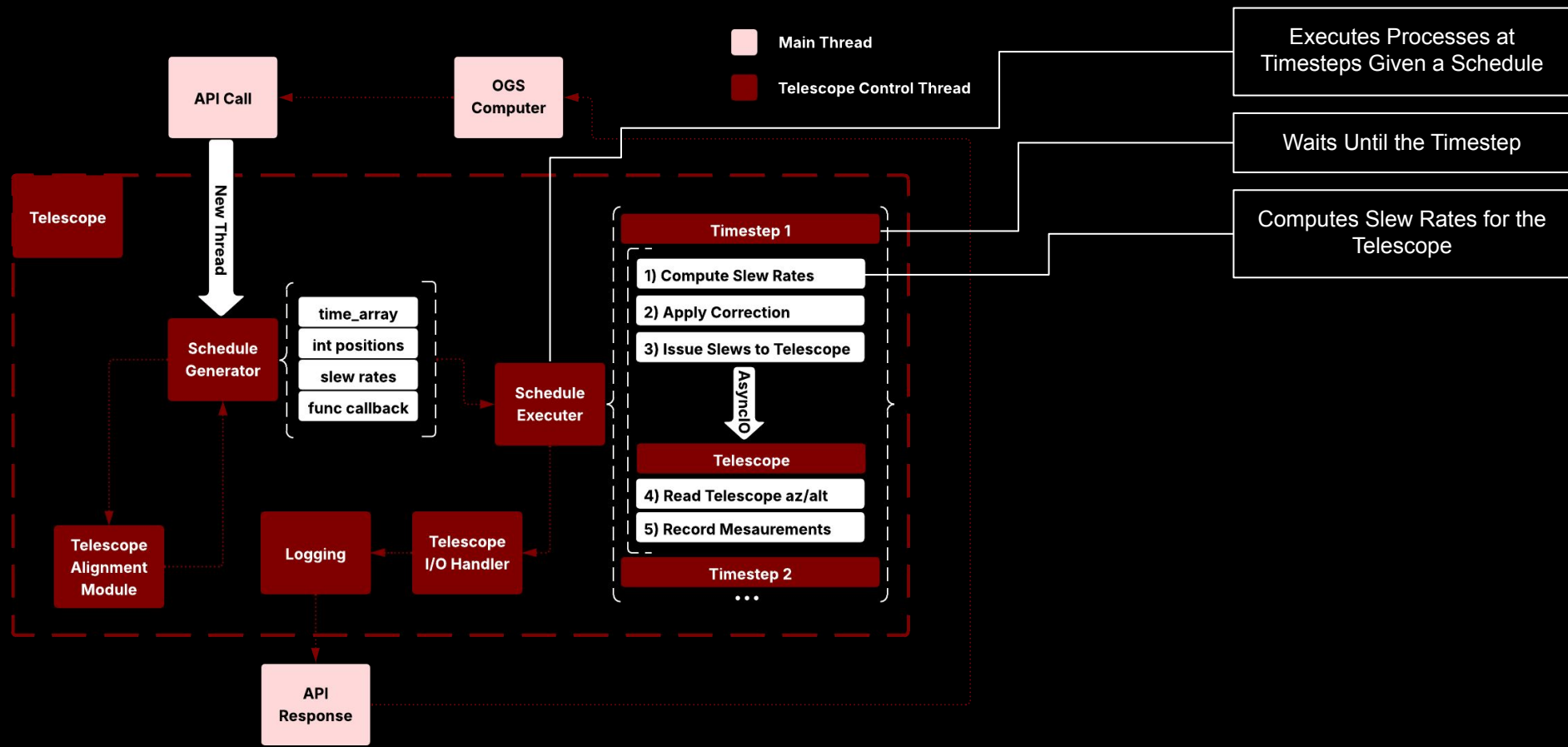


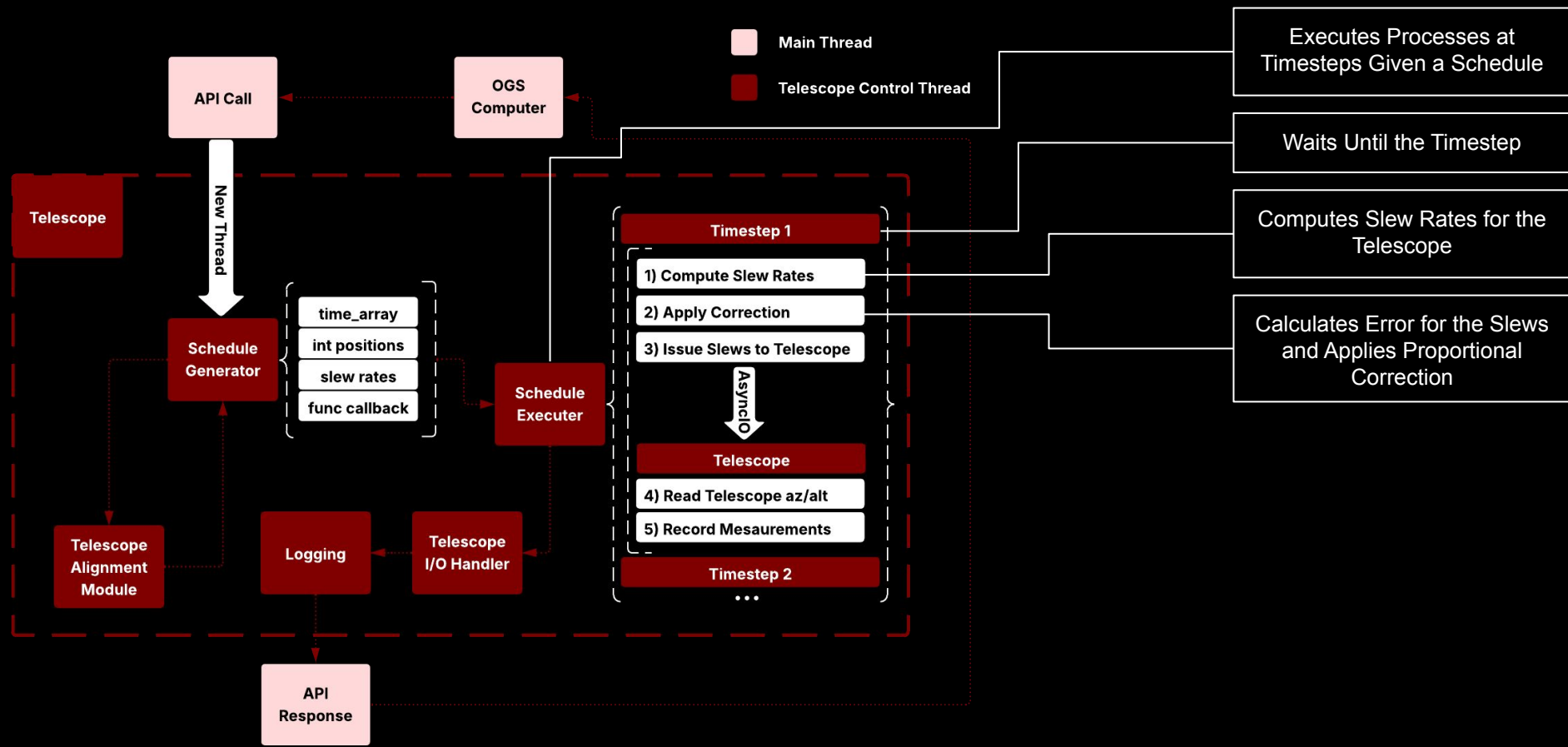


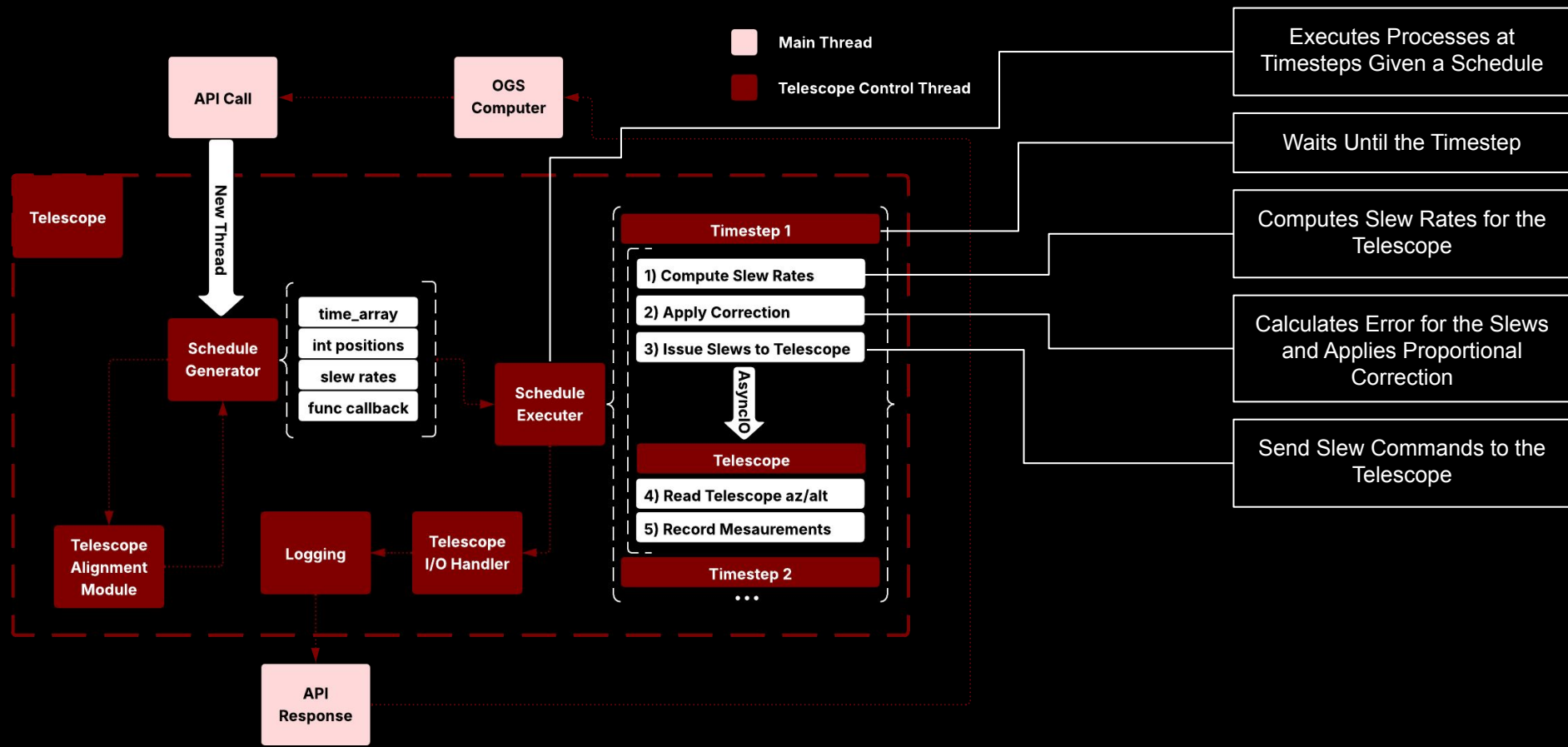


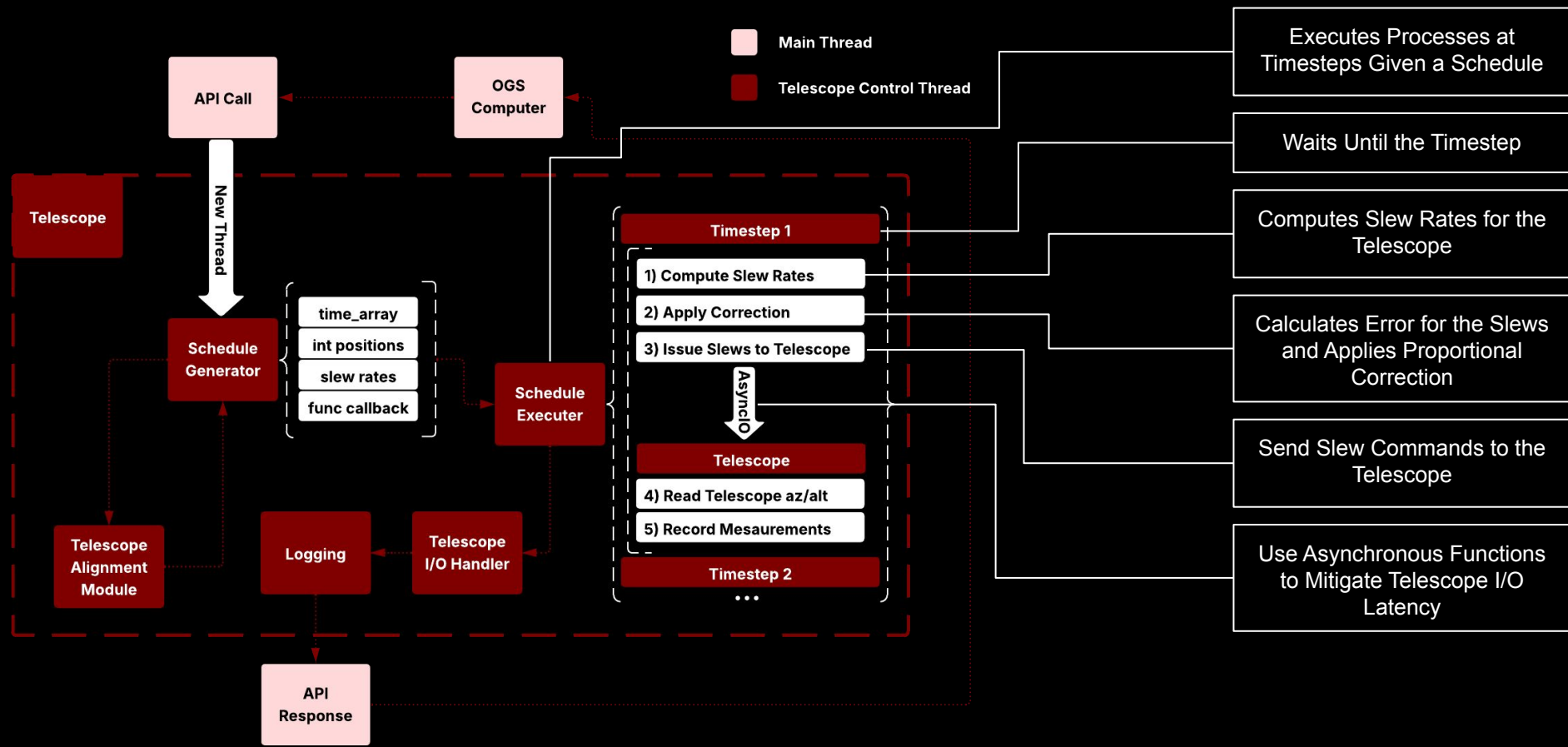


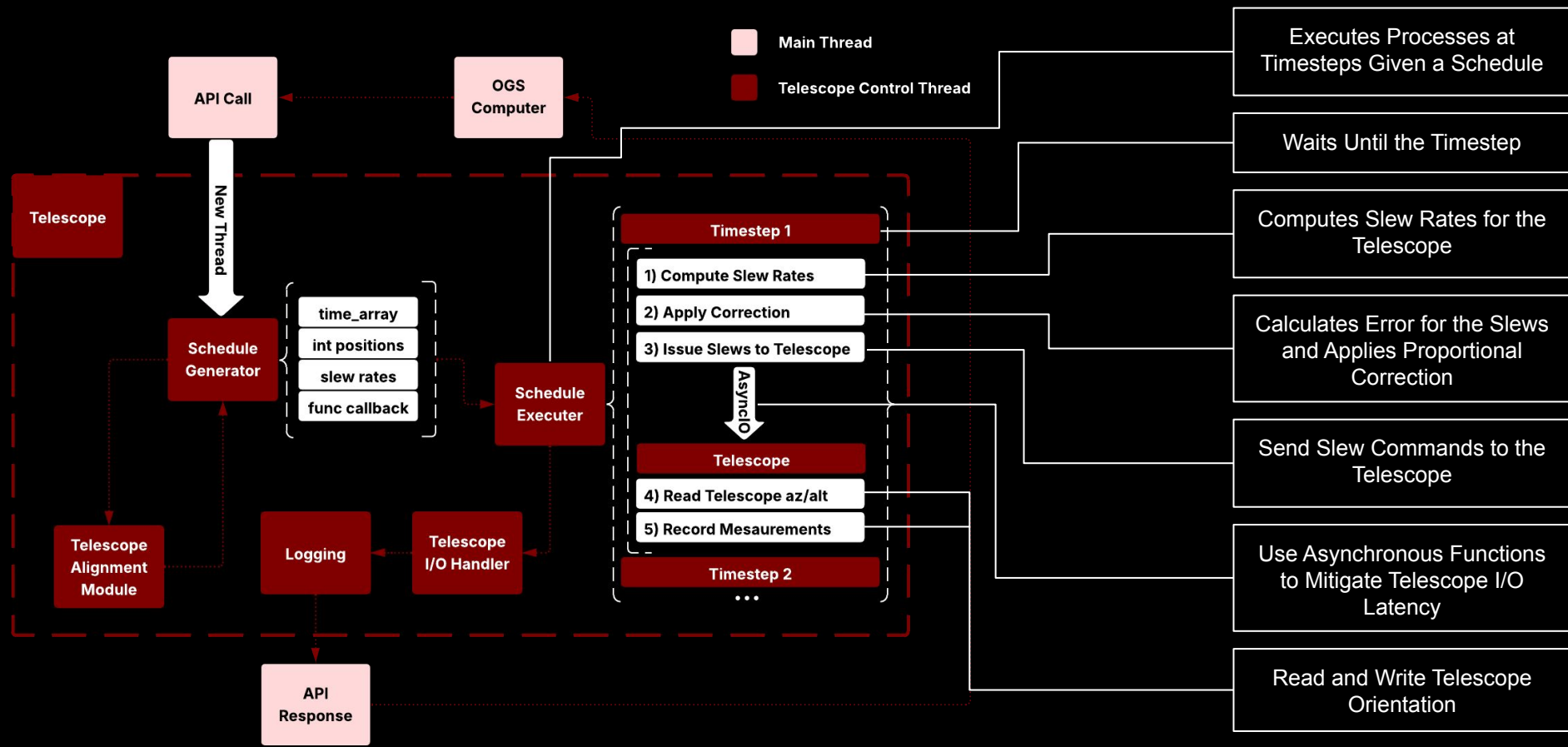


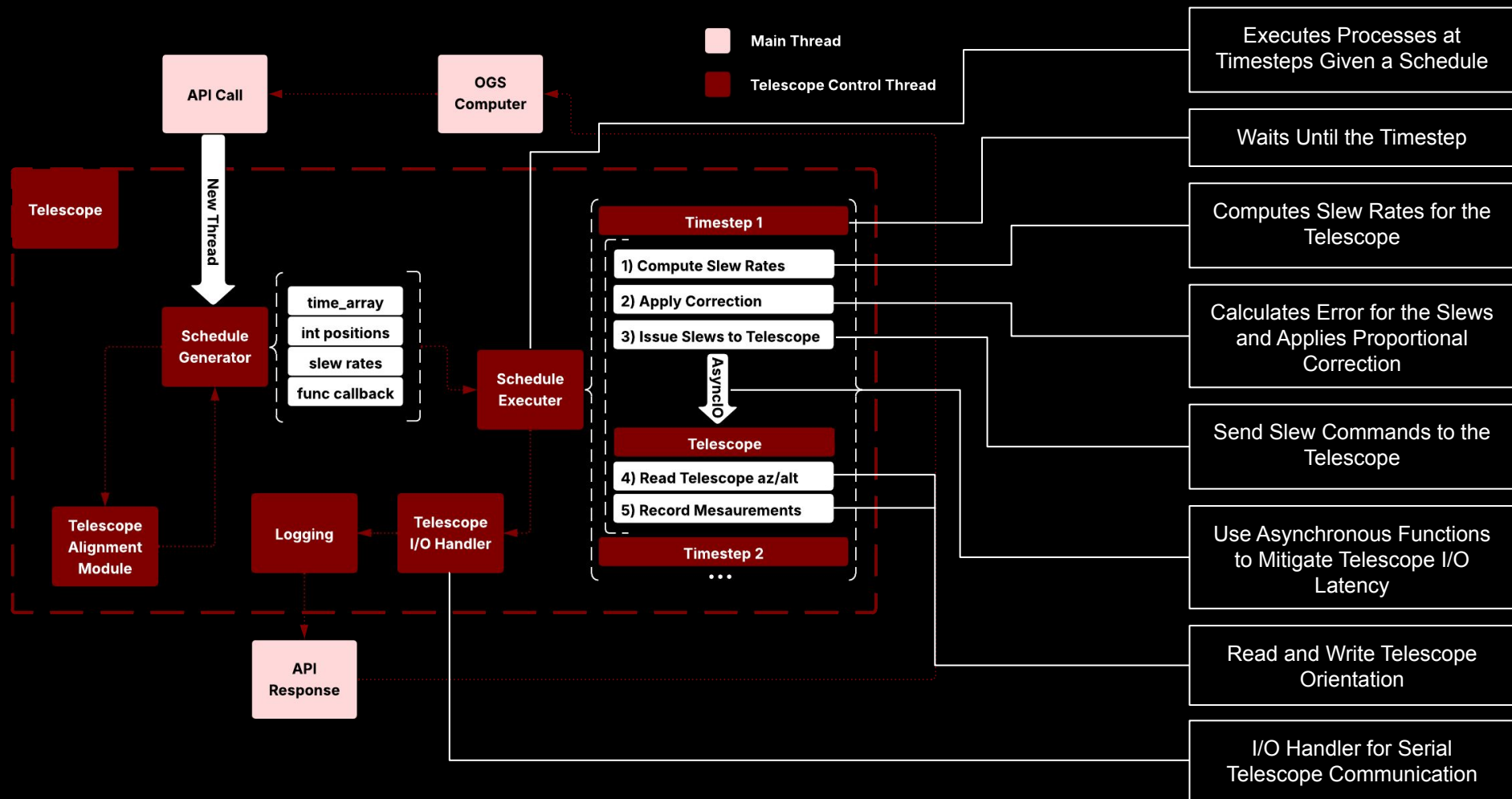




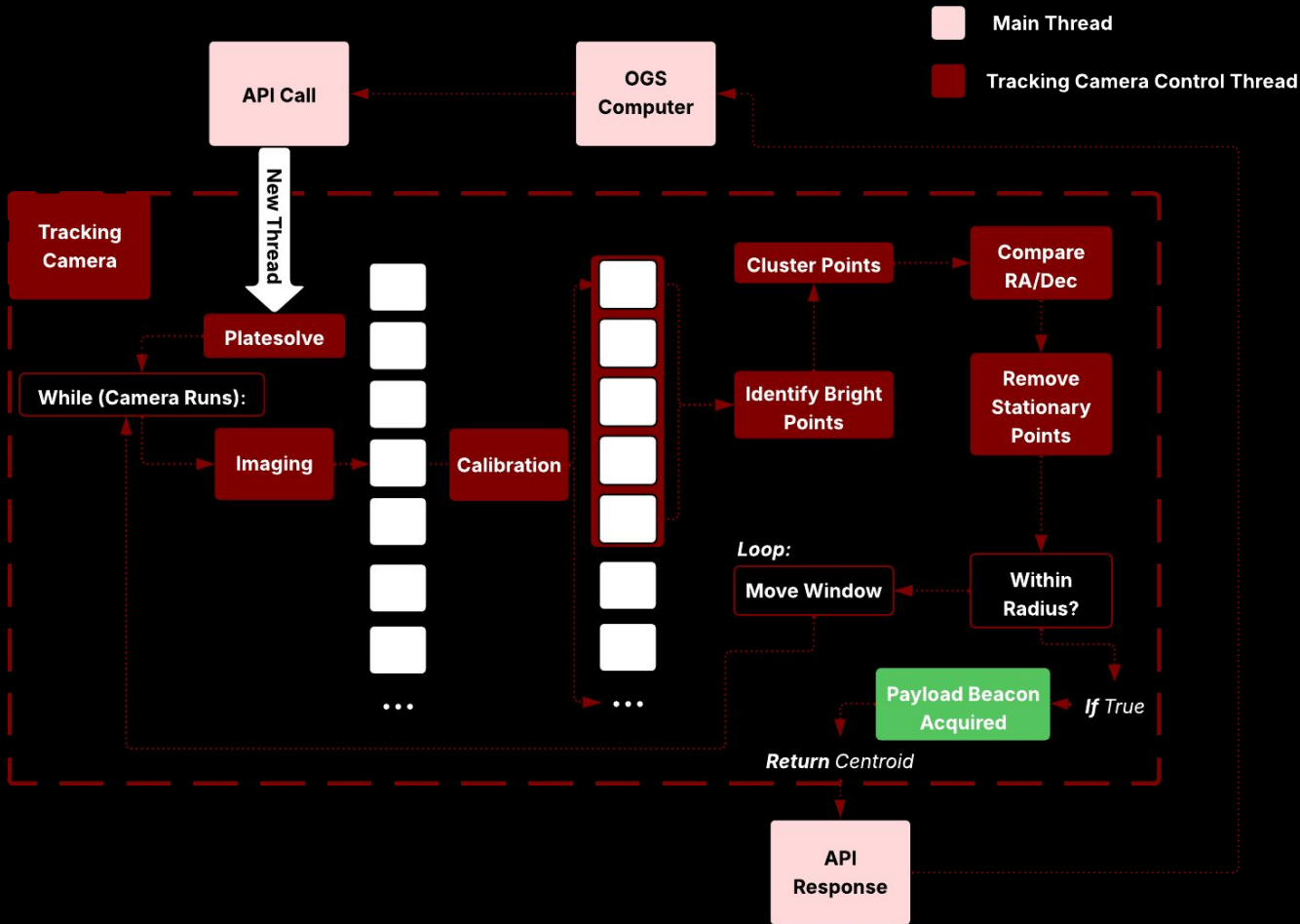


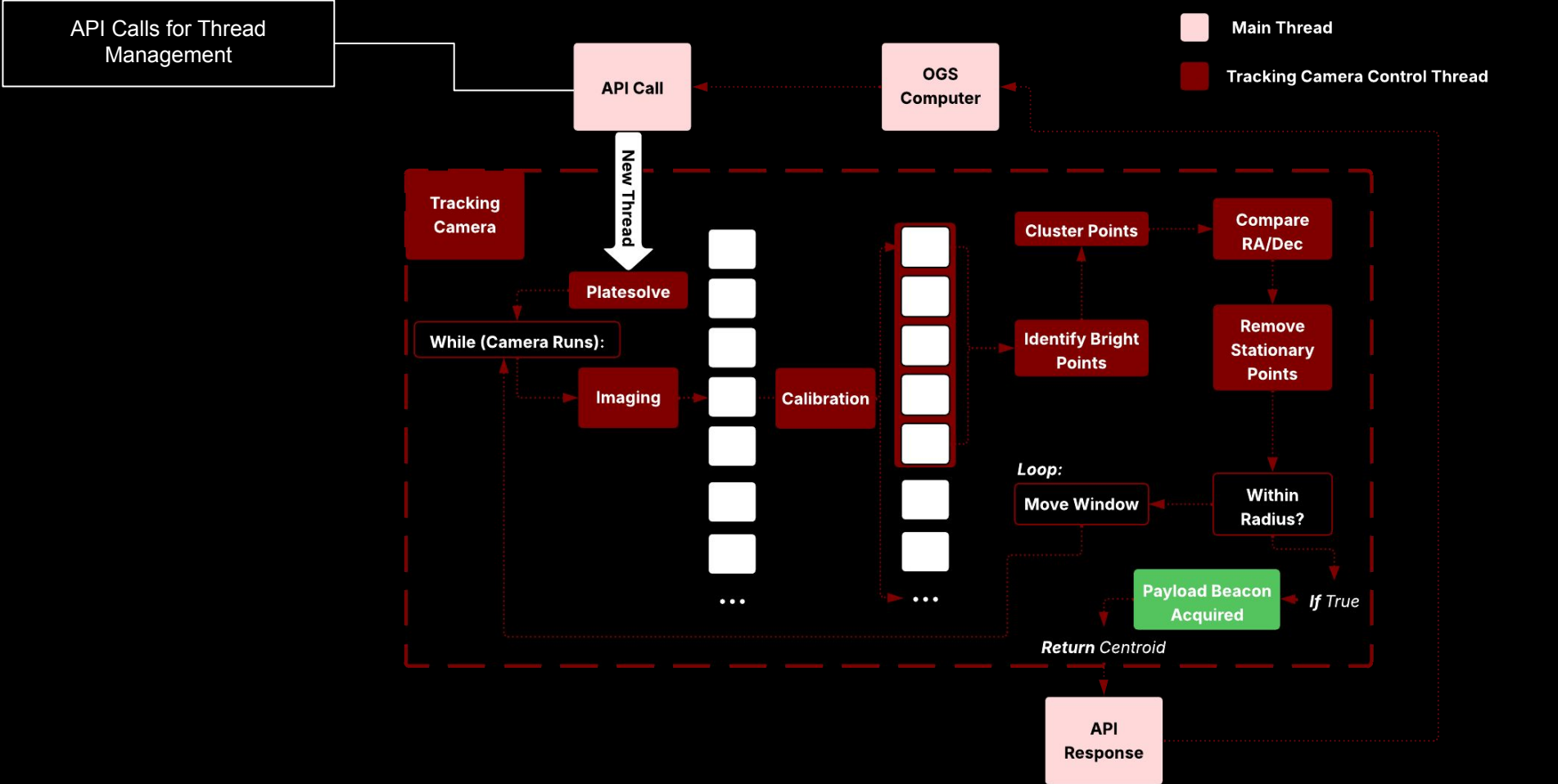


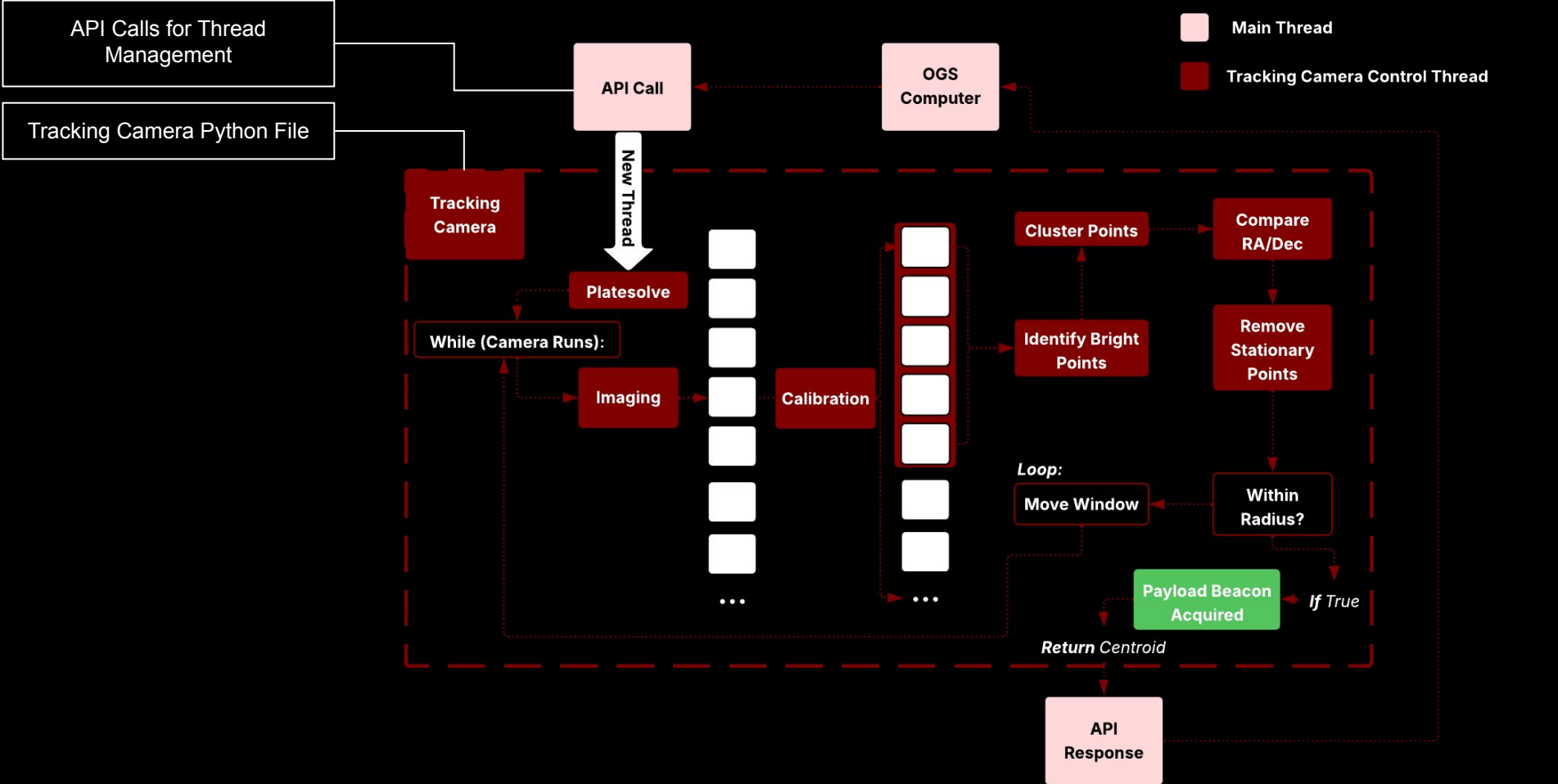


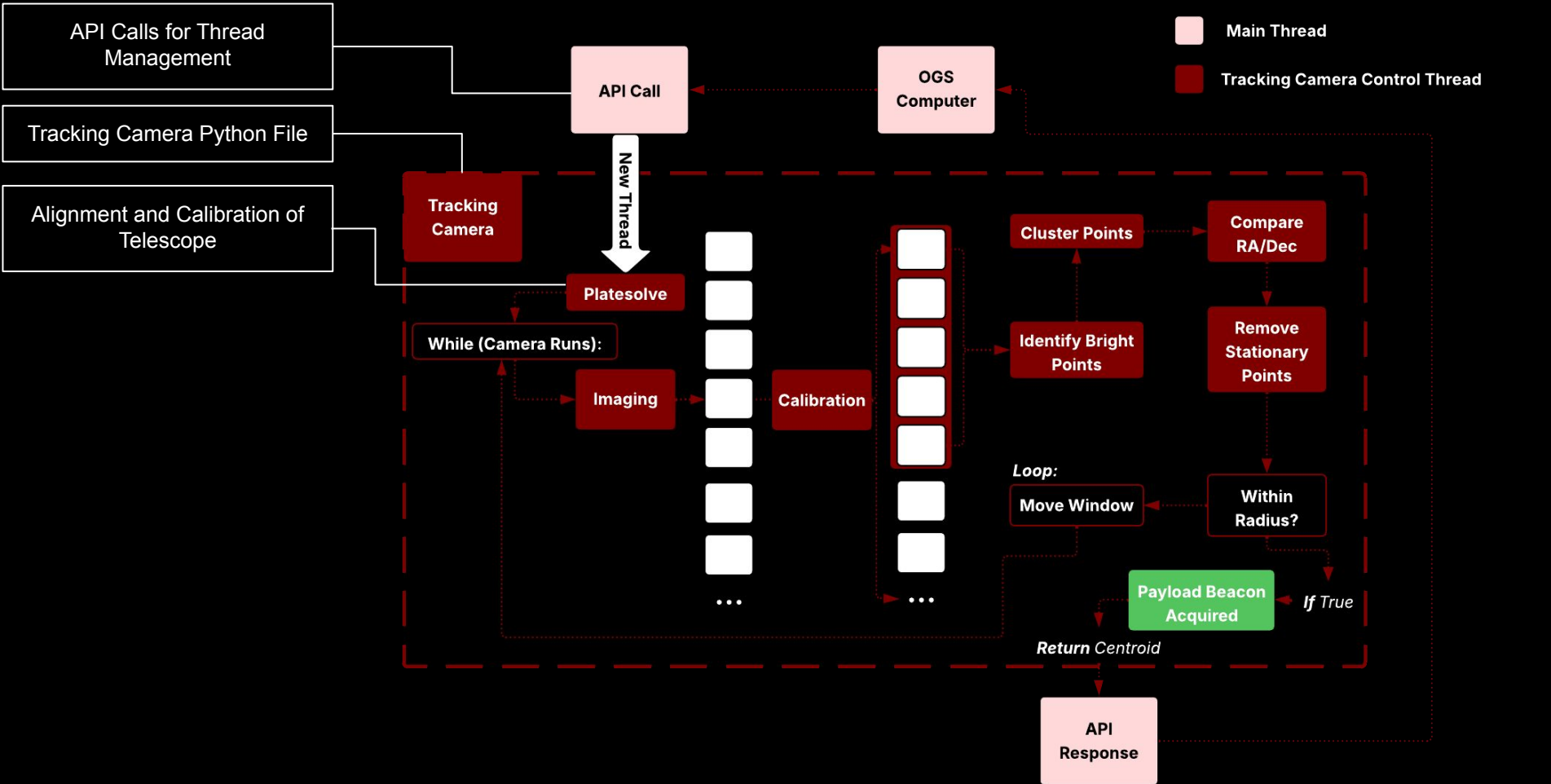


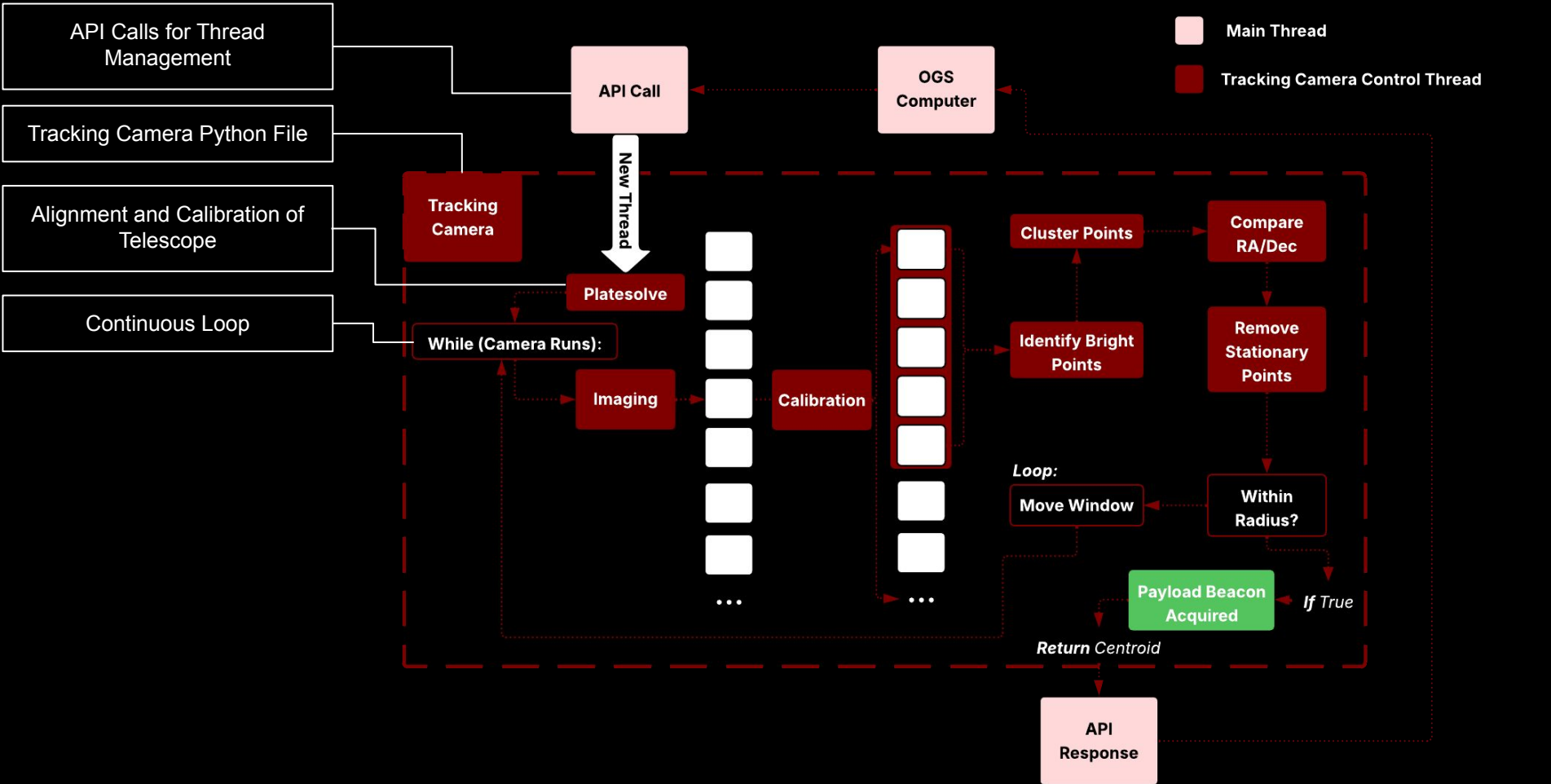
Ground Control: Tracking Camera

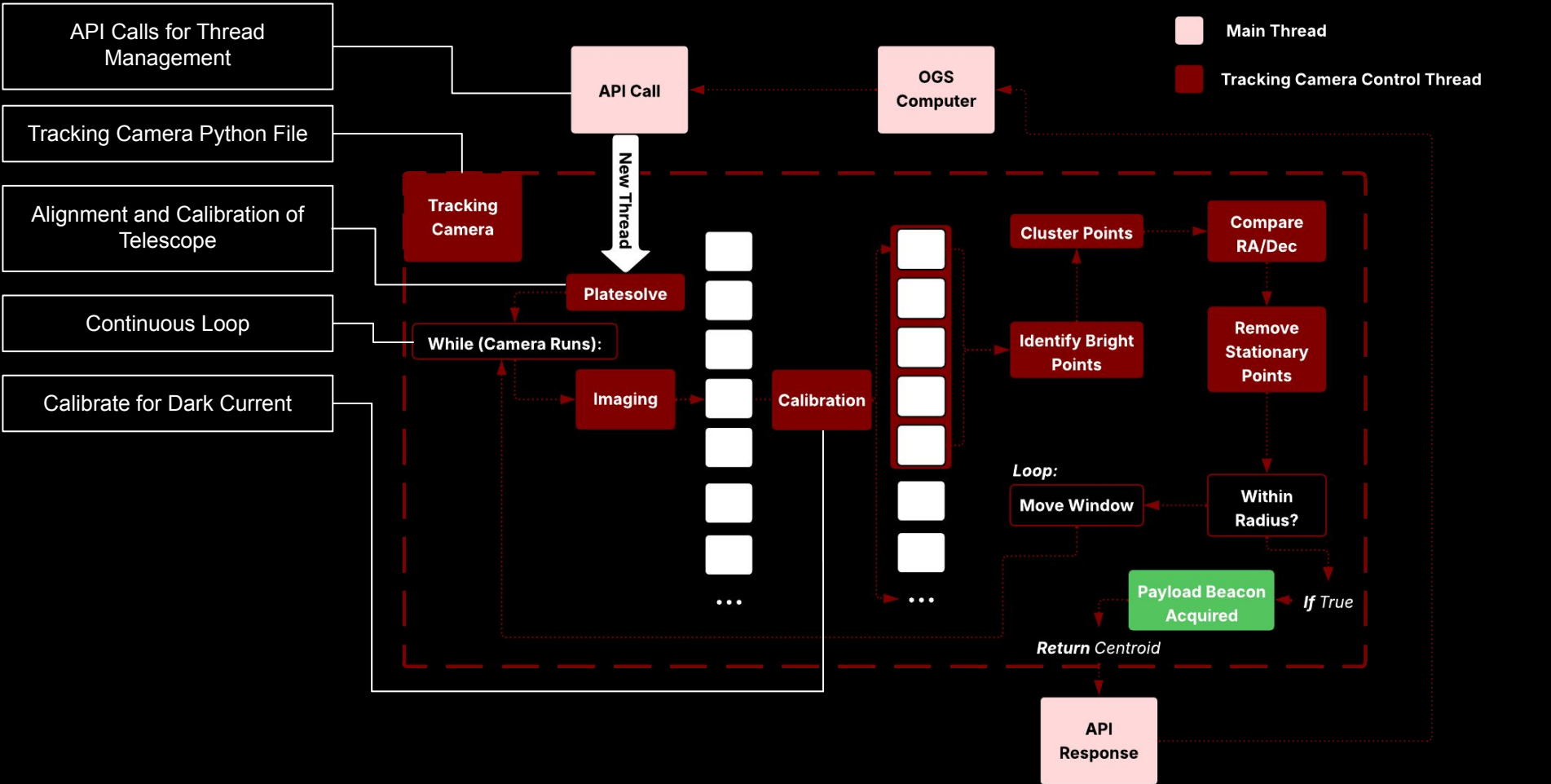


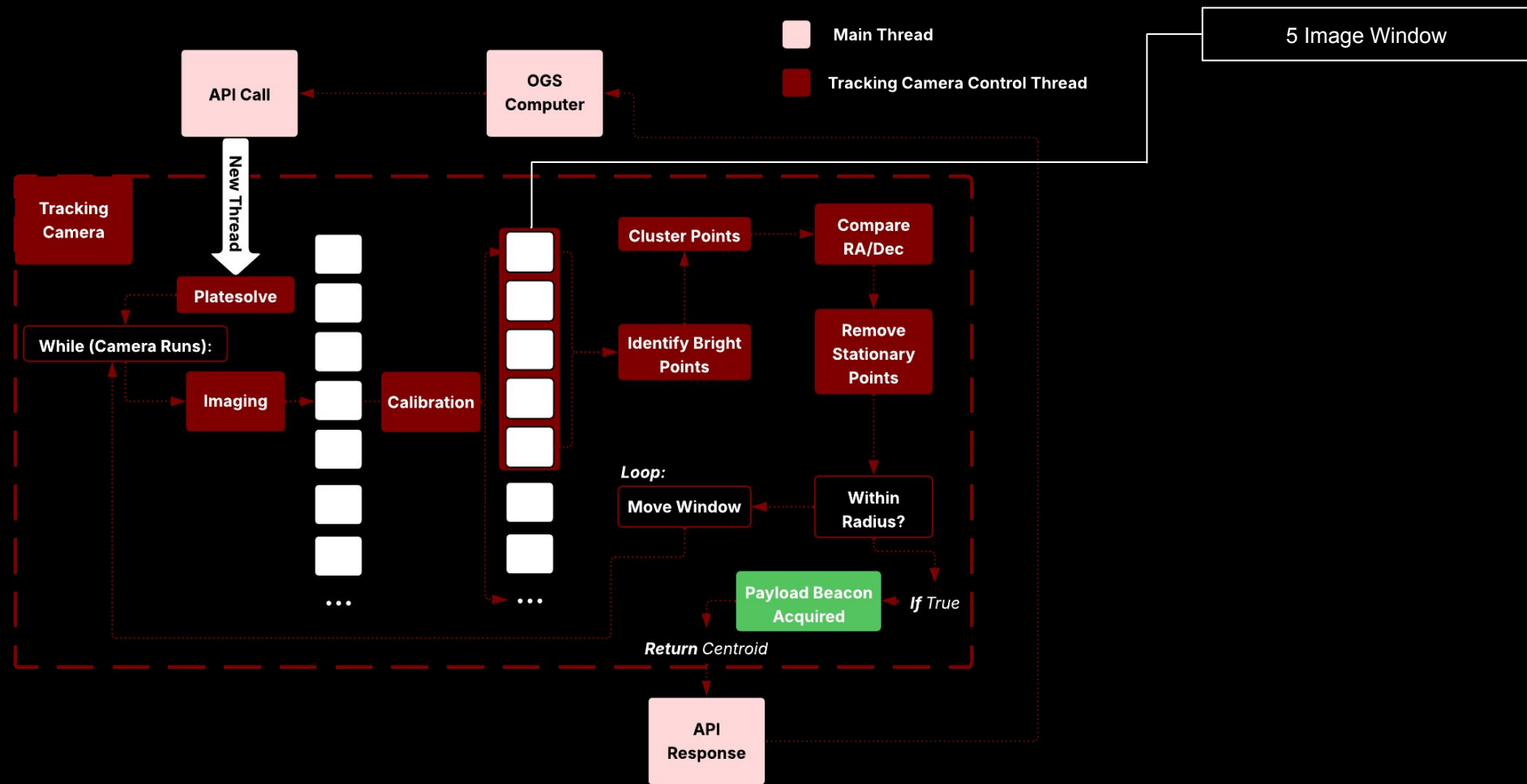


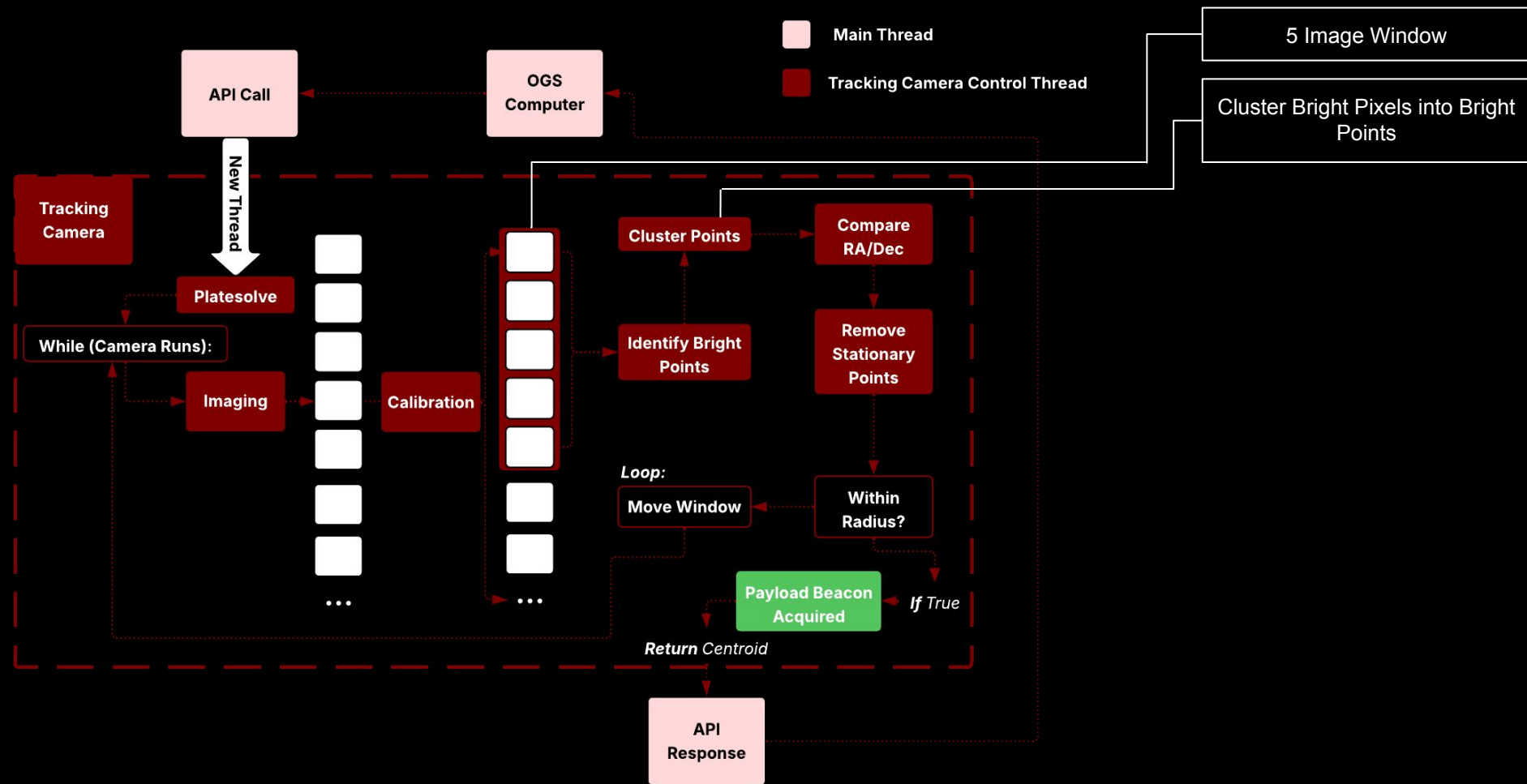


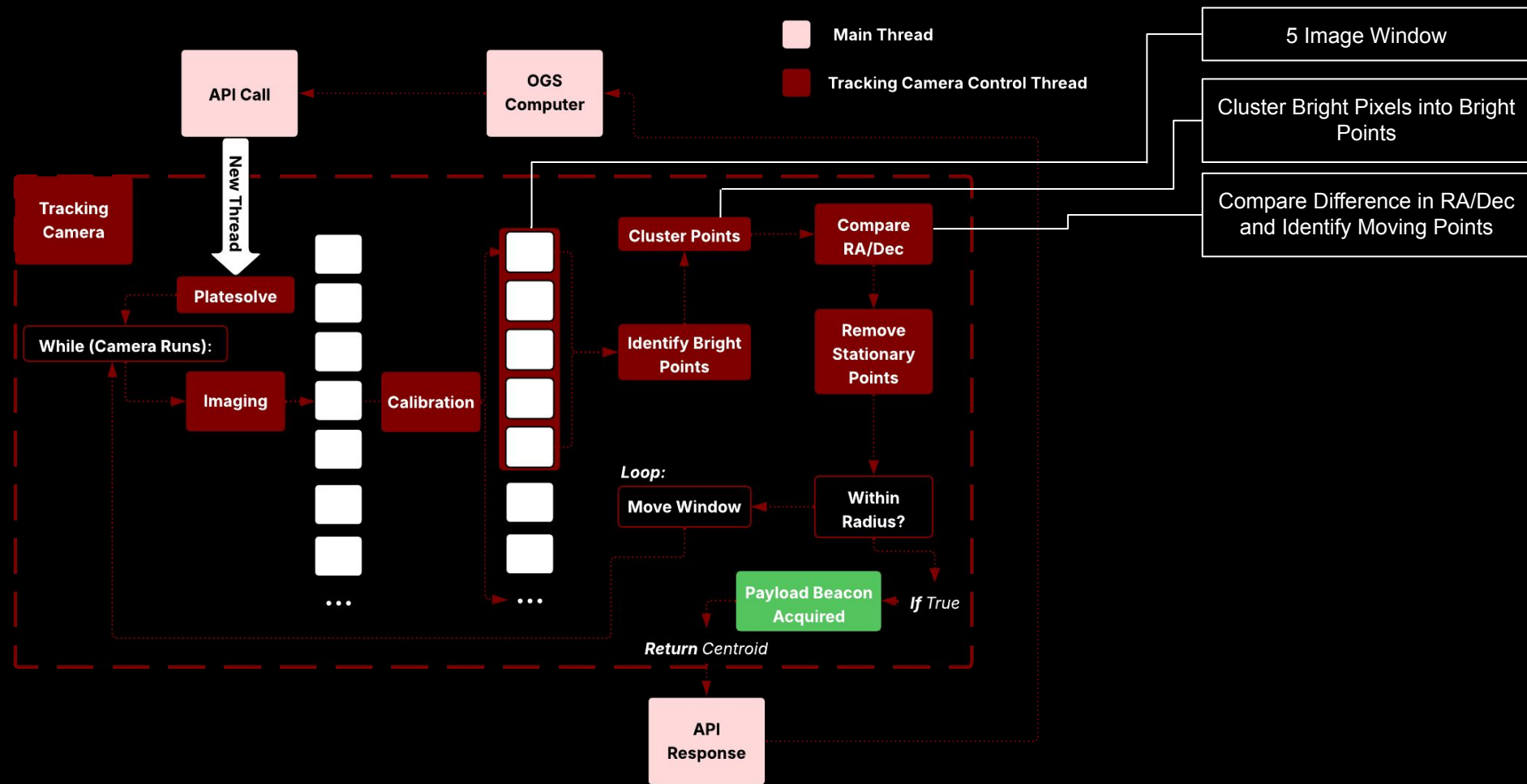


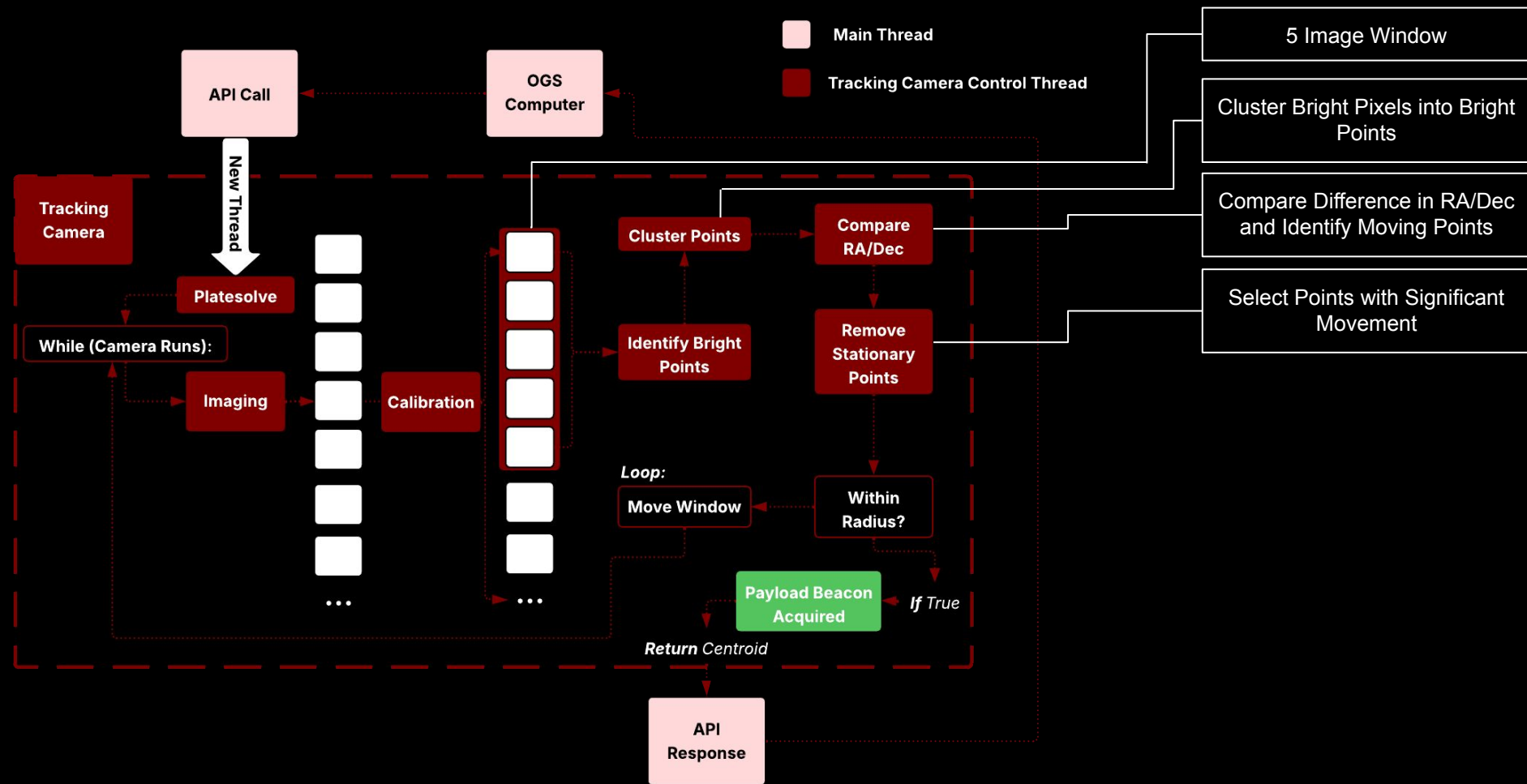


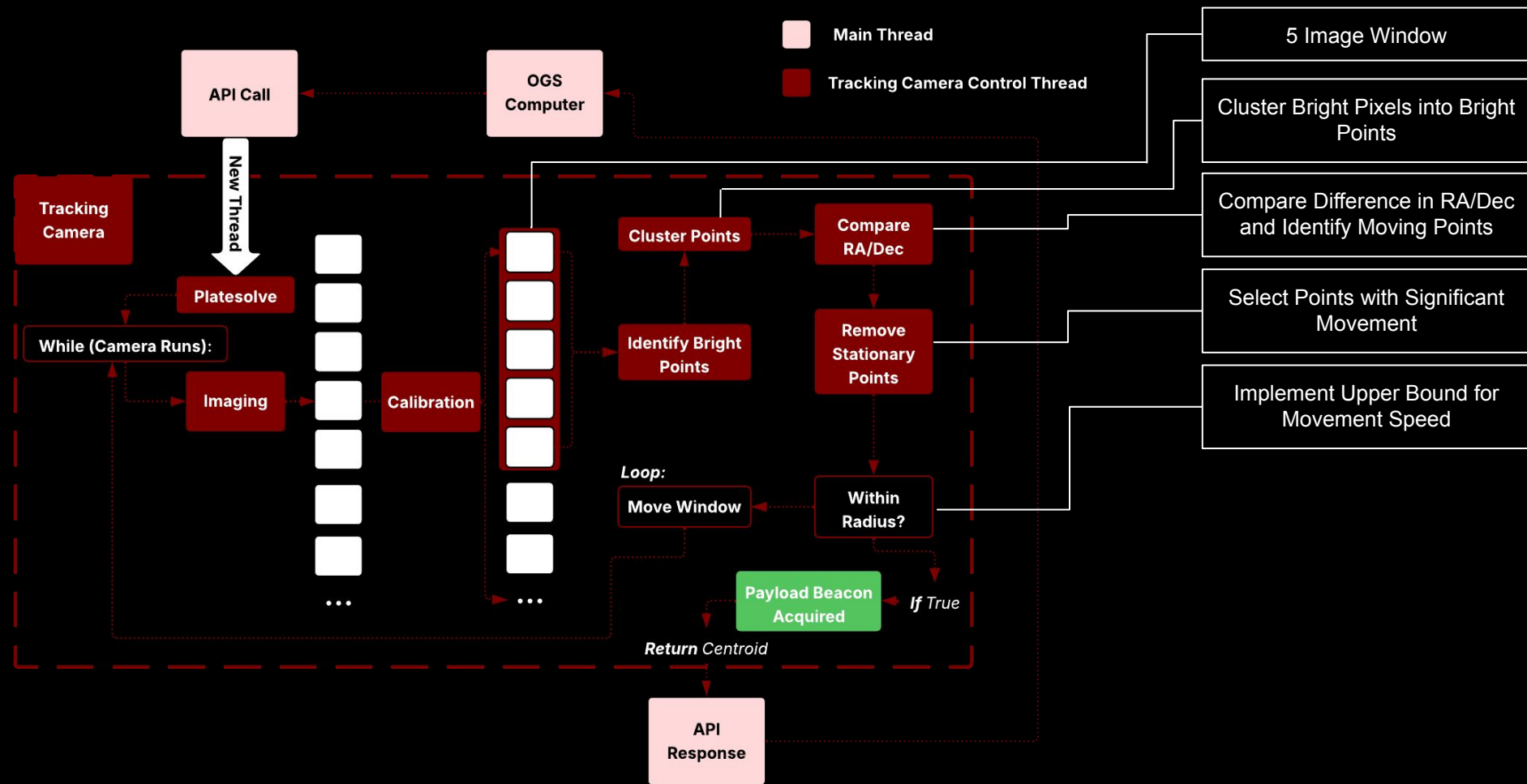


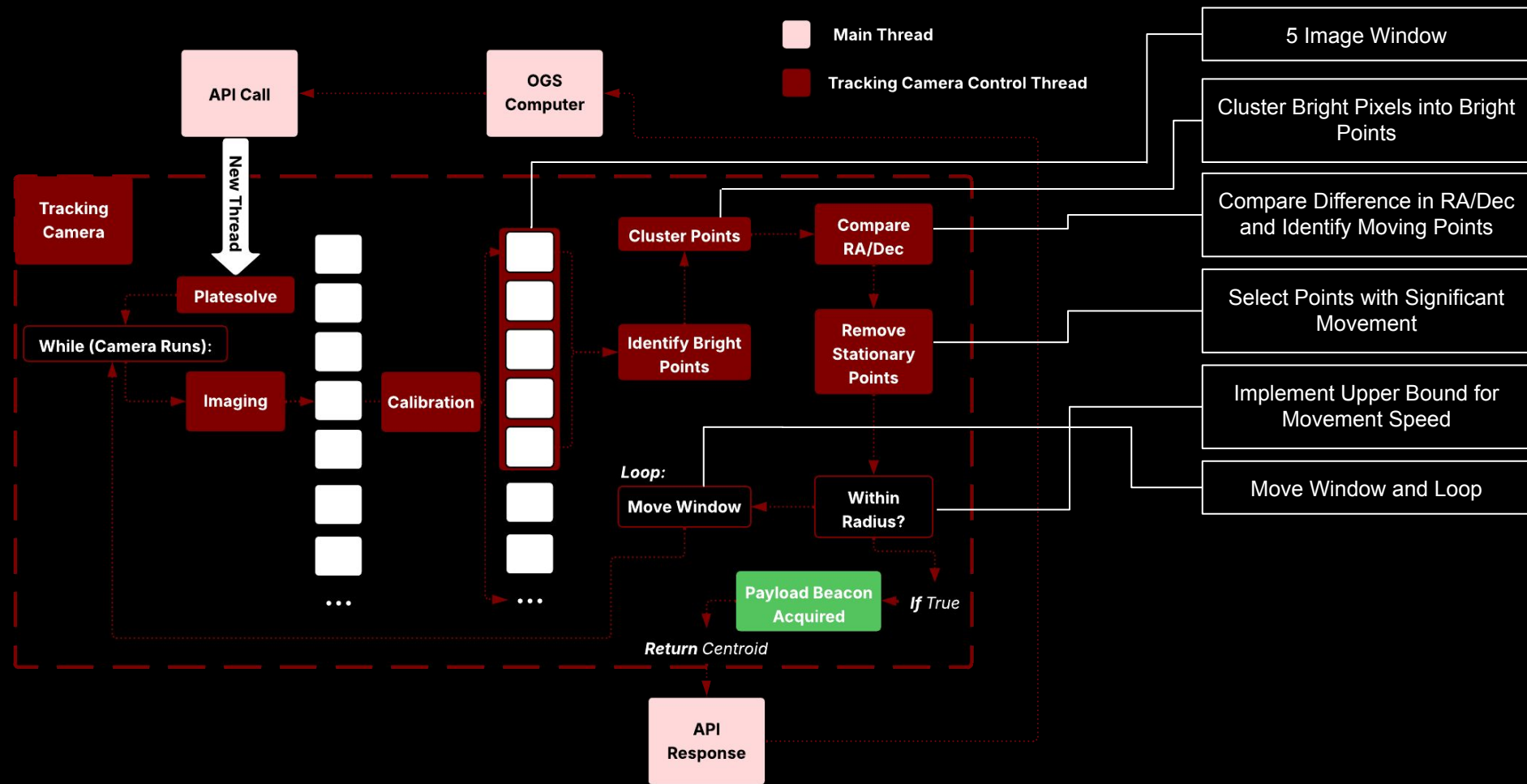


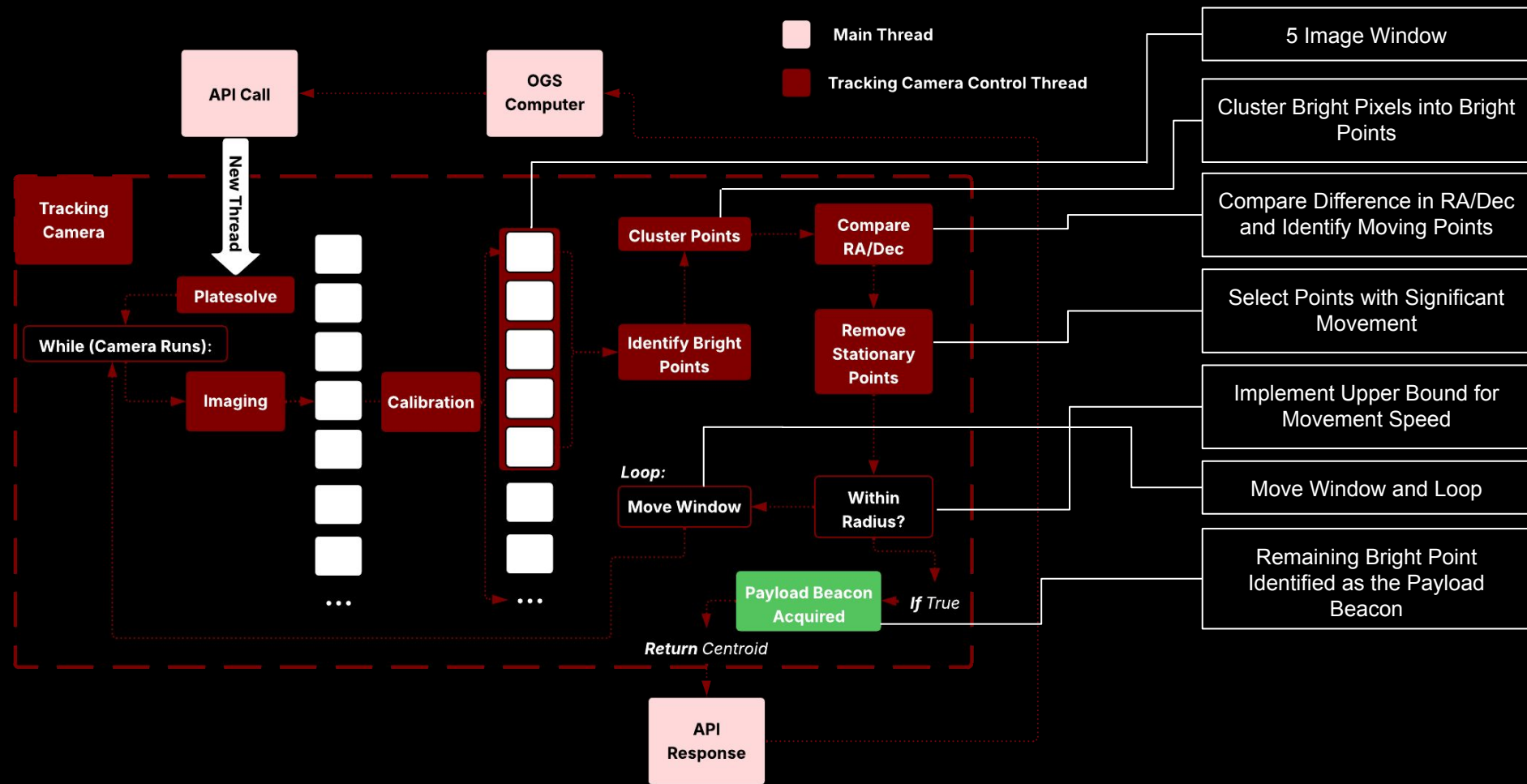


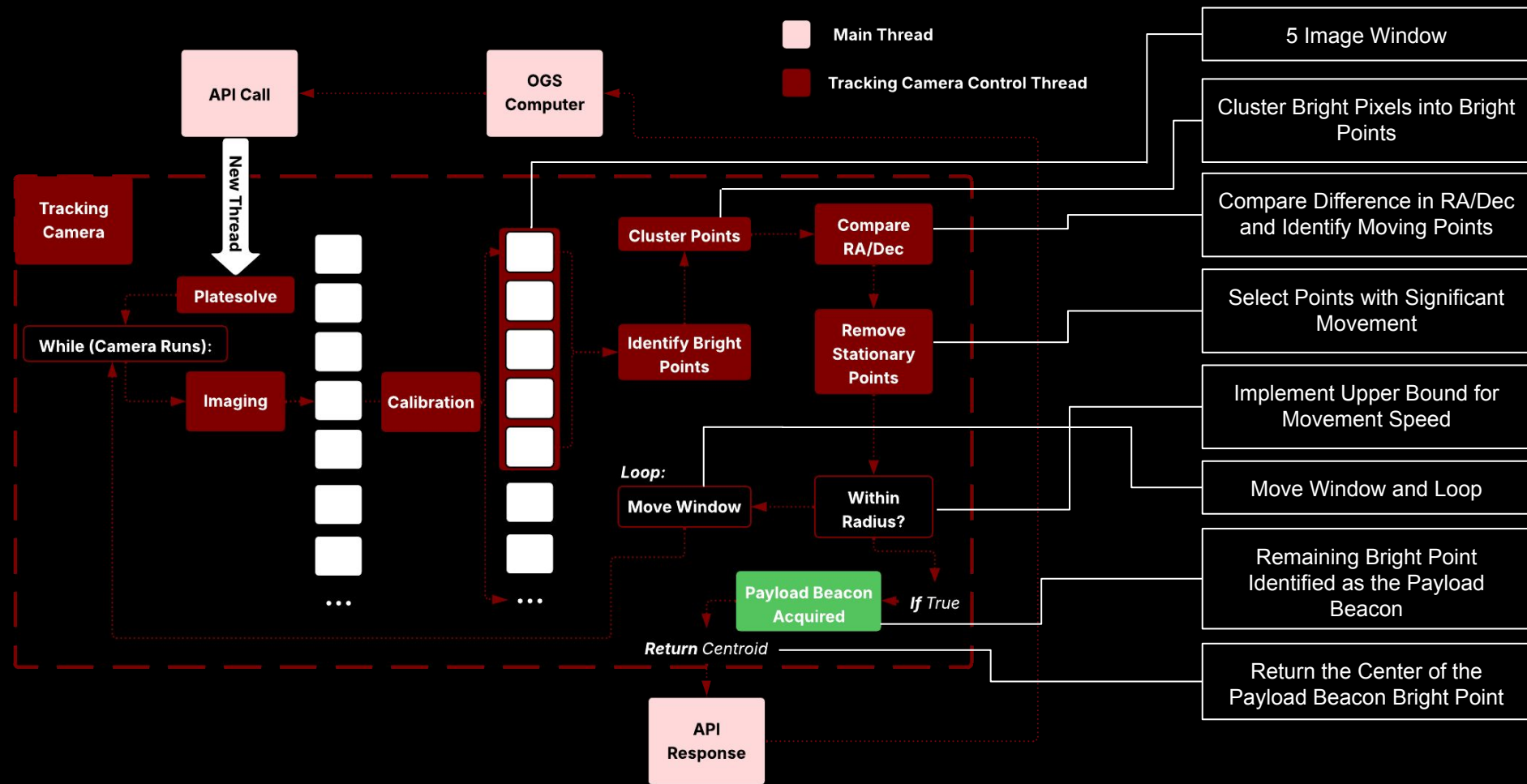


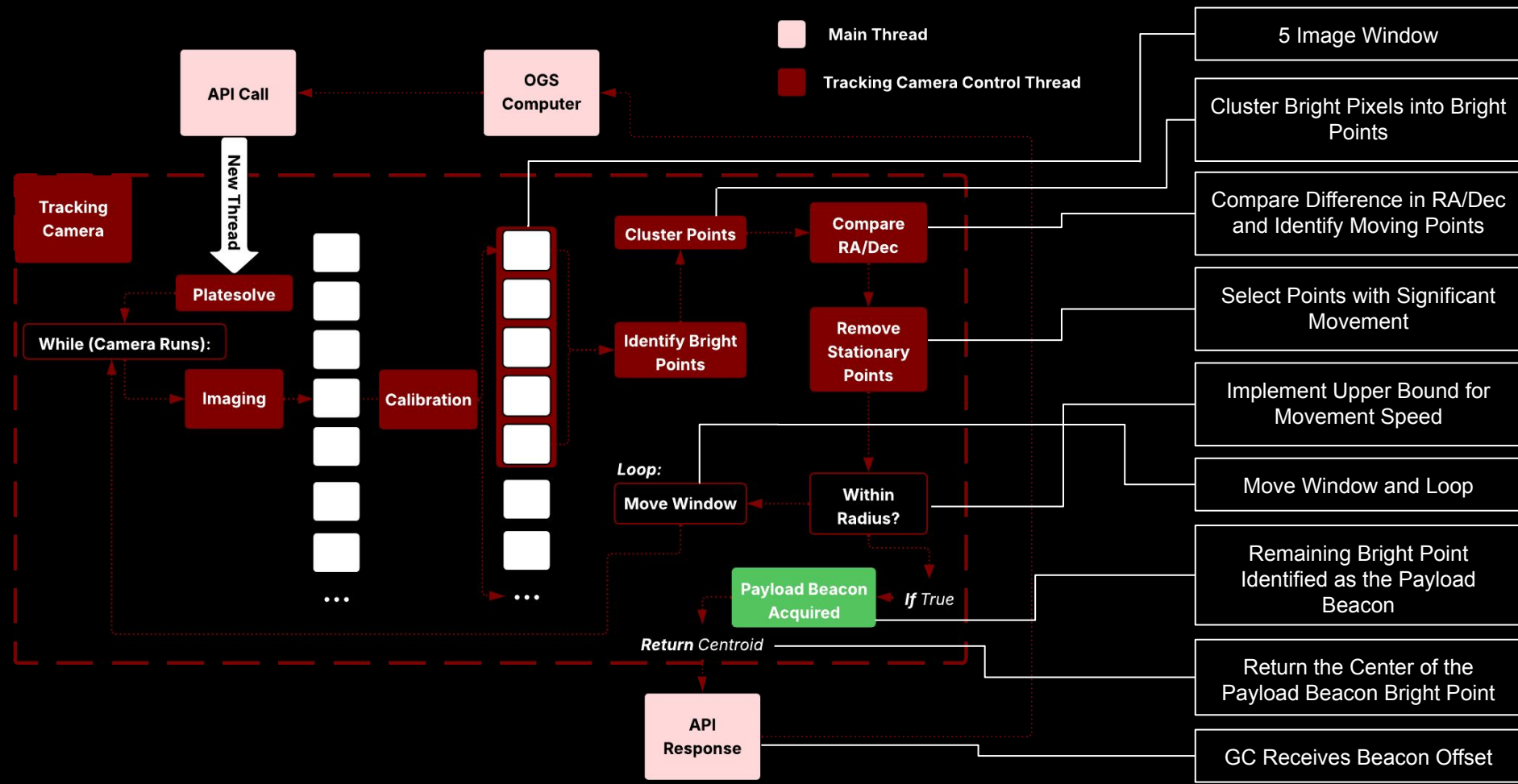


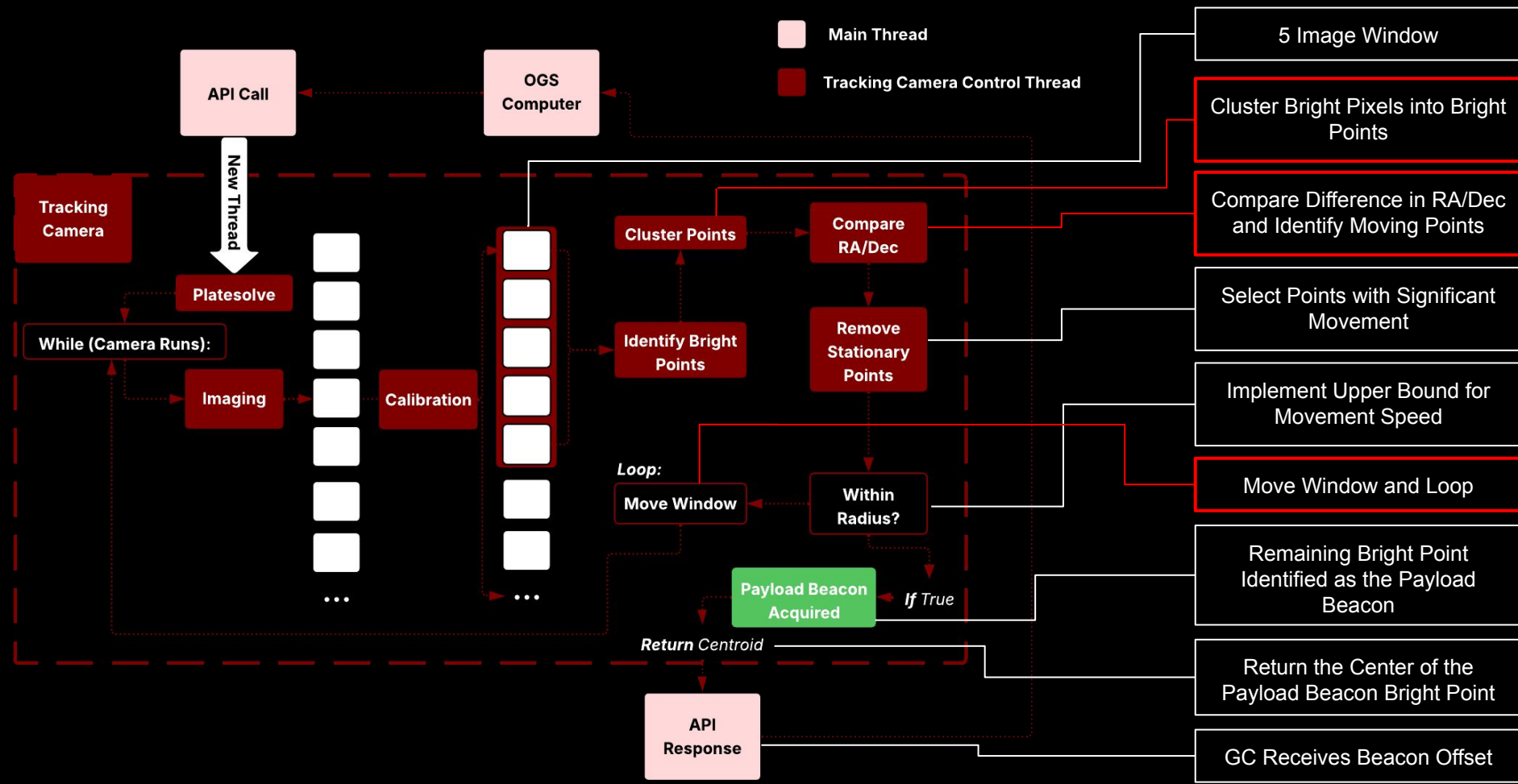












Ground Control: Graphical User Interface

Menu controlling which view is displayed

PULSE-A Interface

PULSE-A Ground Control

Telemetry Viewer

Data Plotter

Telescope Controls

Command Center

Status: Connected

Last Message: 2025-11-15 14:57:39.320

	Timestamp	Parameter	Value
1	2025-11-15 14:57:39.320	Sensor_B	91.790
2	2025-11-15 14:57:38.319	Sensor_B	14.563
3	2025-11-15 14:57:37.319	Sensor_C	72.213
4	2025-11-15 14:57:36.319	Temp	87.061
5	2025-11-15 14:57:35.320	Sensor_B	73.165
6	2025-11-15 14:57:33.320	Temp	34.862
7	2025-11-15 14:57:31.320	Sensor_C	33.511
8	2025-11-15 14:57:30.319	Sensor_C	56.554
9	2025-11-15 14:57:29.319	Sensor_A	88.754
10	2025-11-15 14:57:28.320	Sensor_B	83.375
11	2025-11-15 14:57:27.319	Temp	33.206
12	2025-11-15 14:57:26.320	Sensor_B	33.382
13	2025-11-15 14:57:25.320	Sensor_C	2.202
14	2025-11-15 14:57:24.320	Sensor_C	59.189
15	2025-11-15 14:57:23.320	Sensor_C	60.784
16	2025-11-15 14:57:22.320	Temp	68.908
17	2025-11-15 14:57:21.320	Sensor_A	20.189
18	2025-11-15 14:57:20.320	Sensor_C	27.123
19	2025-11-15 14:57:17.320	Temp	6.612
20	2025-11-15 14:57:15.319	Sensor_C	14.635
21	2025-11-15 14:57:14.319	Sensor_C	44.219
22	2025-11-15 14:57:13.320	Temp	74.227
23	2025-11-15 14:57:11.320	Sensor_C	36.990
24	2025-11-15 14:57:10.319	Sensor_A	24.209
25	2025-11-15 14:57:09.319	Sensor_A	9.383
26	2025-11-15 14:57:08.320	Sensor_B	42.943
27	2025-11-15 14:57:07.320	Sensor_B	61.287
28	2025-11-15 14:57:06.320	Sensor_C	56.107
29	2025-11-15 14:57:05.320	Temp	46.964

Menu controlling which view is displayed

Timestamped live data collected from each sensor

PULSE-A Interface

PULSE-A Ground Control

Telemetry Viewer

Data Plotter

Telescope Controls

Command Center

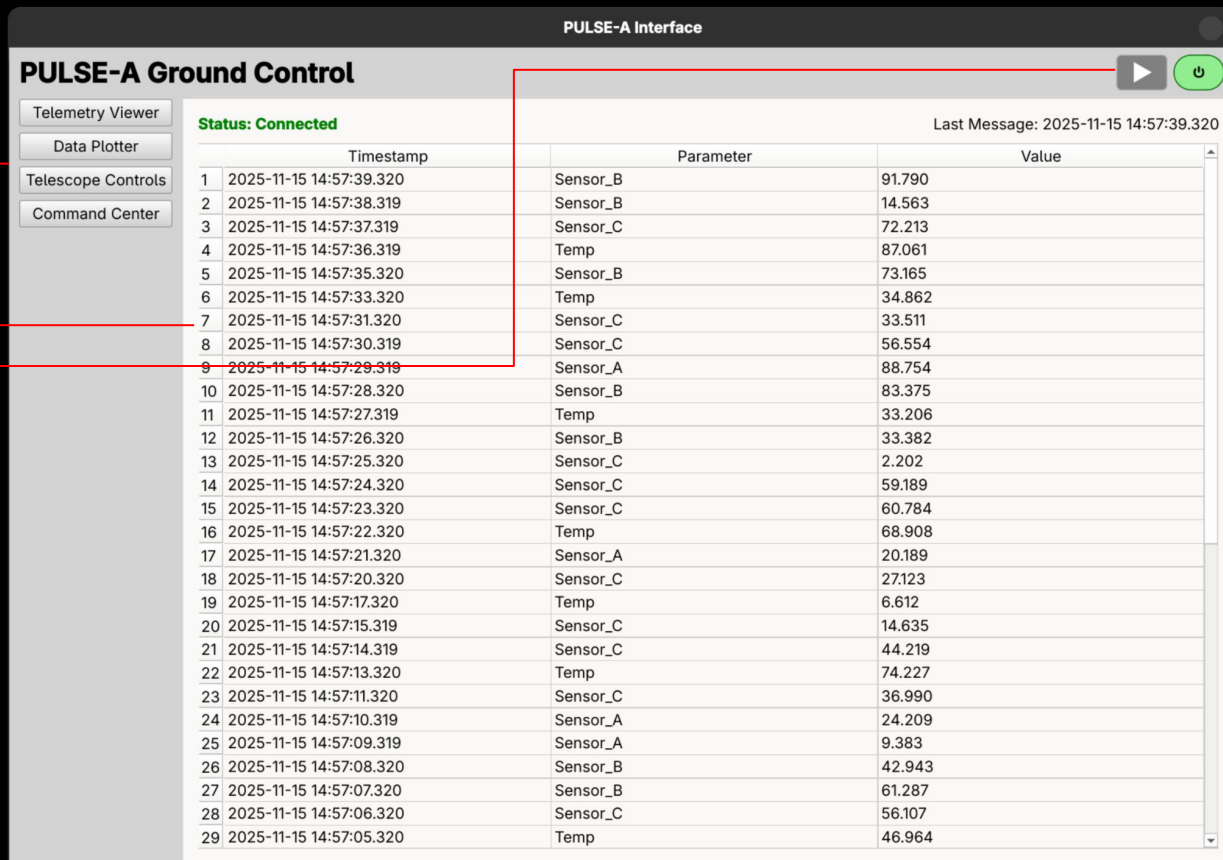
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10	2025-11-15 14:57:28.320	Sensor_B	83.375
11	2025-11-15 14:57:27.319	Temp	33.206
12	2025-11-15 14:57:26.320	Sensor_B	33.382
13	2025-11-15 14:57:25.320	Sensor_C	2.202
14	2025-11-15 14:57:24.320	Sensor_C	59.189
15	2025-11-15 14:57:23.320	Sensor_C	60.784
16	2025-11-15 14:57:22.320	Temp	68.908
17	2025-11-15 14:57:21.320	Sensor_A	20.189
18	2025-11-15 14:57:20.320	Sensor_C	27.123
19	2025-11-15 14:57:17.320	Temp	6.612
20	2025-11-15 14:57:15.319	Sensor_C	14.635
21	2025-11-15 14:57:14.319	Sensor_C	44.219
22	2025-11-15 14:57:13.320	Temp	74.227
23	2025-11-15 14:57:11.320	Sensor_C	36.990
24	2025-11-15 14:57:10.319	Sensor_A	24.209
25	2025-11-15 14:57:09.319	Sensor_A	9.383
26	2025-11-15 14:57:08.320	Sensor_B	42.943
27	2025-11-15 14:57:07.320	Sensor_B	61.287
28	2025-11-15 14:57:06.320	Sensor_C	56.107
29	2025-11-15 14:57:05.320	Temp	46.964

Timestamped live data collected from
each sensor

Controls to start/stop recording incoming data



Menu controlling which view is displayed

Timestamped live data collected from each sensor

Controls to start/stop recording incoming data

Controls to turn the OGS assembly on/off

PULSE-A Interface

PULSE-A Ground Control

Telemetry Viewer

Data Plotter

Telescope Controls

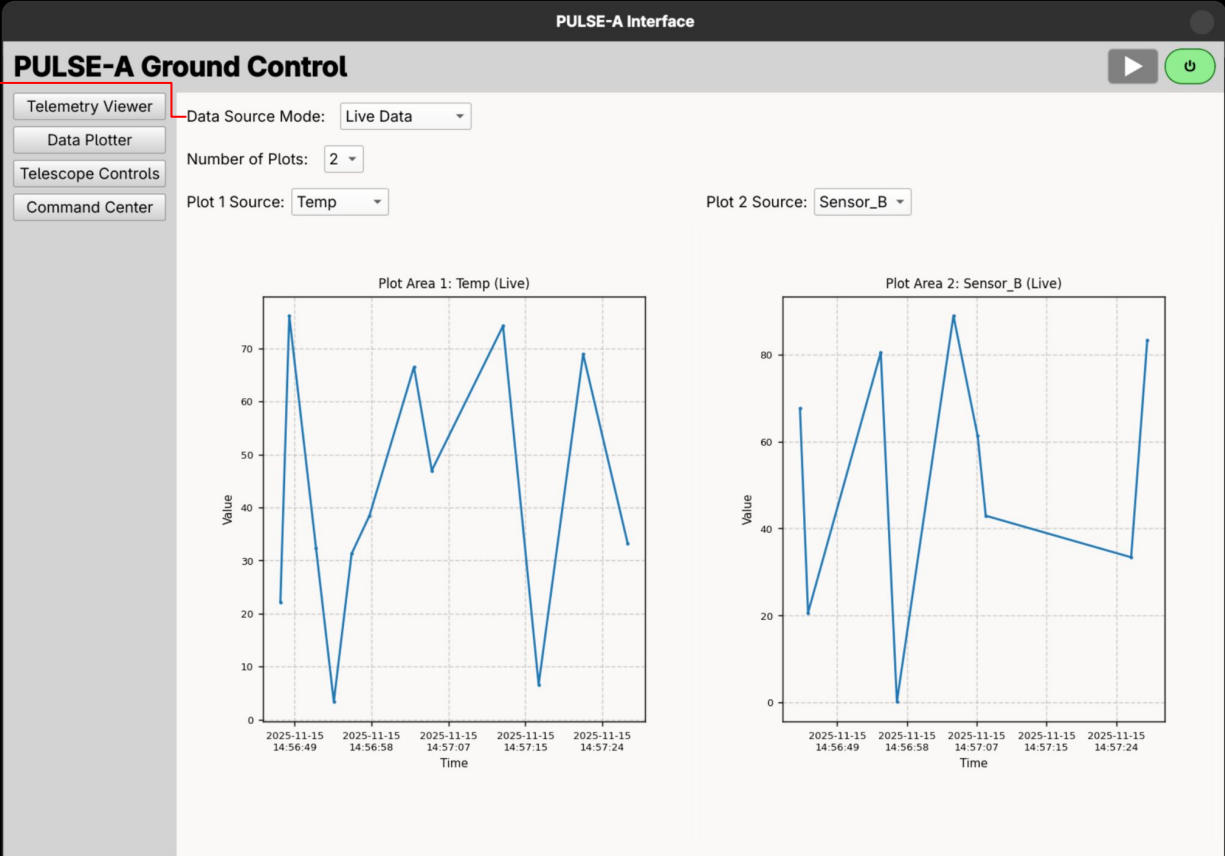
Command Center

Status: Connected

Last Message: 2025-11-15 14:57:39.320

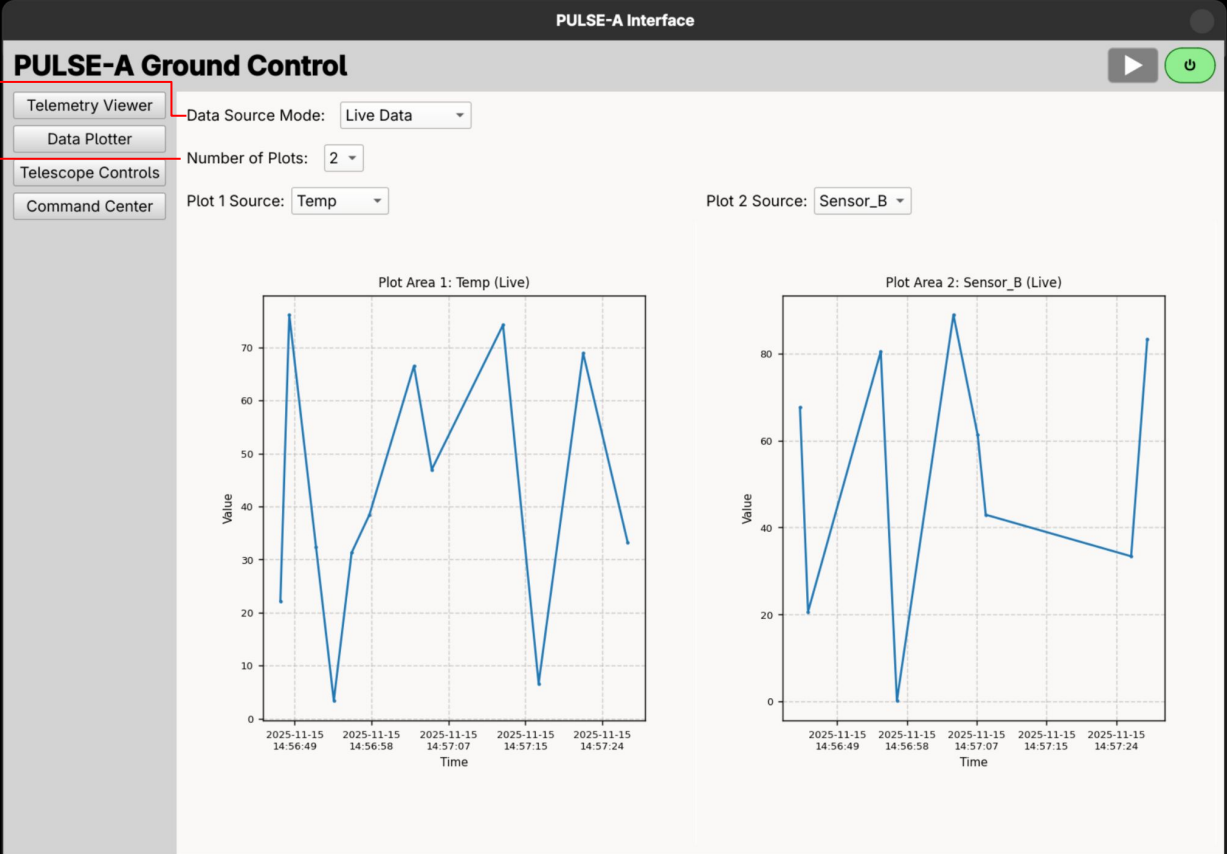
	Timestamp	Parameter	Value
1	2025-11-15 14:57:39.320	Sensor_B	91.790
2	2025-11-15 14:57:38.319	Sensor_B	14.563
3	2025-11-15 14:57:37.319	Sensor_C	72.213
4	2025-11-15 14:57:36.319	Temp	87.061
5	2025-11-15 14:57:35.320	Sensor_B	73.165
6	2025-11-15 14:57:33.320	Temp	34.862
7	2025-11-15 14:57:31.320	Sensor_C	33.511
8	2025-11-15 14:57:30.319	Sensor_C	56.554
9	2025-11-15 14:57:29.319	Sensor_A	88.754
10	2025-11-15 14:57:28.320	Sensor_B	83.375
11	2025-11-15 14:57:27.319	Temp	33.206
12	2025-11-15 14:57:26.320	Sensor_B	33.382
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Allows for live data or archived data plotting



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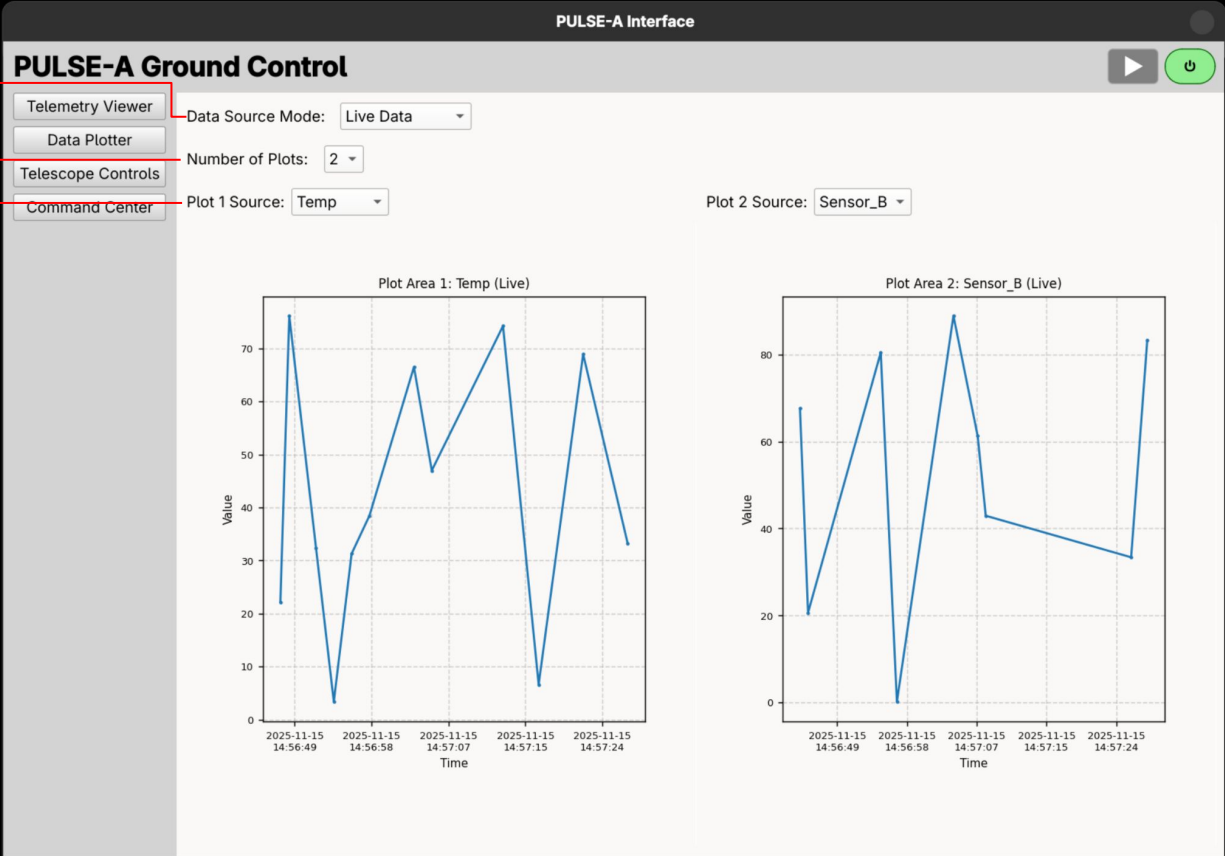
Controls quantity of the plots displayed



Allows for live data or archived data plotting

Controls quantity of the plots displayed

Controls the source of the plots displayed

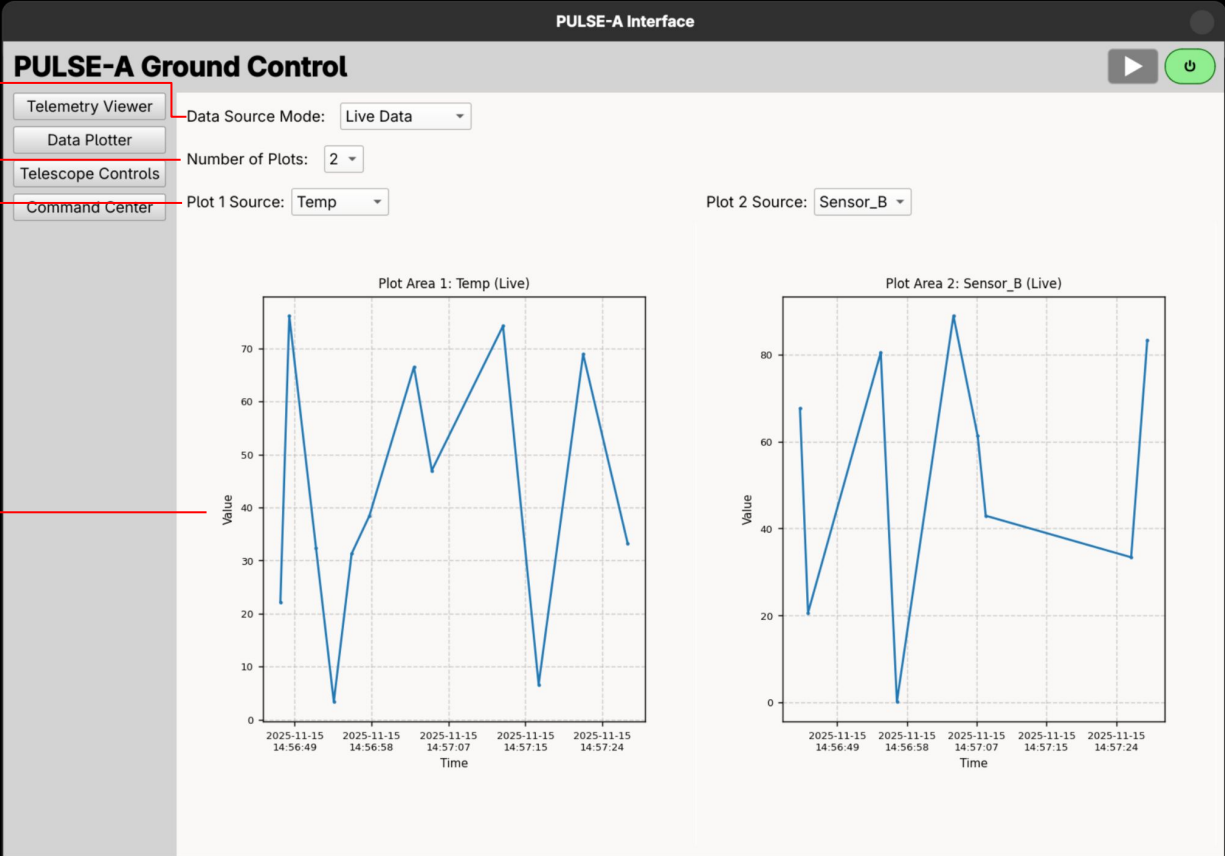


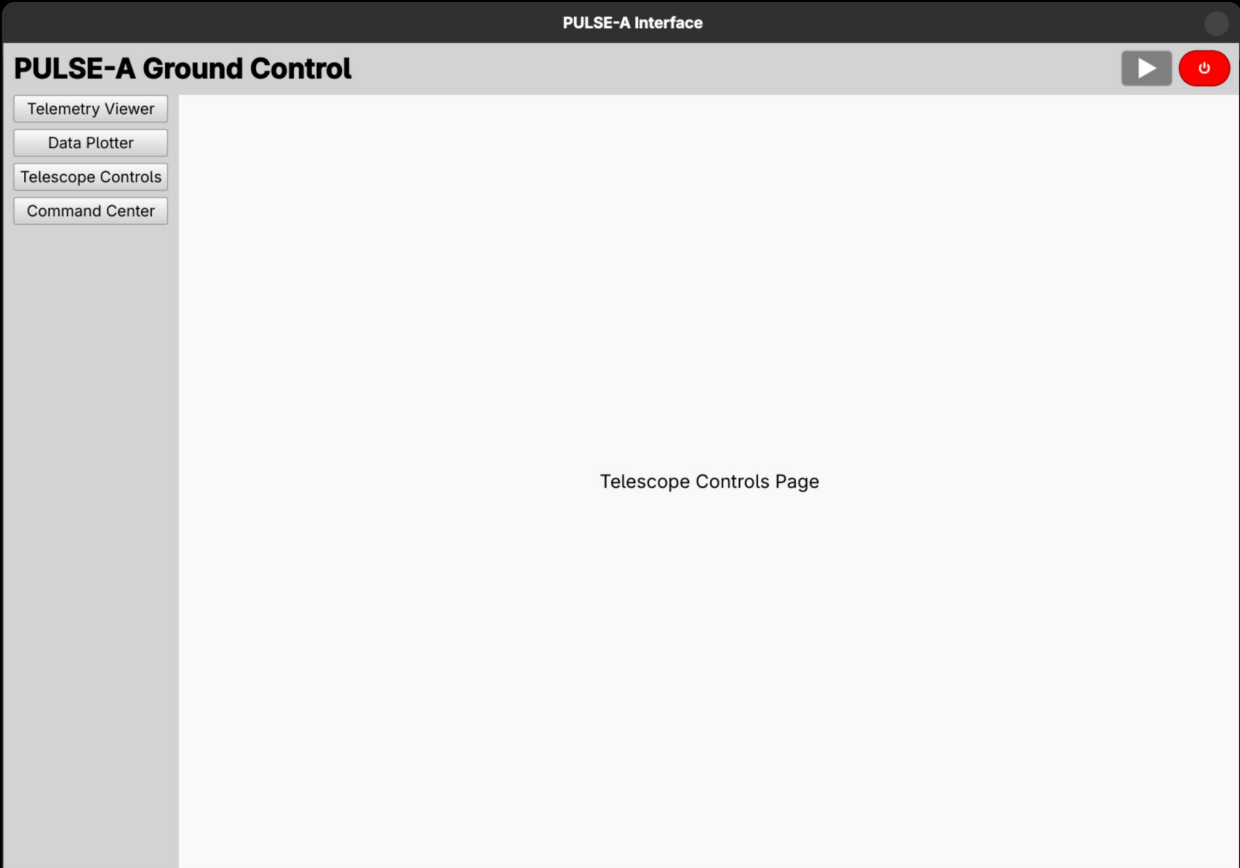
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Controls quantity of the plots displayed

Controls the source of the plots displayed

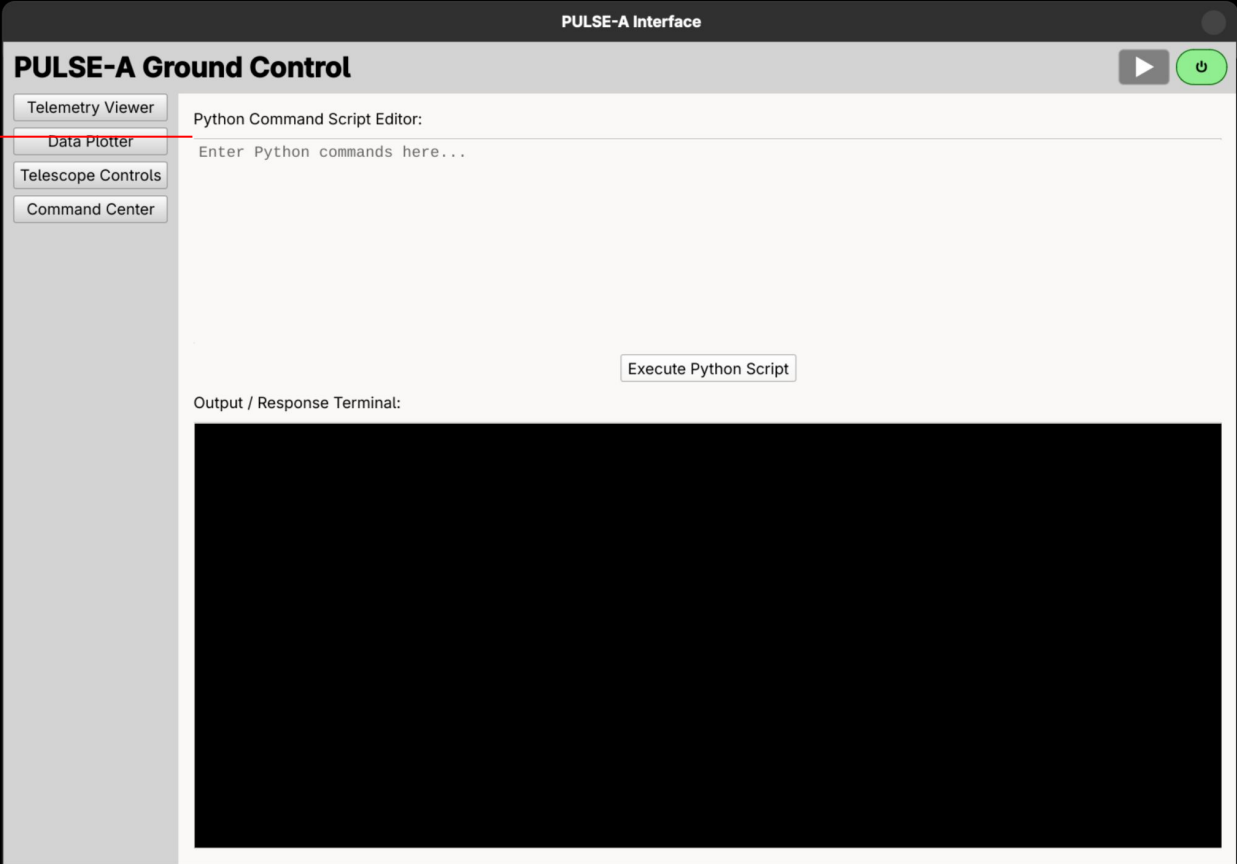
Plots rom each sensor of each data point by timestamp





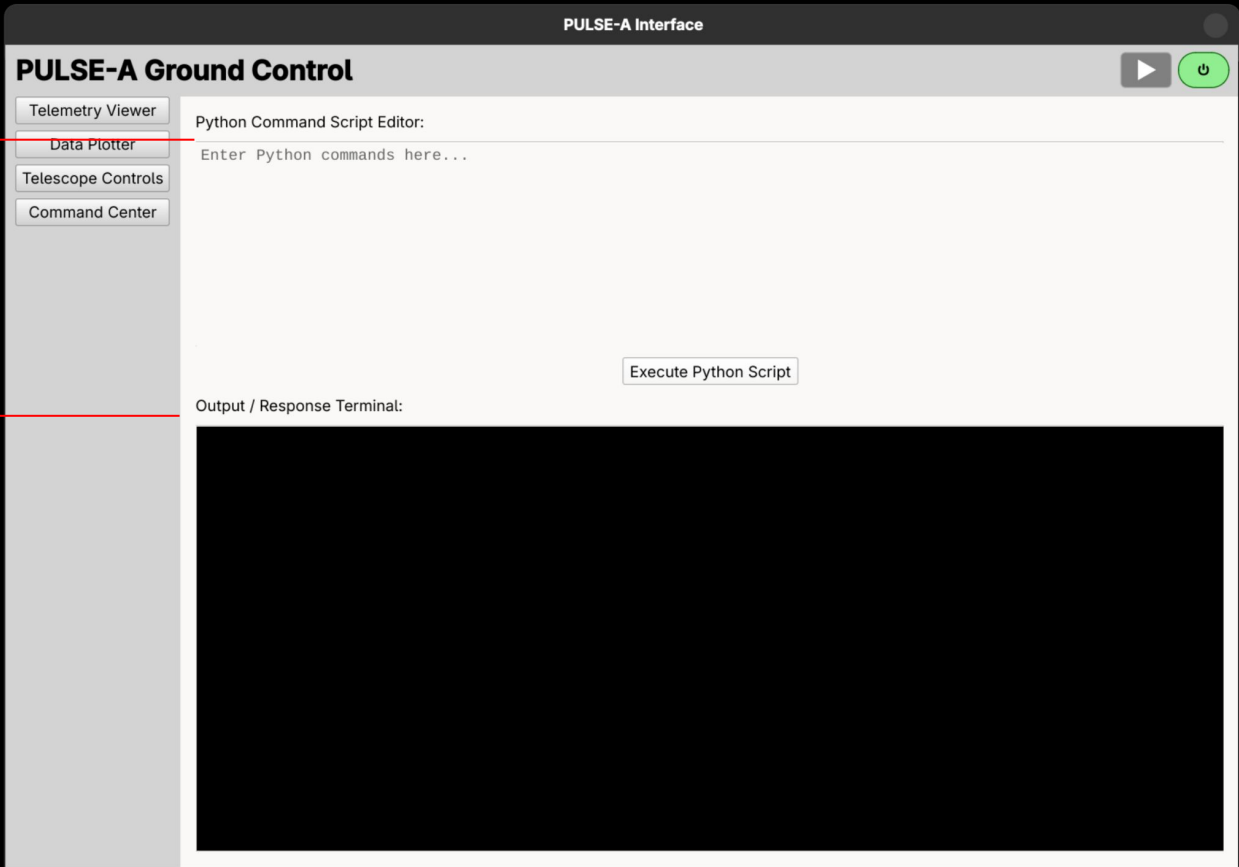
Telescope Controls GUI is currently in progress

Script editor allowing python command to be run on the telescope



Script editor allowing python command to be run on the telescope

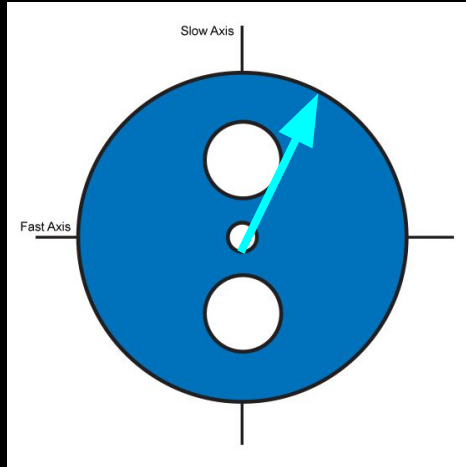
Terminal showing the outputs and responses



Testing & Component Characterization

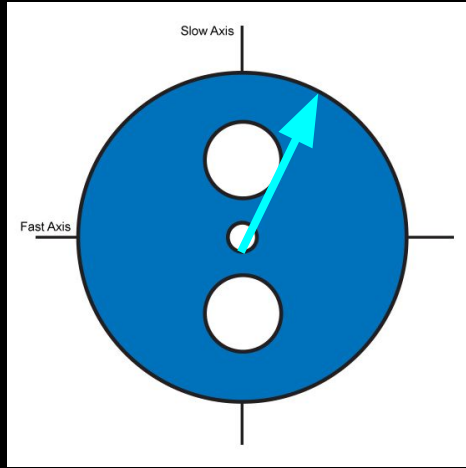
Testing and Component Characterization: **Fiber Experiment**

Fiber Experiment Motivation



- ❑ We were interested in the change of polarization state from linear to elliptical after going through a PM fiber
- ❑ We expect that there will be slight mismatch between the axis of the fiber and the axis of polarization which will cause the polarization to become elliptical.
- ❑ This will be due to imperfections when coupling from lasers to other optical components and shifts during launch.

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Interested in characterizing the effects of mismatch between the linear polarization axis and the axis of the PM fiber.

Mueller Formalism

To characterize polarization states we can use:

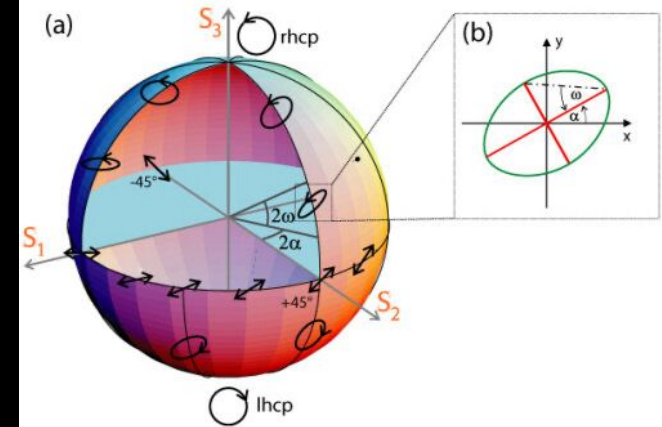
Jones formalism

- ❑ Uses 2×2 complex matrices acting on electric-field amplitude
- ❑ Fully polarized, coherent light only.
- ❑ NO depolarization

Mueller formalism

- ❑ Uses 4×4 real matrices acting on Stokes vectors (intensity + polarization)
- ❑ All Jones can do plus depolarization characterization

$$\begin{bmatrix} S_0' \\ S_1' \\ S_2' \\ S_3' \end{bmatrix}_{\text{out}} = M_{11} \begin{bmatrix} 1 & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix} \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix}_{\text{in}}$$



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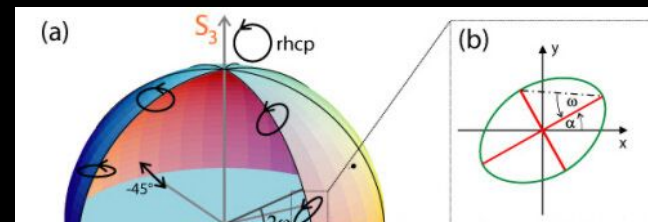
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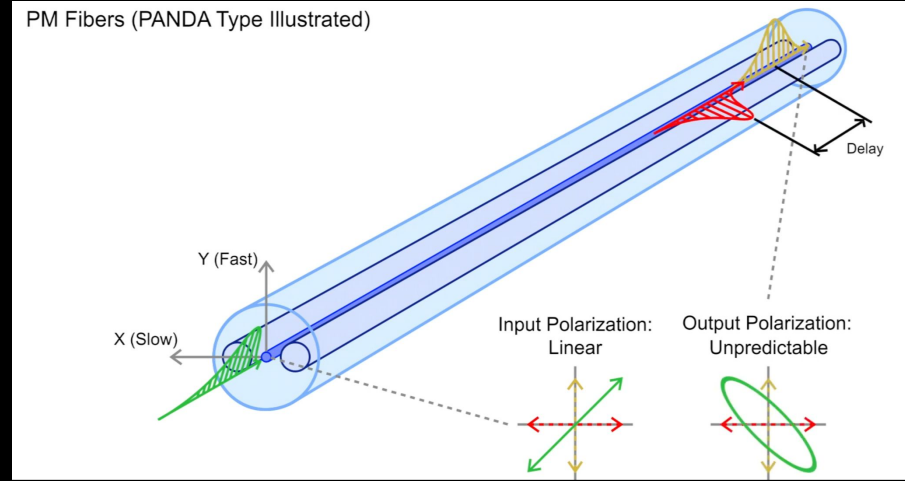
Mueller matrix formalism can be used to describe effects on the polarization states across Payload components.

Disturbances on SOP

Three general effects can be caused by an optical device on the state of polarization:

- ❑ Depolarization
- ❑ Polarization-dependent loss (attenuation)
- ❑ Retardance (phase shift)

We expect that all three effects can be captured by a Mueller matrix that describes the transformation on an initial polarization state.

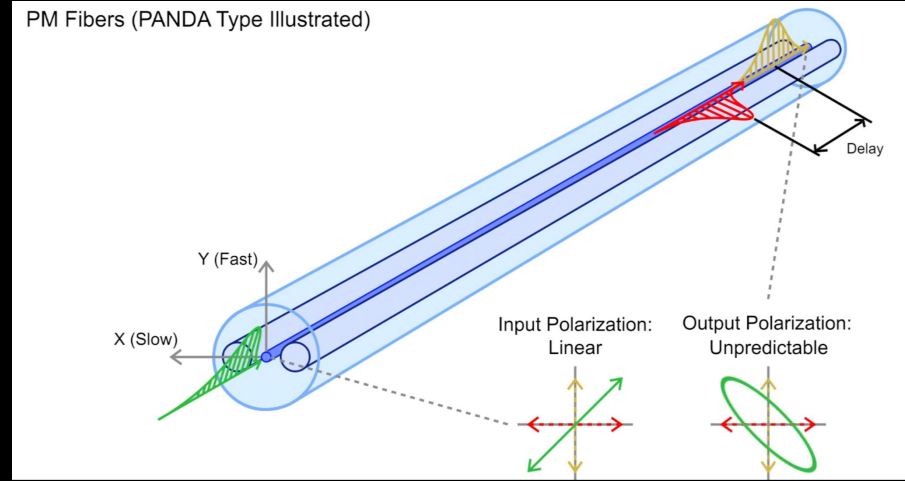


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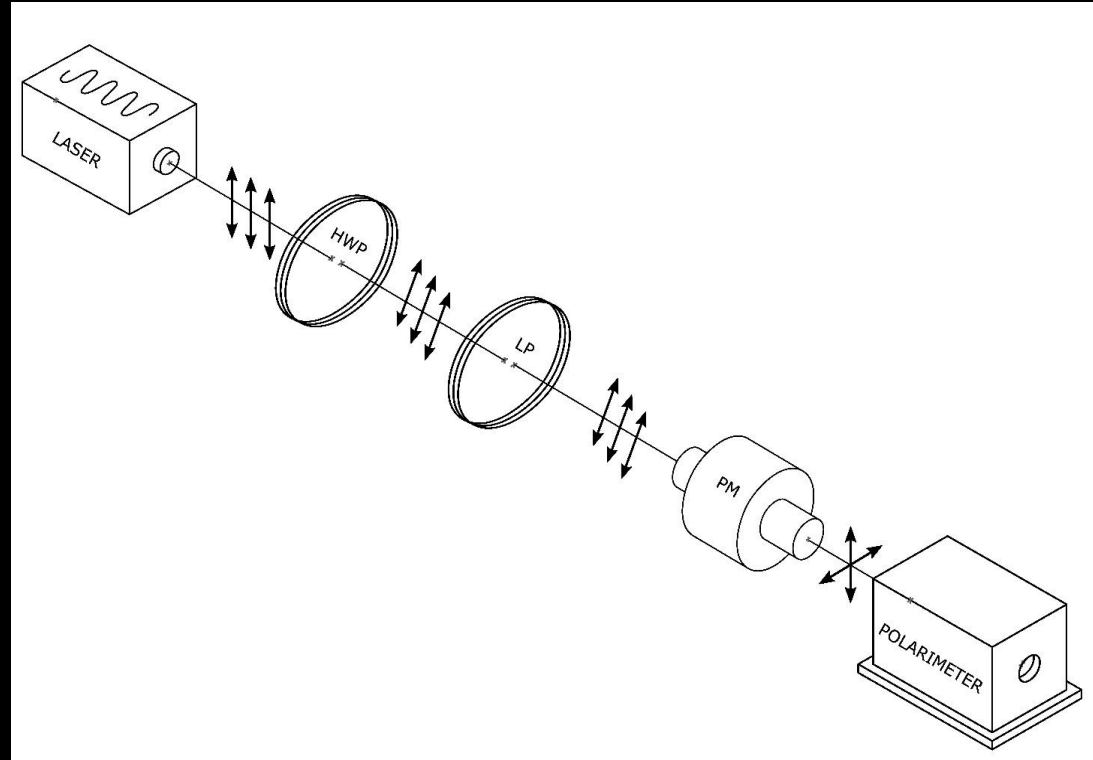
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A single Mueller matrix can describe the transformation of a SOP in our PM fibers.

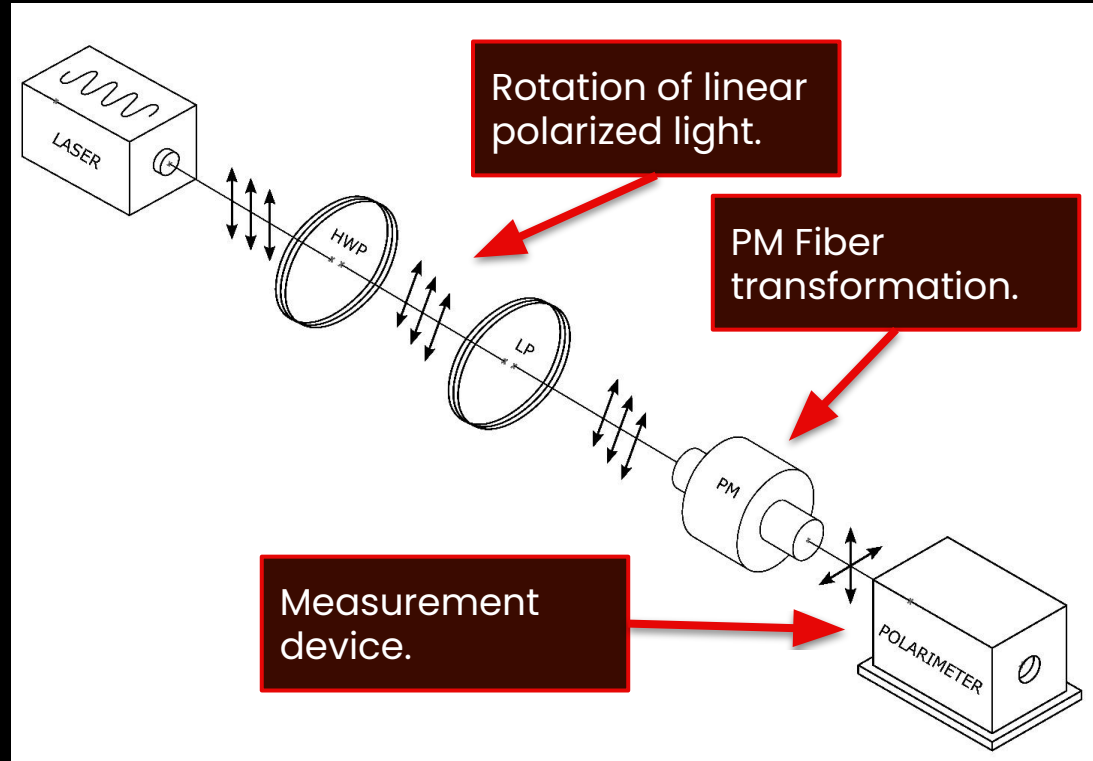
Fiber Experiment Set Up

- ❑ We prepared a general polarized state and obtained a Mueller matrix (transformation) by measuring the Stokes parameters of the final polarization state.
- ❑ We switched to linear polarization and applied the same matrix.
- ❑ We applied a rotation to the initial linear to simulate misalignments.



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Mueller Matrix Transformations

$$\hat{M}^t = R(-\theta) \cdot M^{QWP} \cdot R(\theta), R(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(2\theta) & -\sin(2\theta) & 0 \\ 0 & \sin(2\theta) & \cos(2\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\hat{S}^i = \hat{M}^t \cdot \hat{S}^{vl}$$

$$M^{QWP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

- ❑ $R(\theta)$ is the rotation matrix to align the QWP fast axis with the initial Stokes vectors
- ❑ We perform a change of basis M^{QWP} and the Mueller matrix for QWP with the initial Stokes vectors.
- ❑ The aligned QWP matrix is then multiplied with the input stokes to yield the output stokes vector

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The final result is a series of transformations on the state.

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Mueller Matrix Results

- ❑ Mueller Matrices calculated for various fiber optic components:

Thorlabs PFC1550A Polarization Fiber Combiner:

Best 4 calibration angles: (2, 4, 16, 28)
Lowest NRMSE: 0.040373048829467774

Mueller Matrix:

```
[[ 1.         -0.         -0.         0.         ]
 [ 0.936465 -0.003928  0.01607  -0.031485]
 [-0.346853 -0.006161  0.051512 -0.097144]
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Standard Pandas PM Fiber:

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Pm Fiber Coupler:

Mueller Matrix of Coupler:

```
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Transformation matrix
experimentally determined.

Standard Pandas PM Fiber:

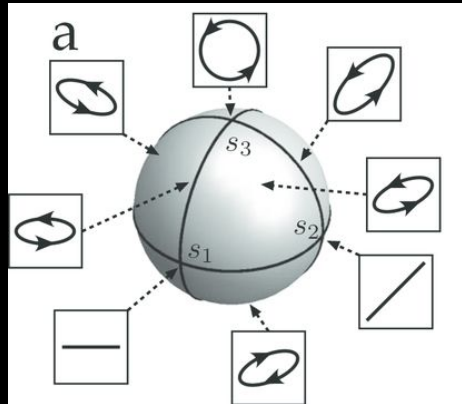
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Fiber Experiment Next Steps

- ❑ Apply Mueller formalism for fiber connectors and find the degree of misalignment generated by the connectors
- ❑ Apply the Mueller formalism for EDFA characterization.
- ❑ Complete the series of matrix transformations for the full Payload.

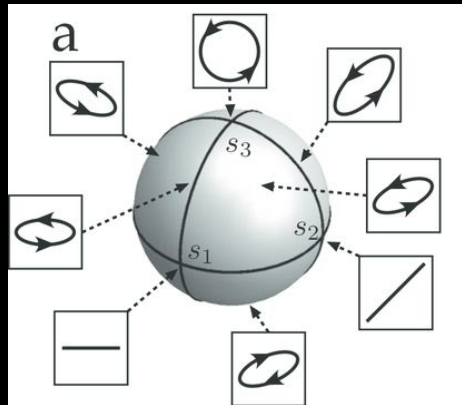


- ❑ Series of transformation can be mapped on a Poincaré sphere
- ❑ Any two polarization states lie on a single great-circle arc of the Poincaré sphere.
- ❑ This means we can treat an optical element as a rotation on the sphere and easily determine its effect!

Fiber Experiment Next Steps

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Optical components can be modeled as transformations on the state of polarization (SOP).



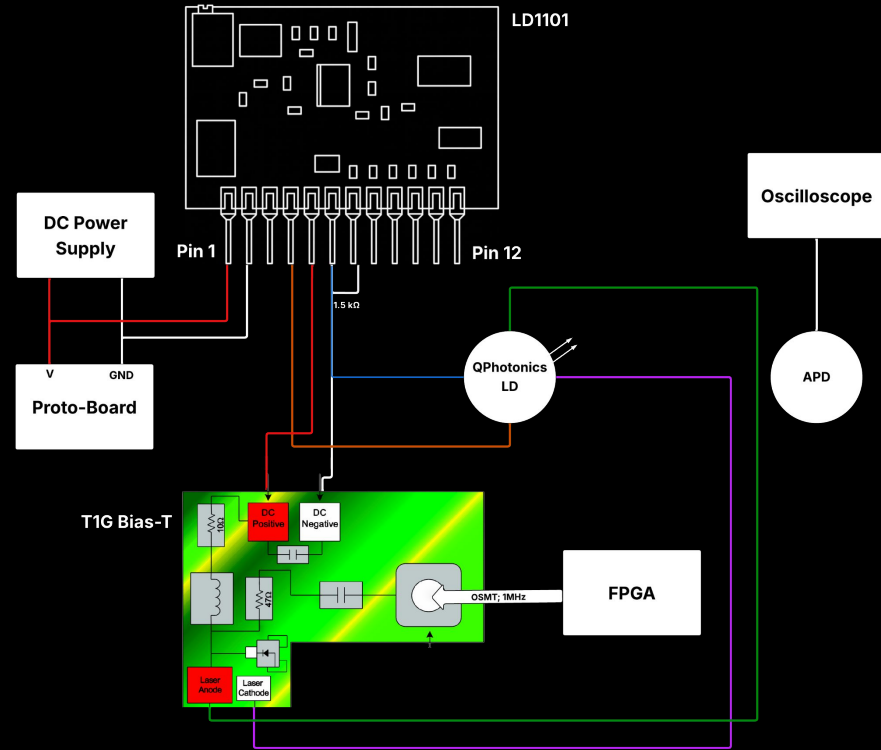
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Testing and Component Characterization: **Laser Driving**

Laser Driving Testing:

General testing overview:

- ❑ Have successfully modulated one Qphotonics LD at 10khz
- ❑ Currently using function generator to modulate instead of FPGA
- ❑ Testing at steady power output of 3mW
- ❑ Next Steps:
 - ❑ Dual laser driving
 - ❑ Replace function generator with FPGA

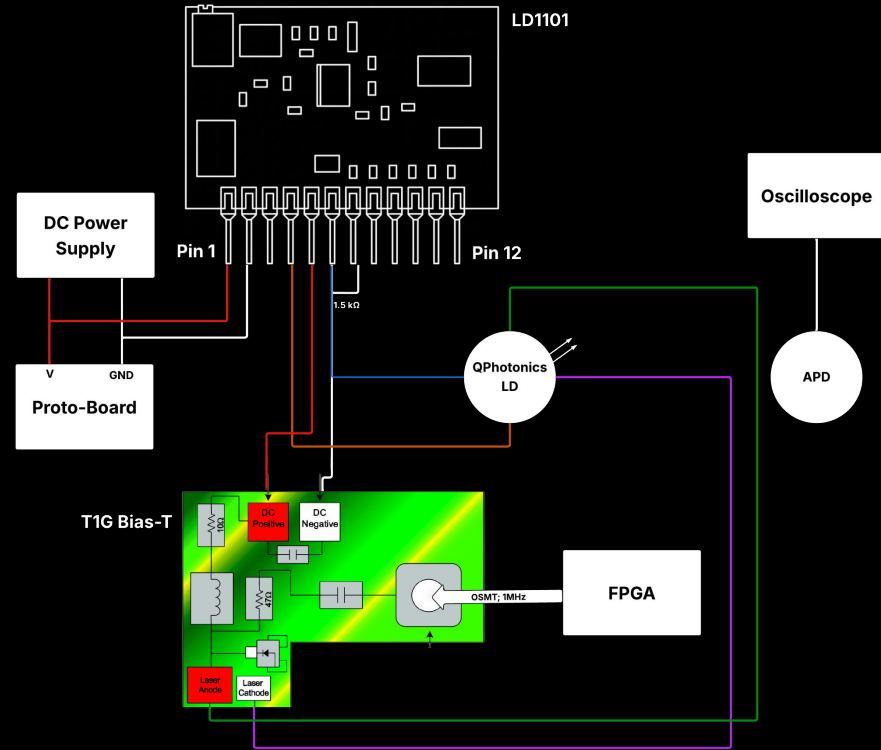


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Successful modulation
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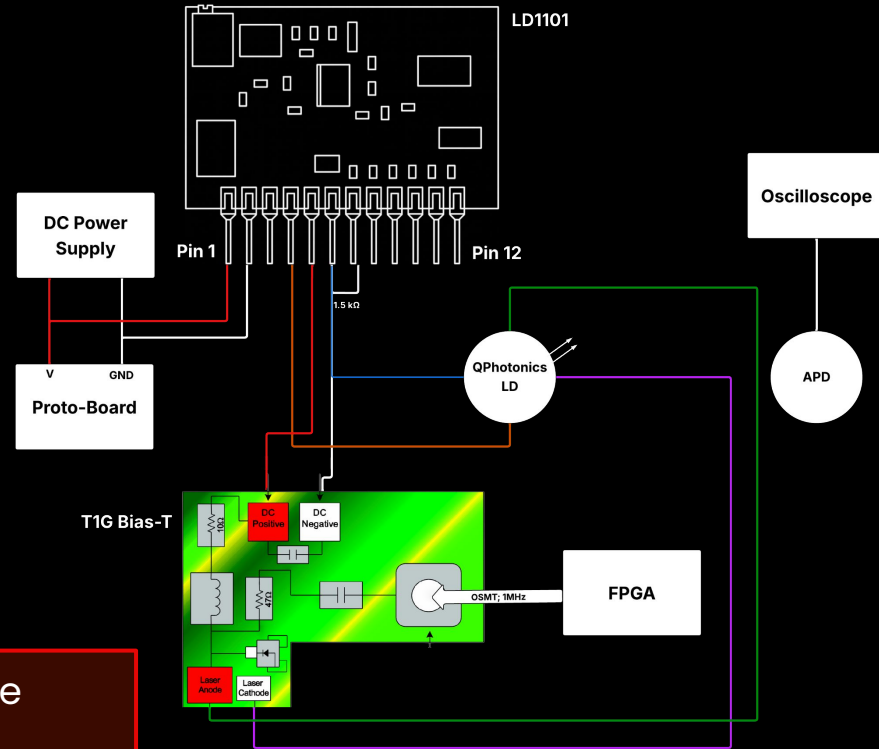
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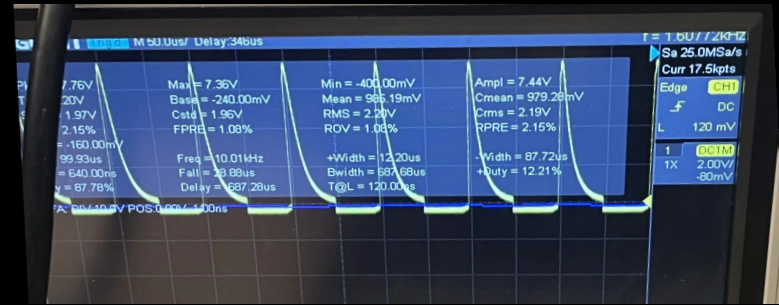
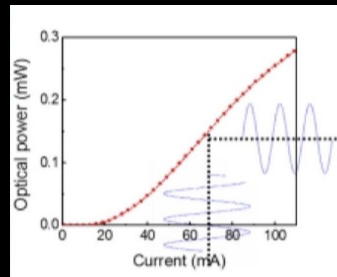
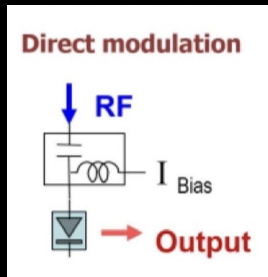
Set up needs to be duplicated for full CPolsk demo.



Laser Driving Testing:

Some attempted results:

- ❑ Some initial results for a square and sine wave shows cutoff at the end.
- ❑ This can be explained by the fact the current I_{square} or I_{sin} is below the threshold current before it reaches the minimum.
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Square wave signal.

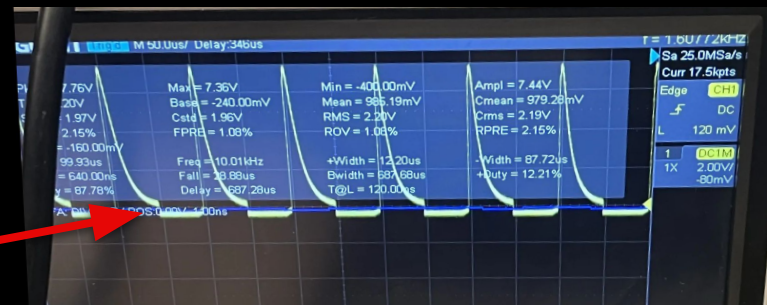


Sine wave signal.

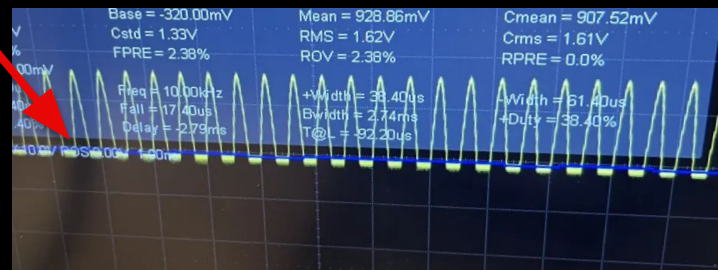
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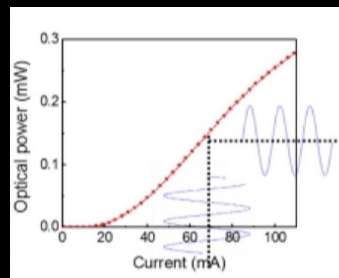
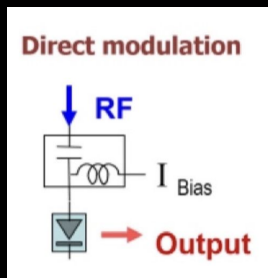
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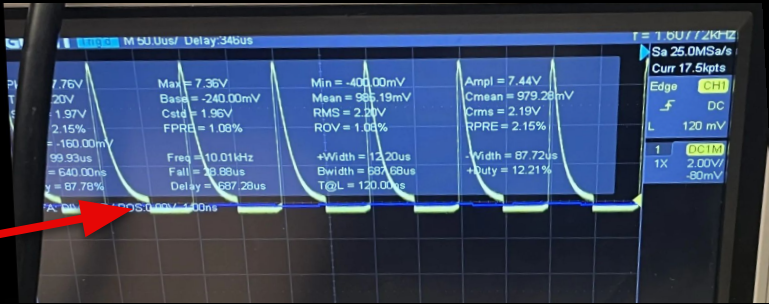
Sine wave signal.



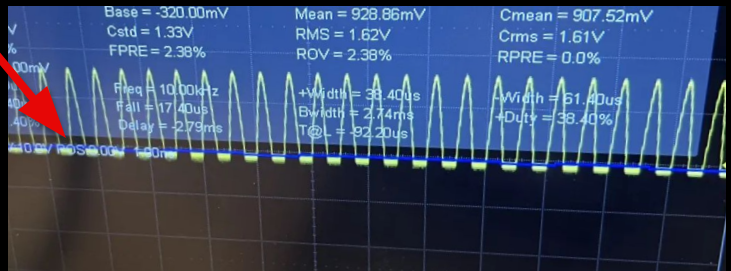
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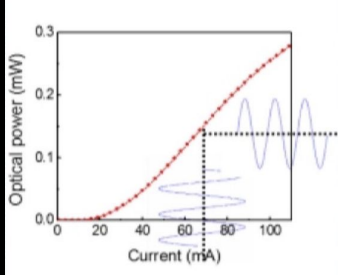
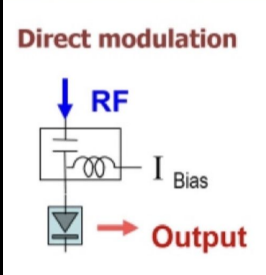


Square wave signal.



Sine wave signal.

Modulation initial results.

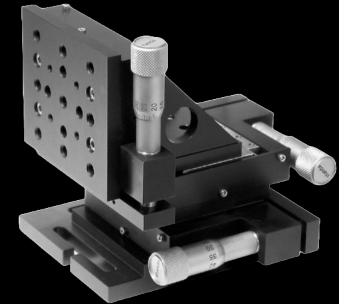
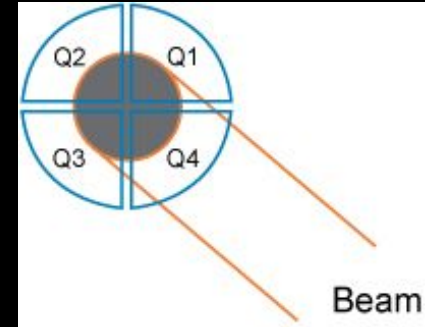


Testing and Component
Characterization:
QPD Characterization Plan

QPD Testing

Plan:

- ❑ QPD mounted on precision X-Y translational stage
- ❑ Four readings (one from each quadrant)
- ❑ Collimated 1550 nm laser is attenuated to pW level and focused onto the photodiode using a three-lens system (30 mm, 8 mm, and 9 mm lenses)
- ❑ Beam is aligned to hit center of detector and baseline voltage output from each quadrant are recorded
- ❑ Scan in small increments along X + Y + Z axes and record change in voltage distribution across quadrants
- ❑ Use data to compute normalized coordinates

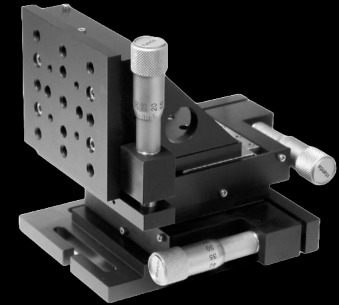
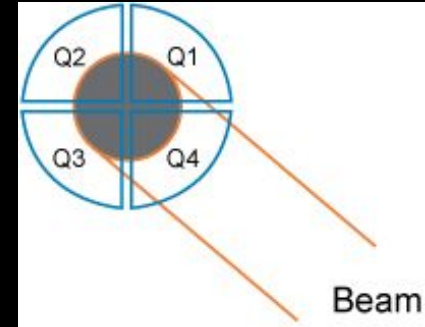


QPD Testing

Simple set up that simulates a collimated beam on the QPD.

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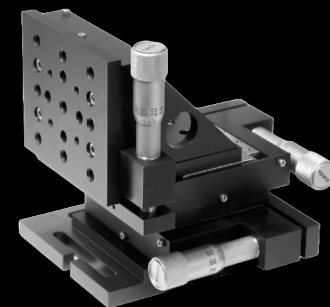
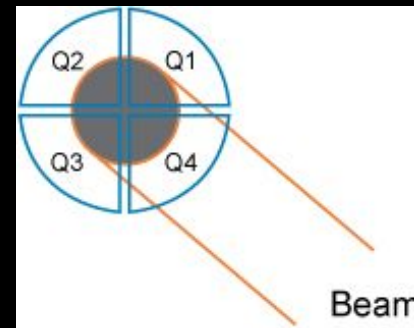


QPD Testing

Simple set up that simulates a collimated beam on the QPD.

Plan:

- ❑ QPD mounted on precision X-Y translational stage
- ❑ Four readings (one from each quadrant)
- ❑ Collimated 1550 nm laser is attenuated to pW level and focused onto the photodiode using a three-lens system (30 mm, 8 mm, and 9 mm lenses)
- ❑ Beam is aligned to hit center of detector and baseline voltage output from each quadrant are recorded
- ❑ Scan in small increments along X +Y + Z axes and record change in voltage distribution across quadrants
- ❑ Use data to compute normalized coordinates



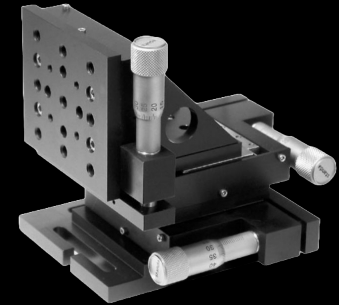
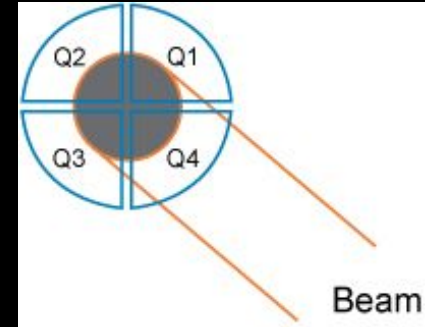
Computing coordinates using results.

QPD Testing

Measuring the QPD's real electrical response to a controlled beam so we can validate its behavior.

Plan:

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- ❑ Four readings (one from each quadrant)
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Optimal QPD Position

- ❑ One can derive the beam displacement given a position on the QPD with trigonometry for a collimated beam.

- ❑ Spot

$$\alpha = \tan^{-1}\left(\frac{x}{f}\right)$$

- ❑ For a defocused (not collimated beam) the QPD no longer measures only an angle:
 - ❑ The incoming wavefront is curved.
 - ❑ The spot position depends on both **angle** and **distance** from focus.
 - ❑ The quad-cell sees a mixture of lateral offset + defocus

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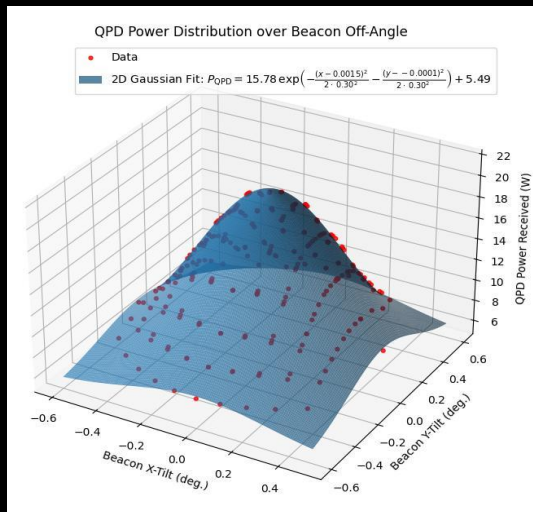
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QPD at a defocusing plane will measure effectively:
angle + lateral displacement + chromatic shift

Recall: Zemax Simulations Plan for Full Characterization

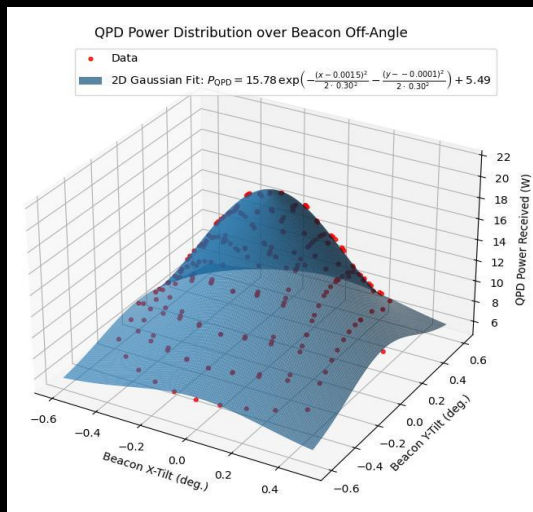
- ❑ **Model of the QPD:** at a given defocusing plane. Adjust QPD image plane along optical axis (8–10 mm behind focus).
- ❑ **Field-angle mapping:** tilt the incoming beam by small angles (0° , 0.005° , 0.025°) and measure the lateral spot shift at the QPD $\rightarrow$ get angle $\leftrightarrow$ displacement relationship.
- ❑ **PSF generation:** Generate the PSF (irradiance map) at the QPD plane on a pixel grid.



- ❑ **Angular scaling:** Convert the grid of x - y positions in the detector plane (PSF grid in mm) to equivalent angular coordinates using the field-angle relationship.
- ❑ **QPD geometry:** Mark the cross-shaped dead zones (QPD gaps) as a binary mask.

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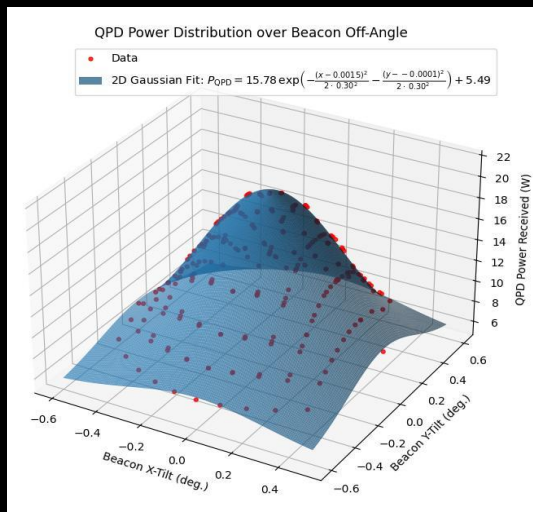


This procedure was inspired by CLICK's mission and their QPD characterization.

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Relating displacement to angle mapping of the QPD.

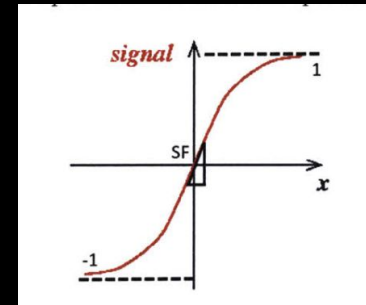
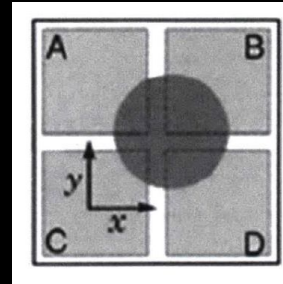
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$$X = \frac{(Q2 + Q3) - (Q1 + Q4)}{(Q1 + Q2 + Q3 + Q4)} = \frac{X_{Diff}}{SUM}$$

$$Y = \frac{(Q1 + Q2) - (Q3 + Q4)}{(Q1 + Q2 + Q3 + Q4)} = \frac{Y_{Diff}}{SUM}$$



- ❑ **Build S-curve:** Translate the PSF laterally across the QPD; for each displacement/angle, compute x. The collection of points x vs spot position (or angle) is the **S-curve**.
- ❑ **Slope Factor (SF):** Fit a straight line to the central, linear region of the S-curve (after converting displacement to angle). The slope gives SF in units of 1/rad.

Recall: Zemax Simulations Plan for Full Characterization

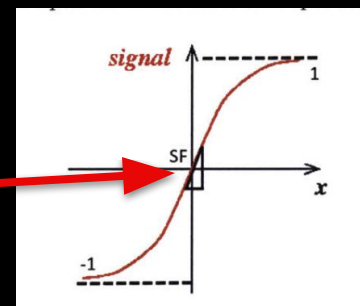
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The Slope Factor quantifies the linear sensitivity of the QPD—how much the signal changes per unit of input angle.



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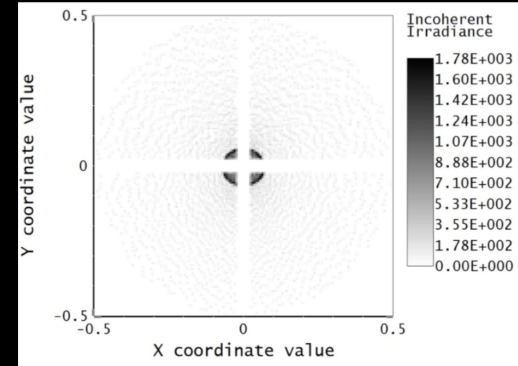
- ❑ **Effective SNR:** Compute how much PSF power falls into the QPD dead zones; use the remaining fraction to scale the optical SNR and obtain the effective SNR (assuming we have determined an SNR for our QPD).

$$\text{Effective SNR} = \eta \text{ SNR}_0$$

η = fraction of the optical power that actually lands on the QPD.

- ❑ **Noise-Equivalent Angle:** Combine SF and effective SNR to get:

$$\text{NEA} = \frac{1}{\text{SF} \cdot \text{SNR}_0}$$



Recall: Zemax Simulations Plan for Full Characterization

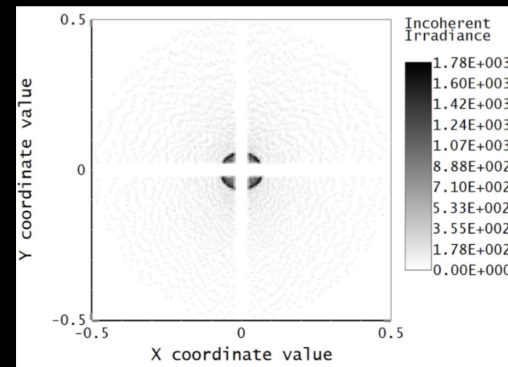
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NEA will help determine the ideal spot size at the QPD and hence its distance from the focus.

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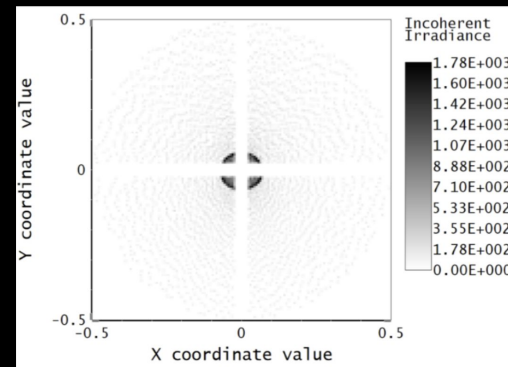
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This calculation needs to be done for the every defocusing plane to find the optimal position of QPD.

Testing and Component
Characterization:
Telescope Experiment

Telescope Experiment Motivation

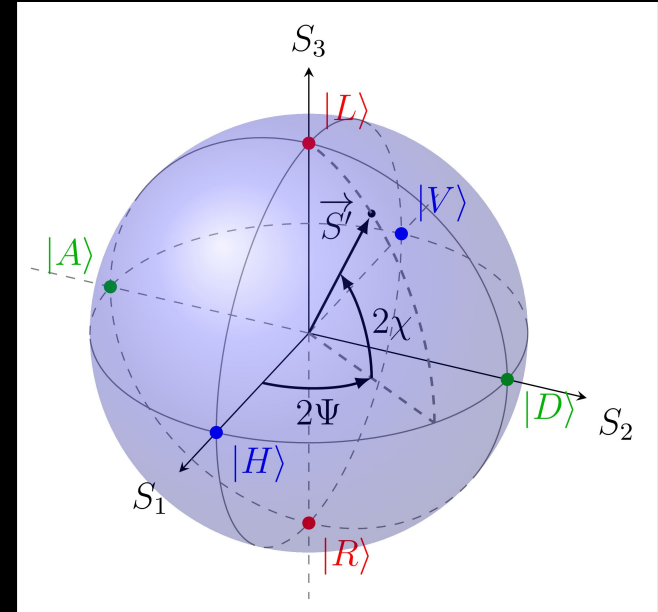
- ❑ The purpose of this experiment is to quantify how the telescope alters the polarization state of a circular polarized optical beam by measuring the stokes parameters before and after the telescope under a controlled set of polarization states.
- ❑ Optical elements in the telescope can introduce changes to the polarization state which modify the measured Stokes vector (S_0, S_1, S_2, S_3)
- ❑ The goal is to determine whether the telescope induces small, stable polarization changes or any significant deviations that could affect downstream optical performance

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- ❑ Optical elements in the telescope change the polarization state. Goal: Characterization of telescope polarization transformations.
- ❑ The goal is to determine whether the telescope induces small, stable polarization changes or any significant deviations that could affect downstream optical performance.

Polarization Characterization Using Stokes Parameters

- ❑ Similar to the Fiber Experiment, the Telescope Experiment characterizes polarization using Stokes vectors.
- ❑ The polarimeter directly measures these parameters after the beam passes through the telescope, enabling evaluation of:
 - ❑ Retardance
 - ❑ Ellipticity
 - ❑ Degree of Circular Polarization (DoCP)
 - ❑ Degree of Polarization (DoP)
- ❑ Differences between initial and final Stokes vectors quantify how the telescope modifies circular polarized light.



Telescope Experiment Setup

- ❑ A 1550 nm laser is passed through a linear polarizer and a quarter-wave plate (QWP) to generate a controlled sequence of polarization states.
- ❑ The QWP is rotated in 10° increments, beginning at a circularly polarization state and returning to circular polarization near 90° , producing a wide range of elliptical and linear polarization conditions.
- ❑ For each QWP angle, the initial Stokes vector is recorded before the telescope, and the final Stokes vector is recorded after it.

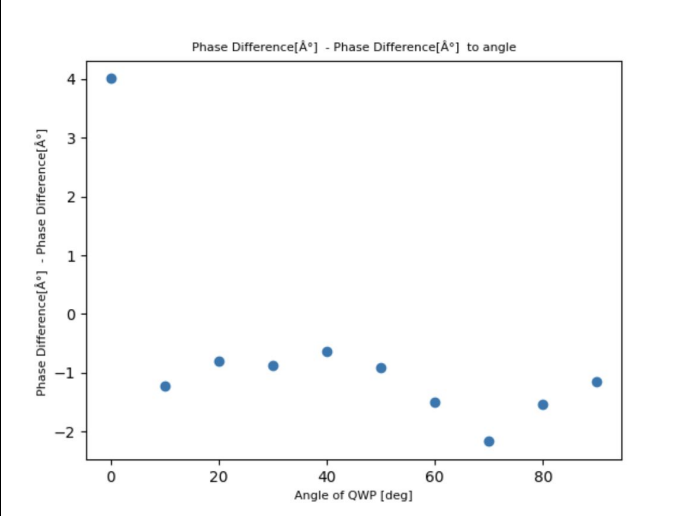
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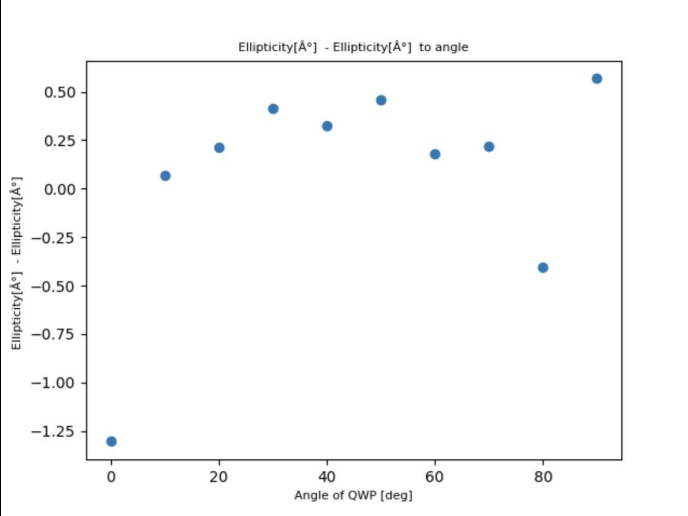
Initial and final stokes vectors recorded for multiple elliptically polarized states.

Telescope Experiment Results

Initial – Final Phase Difference vs Angle

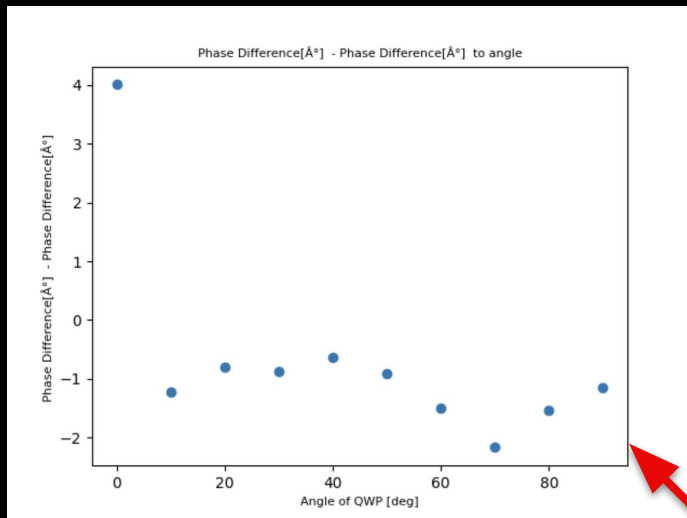


Initial – Final Ellipticity vs Angle

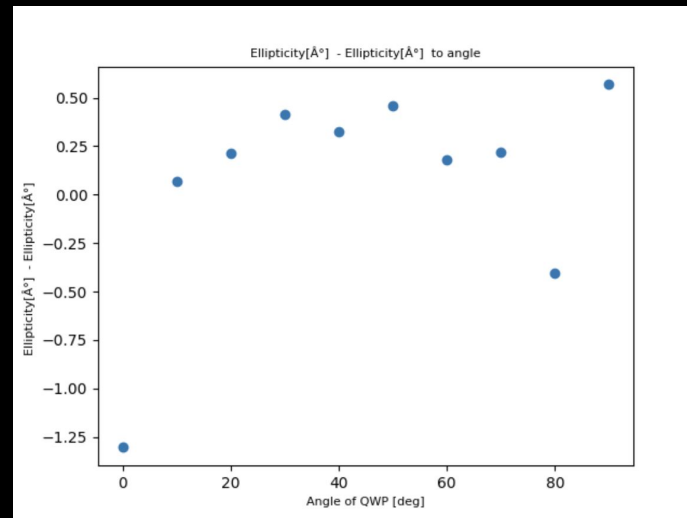


Telescope Experiment Results

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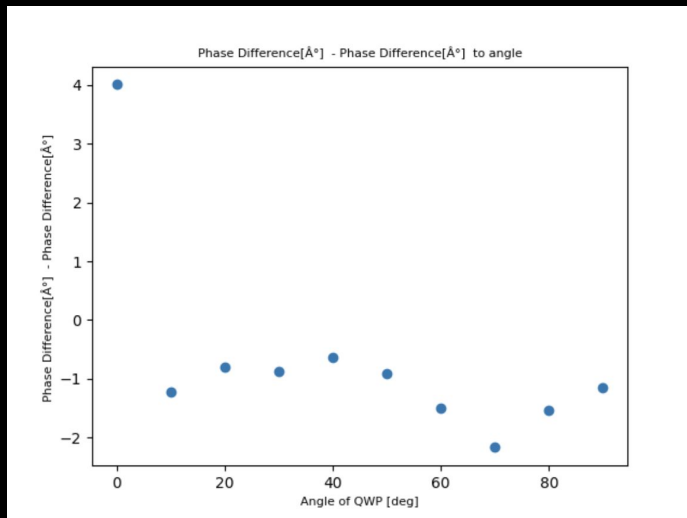
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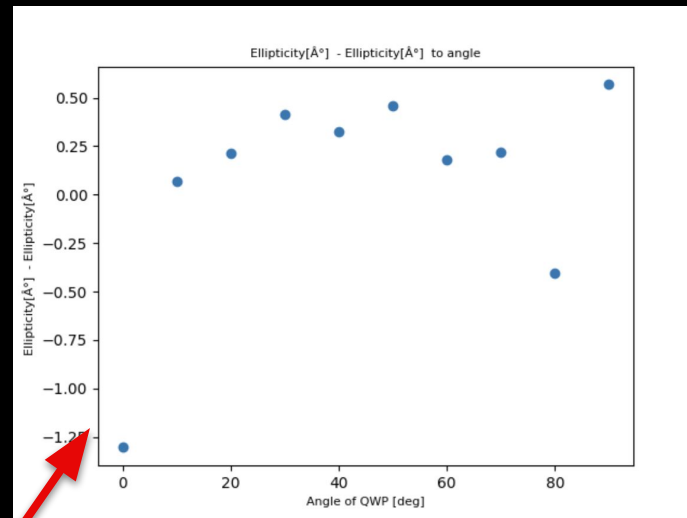
The initial vs. final phase difference remains close to 0 across all QWP angles. Small fluctuations indicate only minor retardance introduced by the telescope.

Telescope Experiment Results

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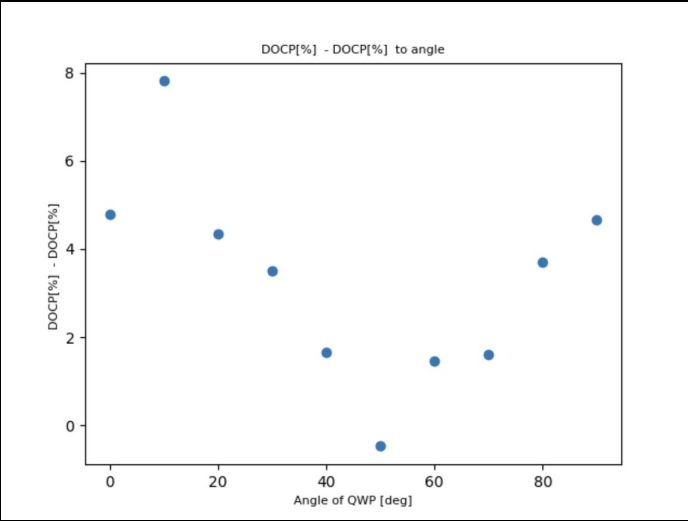
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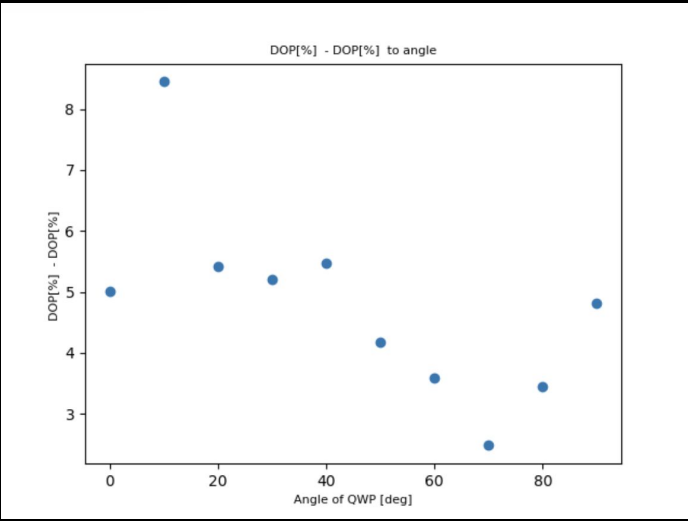
The initial vs. final ellipticity difference remains low, with most values within approximately $\pm 0.5^\circ$. A few larger deviations at circular polarized light.

Telescope Experiment Results

Initial - Final DOCP vs Angle

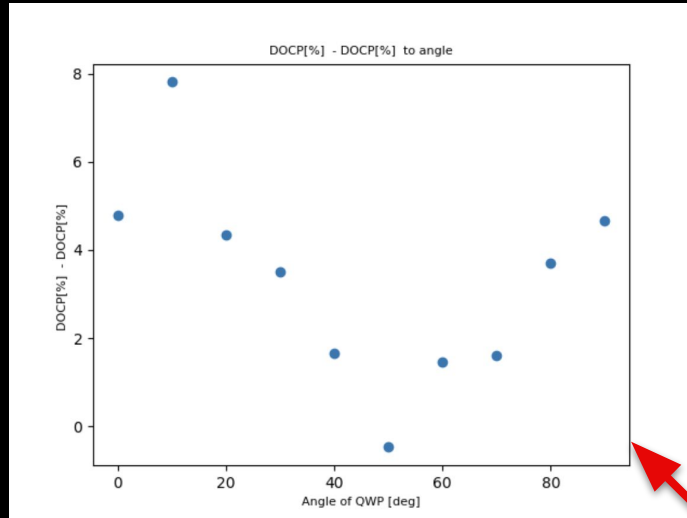


Initial - Final DOP vs Angle

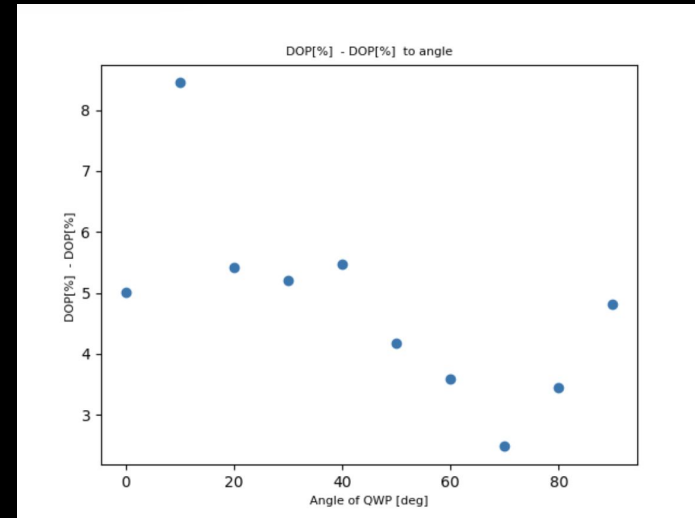


Telescope Experiment Results

Initial - Final DOCP vs Angle



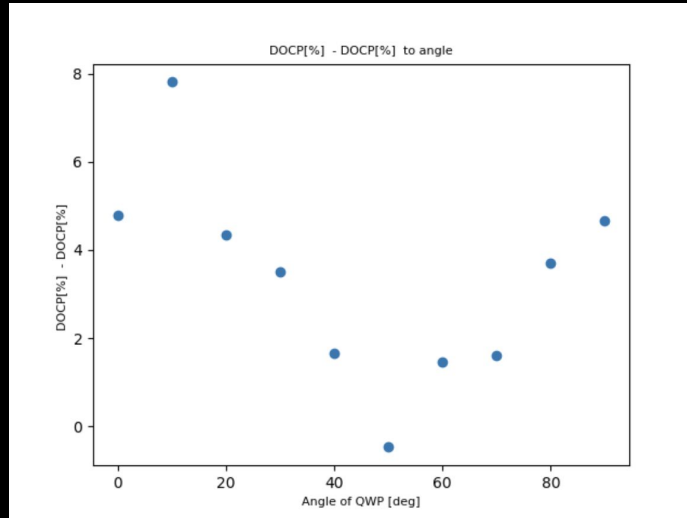
Initial - Final DOP vs Angle



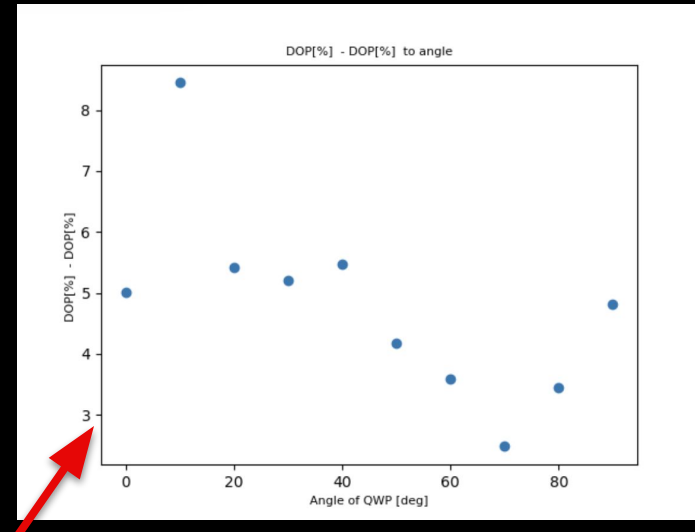
The differences in DOCP is generally small (around 5%), with a dip at angles corresponding to linearly polarized light, where DOCP is ≈ 0 by definition. As such, the telescope preserves circularity well, with no large deviations across the angles.

Telescope Experiment Results

Initial - Final DOCP vs Angle



Initial - Final DOP vs Angle



The differences in DOP shows a similar graph to DOCP, indicating a 5% depolarization introduced by the telescope. No strong angular dependence is observed, implying consistent depolarization across input states.

Pointing, Acquisition, and Tracking (PAT)

Pointing, Acquisition and Tracking: **Coarse Pointing**

Pointing Requirement Assumptions

- ❑ The OGS is assumed to be capable of maintaining the satellite within its FOV
 - ❑ FOV of OGS ~ 0.15 degrees
 - ❑ Telescope pointing error < 0.1 degrees derived experimentally (~ 1.5 arcmin or 0.025 degrees - Riesling 2018)
 - ❑ The change in flux due to this pointing error is negligible

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OGS is assumed to “point perfectly”.
Pointing is then defined with the OGS
beacon as the origin.

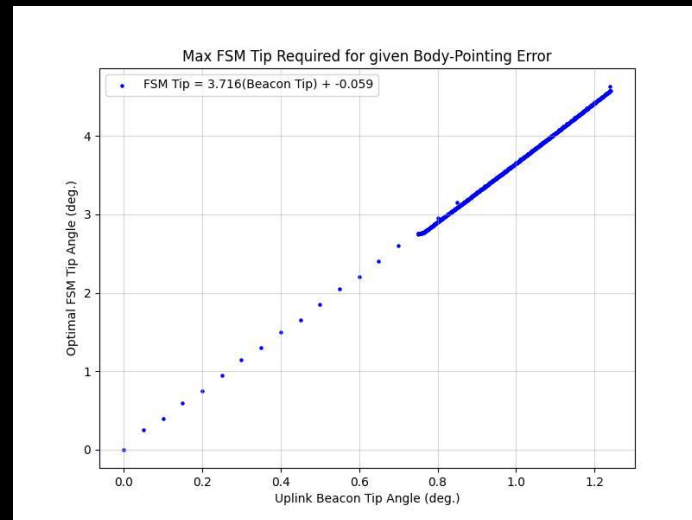
Pointing, Acquisition and Tracking: **Fine Pointing**

Recall: Payload Pointing Capabilities

- ❑ The Payload can correct for a range of < 0.7 of angles (with respect to the optical axis of our system).
- ❑ This result uses 60% of tilt (expected linear range)
- ❑ More than 60% of the tilt is expected non-linear range for FSM response.

Conclusion:

- ❑ **0.7 degrees is strict cutoff** for the allowed angle between the OGS beacon and the Payload optical axis at the aperture which the Payload can correct for.



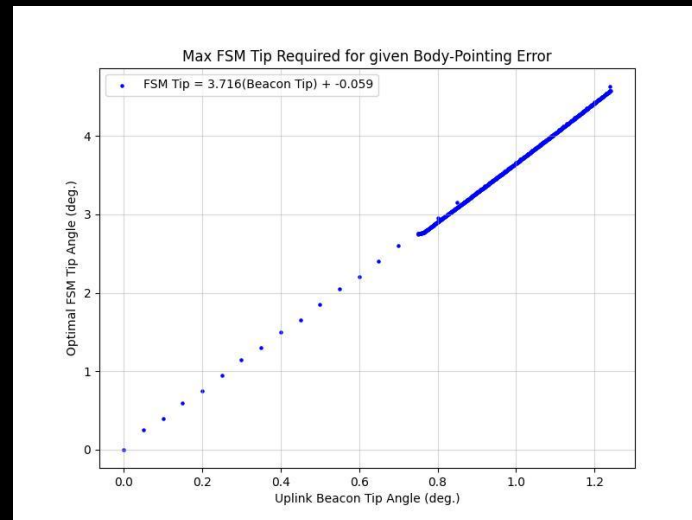
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Body-pointing + Extra Expected Misalignments < 0.7 degrees



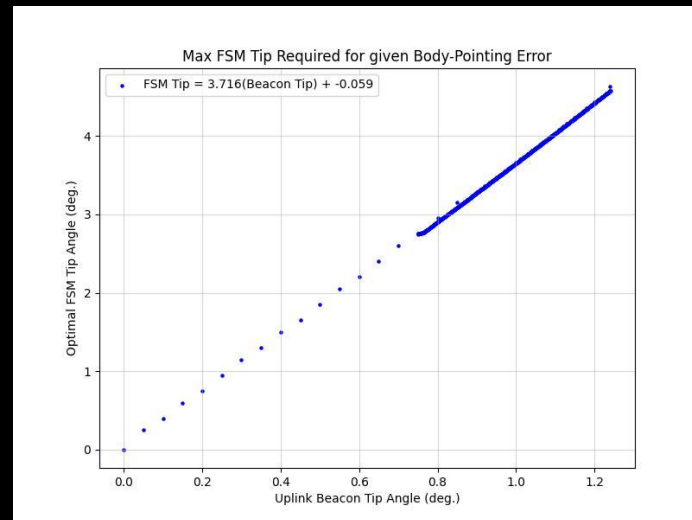
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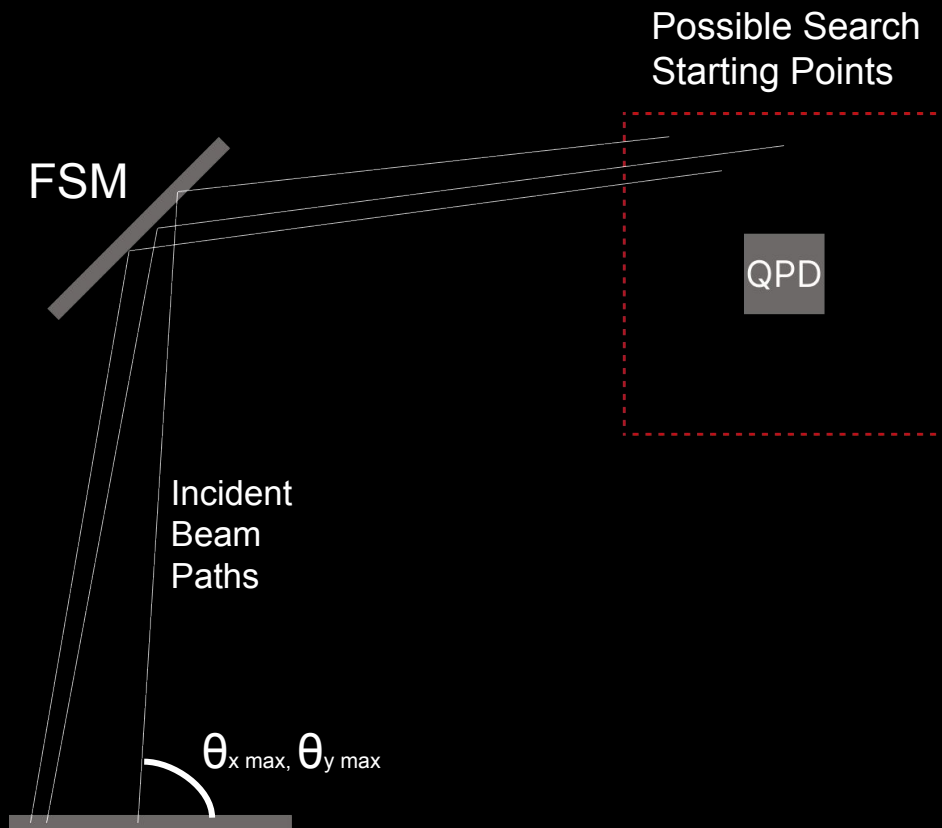
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Final expected misalignment between OGS beacon and transmission beam discussed in the Optical Analysis section



FSM Spiral Search

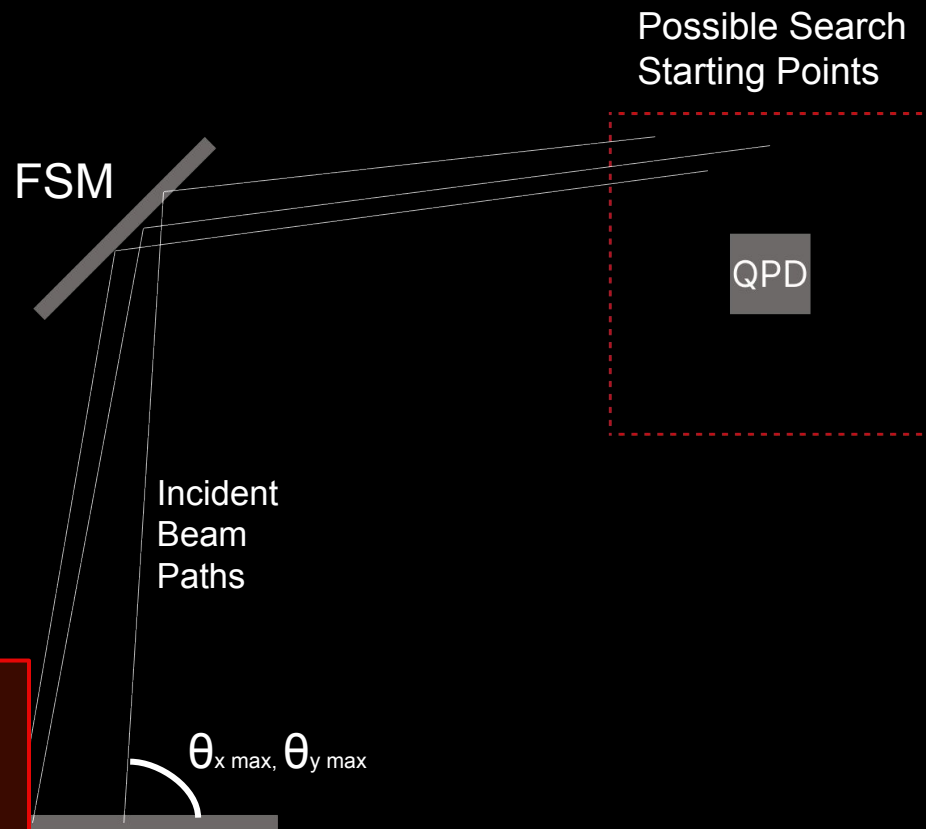
- ❑ FSM first needs to mark a signal at the QPD field of view.
- ❑ There exists a window of possible beam start points centered around the QPD, determined by maximum incident beam angles and FSM displacement.
- ❑ Spiral search protocol with the FSM to locate the QPD by starting from any point in the window.



FSM Spiral Search

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- ❑ Spiral search protocol with the FSM to locate the QPD by starting from any point in the window.

OGS beacon, even with correct pointing, will remain outside QPD field of view. Spiral search is required to detect the beam.



FSM Spiral Search

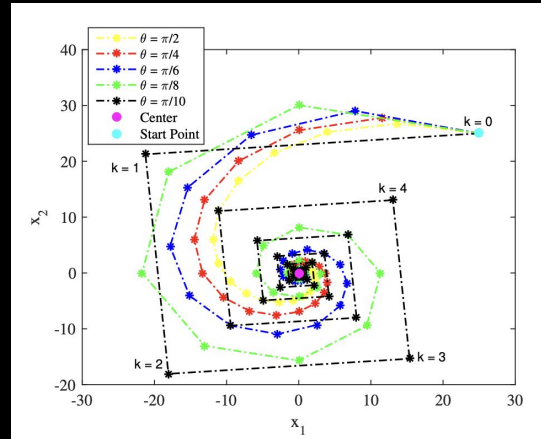
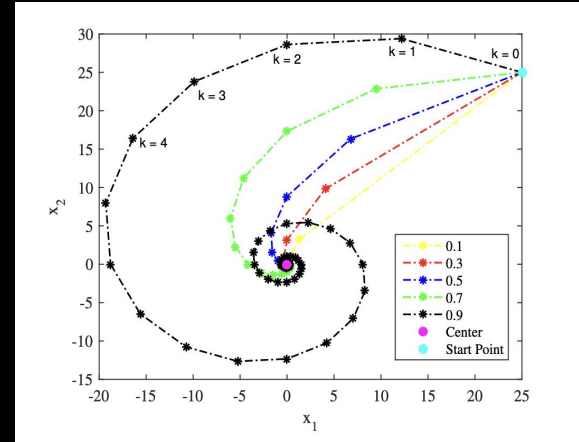
- Search protocol will use a logarithmic spiral with parameters r , θ .

$$x_{k+1} = rR^{(n)}(\theta)x_k - (rR^{(n)}(\theta) - I_n)x^*,$$

$$R^{(2)}(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

- Random disturbances will be injected into the spiral to avoid convergence without finding the QPD.

Spiral Trajectories for various $r \in [-1, 1]$ and $\theta = \pi/8$



Spiral Trajectories for various $\theta \in [-\pi/2, \pi/2]$ and $r = 0.85$

FSM Spiral Search

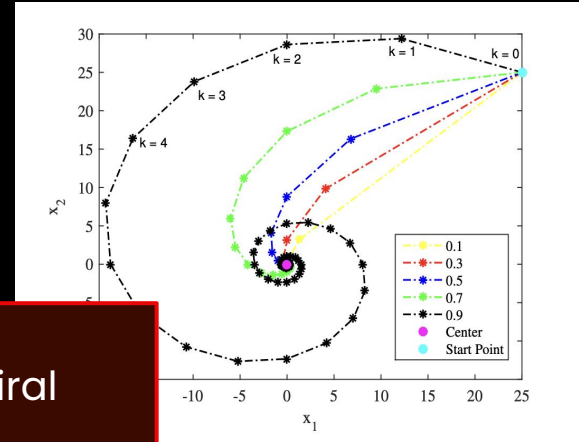
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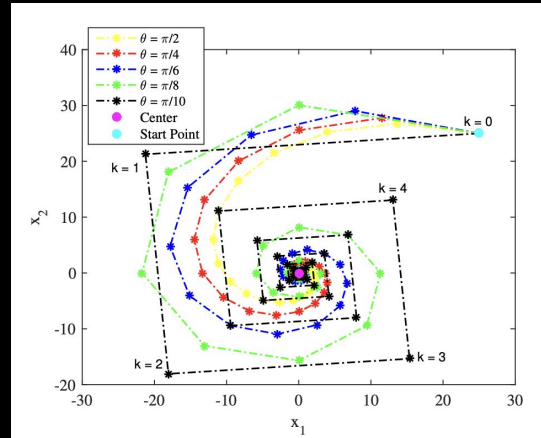
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Expected beam trajectory for spiral search.



Spiral Trajectories for various $\theta \in [-\pi/2, \pi/2]$ and $r = 0.85$

Sources of Errors in Fine Pointing to be Estimated

- ❑ Mechanical / Structural
 - ❑ Micro-vibrations from reaction wheels, magnetorquers, etc.
 - ❑ Jitter induced by structural resonances.
 - ❑ Launch-induced misalignments that shift optical components from their nominal positions.
 - ❑ Connector or mount looseness.
- ❑ Thermal
 - ❑ Thermal expansion / contraction of the optical bench or mounts → line-of-sight drift.
 - ❑ Thermal gradients across lenses or mirrors causing bending or surface deformation.
 - ❑ Refractive index changes in transmissive optics with temperature.
 - ❑ Athermalization errors when components don't expand uniformly.

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Mechanics and thermal gradients expect to be the most impactful ones.

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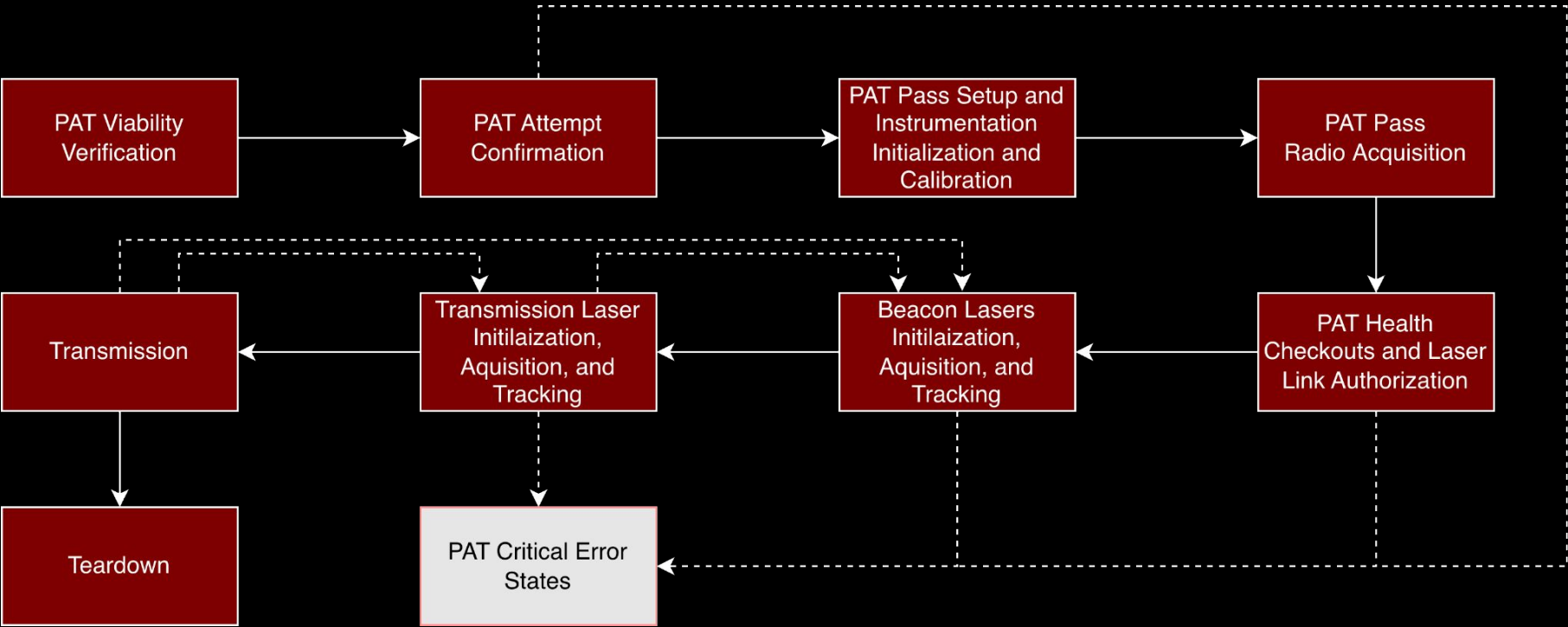
Uniform thermal expansion effects shown in Optical Analysis section.

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Pointing, Acquisition and Tracking: **Sequence Rundown**

PAT Sequence Overview



PAT Viability Verification & Attempt Confirmation

- ❑ Viability Verification
 - ❑ Verifying that overhead pass is “optically viable” previous to operations
 - ❑ Optically viable passes defined by overhead pass between 30 to 80 degrees from the horizon in favorable operating conditions
- ❑ Attempt is confirmed with satellite through RF beacon in pass prior to optically viable pass(es)

PAT Setup and Calibration

- ❑ Instruments calibrated and warmup commenced:
 - ❑ Payload laser warmup and current stabilization
 - ❑ FSM calibration on Payload
 - ❑ OGS RA/Dec calibration / plate solving with star camera
- ❑ Satellite and Telescope coarse pointing activated
 - ❑ Satellite body pointing and ground coordinates commences
 - ❑ Telescope pointed at predicted sunrise location 30 deg above horizon and tracking schedule generated

PAT Radio Acquisition and Health Checkouts

- ❑ Establish RF Contact between RFGS and Satellite
- ❑ Satellite collects and sends payload and bus health data
- ❑ Ground Station health data collected
- ❑ PAT Pass operator reviews health data
- ❑ PAT start command sent to Satellite upon successful health check
- ❑ Satellite receives PAT start command and switches telemetry to PAT packets

PAT Beacon Acquisition and Tracking

- ❑ OGS and Payload beacons activated
- ❑ OGS and Payload beacon acquisition timeout begins:
 - ❑ OGS performs FSM search for Payload Beacon
 - ❑ If Payload beacon acquired, OGS begins fine tracking
- ❑ Beacon acquisition timeout ends:
 - ❑ If OGS beacon not acquired, Satellite begins coarse pointing search
 - ❑ Underlying assumption that OGS telescope is pointing correctly
 - ❑ OGS beacon acquired and confirmation relayed to OGS
 - ❑ OGS beacon acquisition timeout restarts if Payload beacon not seen
- ❑ Closed feedback loops with FSMs begin on both Payload and OGS for beacon tracking

PAT Transmission & Reception

- ❑ Tracking established by OGS confirmation sent to Satellite by RFGS
- ❑ Payload turns on transmission laser
- ❑ Transmission laser acquisition at OGS
 - ❑ OGS performs FSM transmission search (spiral search)
 - ❑ If transmission detected at OGS detectors then acquisition confirmation sent to Satellite by RFGS
- ❑ Data transmission commences
 - ❑ UART transmission protocol
 - ❑ RFGS signal sent in case of transmission laser loss at OGS
- ❑ Satellite goes below 30 degrees from the horizon → Teardown begins

PAT Failure States

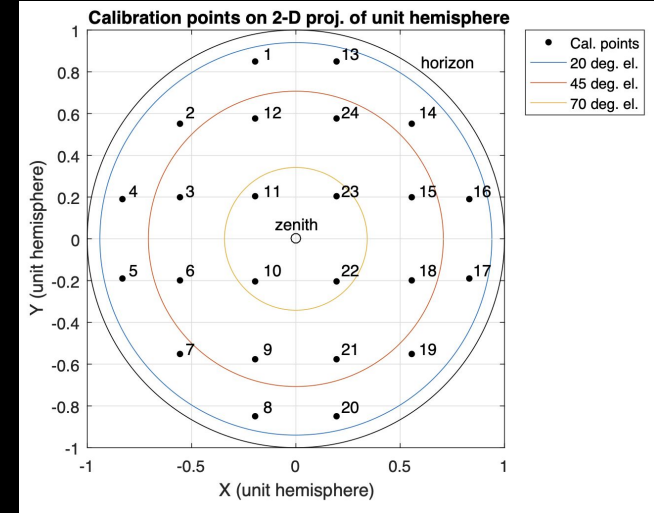
- ❑ RF contact is not acquired
- ❑ Satellite Health Poll Fails
- ❑ Ground Station Health Poll Fails
- ❑ Satellite Coarse Pointing Search Timeout Ends
- ❑ OGS FSM Search Timeout Ends
- ❑ OGS FSM Transmission Search Ends

Pointing, Acquisition and Tracking:
**Telescope Pointing &
Tracking**

Telescope Pointing Model

Before each pass, the OGS will use a star tracking camera (insert model) to take calibration images of the sky over a range of altitude and azimuth values that covers the pass to calibrate the internal pointing model of the CPC 100 telescope. The tracking software then runs during the pass by measuring altitude and azimuth from the telescope.

This pointing model will be tested by slewing to a collection of stars and imaging them without further centering, resulting in a pointing model error measurement for the telescope. This has not been implemented yet.



Calibration image locations from 2018 Reising MIT paper.

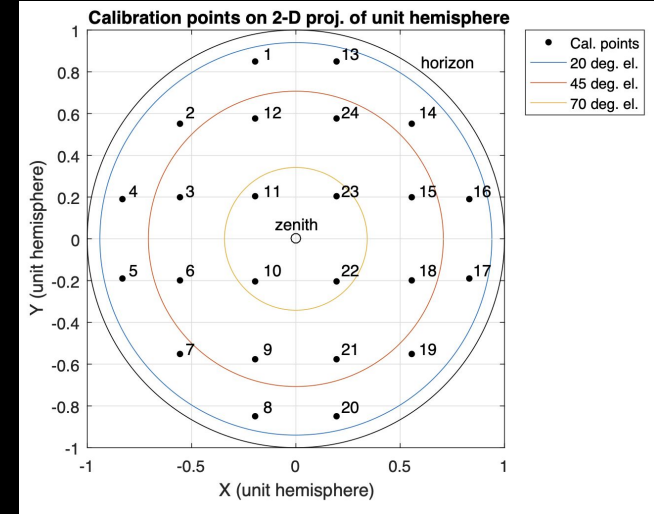
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This pointing model will be tested by slewing to a collection of stars and imaging them without further centering, resulting in a pointing model error measurement for the telescope.

This has not been implemented yet.

By calibrating using a start tracker on the same telescope, MIT recorded a pointing model error RMS of ~60 arcseconds in each coordinate.

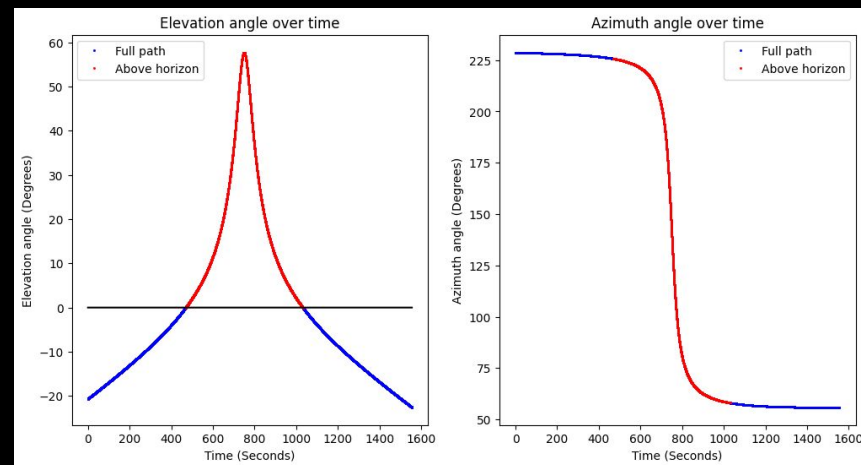


Calibration image locations from 2018 Reising MIT paper.

Tracking Schedule Creation and Coarse Tracking

By using the two-line-element(TLE) for the satellite, and the location of the OGS, the expected pointing angles for the OGS can be computed, which can then be used to create a tracking schedule for the OGS, which follows that schedule to point at the telescope blindly in what we call coarse tracking.

To ensure that the OGS points along the intended path during coarse tracking, the OGS measures its angular position through the telescope pointing model and corrects any errors through a PID loop.



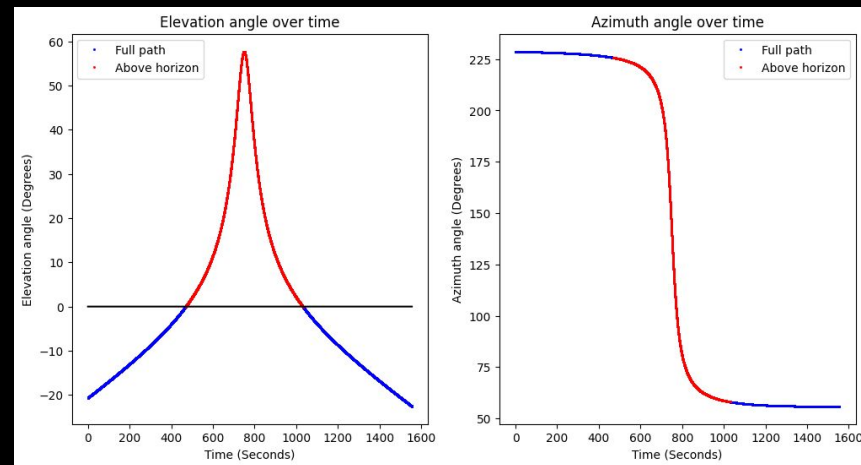
Expected pointing angles for ISS pass on 08/10/2025, with the OGS in Chicago, Illinois.

Tracking Schedule Creation and Coarse Tracking

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To ensure that the OGS points along the intended path during coarse tracking, the OGS measures its angular position through the telescope pointing model and corrects any errors through a PID loop.

Before the payload beacon is acquired on the tracking camera, the OGS points according to a tracking schedule based on the calculated path of the satellite.



Expected pointing angles for ISS pass on 08/10/2025, with the OGS in Chicago, Illinois.

Fine Tracking

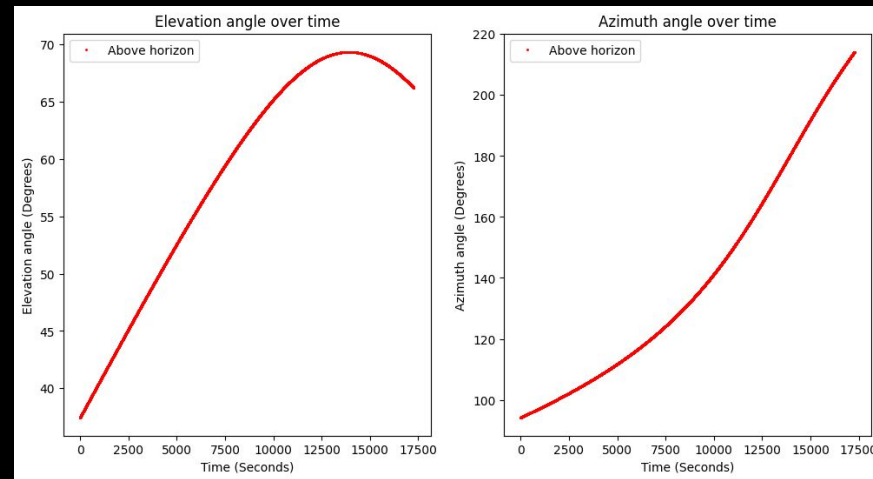
Once the tracking camera has successfully acquired the payload beacon, the telescope and fast steering mirror (FSM) both begin to actively track the satellite based on the tracking camera images.

Because of the difference in command time for the telescope and FSM (0.2S vs fsm_time S), the FSM corrects for any short time scale deviations from the center of the camera, while the telescope corrects for any long time scale errors that have been absorbed into the FSM's pointing angle. MIT has showed this to work in 2018 up to a measured angular error RMS of 5 arcseconds while tracking the ISS. This has not been implemented yet.

Pointing, Acquisition and Tracking: **Telescope Tracking Analysis**

Initial Test of Coarse Tracking

Since plate solving with a star camera has not been implemented, the telescope was calibrated only on Saturn's coordinates and it's internal gears were used to measure relative to the initial calibration. Using Astropy, a coarse tracking schedule was created for Saturn.

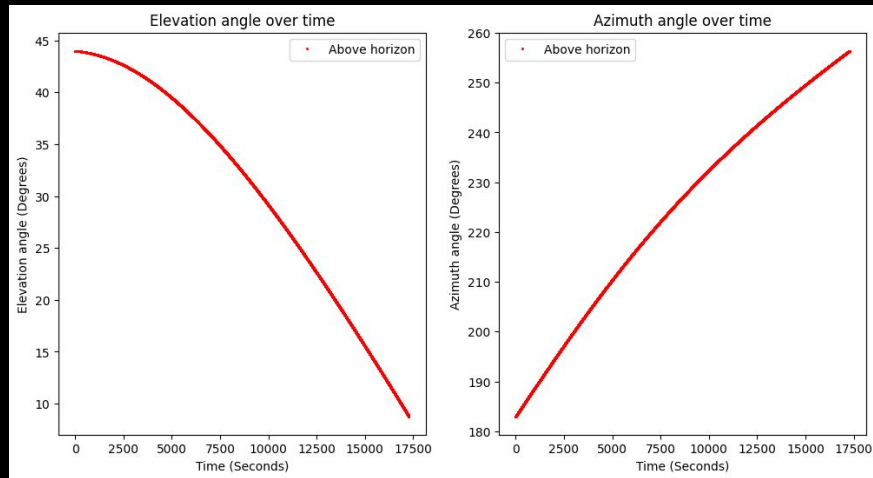


Expected pointing angles for Saturn on 11/16/2025, with the OGS in Chicago, Illinois.

Initial Test of Coarse Tracking

Since plate solving with a star camera has not been implemented, the telescope was calibrated only on a planet's (Jupiter, Saturn) coordinates and it's internal gears were used to measure relative to the initial calibration. Using Astropy, a coarse tracking schedule was created for the planet..

The effectiveness of the coarse tracking was measured by imagining a planet repeatedly using the tracking camera.



Expected pointing angles for Jupiter on 11/16/2025, with the OGS in Chicago, Illinois.

Initial Test of Coarse Tracking

Target Observed:	Jupiter
Date Observed:	November 16, 2025
Duration:	20 minutes

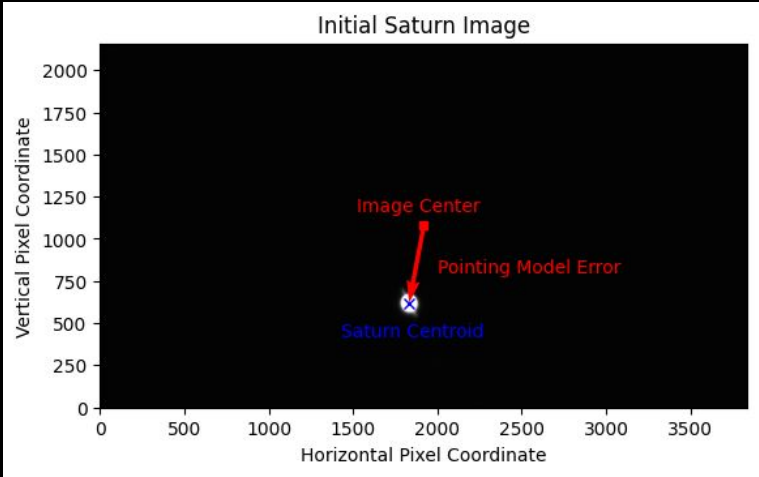
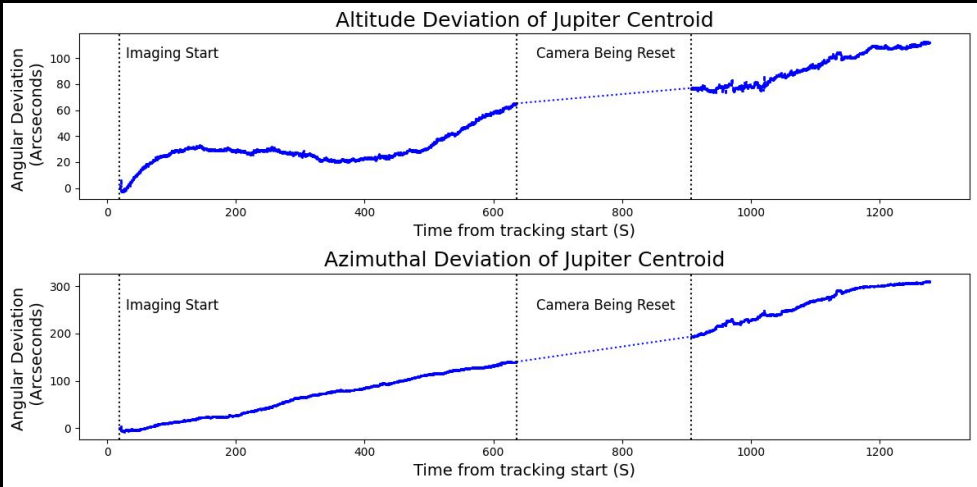


Image of Saturn taken at the start of tracking from separate test,, displaying the error in the simplified pointing model used for this test.

Initial Test of Coarse Tracking



A simplified version of coarse tracking has been tested on Saturn and accrued error at a rate of $0.08''/\text{s}$ in Altitude and $0.24''/\text{s}$ in Azimuth. This corresponds to 1.8% of the total angular speed of Jupiter over the pass. We expect this to improve significantly with the full pointing model.

Analyzing Coarse Tracking Precision

To analyze the tracking precision, the following steps were performed:

1. Derive the timestamps from each frame taken
2. Find the centroid of the planet in each frame, using a weighted average of it's pixel brightness
3. Set our expected coordinate for the planet's position as it's centroid in the first frame (beginning of tracking)
4. Compute residuals between expected centroid in first frame and measured centroid in following frames
5. Rotate coordinate axes from (x,y) on detector to (altitude, azimuth) from telescope field of view
6. Translate pixels to arcseconds with plate scale

Pointing, Acquisition and Tracking: **Sequence Simulations**

PAT Simulations

PAT Simulations are a collaboration between OGS Zemax simulations and flight software's 42 outputs. The purpose of the simulation is to estimate the locations of the beams at both the fiber collimators and the tracking camera as the source beam's angle of incidence varies over time.

To implement this, the Zemax programming interface, ZOS-API, is used in a MATLAB script that can alter components and run analysis methods in the OGS simulation.

The simulation runs with the following structure:

- ❑ **Replace the gaussian source parameters**
 - ❑ Angle of incidence and plane of incidence are translated to "Tilt X" and "Tilt Y"
- ❑ **Run a separate ray trace for each timestamp in the 42 data**
- ❑ **Reproduce the irradiance graphs at the fiber collimators \ tracking camera and output the centroid data**

PAT Simulations

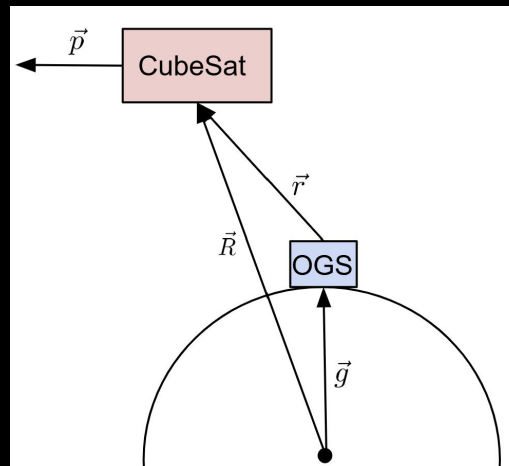
Calculations for the angles needed to change the input parameters of the source beam on Zemax:

α = Angle of incidence with respect to the optical axis

β = Angle of the plane of incidence with respect to the initial plane of incidence

$$\vec{p} \cdot \vec{r} = |\vec{p}| |\vec{r}| \cos(\alpha)$$

$$\vec{n}_0 \cdot \vec{n} = |\vec{n}_0| |\vec{n}| \cos(\beta)$$

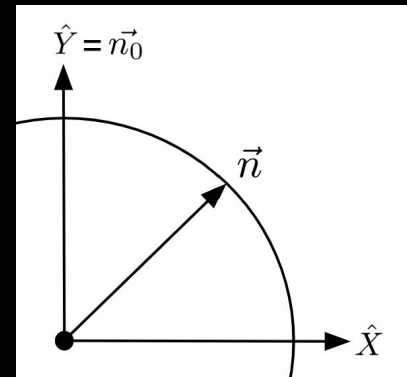


$\vec{p}$:The pointing vector for the satellite

$\vec{r}$:The pointing vector for the OGS

$\vec{R}$:The position of the satellite in respect to the center of the Earth

$\vec{g}$:The position of the ground station in respect to the center of the Earth



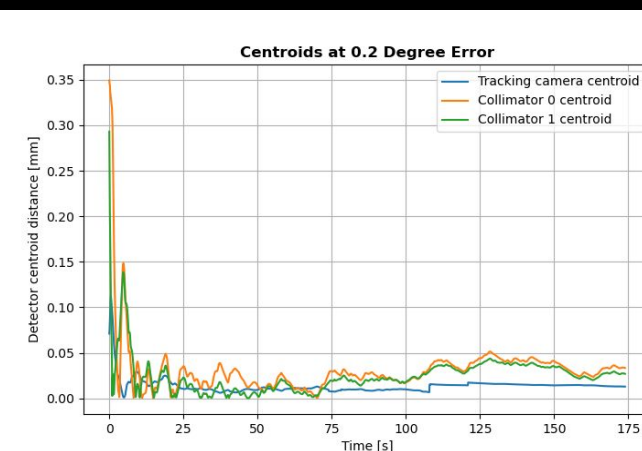
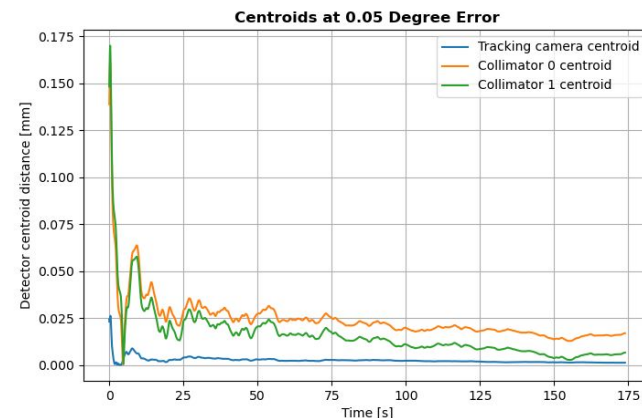
$\vec{n}$ = cross product of the pointing and OGS vectors

$\vec{n}_0$ = cross product of the original pointing and OGS vectors

PAT Simulations

Both figures depict the position shift at the centroids for both fiber collimators and the tracking camera.

- ❑ At 0.05 degrees, the centroid is clustered closer to the optical axis, whereas at 0.2 degrees, it exhibits larger deviations which is an expected outcome due to the increased error.
- ❑ The tracking camera's centroid remains relatively fixed near 0 degrees because the rays that split into the tracking camera travel a shorter optical path, causing minimal shift.

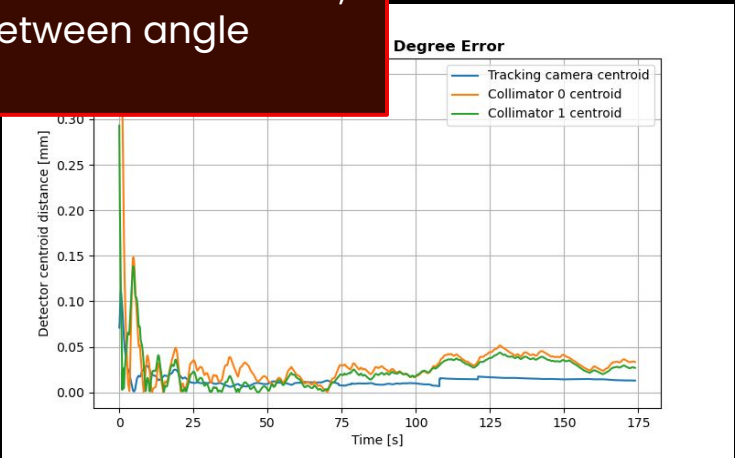
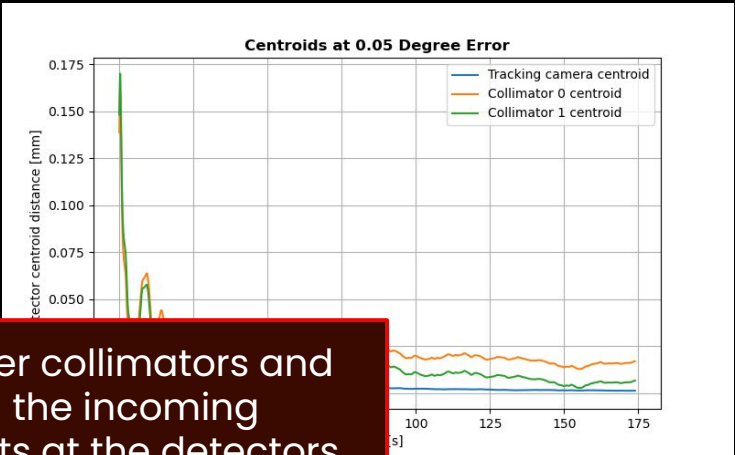


PAT Simulations

Both figures depict the position shift at the centroids for both fiber collimators and the tracking camera.

- ❑ At 0.05 degree error, the optical axis is shifted by a larger deviation due to the incoming beam angle translate directly into shifts at the detectors, validating the expected relationship between angle deviation and beam position.
- ❑ The tracking camera is fixed near 0 degrees, so as the beam angle deviates into the tracking camera travel a shorter optical path, causing minimal shift.

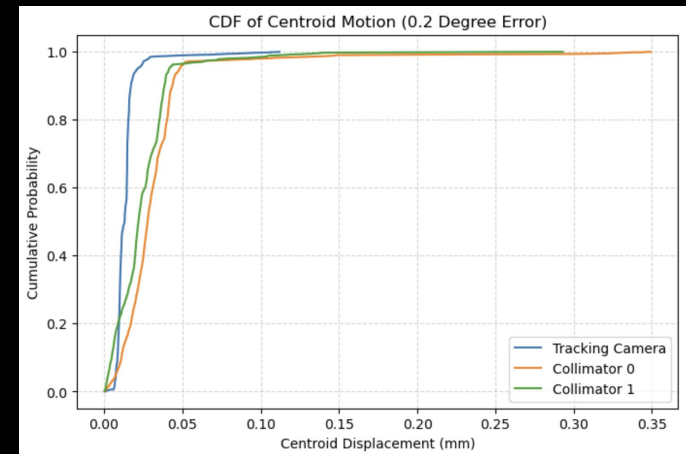
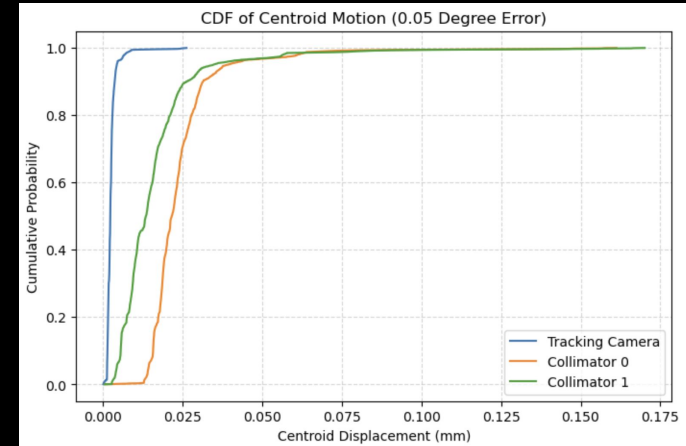
The centroid measurements at the fiber collimators and tracking camera confirm that errors in the incoming beam angle translate directly into shifts at the detectors, validating the expected relationship between angle deviation and beam position.



PAT Simulations

The CDF plots show the probability that the centroid deviation stays below a given distance from the optical center.

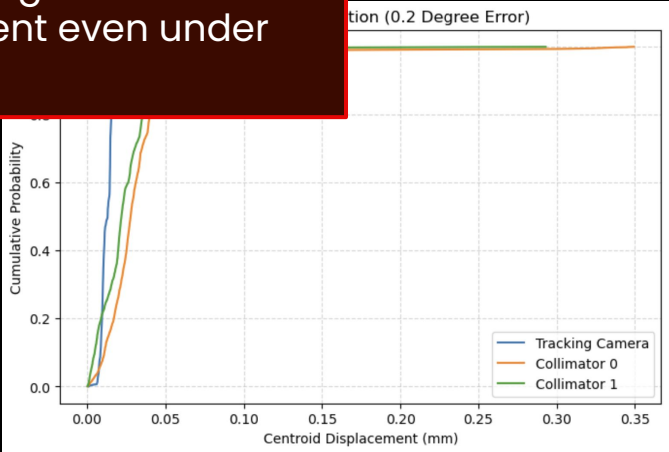
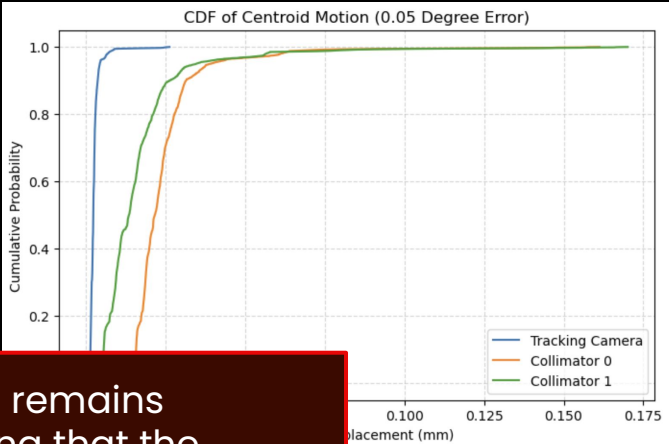
- ❑ At 0.05° error, nearly all centroid motion is contained within a very small radius. The tracking camera remains the most stable, with almost all values within a few microns.
- ❑ At 0.2° error, the distribution spreads to larger deviations in a predictable manner. Even with the increased error, more than 95% of all centroid positions remain within a small displacement from the center.



PAT Simulations

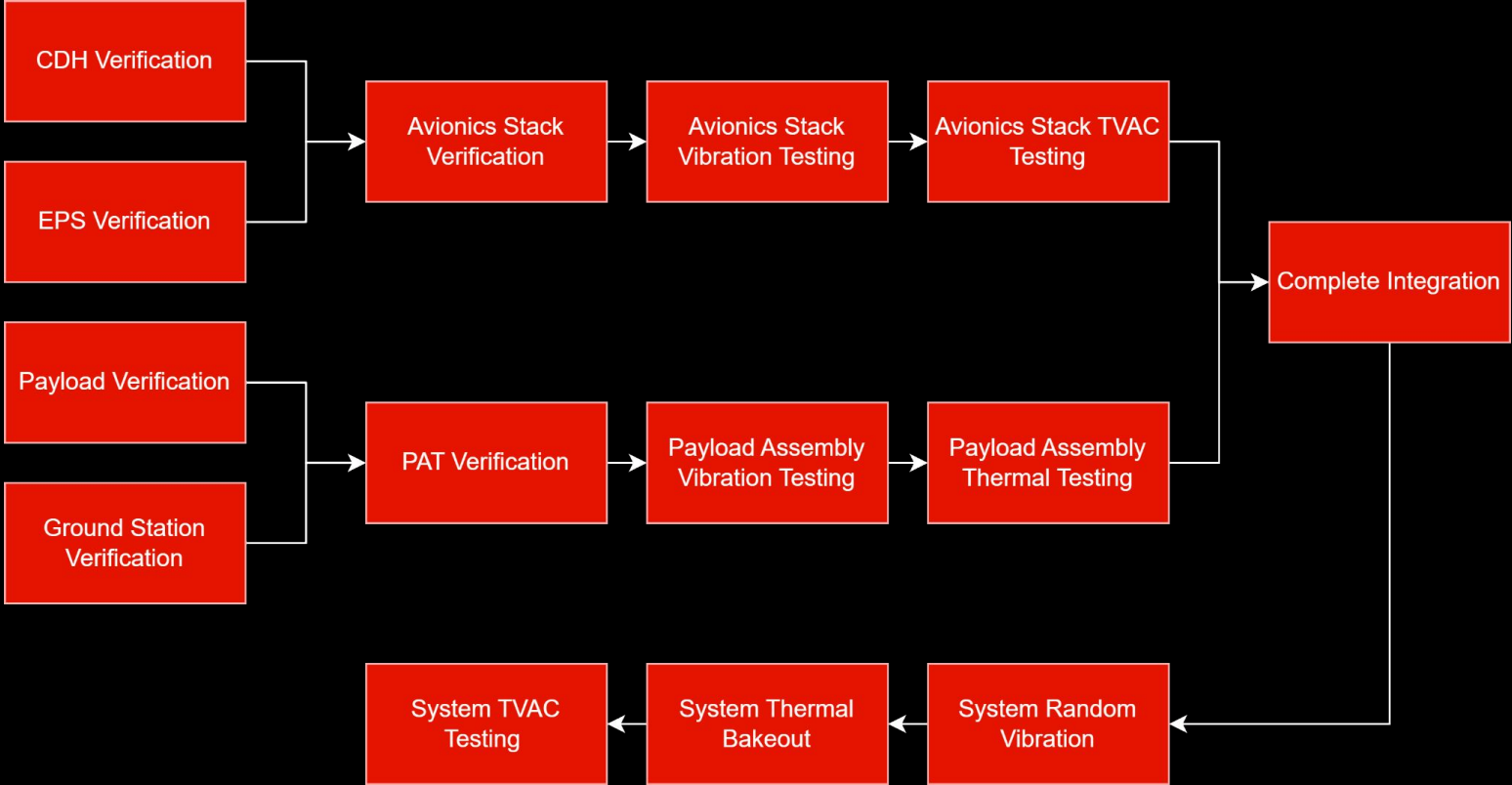
The CDF plots show the probability that the centroid deviation stays below a given distance from the optical center.

- At 0.05° error, the centroid motion remains contained and predictable, demonstrating that the optical system maintains stable alignment even under imperfect conditions.
- At 0.2° error, the centroid motion remains contained and predictable. Even with the increased error, more than 95% of all centroid positions remain within a small displacement from the center.



Assembly, Integration, and Test (AIT)

AIT Process Flow Diagram



Payload Verification Testing

#	Type of Test	Success Criteria
1	Optical Alignment & Path Verification	Beam remains within the expected misalignment range, throughput losses stay within the optical budget, and no element introduces unexpected angular deviation or clipping.
2	Laser Output & Power-Stability Verification	Laser outputs meet expected power levels and wavelength stability, EDFA gain is consistent with design, and no dropouts, spikes, or thermally induced instability appear during extended operation.
3	Optical Fiber & Coupling Efficiency Testing	Total fiber-path losses match design expectations and polarization alignment is preserved within the expected error range across all fiber interfaces.
4	Beacon Acquisition & SNR Verification	Quadcell responds cleanly to beacon signal with SNR above acquisition threshold, minimal false detections under noise, and centered focal-plane alignment that produces expected differential patterns.
5	Fine-Pointing Control (FSM) Verification	FSM responds smoothly across its tilt range, and shows no oscillation, saturation, or control instability at expected command rates.
6	Continuous-Operation & Thermal Drift Testing	No significant drift in FSM, beam alignment, optical power, or polarization state is observed over the full run. Thermal behavior remains predictable and within safe bounds, with no drop in optical performance.

Payload Environmental Testing – Vibration

GEVS 2.4.2.4: Random vibration is applied at the **payload assembly level**, since mechanical loading and optical alignment depend on the complete assembled structure, not individual components

Assembly: Fully assembled payload mounted to a flight-like payload mounting plate (same interface used inside bus).

Random Vibration Test Levels (GEVS Protoflight)

- ❑ **Frequency Range:** 20–2000 Hz
Test Level: Qualification level = flight limit + 3 dB
- ❑ **Axes:** X, Y, Z
- ❑ **Duration:** ~1 minute per axis
- ❑ **Equalization:** Vibration table kept within required tolerance (GEVS-compliant)

Payload Environmental Testing – Thermal Vacuum

TVAC Test Conditions for Assembly Testing (Derived from GEVS + Payload Thermal Model)

- **Vacuum Level:** $\leq 1 \times 10^{-5}$ torr
- **Cycles:** 4 full TVAC cycles
- **Ramp Rate:** ≤ 1 °C/min
- **Dwell Times:** 4 hours cold + 4 hours hot per cycle
- **All components integrated in flight-like configuration**
- **Temperature range:** 0 °C +/- 5 °C (cold plateau) to 60 °C +/- °C (hot plateau)
- **Powering Rules:**
 - Passive optics always unpowered
 - Electronics powered **only during final hot/cold dwells** (and only within vendor safe limits)
- **Component Level Testing:** Test temperatures must remain within component-specific operating envelopes and within payload thermal model ± 10 °C margins

OGS Verification Testing

#	Type of Test	Success Criteria
1	Payload Beacon Tracking	Simulate the apparent motion of a payload beacon through an ISS pass and run the closed-loop tracking algorithm that drives both the telescope and fast steering mirror Verify from recorded tracking-camera frames and control logs that the OGS maintains the beacon near the center of the field of view for the duration of the simulated pass without losing lock. Take images of stars with similar brightness (apparent magnitude) as beacon
2	Polarization Separation and Bit Detection	Use optical bench testing to pass controlled polarization states into the OGS confirm via APD outputs and comparator logic that the two channels cleanly separate the orthogonal polarization states as planned
3	Time-tagging and data rate verification	Operate the optical bench at a simulated 10 MHz signal and confirm that the incoming data is digitized at the required rates

Risk Assessment and Mitigation

Payload Risks

#	Parent Req	Risk	Effects
1	PAY-01	Optical path blocked or contaminated by particulates post-integration.	Optical throughput loss and link failure.
2	PAY-05	Beacon signal not detected due to atmosphere attenuation, scintillation, mispointing	Beacon acquisition error preventing fine pointing.
3	PAY-05	Beacon signal not detected due to scintillation.	Beacon acquisition failure preventing fine-pointing.
4	PAY-08	Beacon optical emission pattern beyond optical tolerances we can correct for due to spacecraft jitter.	Reduced illumination of OGS tracking system.
5	PAY-08	Beacon optical emission pattern beyond optical tolerances we can correct for due to unexpected thermal expansion of mounts and plates and lenses.	Reduced illumination of OGS tracking system.

Payload Risks

#	Parent Req	Risk	Effects
6	PAY-11	EDFA saturation or underdrive limits output range.	Fluctuating signal amplitude during downlink.
7	PAY-10	Beacon output power below expected value.	Beacon visibility reduced at OGS; acquisition difficulty.
8	PAY-13	Depolarization occurs in fiber path, as stress or bending in optical fiber changes polarization state.	Reduced polarization contrast and decoding accuracy.
9	PAY-15	MEMS mirror precision insufficient for fine control.	Beam jitter and link instability.
10	PAY-16	Laser overheats during extended operation.	Automatic shutdown and link interruption.
11	PAY-18	Optical components shift alignment after vibration loads.*	Optical path deviation reduces link efficiency.

*This specifically refers to the FSM, which we are not planning on vibration testing.

Optical Ground Station Risks

#	Parent Req	Risk	Effects
1	OGS-01	OGS may fail to decode the downlink transmission laser, due to a weak signal as a result of atmospheric effects (ie. cloud coverage).	Lost, corrupted, or incomplete data from the Satellite
2	OGS-08	OGS may fail to decode polarization states due to unpredictable effects of anisotropic atmospheric turbulence on polarization states.	Incorrect bit decoding and higher error rates.
3	OGS-16	The OGS may fail to detect the Payload beacon laser at a link margin of $3 \text{ dB} \leq \text{Margin} \leq 6 \text{ dB}$ due to weak beacon power or high atmospheric scattering.	Loss of tracking and pointing accuracy.
4	SAT-06	FAA does not permit the OGS to operate due to aerospace considerations.	Inability for OGS to function during period of restrictions.
5	OGS-11	Extreme weather conditions (ie. snow, heavy rain) prevent functionality of the OGS during specific passes.	Inability for OGS to function for specific passes during these conditions.
6	SAT-06	Availability to operate the OGS is limited due to student schedules.	Inability for OGS to operate during specific passes.

Conclusion

IDR Exit Criteria

- ❑ The current detailed design is expected to meet the requirements at an acceptable level of risk.
- ❑ The items remaining to be designed are being considered appropriately and the direction being chosen for their design is expected to meet the requirements at an acceptable level of risk.
- ❑ Definition of the technical interfaces is consistent with the overall technical maturity and provides an acceptable level of risk.
- ❑ Adequate technical interfaces are consistent with the overall technical maturity and provide an acceptable level of risk.
- ❑ Adequate technical margins exist with respect to TPMs.
- ❑ The project risks are understood and have been credibly assessed. A plan, a process, and resources exist to effectively manage them.
- ❑ The operational concept is technically sound, includes (where appropriate) human factors, and is derived from the flow down of requirements for its execution.

Team Members

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UCHICAGO SPACE PROGRAM

THANK YOU! INTERMEDIATE DESIGN REVIEW FLIGHT SYSTEMS

NOVEMBER 23, 2025

